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ACCA

Financial Management (FM)

Study Text



ACCA

Applied Skills

Financial Management (FM)

Study Text

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LINGUISTIC DIVERSITY, EQUALITY AND INCLUSION

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We will seek to devise simple measures that can be used by independent assessors to randomly check our success in the implementation of our Linguistic Equality, Diversity and Inclusion Policy.

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Contents

	Page
Chapter 1 The financial management function	1
Chapter 2 Basic investment appraisal techniques	35
Chapter 3 Investment appraisal – Discounted cash flow techniques	55
Chapter 4 Investment appraisal – Further aspects of discounted cash flows	93
Chapter 5 Asset investment decisions and capital rationing	123
Chapter 6 Investment appraisal under uncertainty	147
Chapter 7 Working capital management	179
Chapter 8 Working capital management – Inventory control	217
Chapter 9 Working capital management – Accounts receivable and payable	241
Chapter 10 Working capital management – Cash and funding strategies	279
Chapter 11 The economic environment for business	319
Chapter 12 Financial markets and the treasury function	345
Chapter 13 Foreign exchange risk	383
Chapter 14 Interest rate risk	425
Chapter 15 Sources of finance	447
Chapter 16 Dividend policy	493
Chapter 17 The cost of capital	505
Chapter 18 Capital structure	559
Chapter 19 Financial ratios	593
Chapter 20 Business valuations and market efficiency	621
Chapter 21 Employability and technology skills	659
Chapter 22 Questions and Answers	667
Index	I.1



Introduction

How to use the Materials

These Kaplan Publishing learning materials have been carefully designed to make your learning experience as easy as possible and to give you the best chances of success in your examinations.

The product range contains a number of features to help you in the study process. They include:

- (1) Detailed study guide and syllabus objectives
- (2) Description of the examination
- (3) Study skills and revision guidance
- (4) Study text
- (5) Question practice

The sections on the study guide, the syllabus objectives, the examination and study skills should all be read before you commence your studies. They are designed to familiarise you with the nature and content of the examination and give you tips on how to best approach your learning.

The **Study text** comprises the main learning materials and gives guidance as to the importance of topics and where other related resources can be found. Each chapter includes

- The **learning objectives**, which have been carefully mapped to the examining body's own syllabus learning objectives or outcomes. You should use these to check you have a clear understanding of all the topics on which you might be assessed in the examination.
- The **chapter diagram** provides a visual reference for the content in the chapter, giving an overview of the topics and how they link together.
- The **content** for each topic area commences with a brief explanation or definition to put the topic into context before covering the topic in detail. You should follow your studying of the content with a review of the illustration/s. These are worked examples, which will help you to understand better how to apply the content for the topic.
- **Test your understanding** sections provide an opportunity to assess your understanding of the key topics by applying what you have learned to short questions. Answers can be found at the back of each chapter.
- **Summary diagrams** complete each chapter to show the important links between topics and the overall content of the examination. These diagrams should be used to check that you have covered and understood the core topics before moving on.
- **Question practice** is provided at the back of each text.

Quality and accuracy are of the utmost importance to us so if you spot an error in any of our products, please send an email to mykaplanreporting@kaplan.com with full details, or follow the link to give feedback in MyKaplan.

Our Quality Coordinator will work with our technical team to verify the error and take action to ensure it is corrected in future editions.

Icon Explanations



Definition – Key definitions that you will need to learn from the core content.



Key point – Identifies topics that are key to success and are often examined.



Test your understanding – Exercises for you to complete to ensure that you have understood the topics just learned.



Illustration – Worked examples help you understand the core content better.



Tricky topic – When reviewing these areas care should be taken and all illustrations and Test your understanding exercises should be completed to ensure that the topic is understood.



Supplementary reading – These sections will help to provide a deeper understanding of core areas. The supplementary reading is NOT optional reading. It is vital to provide you with the breadth of knowledge you will need to address the wide range of topics within your syllabus that could feature in an exam question. **Reference to this text is vital when self-studying.**

On-line subscribers

Our on-line resources are designed to increase the flexibility of your learning materials and provide you with immediate feedback on how your studies are progressing.

If you are subscribed to our on-line resources you will find:

- (1) On-line reference ware: reproduces your Study Text on-line, giving you anytime, anywhere access.
- (2) On-line testing: provides you with additional on-line objective testing so you can practice what you have learned further.
- (3) On-line performance management: immediate access to your on-line testing results. Review your performance by key topics and chart your achievement through the course relative to your peer group.

Ask your local customer services staff if you are not already a subscriber and wish to join.

Syllabus Introduction

Syllabus background

The aim of ACCA FM, Financial management, is to develop the knowledge and skills expected of a financial manager, in relation to investment, financing, and dividend policy decisions.

Main capabilities

- A – Discuss the role and purpose of the financial management function.
- B – Assess and discuss the impact of the economic environment on financial management.
- C – Discuss and apply working capital management techniques.
- D – Carry out effective investment appraisal.
- E – Identify and evaluate alternative sources of business finance.
- F – Discuss and apply principles of business and asset valuations.
- G – Explain and apply risk management techniques in business.
- H – Apply employability and technology skills.

Core areas of the syllabus

- Financial management function (A).
- Financial management environment (B).
- Working capital management (C).
- Investment appraisal (D).
- Business finance (E).
- Business valuations (F).
- Risk management (G).
- Employability and technology skills (H).

ACCA performance objectives

All ACCA students are required to complete their exams, take the Ethics and Professional Skills module and complete the PER (Practical Experience Requirement) before they can become an ACCA member.

Performance objectives are benchmarks of effective performance that describe the types of work activities students and affiliates will be involved in as trainee accountants.

Trainees are required to achieve nine performance objectives in total – all five Essentials performance objectives and any four Technical performance objectives.

The financial management syllabus relates to four of the technical performance objectives:

P09 – Evaluate investment and financing decisions

P10 – Manage and control working capital

P11 – Identify and manage financial risk

P22 – Data analysis and decision support

You can find more information on PER on the ACCA website:

<https://www.accaglobal.com/uk/en/student/practical-experience-per.html>

Progression

There are two elements of progression that we can measure: first how quickly students move through individual topics within a subject; and second how quickly they move from one course to the next. We know that there is an optimum for both, but it can vary from subject to subject and from student to student. However, using data and our experience of student performance over many years, we can make some generalisations.

A fixed period of study set out at the start of a course with key milestones is important. This can be within a subject, for example 'I will finish this topic by 30 June', or for overall achievement, such as 'I want to be qualified by the end of next year'.

Your qualification is cumulative, as earlier papers provide a foundation for your subsequent studies, so do not allow there to be too big a gap between one subject and another. We know that exams encourage techniques that lead to some degree of short-term retention, the result being that you will simply forget much of what you have already learned unless it is refreshed (look up Ebbinghaus Forgetting Curve for more details on this). This makes it more difficult as you move from one subject to another: not only will you have to learn the new subject, you will also have to relearn all the underpinning knowledge as well. This is very inefficient and slows down your overall progression, which makes it more likely you may not succeed at all.

In addition, delaying your studies slows your path to qualification, which can have negative impacts on your career, postponing the opportunity to apply for higher level positions and therefore higher pay.

You can use the following diagram showing the whole structure of your qualification to help you keep track of your progress.



Syllabus objectives

We have reproduced the ACCA's study guide below, showing where the objectives are explored within this book.

Within the chapters, we have broken down the extensive information found in the study guide into easily digestible and relevant sections, called Content Objectives. These correspond to the objectives at the beginning of each chapter.

Syllabus learning objective	Chapter reference
A FINANCIAL MANAGEMENT FUNCTION	
1 The nature and purpose of financial management	
(a) Explain the nature and purpose of financial management. ^[1]	1
(b) Explain the relationship between financial management and financial and management accounting. ^[1]	1
2 Financial objectives and the relationship with corporate strategy	
(a) Discuss the relationship between financial objectives, corporate objectives and corporate strategy. ^[2]	1
(b) Identify and describe a variety of financial objectives, including: ^[2]	1
(i) shareholder wealth maximisation	
(ii) profit maximisation	
(iii) earnings per share growth.	
3 Stakeholders and impact on corporate objectives	
(a) Identify the range of stakeholders and their objectives. ^[2]	1
(b) Discuss the possible conflict between stakeholder objectives. ^[2]	1
(c) Discuss the role of management in meeting stakeholder objectives, including the application of agency theory. ^[2]	1
(d) Describe and apply ways of measuring achievement of corporate objectives including: ^[2]	19
(i) ratio analysis using appropriate ratios such as return on capital employed, return on equity, earnings per share and dividend per share	
(ii) changes in dividends and share prices as part of total shareholder return.	

Syllabus learning objective

Chapter reference

(e) Explain ways to encourage the achievement of stakeholder objectives, including: ^[2]		
(i) managerial reward schemes such as share options and performance-related pay		1
(ii) regulatory requirements such as corporate governance codes of best practice and stock exchange listing regulations		
4	Financial and other objectives in not-for-profit organisations	
(a)	Discuss the impact of not-for-profit status on financial and other objectives. ^[2]	1
(b)	Discuss the nature and importance of Value for Money as an objective in not-for-profit organisations. ^[2]	1
(c)	Discuss ways of measuring the achievement of objectives in not-for-profit organisations. ^[2]	1
B	FINANCIAL MANAGEMENT ENVIRONMENT	
1	The economic environment for business	
(a)	Identify and explain the main macroeconomic policy targets. ^[1]	11
(b)	Define and discuss the role of fiscal, monetary, interest rate and exchange rate policies in achieving macroeconomic policy targets. ^[1]	11
(c)	Explain how government economic policy interacts with planning and decision-making in business. ^[2]	11
(d)	Explain the need for, and the interaction with, planning and decision-making in business of:	11
(i)	competition policy ^[1]	
(ii)	government assistance for business ^[1]	
(iii)	green policies and sustainability issues ^[1]	
(iv)	corporate governance regulation. ^[2]	

Syllabus learning objective

Chapter reference

2 Management of inventories, accounts receivable, accounts payable and cash

- (a) Explain the cash operating cycle and the role of accounts payable and accounts receivable.^[2] 7
- (b) Explain and apply relevant accounting ratios, including: ^[2] 7
 - (i) current ratio and quick ratio
 - (ii) inventory turnover ratio, average collection period and average payable period
 - (iii) sales revenue/net working capital ratio.
- (c) Discuss, apply and evaluate the use of relevant techniques in managing inventory, including the Economic Order Quantity model and Just-in-Time techniques.^[2] 8
- (d) Discuss, apply and evaluate the use of relevant techniques in managing accounts receivable, including: 9
 - (i) assessing creditworthiness^[1]
 - (ii) managing accounts receivable^[1]
 - (iii) collecting amounts owing^[1]
 - (iv) offering early settlement discounts^[2]
 - (v) using factoring and invoice discounting^[2]
 - (vi) managing foreign accounts receivable.^[2]
- (e) Discuss and apply the use of relevant techniques in managing accounts payable, including: 9
 - (i) using trade credit effectively^[1]
 - (ii) evaluating the benefits of early settlement and bulk purchase discounts^[2]
 - (iii) managing foreign accounts payable.^[1]
- (f) Explain the various reasons for holding cash, and discuss and apply the use of relevant techniques in managing cash, including: ^[2] 10
 - (i) preparing cash flow forecasts to determine future cash flows and cash balances
 - (ii) assessing the benefits of centralised treasury management and cash control
 - (iii) cash management models, such as the Baumol model and the Miller-Orr model
 - (iv) investing short-term

Syllabus learning objective	Chapter reference
3 Determining working capital needs and funding strategies	
(a) Calculate the level of working capital investment in current assets and discuss the key factors determining this level, including: ^[2] (i) the length of the working capital cycle and terms of trade (ii) an organisation's policy on the level of investment in current assets (iii) the industry in which the organisation operates.	7
(b) Evaluate and discuss the key factors in determining working capital funding strategies, including: (i) the distinction between permanent and fluctuating current assets.^[2] (ii) the relative cost and risk of short-term and long-term finance.^[2] (iii) the matching principle.^[2] (iv) the relative costs and benefits of aggressive, conservative and matching funding policies.^[2] (v) management attitudes to risk, previous funding decisions and organisation size: ^[1]	10
D INVESTMENT APPRAISAL	
1 Investment appraisal techniques	
(a) Identify and calculate relevant cash flows for investment projects. ^[2]	2
(b) Calculate payback period and discuss its usefulness as an investment appraisal method. ^[2]	2
(c) Calculate discounted payback and discuss its usefulness as an investment appraisal method. ^[2]	6
(d) Calculate return on capital employed (accounting rate of return) and discuss its usefulness as an investment appraisal method. ^[2]	2
(e) Calculate net present value and discuss its usefulness as an investment appraisal method. ^[2]	3
(f) Calculate internal rate of return and discuss its usefulness as an investment appraisal method. ^[2]	3
(g) Discuss the superiority of discounted cash flow (DCF) methods over non-DCF methods. ^[2]	3
(h) Discuss the relative merits of NPV and IRR. ^[2]	3

Syllabus learning objective	Chapter reference
2 Allowing for inflation and taxation in DCF	
(a) Apply and discuss the real-terms and nominal-terms approaches to investment appraisal. ^[2]	4
(b) Calculate the taxation effects of relevant cash flows, including the tax benefits of tax-allowable depreciation and the tax liabilities of taxable profit. ^[2]	4
(c) Calculate and apply before- and after-tax discount rates. ^[2]	17
3 Adjusting for risk and uncertainty in investment appraisal	
(a) Describe and discuss the difference between risk and uncertainty in relation to probabilities and increasing project life. ^[2]	6
(b) Apply sensitivity analysis to investment projects and discuss the usefulness of sensitivity analysis in assisting investment decisions. ^[2]	6
(c) Apply probability analysis to investment projects and discuss the usefulness of probability analysis in assisting investment decisions. ^[2]	6
(d) Apply and discuss other techniques of adjusting for risk and uncertainty in investment appraisal, including	6
(i) simulation ^[1]	
(ii) adjusted payback ^[1]	
(iii) risk-adjusted discount rates. ^[2]	
4 Specific investment decisions (Lease or buy, asset replacement, capital rationing)	
(a) Evaluate leasing and borrowing to buy using the before- and after-tax costs of debt. ^[2]	5
(b) Evaluate asset replacement decisions using equivalent annual cost and equivalent annual benefit. ^[2]	5
(c) Evaluate investment decisions under single-period capital rationing, including: ^[2]	5
(i) the calculation of profitability indexes for divisible investment projects	
(ii) the calculation of the NPV of combinations of non-divisible investment projects	
(iii) a discussion of the reasons for capital rationing.	

Syllabus learning objective

Chapter reference

E BUSINESS FINANCE

1 Sources of, and raising, business finance

- | | |
|--|----|
| (a) Identify and discuss the range of short-term sources of finance available to businesses, including: ^[2] | 15 |
| (i) overdraft | |
| (ii) short-term loan | |
| (iii) trade credit | |
| (iv) lease finance. | |
| (b) Identify and discuss the range of long-term sources of finance available to businesses, including: ^[2] | 15 |
| (i) equity finance | |
| (ii) debt finance | |
| (iii) lease finance | |
| (iv) venture capital. | |
| (c) Identify and discuss methods of raising equity finance, including: ^[2] | 15 |
| (i) rights issue | |
| (ii) placing | |
| (iii) public offer | |
| (iv) stock exchange listing. | |
| (d) Identify and discuss methods of raising short- and long-term Islamic finance including: ^[1] | 15 |
| (i) major difference between Islamic finance and other forms of business finance | |
| (ii) The concept of riba (interest) and how returns are made by Islamic financial securities | |
| (iii) Islamic financial instruments available to businesses including: | |
| (i) murabaha (trade credit) | |
| (ii) Ijara (lease finance) | |
| (iii) mudaraba (equity finance) | |
| (iv) sukuk (debt finance) | |
| (v) musharaka (venture capital) | |
| (note calculations are not required) | |

Syllabus learning objective**Chapter
reference**

(e)	Identify and discuss internal sources of finance, including: ^[2]	15
(i)	retained earnings	15
(ii)	increasing working capital management efficiency.	15
(iii)	the relationship between dividend policy and the financing decision	16
(iv)	the theoretical approaches to, and the practical influences on, the dividend decision, including legal constraints, liquidity, shareholder expectations and alternatives to cash dividends	16
2	Estimating the cost of capital	
(a)	Estimate the cost of equity, including: ^[2]	17
(i)	Application of the dividend growth model, its assumptions, advantages and disadvantages	
(ii)	Explanation and discussion of systematic and unsystematic risk	
(iii)	Relationship between portfolio theory and the capital asset pricing model (CAPM)	
(iv)	Application of the CAPM, its assumptions, advantages and disadvantages.	
(b)	Estimating the cost of debt: ^[2]	17
(i)	irredeemable debt	
(ii)	redeemable debt	
(iii)	convertible debt	
(iv)	preference shares	
(v)	bank debt.	
(c)	Estimating the overall cost of capital, including: ^[2]	17
(i)	Distinguishing between average and marginal cost of capital	
(ii)	Calculating the weighted average cost of capital (WACC) using book value and market value weightings.	

Syllabus learning objective	Chapter reference
3 Sources of finance and their relative costs	
(a) Describe the relative risk-return relationship and the relative costs of equity and debt. ^[2]	17
(b) Describe the creditor hierarchy and its connection with the relative costs of sources of finance. ^[2]	17
(c) Identify and discuss the problem of high levels of gearing. ^[2]	18
(d) Assess the impact of sources of finance on financial position, financial risk and shareholder wealth using appropriate measures, including: ^[2]	19
(i) ratio analysis using statement of financial position gearing, operational and financial gearing, interest coverage ratio and other relevant ratios	
(ii) cash flow forecasting.	
(iii) leasing or borrowing to buy.	
(e) Impact of cost of capital on investments, including: ^[2]	17
(i) the relationship between company value and cost of capital	
(ii) the circumstances under which WACC can be used in investment appraisal	
(iii) the advantages of the CAPM over WACC in determining a project-specific cost of capital	
(iv) the application of CAPM in calculating a project-specific discount rate.	
4 Capital structure theories and practical considerations	
(a) Describe the traditional view of capital structure and its assumptions. ^[2]	18
(b) Describe the views of Miller and Modigliani on capital structure, both without and with corporate taxation, and their assumptions. ^[2]	18
(c) Identify a range of capital market imperfections and describe their impact on the views of Miller and Modigliani on capital structure. ^[2]	18
(d) Explain the relevance of pecking order theory to the selection of sources of finance. ^[1]	

Syllabus learning objective		Chapter reference
5	Finance for small and medium-sized entities (SMEs)	
(a)	Describe the financing needs of small businesses. ^[2]	15
(b)	Describe the nature of the financing problem for small businesses in terms of the funding gap, the maturity gap and inadequate security. ^[2]	15
(c)	Explain measures that may be taken to ease the financing problems of SMEs, including the responses of government departments and financial institutions. ^[1]	15
(d)	Identify and evaluate the financial impact of sources of finance for SMEs, including sources already referred to in syllabus section E1 and also: ^[2]	15
	(i) Business angel financing	
	(ii) Government assistance	
	(iii) Supply chain financing	
	(iv) Crowdfunding/peer-to-peer funding	
F	BUSINESS VALUATIONS	
1	Nature and purpose of the valuation of business and financial assets	
(a)	Identify and discuss reasons for valuing businesses and financial assets. ^[2]	20
(b)	Identify information requirements for valuation and discuss the limitations of different types of information. ^[2]	20
2	Models for the valuation of shares	
(a)	Discuss and apply asset-based valuation models, including: ^[2]	20
	(i) net book value (statement of financial position) basis.	
	(ii) net realisable value basis.	
	(iii) net replacement cost basis.	
(b)	Discuss and apply income-based valuation models, including: ^[2]	20
	(i) price/earnings ratio method	
	(ii) earnings yield method	
(c)	Discuss and apply cash flow-based valuation models, including: ^[2]	20
	(i) dividend valuation model and the dividend growth model	
	(ii) discounted cash flow basis.	

Syllabus learning objective	Chapter reference
2 Causes of exchange rate differences and interest rate fluctuations	
(a) Describe the causes of exchange rate fluctuations, including:	13
(i) balance of payments ^[1]	
(ii) purchasing power parity theory ^[2]	
(iii) interest rate parity theory ^[2]	
(iv) four-way equivalence ^[2]	
(b) Forecast exchange rates using: ^[2]	13
(i) purchasing power parity	
(ii) interest rate parity.	
(c) Describe the causes of interest rate fluctuations, including: ^[2]	14
(i) structure of interest rates and yield curves	
(ii) expectations theory	
(iii) liquidity preference theory	
(iv) market segmentation.	
3 Hedging techniques for foreign currency risk	
(a) Discuss and apply traditional and basic methods of foreign currency risk management, including:	13
(i) currency of invoice ^[1]	
(ii) netting and matching ^[2]	
(iii) leading and lagging ^[2]	
(iv) forward exchange contracts ^[2]	
(v) money market hedging ^[2]	
(vi) asset and liability management. ^[1]	
(b) Compare and evaluate traditional methods of foreign currency risk management. ^[2]	13
(c) Identify the main types of foreign currency derivatives used to hedge foreign currency risk and explain how they are used in hedging. ^[1]	
(No numerical questions will be set on this topic)	13

Syllabus learning objective	Chapter reference
4 Hedging techniques for interest rate risk (a) Discuss and apply traditional and basic methods of interest rate risk management, including: (i) matching and smoothing ^[1] (ii) asset and liability management ^[1] (iii) forward rate agreements ^[2] (b) Identify the main types of interest rate derivatives used to hedge interest rate risk and explain how they are used in hedging.^[1] (No numerical questions will be set on this topic)	14 14
H EMPLOYABILITY AND TECHNOLOGY SKILLS	
1 Use computer technology to efficiently access and manipulate relevant information	
2 Work on relevant response options, using available functions and technology, as would be required in the workplace	
3 Navigate windows and computer screens to create and amend responses to exam requirements, using the appropriate tools	
4 Present data and information effectively, using the appropriate tools	

The superscript numbers in square brackets indicate the intellectual depth at which the subject area could be assessed within the examination. Level 1 (knowledge and comprehension) broadly equates with the Knowledge module, Level 2 (application and analysis) with the Skills module and Level 3 (synthesis and evaluation) to the Professional level. However, lower level skills can continue to be assessed as you progress through each module and level.

The examination

Examination format

The syllabus is assessed by a three-hour computer based examination.

All questions are compulsory. The exam will contain both computational and discursive elements.

Some questions will adopt a scenario/case study approach.

Prior to the start of the exam candidates are given an extra 10 minutes to read the exam instructions.

Candidates are provided with a formulae sheet and tables of discount and annuity factors.

Examination tips

Read the examination questions carefully.

- **Divide the time** you spend on questions in proportion to the marks on offer.
- One suggestion **for this examination** is to allocate 1.8 minutes to each mark available (180 minutes/100 marks), so a 20 mark question should be completed in approximately 36 minutes.

FM is divided into three different sections, requiring the application of different skills to be successful.

Section A

This section comprises 15 objective testing (OT) questions worth 2 marks each.

Stick to the timing principle of 1.8 minutes per mark. This means that the 15 OT questions in section A (30 marks) should take 54 minutes.

Work steadily. Rushing leads to careless mistakes and the OT questions are designed to include answers which result from careless mistakes.

If you don't know the answer, eliminate those options you know are incorrect and see if the answer becomes more obvious.

Remember that there is no negative marking for an incorrect answer. After you have eliminated the options that you know to be wrong, if you are still unsure, guess.

Practice section A style questions can be found at the end of each chapter.

OT questions in sections A of the CBEs will be of varying styles. These styles include multiple choice, number entry, pull down list, multiple response, hot area, and drag and drop.

The questions in section A can cover any areas of the syllabus.

Section B

This section comprises 3 case study type questions. Each question will contain a scenario and 5 OT questions worth 2 marks each.

You will need to read the scenario information for each case before attempting the questions related to it.

Each OT question is worth two marks. Therefore, you have 18 minutes (1.8 minutes per mark) to answer the five OT questions relating to each case. It is likely that all of the cases will take the same length of time to answer, although some of the OT questions within a case may be quicker than other OT questions within that same case.

Once you have read through the information, you should first answer any of the OT questions that do not require workings and can be quickly answered. You should then attempt the OT questions that require workings utilising the remaining time for that case.

Practice section B style questions can be found in Chapter 21.

All of the tips for section A are equally applicable to each section B question.

OT questions in sections B of the CBEs will be of varying styles. These styles include multiple choice, number entry, pull down list, multiple response, hot area, and drag and drop.

The questions in section B can cover any areas of the syllabus.

Section C

This section comprises 2 constructed response (CR) questions, each worth 20 marks in total and usually made up of a number of different parts.

The CR questions in section C will require a written response made up of a combination of computational elements and discursive elements.

Therefore, different techniques need to be used to score well.

For computational style questions, you will be provided with a spreadsheet answer space. For discursive style questions, you will be provided with a word processing style answer space.

Unless you know exactly how to answer the question, spend some time planning your answer. Stick to the question and tailor your answer to what you are asked. Pay particular attention to the verbs in the question e.g. 'Calculate', 'State', 'Explain'.

If you get **completely stuck** with a question, flag the question and return to it later.

If you do not understand what a question is asking, state your assumptions. Even if you do not answer in precisely the way the examining team hoped, you should be given some credit, provided that your assumptions are reasonable.

You should do everything you can to make things easy for the marker. The marker will find it easier to identify the points you have made if your answers are legible and labelled.

Computations: It is essential to include all your workings in your answers. Many computational questions require the use of a standard format. Be sure you know these formats thoroughly before the examination and use the layouts that you see in the answers given in this book and in model answers.

Adopt a logical approach and cross reference workings to the main computation to keep your answers tidy.

Practice section C style questions can be found in Chapter 21.

The questions in section C will mainly come from the working capital management, investment appraisal and business finance areas of the syllabus (although other areas may be texted here for a few marks).

All sections

Don't skip parts of the syllabus. The FM exam has 32 different questions so the examination can cover a very broad selection of the syllabus each sitting.

Spend time learning the rules and definitions.

There are many formulae that are not provided in the formulae sheets available during the exam. Ensure that you learn all necessary formulae.

Practice plenty of questions to improve your ability to apply the techniques and perform the calculations.

Spend the last five minutes reading through your answers and making any additions or corrections.

Method of Examination

Computer-based examinations (CBEs) are the method of examination for the ACCA Applied Skills Level examinations.

If you would like further information on sitting a CBE FM examination, please contact either Kaplan, or the ACCA.

Computer-based examination (CBE) – Tips

Be sure you understand how to use the software before you start the exam. If in doubt, ask the assessment centre staff to explain it to you.

Questions are **displayed on the screen** and answers are entered using keyboard and mouse.

The CBE exam will not only examine multiple choice questions but could include questions that require data entry or a multiple response.

Do not attempt a CBE until you have **completed all study material** relating to it. **Do not skip any of the material** in the syllabus.

Read each question very carefully.

Double-check your answer before committing yourself to it.

Answer every question – if you do not know an answer, you don't lose anything by guessing. Think carefully before you **guess**.

You will be able to flag questions that you would like to come back to later. This is a useful function for questions where you are unsure of the answer or that you feel will take a long time to answer.

The CBE question types are as follows:

- Multiple choice – where you are required to choose one answer from a list of options provided by clicking on the appropriate 'radio button'
- Multiple response – where you are required to select more than one response from the options provided by clicking on the appropriate tick boxes (typically choose two or three options from the available list)
- Fill in the blank – where you are required to type an answer into a box (which may be numerical or text)
- Drag and drop – where you are required to drag an answer and drop it into the correct place. There may be more than one answer to drag and potentially more answer choices than response areas
- Drop down list – where you are required to select an answer from a drop down list, of which there may be more than one
- Hot spot – where you are required to select a point on an image as your answer.
- Hot area – where you are required to select one or more areas in an image.

With an objective test question, it may be possible to eliminate first those answers that you know are wrong. Then choose the most appropriate answer(s) as required from those that are left. This could be a single answer (e.g. multiple choice) or more than one response (e.g. multiple response and multiple response – matching).

After you have eliminated the ones that you know to be wrong, if you are still unsure, guess. But only do so after you have double-checked that you have only eliminated answers that are definitely wrong.

Don't panic if you realise you've answered a question incorrectly. Try to remain calm, continue to apply examination technique and answer all questions required within the time available.

ACCA support

For additional support with your studies, please also refer to the ACCA Global website.

Study skills and revision guidance

This section aims to give guidance on how to study for your ACCA exams and to give ideas on how to improve your existing study techniques.

Preparing to study

Set your objectives

Before starting to study decide what you want to achieve – the type of pass you wish to obtain. This will decide the level of commitment and time you need to dedicate to your studies.

Devise a study plan

Determine which times of the week you will study.

Split these times into sessions of at least one hour for study of new material. Any shorter periods could be used for revision or practice.

Put the times you plan to study onto a study plan for the weeks from now until the exam and set yourself targets for each period of study – in your sessions make sure you cover the course, course assignments and revision.

If you are studying for more than one examination at a time, try to vary your subjects as this can help you to keep interested and see subjects as part of wider knowledge.

When working through your course, compare your progress with your plan and, if necessary, re-plan your work (perhaps including extra sessions) or, if you are ahead, do some extra revision/practice questions.

Effective studying

Active reading

You are not expected to learn the text by rote, rather, you must understand what you are reading and be able to use it to pass the exam and develop good practice. A good technique to use is SQ3Rs – Survey, Question, Read, Recall, Review:

- (1) **Survey** the chapter – look at the headings and read the introduction, summary and objectives, so as to get an overview of what the chapter deals with.
- (2) **Question** – whilst undertaking the survey, ask yourself the questions that you hope the chapter will answer for you.
- (3) **Read** through the chapter thoroughly, answering the questions and making sure you can meet the objectives. Attempt the exercises and activities in the text, and work through all the examples.
- (4) **Recall** – at the end of each section and at the end of the chapter, try to recall the main ideas of the section/chapter without referring to the text. This is best done after a short break of a couple of minutes after the reading stage.
- (5) **Review** – check that your recall notes are correct. You may also find it helpful to re-read the chapter to try to see the topic(s) it deals with as a whole.

Note-taking

Taking notes is a useful way of learning, but do not simply copy out the text. The notes must:

- be in your own words
- be concise
- cover the key points
- be well-organised
- be modified as you study further chapters in this text or in related ones.

Three ways of taking notes:

Summarise the key points of a chapter.

Make linear notes – a list of headings, divided up with subheadings listing the key points. If you use linear notes, you can use different colours to highlight key points and keep topic areas together. Use plenty of space to make your notes easy to use.

Try a diagrammatic form – the most common of which is a mind-map. To make a mind-map, put the main heading in the centre of the paper and put a circle around it. Then draw short lines radiating from this to the main sub-headings, which again have circles around them. Then continue the process from the sub-headings to sub-sub-headings, advantages, disadvantages, etc.

Highlighting and underlining

You may find it useful to underline or highlight key points in your study text – but do be selective. You may also wish to make notes in the margins.

Revision

The best approach to revision is to revise the course as you work through it. Also, try to leave four to six weeks before the exam for final revision. Make sure you cover the whole syllabus and pay special attention to those areas where your knowledge is weak. Here are some recommendations:

Read through the text and your notes again and condense your notes into key phrases. It may help to put key revision points onto index cards to look at when you have a few minutes to spare.

Review any assignments you have completed and look at where you lost marks – put more work into those areas where you were weak.

Practise exam standard questions under timed conditions. If you are short of time, list the points that you would cover in your answer and then read the model answer, but do try to complete at least a few questions under exam conditions.

In addition, **practise producing answer plans** and comparing them to the model answer.

If you are **stuck** on a topic, find somebody (a tutor) to explain it to you.

Read good newspapers and professional journals, especially ACCA's Student Accountant – this can give you an advantage in the exam.

Ensure you know the structure of the exam – how many questions and of what type you will be expected to answer. During your revision, attempt all the different styles of questions you may be asked.

Further reading

You can find further reading and technical articles under the student section of ACCA's website.

The following publications may also support your studies for this, and other, ACCA examinations:

A Student's Guide to Corporate Finance and Financial Management by David Evans – Kaplan Publishing

Technical update

This text has been updated to reflect Examinable Documents September 2025 to June 2026 issued by ACCA.

FORMULAE AND TABLES

Economic Order Quantity

$$= \sqrt{\frac{2C_0D}{C_H}}$$

Miller-Orr Model

Return point = Lower limit + ($\frac{1}{3}$ × spread)

$$\text{Spread} = 3 \left[\frac{\frac{3}{4} \times \text{transaction cost} \times \text{variance of cash flows}}{\text{interest rate}} \right]^{1/3}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The Asset Beta Formula

$$\beta_a = \left(\frac{V_e}{(V_e + V_d(1-T))} \beta_e \right) + \left(\frac{V_d(1-T)}{(V_e + V_d(1-T))} \beta_d \right)$$

The Growth Model

$$P_0 = \frac{D_0(1+g)}{(r_e - g)} \quad r_e = \frac{D_0(1+g)}{(P_0)} + g$$

Note: $D_0(1+g)$ may be replaced by D_1

Gordon's Growth Approximation

$$g = br_e$$

The Weighted Average Cost of Capital

$$\text{WACC} = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1-T)$$

The Fisher Formula

$$(1+i) = (1+r)(1+h)$$

Purchasing Power Parity and Interest Rate Parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Present value table

Present value of 1, i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Periods (n)	Discount rate (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239
Periods (n)	Discount rate (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065

Annuity table

Present value of an annuity of 1, i.e. $\frac{1-(1+r)^{-n}}{r}$

Where r = discount rate

n = number of periods until payment

Periods (n)	Discount rate (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606

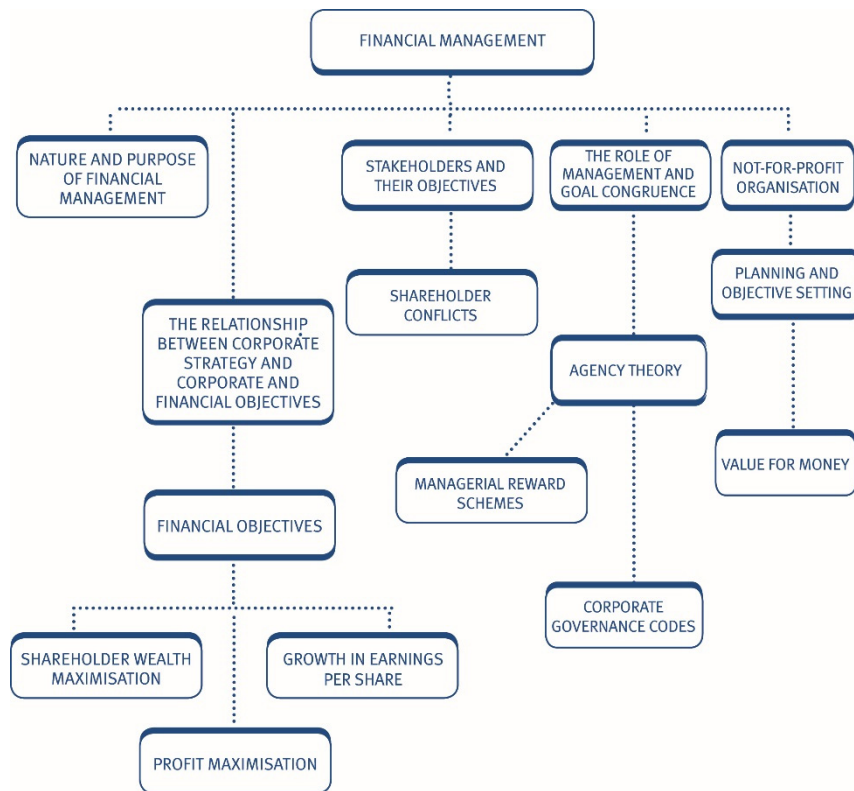
Periods (n)	Discount rate (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.968	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675

The financial management function

Chapter learning objectives

Upon completion of this chapter, you will be able to:

- explain the nature and purpose of financial management
- distinguish between financial management and financial and management accounting
- discuss the relationship between financial objectives, corporate objectives and corporate strategy
- identify and describe a variety of financial objectives, including:
 - shareholder wealth maximisation
 - profit maximisation
 - earnings per share growth
- identify stakeholders, their objectives and possible conflicts
- discuss the possible conflict between stakeholder objectives
- discuss the role of management in meeting stakeholder objectives, including the use of agency theory
- explain ways to encourage the achievement of stakeholder objectives, including:
 - managerial reward schemes
 - regulatory requirements
- discuss the impact of not-for-profit status on financial and other objectives
- discuss the nature and importance of Value for Money as an objective in not-for-profit organisations
- discuss ways of measuring the achievement of objectives in not-for-profit organisations.



1 The nature and purpose of financial management

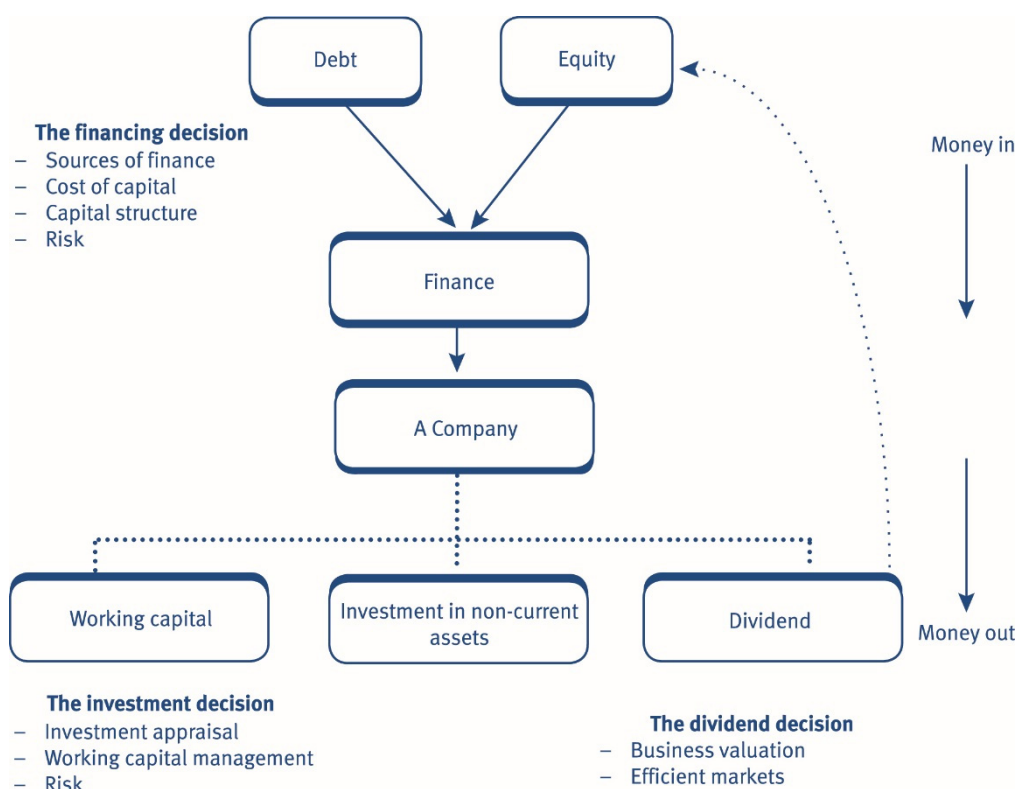


Financial management is concerned with the efficient acquisition and deployment of both short- and long-term financial resources, to ensure the objectives of the enterprise are achieved.

Decisions must be taken in three key areas:

- investment – both long-term investment in non-current assets and short-term investment in working capital.;
- finance – from what sources should funds be raised?
- dividends – how should cash funds be allocated to shareholders and how will the value of the business be affected by this?

These decisions are influenced by each other. For example, increasing the amount of cash allocated for investment projects means that either less cash is available to pay dividends, or the business's available cash needs to be increased by raising new funding.



An understanding of these three key areas is fundamental for the examination.

In taking these decisions, the financial manager will need to take account of:

- the organisation's commercial and financial objectives
- the broader economic environment in which the business operates
- the potential risks associated with the decision and methods of managing that risk.

The FM syllabus covers all these key aspects of financial management, which will also be relevant for those who choose to study for the advanced financial management exam.



The investment decision

To operate, all businesses will need finance and part of the financial manager's role is to ensure this finance is used efficiently and effectively to ensure the organisation's objectives are achieved. This can be further broken down into two elements:

- Investment appraisal considers the long-term plans of the business and identifies the right projects to adopt to ensure financial objectives are met. The projects undertaken will nearly always involve the purchase of non-current assets at the start of the process.

- For a business to be successful, as well as identifying and implementing potentially successful projects, it must survive day to day. Working capital management is concerned with the management of liquidity – ensuring debts are collected, inventory levels are kept at the minimum level compatible with efficient production, cash balances are invested appropriately and payables are paid on a timely basis.



The financing decision

Before a business can invest in anything, it needs to have some finance. A key financial management decision is the identification of the most appropriate sources (both long- and short-term), taking into account the requirements of the company, the likely demands of the investors and the amounts likely to be made available.



The dividend decision

Having invested wisely, a business will hopefully be profitable and generate cash. The final key decision for the financial manager is whether to return any of that cash to the owners of the business (in the form of dividends) and if so, how much should be distributed.

Cash not paid out as dividends is retained in the business, where it can be invested again to earn further returns. This decision is therefore closely linked to the financing decision.

The decision on the level of dividends to be paid can affect the value of the business as a whole, as well the ability of the business to raise further finance in the future.



Financial management should be distinguished from other important financial roles:

- management accounting – concerned with providing information for the more day-to-day functions of control and decision making
- financial accounting – concerned with providing information about the historical results of past plans and decisions.



Financial roles

Management accounting and financial management are both concerned with the use of resources to achieve a given target. Much of the information used and reported is common to both functions.

The main difference is in the time scales. **Financial management** is concerned with the **long-term raising of finance** and the **allocation and control of resources**; it involves targets, or objectives, that are generally long-term by nature, whilst management accounting usually operates within a 12-month time horizon.

Management accounting is concerned with **providing information** for the more **day-to-day** functions of **control and decision-making**. This will involve budgeting, cost accounting, variance analysis, and evaluation of alternative uses of short-term resources.

Financial accounting is not directly involved in the day-to-day planning, control and decision making of an organisation. Rather, it is concerned with **providing information** about the **historical results** of past plans and decisions. Its purpose is to keep the owners (shareholders) and other interested parties informed of the overall financial performance and position of the business, and it will not be concerned with the detailed information used internally by management accountants and financial managers.

The following table illustrates the distinction between some of the tasks carried out by each of these financial roles:

	Management accounting	Financial management	Financial accounting
Review of overtime spending	✓		
Depreciation of non-current assets			✓
Establishing dividend policy		✓	
Evaluating proposed expansion plans		✓	
Apportioning overheads to cost units	✓		
Identifying accruals and prepayments			✓

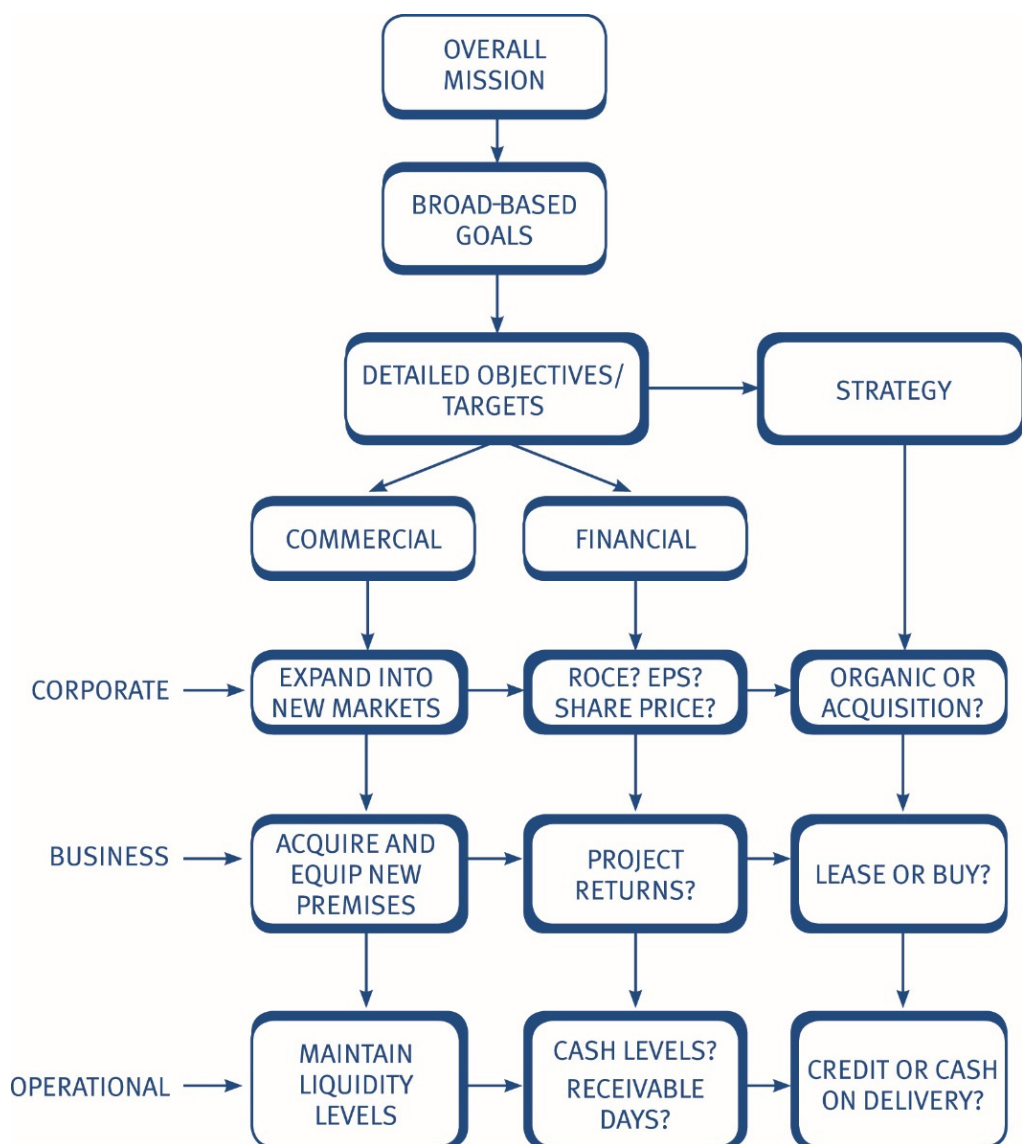
2 The relationship between corporate strategy and corporate and financial objectives



Objectives/targets define **what** the organisation is trying to achieve. Strategy considers **how** to go about it.



Objectives and strategy



The diagram above is key to understanding how financial management fits into overall business strategy

The business should recognise its overriding purpose or mission and develop broad-based goals for the business to pursue to ensure it fulfils that purpose.

Each goal is then further broken down into detailed commercial and financial objectives, each of which should have appropriate identifiable, measurable targets so that progress towards them can be monitored.

The distinction between 'commercial' and 'financial' objectives is to emphasise that not all objectives can be expressed in financial terms and that some objectives derive from commercial marketplace considerations.

These are then cascaded down throughout the organisation through the setting of targets so that all parts of the business are working to achieve the same overall goal. For example, the accounts receivable collection period (cash collection) target of the credit control department should be linked to the cash needs of the investment projects, and the projects should be selected to achieve the overall corporate aim such as improving the share price. This in turn then satisfies the shareholders by increasing their wealth.

Once objectives and targets are set, the enterprise must then work to achieve them by developing and implementing appropriate strategies. Strategies will be developed at all levels of the business.

- **Corporate strategy** concerns the decisions made by senior management about matters such as the particular business the company is in, whether new markets should be entered or whether to withdraw from current markets. Such decisions can often have important financial implications. If, for example, a decision is taken to enter a new market, an existing company in that market could be bought, or a new company could be started from scratch.
- **Business strategy** concerns the decisions to be made by the separate strategic business units within the group. Each unit will try to maximise its competitive position within its chosen market. This may involve, for example, choosing whether to compete on the basis of quality or cost.
- **Operational strategy** concerns how the different functional areas within a strategic business unit plan their operations to satisfy the corporate and business strategies being followed. We are, in this syllabus, most interested in the decisions facing the finance function. These day-to-day decisions include all aspects of working capital management.

Almost all strategies developed by the business will have financial implications and the financial manager has a key role to play in helping business strategies succeed.



Additional question – Objectives and strategy

The following list contains some commercial objectives/targets, some financial objectives/targets and some strategies, at each level of the business. Place one in each box of the table, as appropriate.

- Implement a Just-In-Time (JIT) inventory system.
- Increase earnings per share (EPS) by 5% on prior year.
- Acquire a rival in a share-for-share purchase.
- Buy four new cutting machines for \$250,000 each.
- Achieve returns of 15% on new manufacturing investment.
- Improve the ratio of current assets to current liabilities from 1.7 to 1.85.
- Reduce unsold inventory items by 12%.
- Update manufacturing capacity to incorporate new technology.
- Improve brand awareness within the UK.

	Commercial objectives/targets	Financial objectives/targets	Strategies
Corporate level			
Business level			
Operational level			

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

3 Financial objectives

Shareholder wealth maximisation



Shareholder wealth maximisation is a fundamental principle of financial management. You should seek to understand the different aspects of the syllabus (e.g. finance, dividend policy, investment appraisal) within this unifying theme.

Many other objectives are also suggested for companies including:

- profit maximisation
- growth
- market share
- social responsibilities



Financial objectives

Shareholder wealth maximisation

If strategy is developed in response to the need to achieve objectives, it is obviously important to be clear about what those objectives are.

Most companies are owned by shareholders and originally set up to make money for those shareholders. The primary objective of most companies is thus to maximise shareholder wealth. (This could involve increasing the share price and/or dividend pay-out.)

One method of measuring this is to calculate the total shareholder return (TSR). The detail in relation to this calculation can be found in the later chapter on Financial Ratios.

Profit maximisation

In financial management, we assume that the objective of the business is to maximise shareholder wealth. This is not necessarily the same as maximising profit.

Firms often find that share prices bear little relationship to reported profit figures (e.g. Tesla's share price grew from an average of \$4.67 in 2010 to \$63.4 in 2018 without recording anything other than annual losses during this time).

There are a number of potential problems with adopting an objective of profit maximisation:

- Long run versus short run issues: In any business, it is possible to boost short-term profits at the expense of long-term profits. For example, discretionary spending on training, advertising, maintenance and research and development (R&D) may be cut. This will improve reported profits in the short-term but damage the long-term prospects of the business. The stock exchange will normally see through such a tactic and share prices will fall.
- Quality (risk) of earnings: A business may increase its reported profits by taking a high level of risk. However, the risk may endanger the returns available to shareholders. The stock exchange will then generally regard these earnings as being of a poor quality and the more risk-averse shareholders may sell. Once again, the share price could fall.
- Cash: Accounting profits are just a paper figure. Dividends are paid with cash. Investors will therefore consider cash flow as well as profit

Earnings per share (EPS) growth

A widely used measure of corporate success is EPS (calculated as profit after tax and preference share dividends divided by the number of shares in issue), as it provides a measure of the return to equity holders. EPS growth is therefore a commonly pursued objective.

However, it is a measure of profitability, not wealth generation, and it is therefore open to the same criticisms as profit maximisation above.

The disadvantage of EPS is that it does not represent income of the shareholder. Rather, it represents that investor's share of the income generated by the company according to an accounting formula.

Whilst there is obviously a correlation between earnings and the wealth received by individual shareholders, they are not the same.



Maximising and satisficing

A distinction must be made between maximising and satisficing:

- maximising – seeking the maximum level of returns, even though this might involve exposure to risk and much higher management workloads.
- satisficing – finding a merely adequate outcome, holding returns at a satisfactory level, avoiding risky ventures and reducing workloads.

Within a company, management might seek to maximise the return to some groups (e.g. shareholders) and satisfy the requirements of other groups (e.g. employees). The discussion about objectives is really about which group's returns management is trying to maximise.

4 Stakeholder objectives and conflicts



A stakeholder group is one with a vested interest in the company.

Typical stakeholders for an organisation would include:

Internal

- Company employees
- Company managers/directors

Connected

- Equity investors (ordinary shareholders)
- Customers
- Suppliers
- Finance providers (debt holders/bankers)
- Competitors

External

- The government
- The community at large
- Pressure groups
- Regulators.

Many argue that managers should balance the needs and objectives of all stakeholders.



Conflict between and within groups of stakeholders and the need for management to balance the various interests is a key issue.



The stakeholder view

We've already stated that the primary objective of a company is to maximise the wealth of shareholders. However, modern organisations vary in type, and talk of pursuing single objectives is perhaps a little simplistic. Many argue that a business must adopt the **stakeholder view**, which involves balancing the competing claims of a wide range of stakeholders, and taking account of broader economic and social responsibilities.

Professor Charles Handy is a prominent advocate of this view, arguing that maximisation of shareholder wealth, while important, cannot be the single overall objective of organisations, and account must be taken of broader economic and social responsibilities. Increasing globalisation, and the fact that some multinational companies have revenue in excess of the incomes of small countries, put these responsibilities sharply into focus. Typical stakeholders for an organisation might include the following.

- Equity investors (ordinary shareholders) – within any economic system, the equity investors provide the risk finance. There is a very strong argument for maximising the wealth of equity investors. In order to attract funds, the company has to compete with risk-free investment opportunities, e.g. government securities. The attraction is the accrual of any surplus earned by the business to the equity investors. In effect, this is the risk premium (the reward to the equity investors for taking on the risk of investing in the company rather than in safer investments), which is essential to encourage the allocation of resources to relatively risky investments, such as company shares.
- Company managers/directors – such senior employees are in an ideal position to follow their own aims at the expense of other stakeholders. Their goals will be both long-term (defending against takeovers, sales maximisation) and short-term (profit margins leading to increased bonuses).

- Company employees – obviously, many trade unionists would like to see their members as the residual beneficiaries of any surplus the company creates. Certainly, there is no measurement problem: returns = wages or salaries and bonuses. However, maximising the returns to employees does assume that risk finance can be raised purely on the basis of satisficing, i.e. providing no more than an adequate return to shareholders.
- Customers – satisfaction of customer needs will be achieved through the provision of value-for-money products and services. There remains, of course, the requirement for organisations to accurately identify precisely what those needs are.
- Suppliers – suppliers to the organisation will have short-term goals, such as prompt payment terms, alongside long-term requirements including contracts and regular business. The importance of the needs of suppliers will depend upon both their relative size and the number of suppliers.
- Finance providers – providers of finance (banks, loan creditors) will primarily be interested in the ability of the organisation to repay the finance, including interest. As a result, it will be the organisation's ability to generate cash, in both the long- and short-term, that will be the basis of interest to these providers.
- The government – the government will have political and financial interests in the organisation. Politically they will wish to increase exports and decrease imports whilst monitoring companies via a regulator, such as the Competition and Markets Authority in the UK. Financially, governments require businesses to earn long-term profits to maximise their own taxation income. Equally importantly, the government, via its own agencies or via the legal system, will seek to ensure that organisations observe health and safety, planning and minimum wage legislation. Also, on behalf of the community at large, the government must consider modifying the behaviour of both individuals and organisations for environmental/health reasons, e.g. banning smoking in certain public places and restricting the advertising of tobacco companies.
- The community at large – the community will have, for instance, environmental expectations from public sector, private sector and regulated organisations, such as good access to healthcare, trains that run on time and safe foods. For organisations there are problems of measurement – what are returns to the community at large? The goals of the community will be broad but will include such aspects as legal and social responsibilities, pollution control and employee welfare.

Critics of the stakeholder view argue that while the interests of these other groups must obviously be balanced and managed, only shareholders have a relationship with organisations that is one of risk and return, and so practically, the long-term financial objective of a private sector organisation is to maximise the wealth of equity investors. The debate will continue, but there seems to be an acceptance that organisations need to have a greater awareness of the societies in which they operate.



Additional question – Stakeholder conflicts

Suggest the potential conflicts in objectives which could arise between the following groups of stakeholders in a company.

Stakeholders	Potential conflict
Employees ⇔ Shareholders	
Customers ⇔ Community at large	
Shareholders ⇔ Finance providers	
Customers ⇔ Shareholders/managers	
Government ⇔ Shareholders	
Shareholders ⇔ Managers	

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

5 The role of management and goal congruence

Agency theory

Agency theory is often used to describe the relationships between the various interested parties in a firm and can help to explain the various duties and conflicts that occur:



Agency relationships occur when one party, **the principal**, employs another party, **the agent**, to perform a task on their behalf. In particular, directors (agents) act on behalf of shareholders (principals).

How to reduce the problems caused by agency relationships

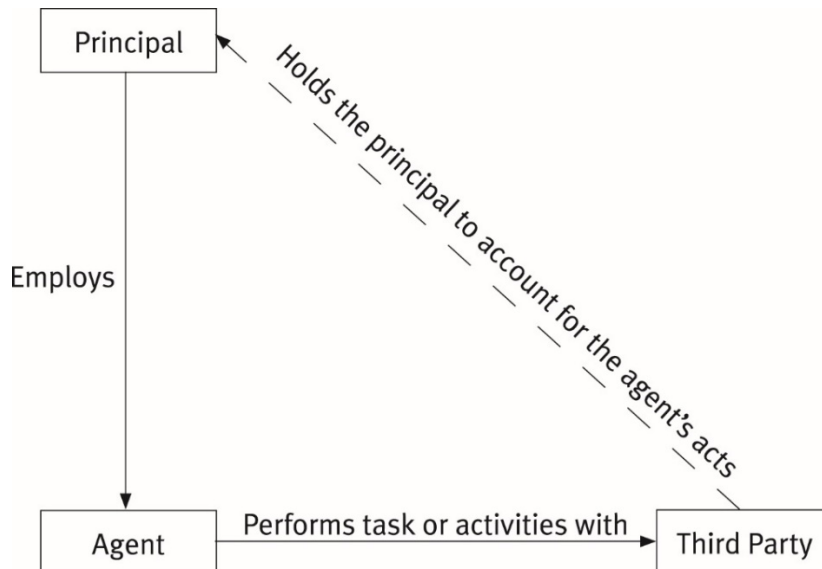


Finding ways to reduce the problems of the agency relationship and ensure that managers take decisions which are consistent with the objectives of shareholders is a key issue



Agency theory

Agency theory can help to explain the actions of the various interest groups in the corporate governance debate.



For example, managers can be seen as the agents of shareholders, employees as the agents of managers, managers and shareholders as the agents of long- and short-term creditors, etc. In most of these principal-agent relationships, conflicts of interest will exist.

The problem lies in the fact that once the agent has been appointed, they are able to act in their own selfish interests rather than pursuing the objectives of the principal.

The divorce of ownership and control

By far the most important conflict of those mentioned above is that between the interests of shareholders who own the company and the directors/managers who run it.

- Shareholders are reliant upon the management of the company to understand and pursue the objectives set for them.
- Although shareholders can intervene via resolutions at general meetings, the managers are usually left alone on a day-to-day basis.
- Management are uniquely placed to make decisions to maximise their own wealth or happiness rather than the wealth of the shareholders.

During the past few decades, and particularly following such scandals as Maxwell, Enron and WorldCom, attention has focused increasingly on the activities of directors. There has been an allegation that directors have been making corporate decisions in their own interests rather than for the benefit of the company as a whole and of the shareholders in particular. Examples include the following:

Remuneration

The UK recession in 2008 – 2009 led to many companies reporting sharply reduced profits and laying off staff or cutting levels of employee pay. At the same time, many directors were awarded large increases in pay or large bonuses. The greatest media attention focused on the companies who received government assistance as part of a bail-out package but continued to reward their top executives highly.

Empire building

The high level of corporate takeover activity in the 1980s led to many chief executives believing that building as large a group as possible was a valid aim in itself, an objective described as empire building. Executives gained prestige from successful bids and from being in charge of large conglomerates, but the returns to shareholders were often disappointing.

Creative accounting

The directors are responsible for selecting the accounting policies to be used by their company, subject to accounting standards and the opinion of the auditors. Despite the constraints upon them, the directors are still free to use creative accounting techniques to flatter their published accounts and perhaps artificially boost the share price. Examples of such techniques are: capitalising expenses on the statement of financial position (e.g. development expenditure, advertising expenditure), not depreciating non-current assets, maximising the value of intangibles on the balance sheet (e.g. putting a value to brands), recognising revenue on long-term contracts at the earliest possible time, and other forms of off-balance sheet financing (see below).

Those charged with setting accounting standards have a continuing work programme that aims to cut out creative accounting practices as much as is practically possible.

Off-balance-sheet finance refers to ways of financing assets where the method of funding is not recorded on the balance sheet (statement of financial position).

One example is the use of quasi-subsidaries into which financial liabilities are moved so that they don't appear in the parent company's balance sheet. The introduction of FRS 5 aimed to restrict off-balance sheet finance, for example by requiring quasi-subsidaries to be consolidated into the group accounts. The collapse of Enron in the US in late 2001 put the spotlight on the various forms of off-balance-sheet finance used.

Takeover bids

Boards of directors often spend considerable amounts of time and money attempting to defend their companies against takeover bids, even when it appears that the takeover would be in the best interests of the target company's shareholders. These directors are accused of trying to protect their own jobs, fearing that they will be retired if their company is taken over. Directors of public companies must now comply with the City Code on Takeovers and Mergers during a bid period.

Unethical activities

Unethical activities might not be prohibited by the Companies Acts or stock exchange regulations, but are believed by many to be undesirable to society as a whole. Examples are trading with countries ruled by dictatorships, testing products on animals, emitting pollution or carrying out espionage against competitors. The importance of **good business ethics** and **corporate social responsibility** (CSR) has been recognised in recent years and it is hoped that further progress is being made.

Managerial reward schemes

One way to help ensure that managers take decisions which are consistent with the objectives of shareholders is to introduce carefully designed remuneration packages. The schemes should:

- Be clearly defined, impossible to manipulate and easy to monitor
- Link rewards to changes in shareholder wealth
- Match managers' time horizons to shareholders' time horizons
- Encourage managers to adopt the same attitudes to risk as shareholders.

Common types of reward schemes include:

- Remuneration linked to:
 - minimum profit levels
 - economic value added (EVA)
 - revenue growth
- executive share option schemes (ESOP).



Managerial reward schemes

Types of remuneration schemes include:

Remuneration linked to minimum profit levels

This scheme would be easy to set up and monitor.

Disadvantages are that the scheme may lead to managers taking decisions that would result in profits being earned in the short-term at the expense of long-term profitability. It could also lead to managers underachieving, i.e. relaxing as soon as the minimum is achieved. The scheme might also tempt managers to use creative accounting to boost the profit figure.

Remuneration linked to economic value added (EVA)

EVA is a measure of the increase in the value of shareholder wealth in the period. Schemes such as these are therefore designed to more closely align the interests of the employee and the shareholder.

A potential disadvantage is that calculating the bonus may be complex.

Remuneration linked to revenue growth

Growth of the business and higher production levels can lead to economies of scale, which in turn can help the business compete more successfully on price.

However, revenue growth could be achieved at the expense of profitability, e.g. by reducing selling prices or by focussing on high revenue product lines, which may not necessarily be the most profitable. Maximising revenue is therefore unlikely to maximise shareholder wealth.

An executive share option scheme (ESOP)

This scheme has the advantage that it will encourage managers to maximise the value of the shares of the company, i.e. the wealth of the shareholders. When an executive is awarded share options, the theory is that it is in their interests for the share price to rise, so they will do whatever possible to improve the share price. Their interests will be best served by working towards a goal that is also in the interests of the shareholders.

Such schemes are normally set up over a relatively long period, thereby encouraging managers to make decisions to invest in positive return projects, which should result in an increase in the price of the company shares. However, efficient managers may be penalised at times when share prices in general are falling.

There are several criticisms of ESOPs.

- When directors exercise their share options, they tend to sell the shares almost immediately to cash in on their profit. Unless they are awarded more share options, their interest in the share price therefore ends when the option exercise date has passed.
- If the share price falls low enough for the options to have no value, they cannot act as an incentive.
- If a company issues large quantities of share options, there could be some risk of excessive dilution of the equity interests of the existing shareholders.
- Directors may distort reported profits (creative accounting) to protect the share price and the value of their share options.



Additional question – Managerial reward schemes

Gretsch Inc, a listed company, has developed a highly successful new product and is thus growing rapidly. However, with this growth the firm is experiencing cash flow problems. Managers are currently awarded bonuses if there is growth in reported earnings per share (EPS).

Comment on the current remuneration scheme.

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

Corporate governance codes

The director/shareholder conflict has also been addressed by the requirements of a number of corporate governance codes. The following key areas relate to this conflict.

- Non-executive directors (NEDs)
 - important presence on the board
 - must give obligation to spend sufficient time with the company
 - should be independent.
 - at least half the board to be independent NEDs
- Executive directors
 - separation of chairman and chief executive officer (CEO)
 - submit for re-election
 - clear disclosure of financial rewards



Corporate governance codes

After a number of high-profile firms collapsed, concerns over how the companies had been run led to a determination to ensure good corporate governance in future. A number of committees met and produced reports containing recommendations on how to improve corporate governance procedures. One of the areas addressed was the conflict between director and shareholder interests. Below is a selection of the current requirements:

NEDs

- At least half of the members of the board, excluding the chairperson, should be independent NEDs. These are directors who do not take part in the running of the business. They attend board meetings, provide advice, listen to what is said and are generally meant to act as a control on the actions of the executive directors. Independence means they are free of any business or other relationship that could materially interfere with the exercise of their independent judgement. Boards should disclose in the annual report which of the NEDs are considered to be independent.
- NEDs should get extra fees for chairing company committees but should not hold share options in the company. There was concern that allowing NEDs to hold share options in a company could encourage corporate excess or wrongdoing.
- One of the independent NEDs should be appointed as senior independent NED and would act as a champion for the interests of shareholders.
- Prospective NEDs should conduct due diligence before accepting the role. They should satisfy themselves that they have the knowledge, skills, experience and time to make a positive contribution to the company board.
- On appointment, the NEDs would also undertake that they have the time to meet their obligations. Company nomination committees would examine their performance, and make an assessment of whether they were devoting enough time to their duties.

Executive directors

- The chairperson and the CEO roles should be separate, and a CEO should not become chairperson of the same company.
- The chairperson should be independent at the time of their appointment. Note that the effect of this, combined with the point about independent NEDs making up at least half of the board, will be to place independent NEDs in a majority on the board.

- All directors should submit themselves for re-election at least every three years.
- There should be clear disclosure of directors' total emoluments and those of the chairperson and highest-paid UK director.
- Boards should set as their objective the reduction of directors' contract periods to one year or less.

Stock exchange listing requirements and other regulations

Although adherence to the principles of the corporate governance codes is voluntary, they are often referred to in the listing requirements of stock exchanges.

6 Measuring achievement of corporate objectives

It is necessary for managers, shareholders and other stakeholders to have ways of measuring the progress of the company towards its objectives. This is commonly done via ratio analysis.



Ratio analysis compares and quantifies relationships between financial variables.

Ratio analysis can be grouped into four main categories:

- Profitability and return
- Debt and gearing
- Liquidity
- Investor

The specific ratios covered in the FM syllabus will be looked at in detail in Chapter 19. Some of them may already be familiar to you from previous exams.

7 Objective setting in not-for-profit organisations

The primary objective of not-for-profit organisations (NFPs or NPOs) is not to make money but to benefit prescribed groups of people.

As with any organisation, NFPs will use a mixture of financial and non-financial objectives.

However, with NFPs the non-financial objectives are often more important and more complex because of the following:

- Most key objectives are very difficult to quantify, especially in financial terms, e.g. quality of care given to patients in a hospital.
- Multiple and conflicting objectives are more common in NFPs, e.g. quality of patient care versus number of patients treated.



Objective setting in NFPs

A number of factors influence the way in which management objectives are determined in NFPs, which distinguish them from commercial businesses:

- wide range of stakeholders
- high level of interest from stakeholder groups
- significant degree of involvement from funding bodies and sponsors
- little or no financial input from the ultimate recipients of the service
- funding often provided as a series of advances rather than as a lump sum
- projects typically have a longer-term planning horizon
- may be subject to government influence/government macroeconomic policy.

For a company listed on the stock market, we can take the maximisation of shareholder wealth as a working objective and know that the achievement of this objective can be monitored with reference to share price and dividend payments.

For an NFP the situation is more complex. There are two questions to be answered:

- in whose interests is it run?
- what are the objectives of the interested parties?

Many such organisations are run in the interests of society as a whole and therefore should seek to attain the position where the gap between the benefits they provide to society and the costs of their operation is the widest (in positive terms).

The cost is relatively easily measured in accounting terms. However, many of the benefits are intangible. For example, the benefits of such bodies as the National Health Service (NHS) or local education authorities (LEAs) are almost impossible to quantify.

Because of the problem of quantifying the non-monetary objectives of such organisations, most public bodies operate under objectives determined by the government (and hence ultimately by the electorate).

Value for money (VFM) and the 3 Es



VFM can be defined as 'achieving the desired level and quality of service at the most economical cost'.



VFM

Because a significant number of NFPs are funded from the public purse, the lack of clear financial performance measures has been seen as a particular problem. It is argued that the public are entitled to reassurance that their money (in the form of taxes for public sector organisations or donations for charities) is being properly spent.

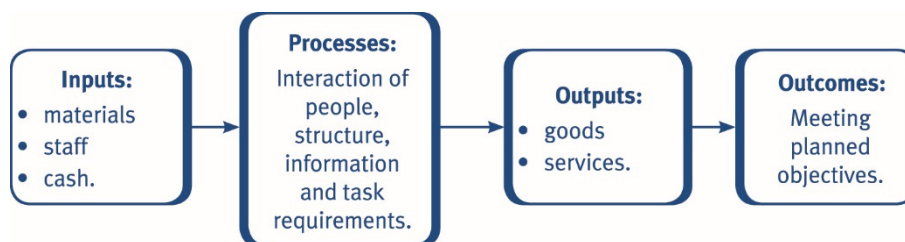
In addition, the complex mix of objectives with no absolute priority has also led to concern that the money may be being directed towards the wrong ends.

These issues, along with a growth in the perceived need for greater accountability among public officials, led to the development of the concept of evaluating VFM in public sector organisations. The principles developed are now widely applied in NFPs.

Systems analysis

A more detailed analysis of what is meant by VFM can be achieved by viewing the organisation as a system set up to achieve its objectives by means of processing inputs into outputs.

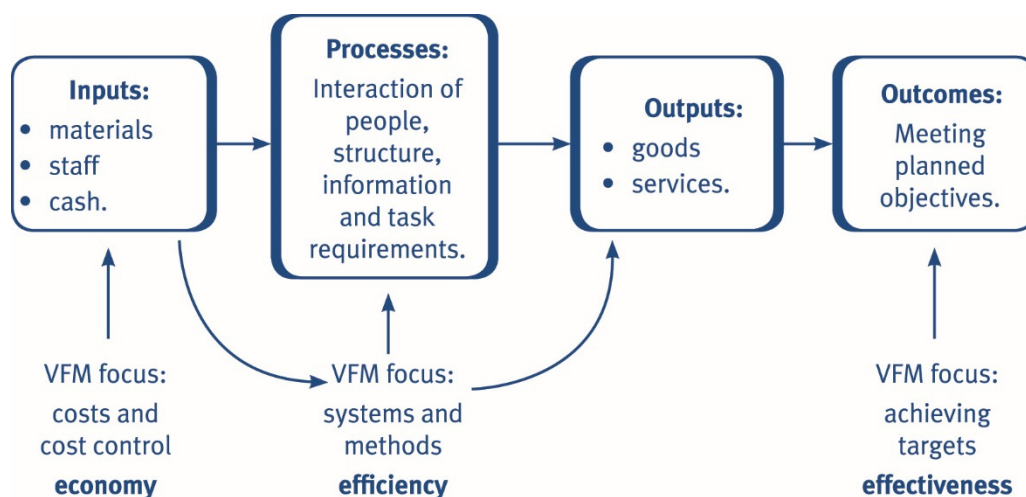
The organisation as a system



The three Es

Assessing whether the organisation provides value for money involves looking at all functioning aspects of the organisation. Performance measures have been developed to permit evaluation of each part separately.

VFM





Economy: Minimising the costs of inputs required to achieve a defined level of output.

Efficiency: Ratio of outputs to inputs – achieving a high level of output in relation to the resources put in (input driven) or providing a particular level of service at reasonable input cost (output driven).

Effectiveness: Whether outputs are achieved that match the predetermined objectives.



Use of the 3 Es as a performance measure and a way to assess VFM is a key issue for examination questions that relate to NFPs and public sector organisations.



3Es

Known as the 3 Es, these measures are fundamental to an understanding of VFM. An organisation achieving economy, efficiency and effectiveness in each part of the system is considered to be providing good VFM.

Public sector organisations are subject to regular VFM (or best value) reviews, and the results have important impacts on future plans and funding decisions.

Economy

Acquiring resources of appropriate quality and quantity at the lowest cost. Note that whilst obtaining low prices is an important consideration it is not the only one. Achieving true economy will include ensuring the purchases are fit for purpose and meet any predetermined standards. For instance, selecting the cheapest (lowest quality) inputs may lead to quality issues that compromise the level of output achievable.

Efficiency

Maximising the useful output from a given level of resources, or minimising the inputs required to produce the required level of output.

Some public services fall within the first definition as they try to provide as much of a service as possible with strictly limited resources and few opportunities to generate further income sources. This is defined as 'input-driven' efficiency. This would include services such as library provision.

However, in many areas there is a statutory obligation to provide a particular standard of service, for example prison services, which cannot be significantly reduced or withdrawn. In this case, the obligation is to provide the service at a reasonable cost and is known as 'output-driven' efficiency. In both areas, the key consideration is whether the resources used were put to good use and the methods and processes carried out represent best practice.

Effectiveness

Ensuring that the output from any given activity is achieving the desired result.

For example, the cheapest site on which to build and run a sports centre may be a disused brownfield site on the edge of town. However, if the council's objectives included reduction in car use and accessible opportunities for health improvement, then the output of the building process – the sports centre, even if built economically and efficiently, would not be considered effective as it failed to meet the stated objectives



Measuring the achievement of objectives in NFPs

Since the services provided are limited primarily by the funds available, key financial objectives for NFPs will be to:

- raise as large a sum as possible
- spend funds as effectively as possible.

Targets may then be set for different aspects of each accounting period's finances such as:

- total to be raised in grants and voluntary income
- maximum percentage of this total that fund-raising expenses represents
- amounts to be spent on specified projects or in particular areas
- maximum permitted administration costs
- meeting budgets
- breaking even in the long run.

The actual figures achieved can then be compared with these targets and control action taken if necessary.



Test your understanding 1 – Not-for-profit organisations

A subsidised college canteen service is to be evaluated by the local council to assess, amongst other things, whether it is financially sound and offers value for money.

Suggest appropriate measures of achievement that could be set for the service.



Test your understanding 2

Which of the following is NOT one of the three main types of decision facing the financial manager in a company?

- A Dividend decision
- B Investment decision
- C Economic decision
- D Financing decision



Test your understanding 3

The agency problem is a driving force behind the growing importance attached to sound corporate governance.

In this context, the 'agents' are the:

- A Customers
- B Shareholders
- C Managers
- D Auditors



Test your understanding 4

Which of the following is an example of a financial objective that a company might choose to pursue?

- A Provision of good wages and salaries
- B Dealing honestly and fairly with customers on all occasions
- C Producing environmentally friendly products
- D Restricting the level of gearing to below a specified target level



Test your understanding 5

In the context of managing performance in 'not-for-profit' organisations, which of the following definitions is incorrect?

- A Value for money means providing a service in a way which is economical, efficient and effective
- B Economy means doing things cheaply: not spending \$2 when the same thing can be bought for \$1
- C Efficiency means doing things quickly: minimising the amount of time that is spent on a given activity
- D Effectiveness means doing the right things: spending funds so as to achieve the organisation's objectives

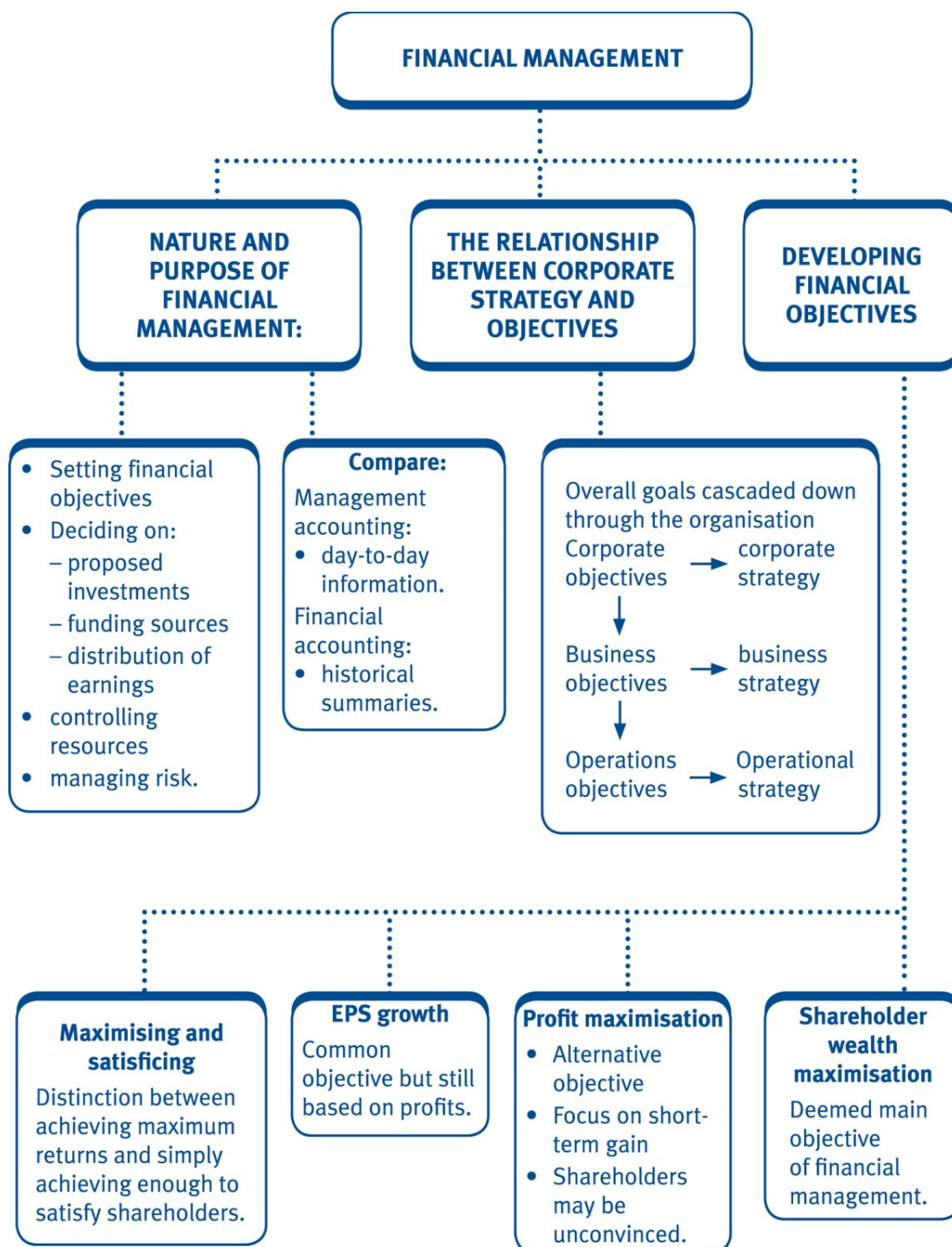


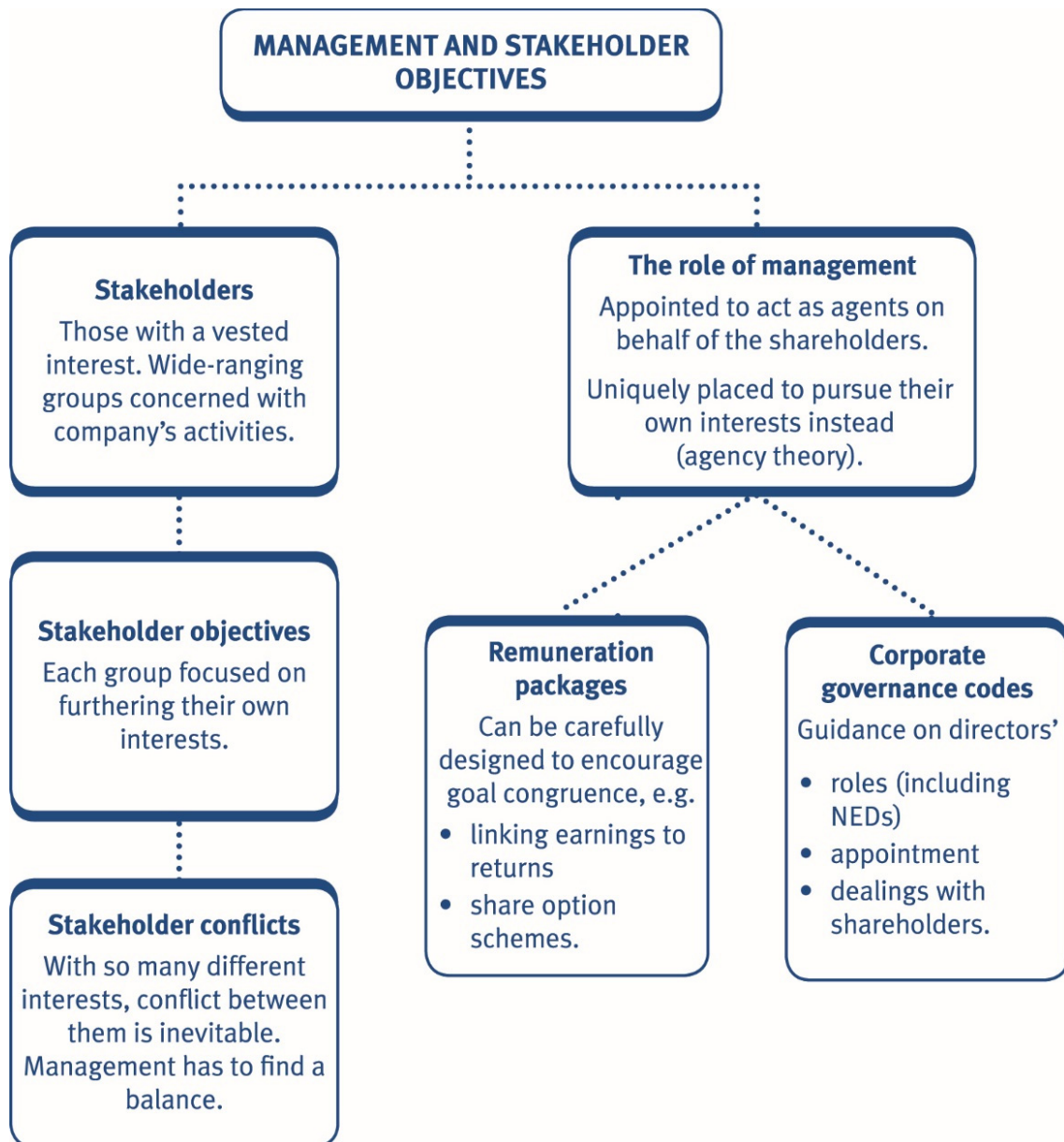
Test your understanding 6

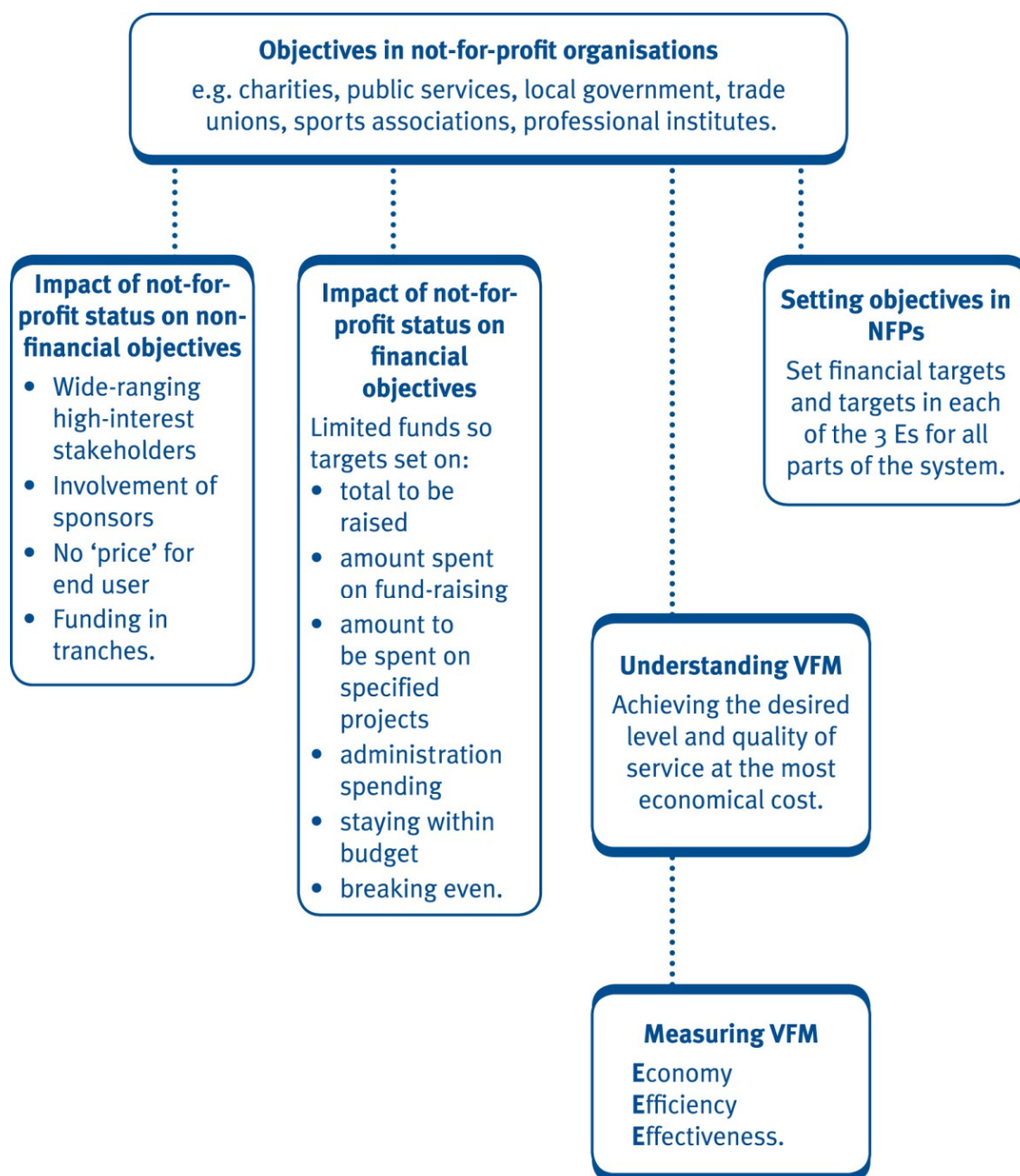
Which of the following is a problem associated with managerial reward schemes?

- A By rewarding performance, an effective scheme creates an organisation focused on continuous improvement
- B Schemes based on shares can motivate employees/managers to act in the long-term interests of the company
- C Self-interested performance may be encouraged at the expense of team work
- D Effective schemes attract and keep the employees valuable to an organisation

8 Chapter summary









Answer to additional question – Objectives and strategy

	Commercial objectives /targets	Financial objectives /targets	Strategies
Corporate level	Improve brand awareness within the UK.	Increase EPS by 5% on prior year.	Acquire rival chain in a share-for-share purchase.
Business level	Update manufacturing capacity to incorporate new technology.	Achieve returns of 15% on new manufacturing investment.	Buy four new cutting machines for \$250,000 each.
Operational level	Reduce unsold inventory items by 12%.	Improve the ratio of current assets to current liabilities from 1.7 to 1.85.	Implement a JIT inventory system.



Answer to additional question – Stakeholder conflicts

Stakeholders	Potential conflict
Employees ⇔ Shareholders	Employees may resist the introduction of automated processes that would improve efficiency but cost jobs. Shareholders may resist wage rises demanded by employees.
Customers ⇔ Community at large	Customers may demand lower prices and greater choice, but in order to provide them a company may need to squeeze vulnerable suppliers or import products at great environmental cost.

Stakeholders	Potential conflict
Shareholders ⇔ Finance providers	Shareholders may encourage management to pursue risky strategies in order to maximise potential returns, whereas finance providers prefer stable lower-risk policies that ensure liquidity for the payment of debt interest.
Customers ⇔ Shareholders/managers	Customers may require higher service levels (such as 24 rather than 48 hour delivery) which are resisted by shareholders as too expensive, or by management due to increased workload.
Government ⇔ Shareholders	Government will often insist upon levels of welfare (such as the minimum wage and health and safety practices) which would otherwise be avoided as an unnecessary expense.
Shareholders ⇔ Managers	Shareholders are concerned with the maximisation of their wealth. Managers may instead pursue strategies focused on growth as these may bring the greatest personal rewards.

Note: You may have come up with different suggestions. The point is to recognise that there is a huge range of potential conflicts of interest and senior management will need to work to achieve a balance.



Answer to additional question – Managerial reward schemes

Advantages

- Goal congruence – managers will work to achieve growth in EPS, which will make shareholders feel that their wealth is increasing.
- The figure is difficult (but not impossible!) to manipulate from one period to another as it will be audited.

Disadvantages

- There is little incentive for managers to control working capital and cash flow – a pressing problem. Growth may be at the expense of liquidity and ultimately compromise the firm's future survival.
- Managers may gain bonuses simply because of the popularity of the products concerned rather than their own efforts.
- Long-term shareholder value and EPS are not well correlated.

Test your understanding answers



Test your understanding 1 – Not-for-profit organisations

Financial measures:

- proportion of overall funds spent on administration costs
- ability to stay within budget/break even
- revenue targets met.

Economy targets:

- costs of purchasing provisions of suitable nutritional quality
- costs of negotiating for and purchasing equipment
- negotiation of bulk discounts
- pay rates for staff of appropriate levels of qualification.

Efficiency targets:

- numbers of portions produced
- cost per meal sold
- levels of wastage of unprepared and of cooked food
- staff utilisation
- equipment life.

Effectiveness targets:

- numbers using the canteen
- customer satisfaction ratings
- nutritional value of meals served.



Test your understanding 2

Answer C

Financial management aims to ensure that the money is available to finance profitable projects and to select those projects that the company should undertake. Once profits have been made, the decision then needs to be taken about how much to distribute to the owners and how much to re-invest for the future.



Test your understanding 3

Answer C

The separation of ownership and control creates a situation where managers act as the agents of the owners (shareholders).



Test your understanding 4

Answer D

This is a financial objective that relates to the level of risk that the company accepts.



Test your understanding 5

Answer C

Efficiency means doing things well: getting the best use out of what money is spent on.



Test your understanding 6

Answer C

This identifies a typical problem associated with many managerial reward schemes.

Basic investment appraisal techniques

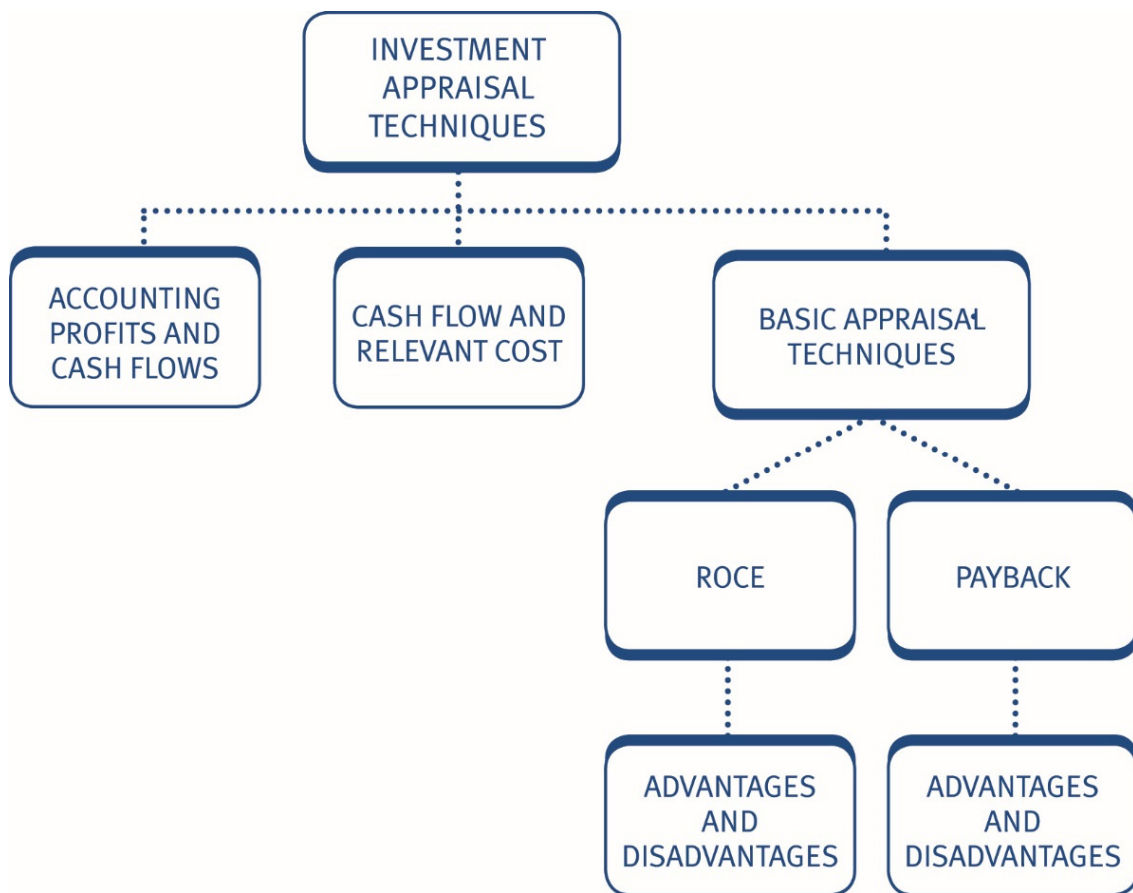
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- define a relevant cash flow (and distinguish it from an accounting profit)
- identify and calculate relevant cash flows in a scenario
- calculate the payback period and use it to appraise an investment
- discuss the usefulness of payback as an investment appraisal method
- calculate return on capital employed (ROCE) (accounting rate of return) and use it to appraise an investment
- discuss the usefulness of ROCE as an investment appraisal method.



One of the PER performance objectives (PO9) is to evaluate investment and financing decisions. You evaluate investment opportunities and their consequences and advise on their costs and benefits to the organisation. Working through this chapter should help you understand how to demonstrate that objective.



1 Investment appraisal process



One stage in the capital budgeting process is **investment appraisal**. This appraisal has the following features:

- assessment of the level of expected returns earned for the level of expenditure made
- estimates of future costs and benefits over the project's life.

Two basic appraisal techniques are covered in this chapter:

- ROCE
- Payback.

More sophisticated methods of investment appraisal are dealt with in the next chapter.

You will also see versions of the ROCE calculation in other exams such as Performance Management and Financial Reporting.



Examination questions may ask you to compare and contrast the use of these two basic techniques.

2 ROCE

This is also known as accounting rate of return (ARR).



$$\text{ROCE} = \frac{\text{Average annual operating profit}}{\text{Initial capital costs}} \times 100\%$$

or alternatively:

$$\text{ROCE} = \frac{\text{Average annual operating profit}}{\text{Average capital investment}} \times 100\%$$

The average investment can be calculated as:

$$\text{Average capital investment} = \frac{\text{Initial investment} + \text{scrap value}}{2}$$

Operating profit represent profit before interest, tax and dividend deductions have been made.

In the exam, you should use the initial capital cost unless you are told otherwise. Which figure to use will be made clear in the question.

Decision rule:

- If the expected ROCE for the investment is greater than the target or hurdle rate (as decided by management) then the project should be accepted.



Test your understanding 1 – ROCE

A project involves the immediate purchase of an item of plant costing \$110,000. It would generate annual cash flows of \$24,400 for five years, starting in Year 1. The plant purchased would have a scrap value of \$10,000 in five years, when the project terminates.

Determine the project's ROCE using:

- initial capital costs
- average capital investment



Test your understanding 2 – ROCE

A project requires an initial investment of \$800,000 and then earns net cash inflows as follows:

Year	1	2	3	4	5	6	7
Cash inflows (\$000)	100	200	400	400	300	200	150

In addition, at the end of the seven-year project the assets initially purchased will be sold for \$100,000.

Determine the project's ROCE using:

- (a) initial capital costs
- (b) average capital investment.



Initial capital cost

The initial capital cost could comprise any or all of the following:

- cost of new assets bought
- net book value (NBV) of existing assets to be used in the project
- investment in working capital
- capitalised R&D expenditure (**NB** ensure this is amortised against profit).

3 Advantages and disadvantages of ROCE

Advantages include:

- simplicity
- links with other accounting measures.

Disadvantages include:

- no account is taken of project life
- no account is taken of timing of cash flows
- it varies depending on accounting policies
- it may ignore working capital
- it does not measure absolute gain
- there is no definitive investment signal.



In the examination, it is important that you can discuss the features of ROCE as an investment appraisal technique, in addition to being able to calculate it.



Advantages and disadvantages of ROCE

Advantages

Simplicity – being based on widely reported measures of return (profits) and assets (statement of financial position values), it is easily understood and easily calculated. It is therefore a widely used measure. It is expressed in percentage terms, with which both managers and accountants are familiar.

Links with other accounting measures – it can be used alongside operating margin and asset turnover figures for analysis.

Disadvantages

ROCE does include numbers from the whole life of the project, but doesn't take account of the timings within it and would not differentiate between projects of different lengths as long as the same average profits were earned.

In failing to take account of the timing of cash flows within the project, it therefore ignores the time value of money (see next chapter).

It will vary with specific accounting policies, and the extent to which project costs are capitalised. Profit measurement is thus 'subjective', and ROCE figures for identical projects would vary from business to business.

It might ignore working capital requirements.

Like all rate of return measures, it is not a measurement of absolute gain in wealth for the business owners. It does not therefore fully correlate to the objective of maximisation of shareholder wealth.

There is no definite investment signal. The decision to invest or not remains subjective in view of the lack of an objectively set target ROCE.

4 Accounting profits and cash flows



In capital investment appraisal it is more appropriate to evaluate future cash flows than accounting profits, because:

- profits cannot be spent
- profits are subjective
- cash is required to pay dividends.



Profits versus cash flows

Cash flows are a better measure of the suitability of a capital investment because:

- cash is what ultimately counts – profits are only a guide to cash availability: they cannot actually be spent
- profit measurement is subjective – the time period in which income and expenses are recorded, and so on, are a matter of judgement
- cash is used to pay dividends – dividends are the ultimate method of transferring wealth to equity shareholders.

In practice, the cash flow effects of a project are likely to be similar to the project's effects on profits. Major differences in cash and profit flows will be linked to areas such as:

- changes in working capital
- asset purchase and depreciation
- deferred taxation
- capitalisation of research and development expenditure.

5 Relevant cash flows

For all methods of investment appraisal, with the exception of ROCE, only relevant cash flows should be considered. These are:

- future
- incremental
- cash-based.

or

- opportunity costs.

Ignore:

- sunk costs
- committed costs
- non-cash items
- allocated costs.

The total of relevant cash flows for an investment project is the difference between the cash position if the project goes ahead versus the cash position if it doesn't.



Relevant cash flows

Future: Only future cash flows that occur as a result of the decision should be considered, e.g. any future costs or revenues arising because of the project.

- Costs that have already been incurred are not relevant to a current decision. For example, suppose a company makes a non-returnable deposit as a down payment for an item of equipment, and then reconsiders whether it wants the equipment after all. The money that has already been spent cannot be recovered and so is not relevant to the current decision about obtaining the equipment.

Incremental: Only extra cash flows that occur as a result of the decision should be considered, e.g. additional costs or revenues arising because of the project.

- Costs that will be incurred anyway, whether or not a capital project goes ahead, cannot be relevant to a decision about investing in the project. Fixed cost expenditures are an example of non-relevant 'committed costs'. For the purpose of investment appraisal, a project should not be charged with an amount for a share of fixed costs that will be incurred in any event.

However, be aware that if a project causes a fixed cost to increase or reduce, the amount of the increase or reduction is incremental and therefore relevant. For instance, the cost of supervisors paid on a salary basis is a fixed cost, but if a project requires the employment of an additional supervisor, that person's salary cost would be relevant.

- Overheads attributed to projects should, in the examination, be taken as absorbed figures unless specified otherwise. It should be assumed there is no change to the actual overhead paid (unless specified otherwise in the question), and thus no incremental cash flow.

Cash flows: Only cash items are relevant to the decision.

- Non-cash items can never be relevant to an investment appraisal based on relevant cash flows. In particular, the depreciation charges on a non-current asset are not relevant costs because depreciation is not a cash expenditure.

Opportunity costs: These are the cash flows in relation to the next best alternative, e.g. if a machine that was to be sold is used on a project, the lost sale value is an opportunity cost and is relevant to the decision to go ahead with the project.



Test your understanding 3 – Relevant cash flows

A manufacturing company is considering the production of a new type of widget. Each widget will take two hours to make.

Fixed overheads are apportioned on the basis of \$1 per labour hour.

If the new widgets are produced, the company will have to employ an additional supervisor at a salary of \$15,000 per year. The company will produce 10,000 widgets per year.

What are the relevant cash flows?



Additional question – Relevant cash flows

A company is evaluating a proposed expenditure on an item of equipment that would cost \$160,000.

A technical feasibility study has been carried out by consultants, at a cost of \$15,000, into benefits from investing in the equipment.

It has been estimated that the equipment would have a life of four years, and annual profits would be \$8,000, after deducting annual depreciation of \$40,000 and an annual charge of \$25,000 for a share of the current fixed cost of the company.

What are the relevant cash flows for this?

The answer to this question can be found after the chapter summary diagram at the end of this chapter.



Note: In the exam, you may be given profit figures when the answer requires the use of cash flows (e.g. for payback calculations) and cash flows when profits should be used (e.g. for ROCE calculations). Generally, to convert one to the other, the only adjustment required will be for depreciation.

6 Payback method of appraisal



The **payback** period is the time a project will take to pay back the money spent on it. It is based on expected cash flows and provides a measure of liquidity.

Decision rule:

- only select projects that pay back within the specified time period
- choose between options on the basis of the fastest payback

Constant annual cash flows

$$\text{Payback period} = \frac{\text{initial investment}}{\text{annual cash flow}}$$


Test your understanding 4 – Payback with constant annual cash

An expenditure of \$2 million is expected to generate net cash inflows of \$500,000 each year for the next seven years.

What is the payback period for the project?

A payback period may not be for an exact number of years. To calculate the payback in years and months you should multiply the decimal fraction of a year by 12 to get the number of months. Months should be rounded up rather than down. For example, a payback period of 3 years and 8.3 months would be rounded to 3 years and 9 months.


Test your understanding 5 – Payback in years and months

A project will involve spending \$1.8 million now. Annual cash flows from the project would be \$350,000.

What is the expected payback period?

Uneven annual cash flows

In practice, cash flows from a project are unlikely to be constant. Where cash flows are uneven, payback is calculated by working out the cumulative cash flow over the life of the project.


Test your understanding 6 – Payback with uneven cash flows

A project is expected to have the following cash flows:

Year	Cash flow
	\$000
0	(1,900)
1	300
2	500
3	600
4	800
5	500

What is the expected payback period?



Test your understanding 7 – Payback with uneven cash flows

Calculate the payback period in years and months for the following project:

Year	Cash flow \$000
0	(3,100)
1	1,000
2	900
3	800
4	500
5	500



7 Advantages and disadvantages of payback

Advantages include:

- it is simple
- it is useful in certain situations:
 - rapidly changing technology
 - improving investment conditions
- it favours quick return:
 - helps company growth
 - minimises risk
 - maximises liquidity
- it uses cash flows, not accounting profit.

Disadvantages include:

- it ignores returns after the payback period
- it ignores the timings of the cash flows
- it is subjective – no definitive investment signal
- it ignores project profitability.



In the examination, it is important that you can discuss the features of payback as an investment appraisal technique as well as being able to do the calculation.



Advantages and disadvantages of payback

Advantages

Simplicity – as a concept, it is easily understood and is easily calculated.

Rapidly changing technology – If new plant is likely to be scrapped in a short period because of obsolescence, a quick payback is essential.

Improving investment conditions – When investment conditions are expected to improve in the near future, attention is directed to those projects that will release funds soonest, to take advantage of the improving climate.

Payback favours projects with a quick return – It is often argued that these are to be preferred for three reasons:

- (1) Rapid project payback leads to rapid company growth, but in fact such a policy may lead to many profitable investment opportunities being overlooked because their payback period does not happen to be particularly swift.
- (2) Rapid payback minimises risk (the logic being that the shorter the payback period, the less there is that can go wrong). Not all risks are related to time, but payback is able to provide a useful means of assessing time risks (and only time risks). It is likely that earlier cash flows can be estimated with greater certainty.
- (3) Rapid payback maximises liquidity – but liquidity problems are best dealt with separately, through cash forecasting and planning.

Cash flows – Unlike ROCE, it uses relevant cash flows, rather than profits, and so is less likely to produce a figure distorted by assorted accounting conventions.

Disadvantages

Project returns may be ignored – cash flows arising after the payback period are ignored.

Timing ignored – cash flows are effectively categorised as pre-payback or post-payback, but no more accurate measure is made. In particular, the time value of money (discussed in chapter 3) is ignored.

Lack of objectivity – there is no objective measure as to what length of time should be set as the target payback period. Investment decisions are therefore subjective.

Project profitability is ignored – payback takes no account of the effects on business profits and periodic performance of the project, as evidenced in the financial statements. This is critical if the business is to be reasonably viewed by users of the accounts.



Test your understanding 8

Which of the following is an example of a relevant cash flow to be considered in an investment appraisal process for a new project?

- A Market research expenditure already incurred
- B Additional tax that will be paid on extra profits generated
- C Centrally-allocated overheads that are not a consequence of undertaking the project
- D Tax-allowable depreciation



Test your understanding 9

Garfield Co is considering whether to enter into a new project. The machinery which would be used to produce the goods for the contract was purchased seven years ago at a cost of \$80,000, with an estimated life of ten years. Depreciation is on a straight-line basis. The machinery has been idle for some time, and if not used on this contract would be scrapped and sold immediately for an estimated \$5,000. After use on this contract, the machinery would have no value, and would have to be dismantled and disposed of at a cost of \$1,500.

Ignoring the time value of money (covered later in Chapter 3), what is the relevant cost of the machine to the new contract?

- A \$3,500
- B \$5,000
- C \$6,500
- D \$24,500



Test your understanding 10

Which of the following is an advantage of the payback method of investment appraisal?

- A It takes account of the timing of the cash flows within the payback period
- B It uses accounting profits rather than cash flows
- C It takes account of the cash flows after the end of the payback period and therefore the total project return
- D It can be used as a screening device as a first stage in eliminating obviously inappropriate projects prior to more detailed evaluation



Test your understanding 11

Which of the following statements is not true?

- A The return on capital employed method of investment appraisal takes account of the length of the project
- B Focus on an early payback period can enhance liquidity
- C Investment risk is increased if the payback period is longer
- D Shorter term forecasts are likely to be more reliable



Test your understanding 12

Acorn Co is considering purchasing a new machine at a cost of \$110,400 that will be operated for four years, after which time it will be sold for an estimated \$9,600. Acorn uses a straight-line policy for depreciation.

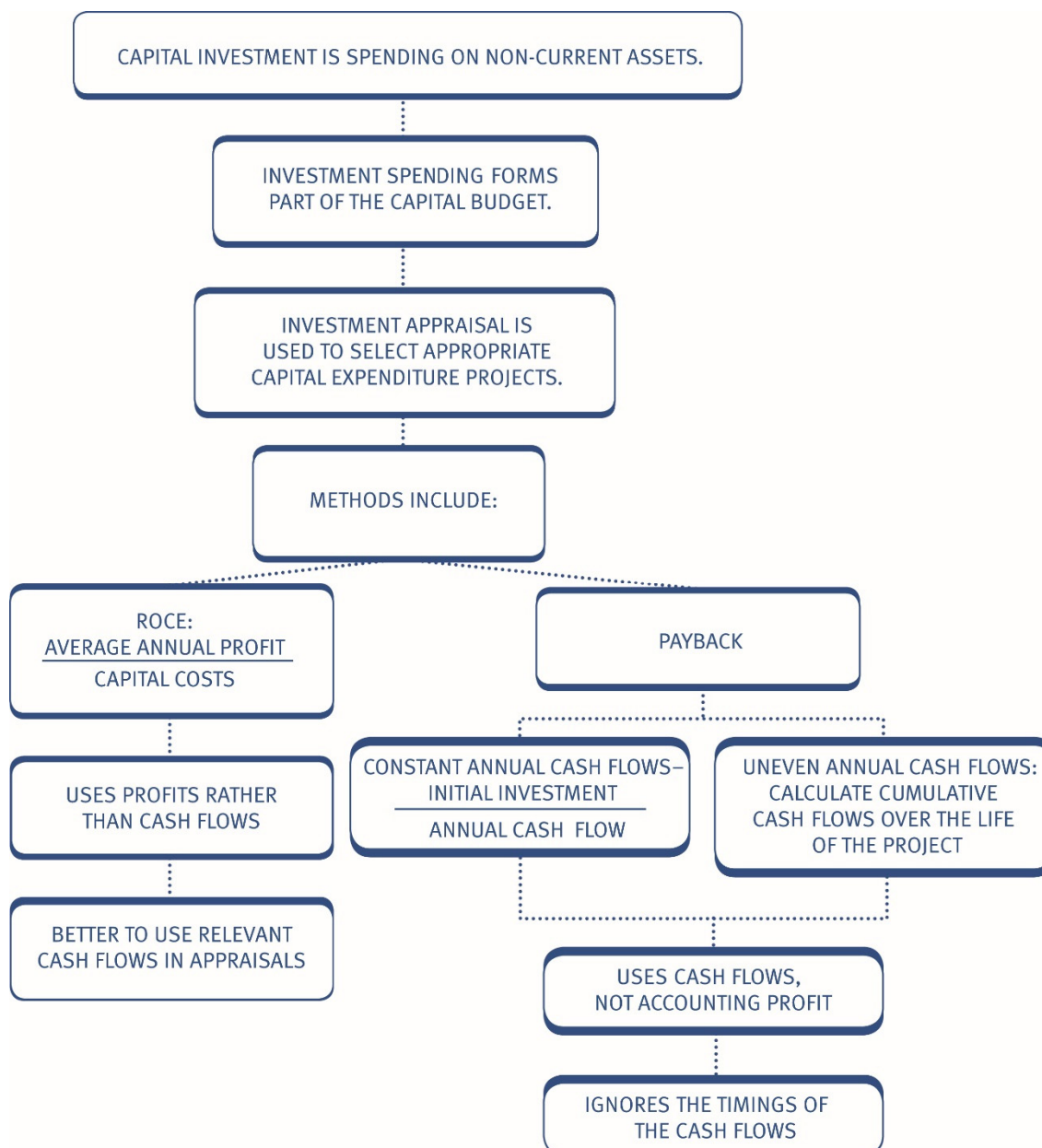
Forecast operating profits to be generated by the machine are as follows:

Year	\$
1	39,600
2	19,600
3	22,400
4	32,400

Select the payback period (PP) and the average return on capital employed (ROCE), calculated as average annual profits divided by the average investment.

- A **PP:** 2.02 years **ROCE:** 47.5%
- B **PP:** 3.89 years **ROCE:** 25.8%
- C **PP:** 3.89 years **ROCE:** 47.5%
- D **PP:** 2.02 years **ROCE:** 25.8%

8 Chapter summary





Answer to additional question – Relevant costs

The \$15,000 already spent on the feasibility study is not relevant, because it has already been spent. (It is a 'sunk cost'). Depreciation and apportioned fixed overheads are not relevant. Depreciation is not a cash flow and apportioned fixed overheads represent costs that will be incurred anyway.

	\$
Estimated profit	8,000
Add back depreciation	40,000
Add back apportioned fixed costs	25,000
Annual cash flows	73,000

The project's cash flows to be evaluated are:

Years	\$
Now (Year 0) Purchase equipment	(160,000)
1–4 Cash flow from profits	73,000 each year

Test your understanding answers

**Test your understanding 1 – ROCE**

Annual cash flows are taken to be profit before depreciation.

Average annual depreciation = $(\$110,000 - \$10,000) \div 5 = \$20,000$

Average annual profit = $\$24,400 - \$20,000 = \$4,400$

Using initial cost:

$$\begin{aligned}\text{ROCE} &= \frac{\text{Average annual profit}}{\text{Initial capital cost}} \times 100\% \\ &= \frac{\$4,400}{\$110,000} \times 100\% = 4\%\end{aligned}$$

Using average capital investment:

Average annual profits (as before) = $\$4,400$

$$\begin{aligned}\text{Average book value of assets} &= \frac{\text{Initial cost} + \text{Final scrap value}}{2} \\ &= \frac{\$110,000 + \$10,000}{2} = \$60,000\end{aligned}$$

$$\begin{aligned}\text{ROCE} &= \frac{\text{Average annual profit}}{\text{Average book value of assets}} \times 100\% \\ &= \frac{\$4,400}{\$60,000} \times 100\% = 7.33\%\end{aligned}$$

**Test your understanding 2 – ROCE**

This uses profits rather than cash flows.

Average annual inflows = $\$1,750,000 \div 7 = \$250,000$

Average annual depreciation = $(\$800,000 - \$100,000) \div 7 = \$100,000$

(A net $\$700,000$ is being written off as depreciation over 7 years.)

Average annual profit = $\$250,000 - \$100,000 = \$150,000$

The average capital invested is $(800,000 + 100,000) \div 2 = \$450,000$

(a)

$$\text{ROCE} = \frac{\text{Average annual profit}}{\text{Initial capital cost}} \times 100 = \frac{\$150,000}{\$800,000} \times 100 = 18.75\%$$

(b)

$$\text{ROCE} = \frac{\text{Average annual profit}}{\text{Average capital investment}} \times 100 = \frac{\$150,000}{\$450,000} \times 100 = 33.33\%$$



Test your understanding 3 – Relevant costs

Only the \$15,000 salary is relevant. The fixed overheads are not incremental to the decision and should be ignored.



Test your understanding 4 – Payback with constant annual cash

$$\text{Payback period} = \frac{\$2,000,000}{\$500,000} = 4 \text{ years}$$



Test your understanding 5 – Payback in years and months

$$\text{Payback} = \frac{\$1,800,000}{\$350,000} = 5.1429 \text{ years}$$

0.1429 of a year $\times$ 12 months = 1.7 months (rounded = 2 months)

The answer can therefore be stated as either:

- 5.1 years
- 5 years 2 months

assuming cash flows occur evenly throughout the year.



Test your understanding 6 – Payback with uneven cash flows

Year	Cash flow	Cumulative cash flow
	\$000	\$000
0	(1,900)	(1,900)
1	300	(1,600)
2	500	(1,100)
3	600	(500)
4	800	300
5	500	800

In the table above a column is added for cumulative cash flows for the project to date. Figures in brackets are negative cash flows.

Each year's cumulative figure is simply the cumulative figure at the start of the year plus the figure for the current year. The cumulative figure each year is therefore the expected position as at the end of that year.

Payback is between the end of Year 3 and the end of Year 4 – that is during Year 4. This is the point at which the cumulative cash flow changes from being negative to positive. If we assume a constant rate of cash flow throughout the year, we could estimate that payback will be three years plus (\$500/800) of Year 4. This is because the cumulative cash flow is minus \$500 at the start of the year and the Year 4 cash flow would be \$800.

$$\$500/800 = 0.625$$

Therefore, payback is after 3.625 years.

Payback in years and months is calculated by multiplying the decimal fraction of a year by 12 months. In this example, 0.625 years = 7.5 months (0.625×12 months), which is rounded to 8 months. So therefore, payback occurs after 3 years 8 months.

Note that if cash flows were deemed to arise at the end of the year then the payback period would be 4 years.



Test your understanding 7 – Payback with uneven cash flows

Year	Cash flow	Cumulative cash flow
	\$000	\$000
0	(3,100)	(3,100)
1	1,000	(2,100)
2	900	(1,200)
3	800	(400)
4	500	100
5	500	600

Payback is between the end of Year 3 and the end of Year 4, in other words during Year 4.

If we assume a constant rate of cash flow through the year, we could estimate that payback will be three years, plus (\$400/500) of Year 4, which is 3.8 years.

$$0.8 \text{ years} = 10 \text{ months } (0.8 \times 12)$$

We could therefore estimate that payback would be after 3 years 10 months.



Test your understanding 8

Answer B

Any cost incurred in the past (A), or any committed costs which will be incurred regardless of whether or not an investment is undertaken (C), are not relevant cash flows. They have occurred, or will occur, whatever investment decision is taken. Any costs which do not represent an actual cash flow (D) should also be ignored.



Test your understanding 9

Answer C

	\$
Loss of opportunity to gain revenue (opportunity cost)	5,000
Pay for dismantle (future, incremental cash flow)	1,500
	6,500



Test your understanding 10

Answer D

The payback method ignores both the timing of the cash flows within the payback period (A) and the cash flows after the end of the payback period (C). The payback method uses cash flows rather than accounting profits (B).



Test your understanding 11

Answer A

ROCE calculates numbers over the whole life of the project, but it does not take into account the project length. For instance, two different projects could have the same ROCE but be of completely different lengths, meaning one may be preferable to the other.



Test your understanding 12

Answer A

PP:

Depreciation must be added back to the annual profit figure to derive the annual cash flows.

Annual depreciation = $(\$110,400 - \$9,600)/4 \text{ years} = \$25,200$

Adding \$25,200 to each year's profit figure produces the following cash flows.

	Cash flow	Cumulative cash flow
	\$	\$
Initial investment	(110,400)	(110,400)
Year 1	64,800	(45,600)
Year 2	44,800	(800)
Year 3	47,600	46,800

Payback period in years = $2 + (800/47,600) = 2.02 \text{ years}$

If you selected a payback period of 3.89 years, you based your calculations on the accounting profits after the deduction of depreciation.

The calculation of the payback period should be based on cash flows.

ROCE:

Average profit = $\$(39,600 + 19,600 + 22,400 + 32,400)/4 = \$28,500$

Average investment = $\$(110,400 + 9,600)/2 = \$60,000$

ROCE = $\$(28,500/60,000) \times 100\% = 47.5\%$

If you selected a ROCE of 25.8%, you calculated the ROCE using the opening investment rather than the average investment.

Investment appraisal – Discounted cash flow techniques

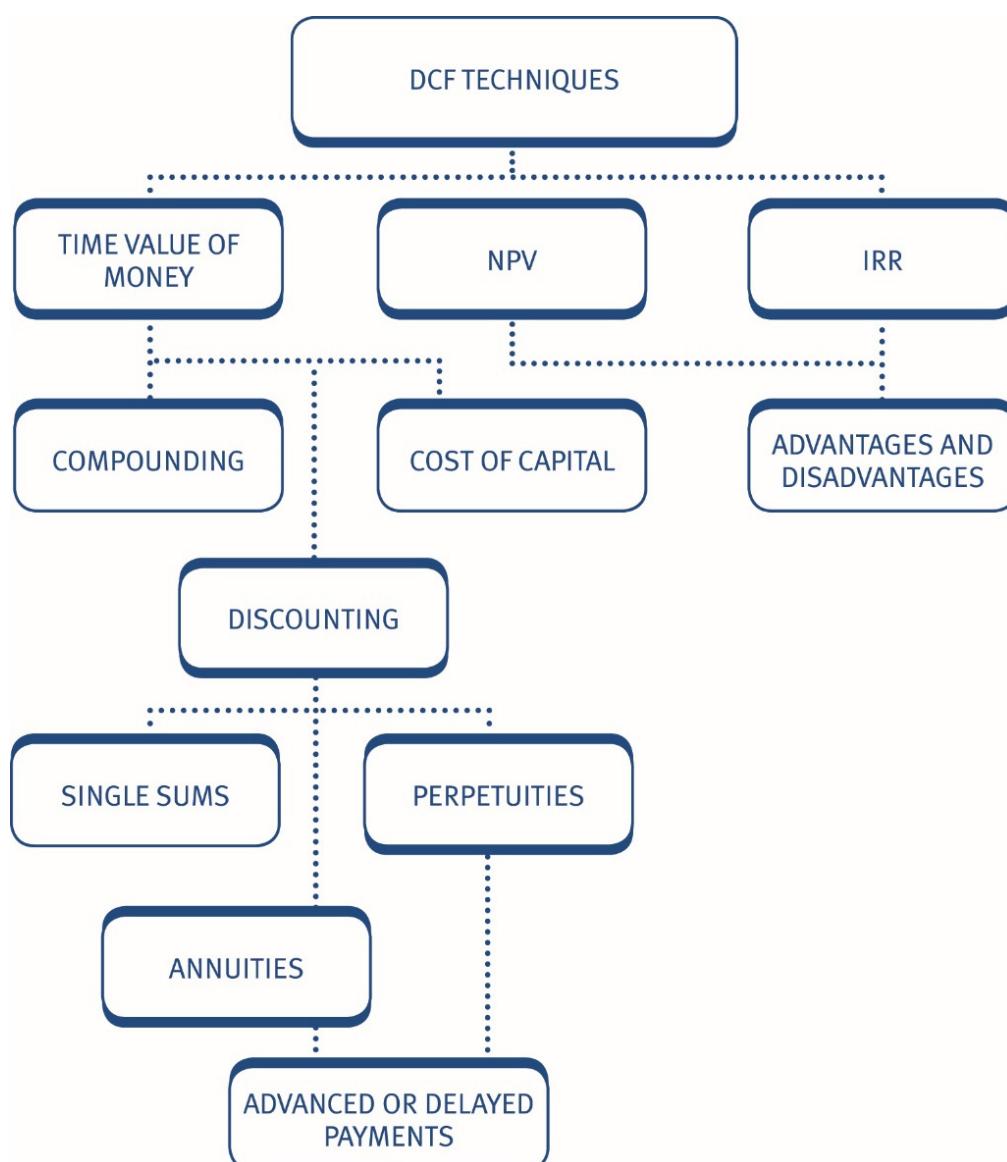
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- explain the concept of the time value of money
- calculate the future value of a sum by compounding
- calculate the present value (PV) of a single sum using a formula
- calculate the PV of a single sum using discount tables
- calculate the PV of an annuity using a formula
- calculate the PV of an annuity using annuity tables
- calculate the PV of a perpetuity using a formula
- calculate the PV of advanced annuities and perpetuities
- calculate the PV of delayed annuities and perpetuities
- explain the basic principle behind the concept of a cost of capital
- calculate the net present value (NPV) of an investment and use it to appraise the proposal
- discuss the usefulness of NPV as an investment appraisal method and its superiority over non-discounted cash flows (DCF) methods
- calculate the internal rate of return (IRR) of an investment and use it to appraise the proposal
- discuss the usefulness of IRR as an investment appraisal method and its superiority over non-DCF methods
- discuss the relative merits of NPV and IRR.



One of the PER performance objectives (PO9) is to evaluate investment and financing decisions. You evaluate investment opportunities and their consequences and advise on their costs and benefits to the organisation. Working through this chapter should help you understand how to demonstrate that objective.



1 The time value of money



Money received today is worth more than the same sum received in the future, i.e. it has a **time value**.

This occurs for three reasons:

- potential for earning interest/cost of finance
- impact of inflation
- effect of risk.



This is a key concept throughout the FM syllabus.

Discounted cash flow (DCF) techniques take account of this time value of money when appraising investments.



The time value of money

Potential for earning interest

If a capital investment is to be justified, it needs to earn at least a minimum amount of profit, so that the return compensates the investor for both the amount invested and also for the length of time before the profits are made. For example, if a company could invest \$80,000 now to earn revenue of \$82,000 in one week's time, a profit of \$2,000 in seven days would be a very good return. However, if it takes four years to earn the money, the return would be relatively much lower.

Therefore, money has a time value. It can be invested to earn interest or profits, so it is better to have \$1 now than in one year's time. This is because \$1 now can be invested for the next year to earn a return, whereas \$1 in one year's time cannot.

Alternatively, in a financing situation, paying out cash immediately leads to an immediate interest charge whereas delaying the payment to a future date delays the charge and reduces the interest cost and therefore the overall cost of making the payment.

Another way of looking at the time value of money is to say that any value of cash at a future date is worth less than the same value now.

Impact of inflation

In most countries and in most years, prices rise as a result of inflation. Therefore, funds received today will buy more than the same amount a year later, as prices will have risen in the meantime. The funds are subject to a loss of purchasing power over time.

Risk

The earlier cash flows are due to be received or paid, the more certain they are – there is less chance that events will prevent payment. Earlier cash flows are therefore considered to be safer and therefore more valuable.

2 Compounding

A sum invested today will earn interest. Compounding calculates the future or terminal value of a given sum invested today for a number of years.

To compound a sum, the figure is increased by the amount of interest it would earn over the period.

Once interest has been earned, it becomes part of the invested balance, allowing it to earn interest itself in subsequent periods.



Illustration 1 – Compounding

An investment of \$100 is to be made today. What is the value of the investment after two years if the interest rate is 10%?

Solution

		\$
Value after one year	$100 \times 1.1 =$	110
Value after two years	$110 \times 1.1 =$	121

The \$100 will be worth \$121 in two years at an interest rate of 10%.

Another way of stating this is that \$121 in two years is worth the same amount as \$100 now when the interest rate applicable over the intervening time period is 10%.

This is a fairly straightforward calculation. However, if the question asked for the value of the investment after 20 years, it would take a lot longer.

To speed up the process, we can use a formula to calculate the future value of a sum invested now. The formula is:

$$F = P(1 + r)^n$$

where F = Future value after n periods

P = Present or Initial value

r = Rate of interest per period

n = Number of periods



The terminal value is the value, in n years' time, of a sum invested now, at an interest rate of r%.



Test your understanding 1 – Compounding

You have \$5,000 to invest now for six years at an interest rate of 5% per year.

What will be the value of the investment after six years?

3 Discounting

In a potential investment project, cash flows will arise at many different points in time. To make a useful comparison of the different flows, they must all be converted to a common point in time, usually the present day, i.e. the cash flows are discounted.



Discounting a single sum



The present value (PV) is the cash equivalent now of money receivable/payable at some future date.

The PV of a future sum can be calculated using the formula:

$$P = \frac{F}{(1+r)^n} = F \times (1+r)^{-n}$$

This is just a re-arrangement of the formula we used for compounding.

$(1 + r)^{-n}$ is called the discount factor (DF).



Test your understanding 2 – Discounting a single sum

What is the PV of \$115,000 receivable in nine years' time if $r = 6\%$?

4 The cost of capital



In the above discussions, we referred to the rate of interest. There are a number of alternative terms used to refer to the rate a firm should use to take account of the time value of money:

- cost of capital
- discount rate
- required return.



Whatever term is used, the rate of interest used for discounting reflects the cost of the finance that will be tied up in the investment. You will learn how to calculate the cost of capital in a later chapter.

5 The Net Present Value (NPV)

To appraise the overall impact of a project using DCF techniques involves discounting all the relevant cash flows associated with the project back to their PV.



If we treat outflows of the project as negative and inflows as positive, the NPV of the project is the sum of the PVs of all flows that arise as a result of doing the project.



The NPV represents the surplus funds (after funding the investment) earned on the project, therefore:

- if the NPV is positive – the project is financially viable
- if the NPV is zero – the project breaks even (just returning enough money to cover the funding costs)
- if the NPV is negative – the project is not financially viable
- if the company has two or more mutually exclusive projects under consideration it should choose the one with the highest NPV
- the NPV gives the impact of the project on shareholder wealth.



What does the NPV actually mean?

Suppose, in an investment problem, we calculate the NPV of certain cash flows at 12% to be – \$97, and at 10% to be zero, and yet at 8% the NPV of the same cash flows is + \$108. Another way of expressing this is as follows.

- If the funds were borrowed at 12% the investor would be \$97 out of pocket – i.e. the investment earns a yield below the cost of capital.
- If funds were borrowed at 10% the investor would break even – i.e. the investment yields a return equal to the cost of capital.
- If funds were borrowed at 8% the investor would be \$108 in pocket – i.e. the investment earns a return in excess of the cost of capital.

In other words, a positive NPV is an indication of the surplus funds available to the investor now as a result of accepting the project.

Assumptions used in discounting

Unless the examiner tells you otherwise, the following assumptions are made about cash flows when calculating the net present value:

- all cash flows occur at the start or end of a year
- initial investments occur at T_0 (today)
- other cash flows start one year after that (T_1).

Also, note you should never include interest payments as cash flows within an NPV calculation as these are taken account of by the cost of capital.

NPV is an important technique, which you will develop further if you choose to study Advanced Financial Management.



Assumptions used in discounting

- All cash flows occur at the start or end of a year.

Although in practice many cash flows accrue throughout the year, for discounting purposes they are all treated as occurring at the start or end of a year. Note also that if today (T_0) is 01/01/20X1, the dates 31/12/20X1 and 01/01/20X2, although technically separate days, can be treated for discounting as occurring at the same point in time, i.e. at T_1 .

- Initial investments occur at once (T_0); most other cash flows start in one year's time (T_1).

In project appraisal, the investment needs to be made before the cash flows can accrue. Therefore, unless the examiner specifies otherwise, it is assumed that investments occur in advance. The first cash flows associated with running the project are therefore assumed to occur one year after the project begins, i.e. at T_1 .



Test your understanding 3 – Net present value

The cash flows for a project have been estimated as follows:

Year	\$
0	(25,000)
1	6,000
2	10,000
3	8,000
4	7,000

The cost of capital is 6%.

Convert these cash flows to present values (PVs).

Calculate the net present value (NPV) of the project to assess whether it should be undertaken.



Test your understanding 4 – Net present value

An organisation is considering a capital investment in new equipment. The estimated cash flows are as follows.

Year	Cash Flow
	\$
0	(240,000)
1	80,000
2	120,000
3	70,000
4	40,000
5	20,000

The company's cost of capital is 9%.

Calculate the NPV of the project to assess whether it should be undertaken.

6 Advantages and disadvantages of using NPV

Advantages



Theoretically, the NPV method of investment appraisal is superior to all others. This is because it:

- considers the time value of money
- is an absolute measure of return
- is based on cash flows, not profits
- considers the whole life of the project
- should lead to maximisation of shareholder wealth.

Disadvantages

- It is difficult to explain to managers
- It requires knowledge of the cost of capital
- It is relatively complex.



Advantages and disadvantages of NPV

When appraising projects or investments, NPV is considered to be superior (in theory) to most other methods. This is because it:

- considers the time value of money – discounting cash flows to PVs takes account of the impact of interest, inflation and risk over time. (See later chapters for detail on how to incorporate the effects of inflation and risk.) These significant issues are ignored by the basic methods of payback and annual rate of return (ARR or ROCE).
- is an absolute measure of return – the NPV of an investment represents the actual monetary surplus raised by the project. This allows a business to plan more effectively. Neither ARR nor payback is an absolute measure.
- is based on cash flows rather than profits – the subjectivity of profits makes them less reliable than cash flows and therefore less appropriate for decision making.
- considers the whole life of the project – methods such as payback only consider the earlier cash flows associated with the project. NPV takes account of all relevant flows associated with the project. Discounting the flows takes account of the fact that later flows are less reliable, which ARR ignores.
- should lead to maximisation of shareholder wealth. If the cost of capital reflects the investors' (shareholders and lenders) required return, then the NPV reflects the theoretical increase in the business wealth (i.e. that of the shareholders) (see later chapter on cost of capital). For a company, this is considered to be the primary objective of the business.

However, there are some potential drawbacks:

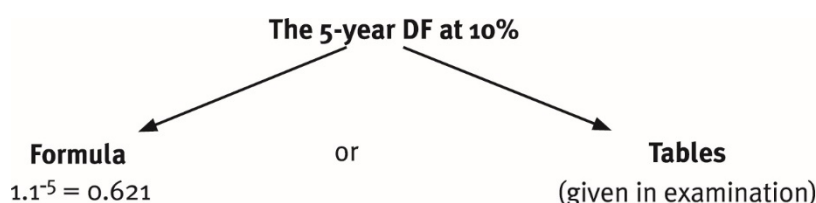
- It is difficult to explain to managers. To understand the meaning of the NPV calculated requires an understanding of discounting. The method is not as intuitive as techniques such as payback.
- It requires knowledge of the cost of capital. As we will see in a later chapter, the calculation of the cost of capital is, in practice, more complex than identifying interest rates. It involves gathering data and making a number of calculations based on that data and some estimates. The process may be deemed too protracted for the appraisal to be carried out.
- It is relatively complex. For the reasons explained above, NPV may be rejected in favour of simpler techniques.

7 Time-saving in the exam

Although you will be able to perform almost any NPV calculation using the above information, there are a number of time-saving techniques that can help you in the exam.

Present Value Tables

$(1 + r)^{-n}$ is called the discount factor (DF). In the exam, you will be provided with a Present Value table that gives the discount factors for various different discount rates over various time period. So, to find the DF, for example if $r = 10\%$ and $n = 5$, you can:



You can simply find the DF from the PV table by locating the DF at the 10% column and the 5-year row, i.e. 0.621



Test your understanding 5 – Discounting using tables

What amount should be invested now to receive \$10,000 in four years' time if $r = 8\%$ per year.

Show your workings using the formula approach and then compare this with using the present value table.

NPV function in the spreadsheet constructed response workspace

Instead of multiplying the total cash flows for each period by the discount factors for that period and then adding the results together to determine the NPV, you may wish to use the NPV spreadsheet function:

`=NPV(discount rate, values from time period 1 onwards)`

Once you have determined the total cash flows for each period, select a cell in which to enter the formula. Enter the discount rate as either a decimal or percentage and include the cash flow values from time period 1 to the end of the project life. The formula will calculate the PV of all cash flows from period 1 onwards. Then deduct the time period 0 cash outflow to complete the NPV calculation.



Discounting annuities

An annuity is a constant annual cash flow for a number of years. The PV of the entire set of annuity cash flows can be determined in a single calculation using an annuity factor.



Illustration 2 – Discounting annuities

A payment of \$1,000 is to be made every year for 3 years, the first payment occurring in one year's time. The interest rate is 10%. What is the PV of the annuity?

Solution

The PV of an annuity could be found by adding the PVs of each payment separately.

Time	Payment	DF @ 10% (from tables)	PV
	\$		\$
T ₁	1,000	0.909	909
T ₂	1,000	0.826	826
T ₃	1,000	0.751	751
		2.486	2,486

However, you can see from the table that the sum of all the DF is 2.486.

Therefore, the PV can be found more quickly:

$$\$1,000 \times 2.486 = \$2,486.$$



The **annuity factor** (AF) is the name given to the sum of the individual DF.

The PV of an annuity can therefore be quickly found using the formula:

$$PV = \text{Annual cash flow} \times AF$$

Like with calculating a discount factor, the AF can be found using a formula.

The formula is:

$$AF = \frac{1 - (1+r)^{-n}}{r}$$



Test your understanding 6 – Discounting annuities

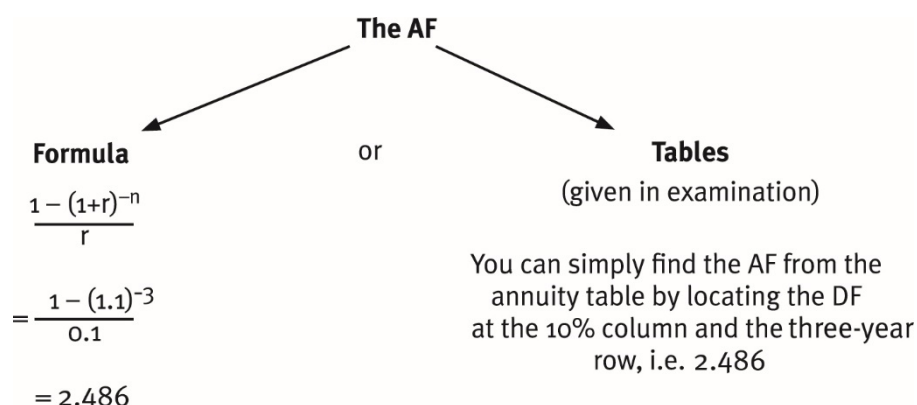
A payment of \$3,600 is to be made every year for seven years, the first payment occurring in one year's time. The interest rate is 8%.

What is the PV of the annuity?

Annuity Tables

Just as you are provided with a Present Value Table in the exam showing pre-calculated discount factors, you will also be provided with an Annuity Table (sometimes referred to as a Cumulative Present Value Table), which provides pre-calculated annuity factors for various different discount rates over various periods.

So again, you have a choice. For example, for a three-year annuity at 10%:



Note: There might be a small difference due to roundings.

The tables should not be used as a substitute for knowing how to use the formula. Remember, the tables only cover a small range of discount rates and time periods and you may be required to calculate a discount factor or annuity factor for variables outside of this range.



Test your understanding 7 – Discounting annuities using tables

A payment of \$11,400 is to be made every year for 13 years, the first payment occurring in one year's time. The interest rate is 5%.

What is the PV of the annuity?

Show your workings using both the formula approach and the annuity table.

Discounting perpetuities



A perpetuity is an annual cash flow that occurs forever.



It is often described by examiners as a cash flow continuing 'for the foreseeable future'.

As with annuities, the PV of the entire set of cash flows can be determined in a single calculation using a perpetuity factor.

The PV of a **perpetuity** is found using the formula

$$PV = \frac{\text{Cash flow}}{r}$$

or

$$PV = \text{cash flow} \times \frac{1}{r}$$

$\frac{1}{r}$ is known as the perpetuity factor.

The PV of a **growing perpetuity** is found using the formula

$$PV = \text{cash flow at } T_1 \times \frac{1}{r - g}$$

$\frac{1}{r - g}$ is known as the perpetuity factor with growth



Test your understanding 8

Calculate the present value of the following, assuming a discount rate of 10%.

- (1) \$3,000 received in one year's time and for ever
- (2) \$3,000 received in one year's time, then growing by 2% per year in perpetuity

Advanced and delayed annuities and perpetuities

The use of annuity factors and perpetuity factors both assume that the first cash flow will be occurring in one year's time. Annuity or perpetuity factors will therefore discount the cash flows back to give the value one year before the first cash flow arose. For standard annuities and perpetuities, this gives the present (T_0) value since the first cash flow started at T_1 .

Be careful: if this is not the case, you will need to adjust your calculation.

Advanced annuities and perpetuities

Some regular cash flows may start now (at T_0) rather than in one year's time (T_1).

Calculate the PV by ignoring the payment at T_0 when considering the number of cash flows and then adding one to the annuity or perpetuity factor.

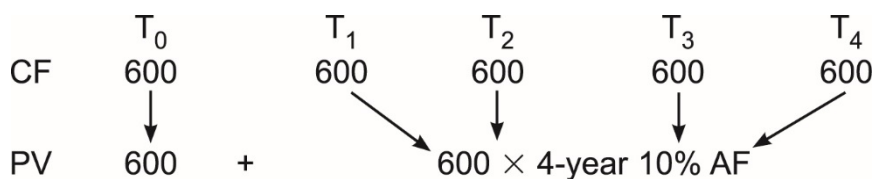


Advanced annuities – An illustration

A 5-year \$600 annuity is starting today. Interest rates are 10%. Find the PV of the annuity.

Solution

This is essentially a standard 4-year annuity with an additional payment at T_0 . The PV could be calculated as follows:



$$PV = 600 + 600 \times 3.17 = 600 + 1902 = \$2,502$$

The same answer can be found more quickly by adding 1 to the four-year AF:

$$PV = 600 \times (1 + 3.17) = 600 \times 4.17 = \$2,502.$$



Advanced perpetuities – An illustration

A perpetuity of \$2,000 is due to commence immediately. The interest rate is 9%. What is the PV?

Solution

This is essentially a standard perpetuity with an additional payment at T_0 .

The PV could be calculated as follows:

T_0	T_1	T_2	T_3	$T_4 \dots$
2,000	$2,000 \rightarrow \infty$			

$$PV(2,000) + (2,000 \times 9\% \text{ perpetuity formula})$$

Again, the same answer can be found more quickly by adding 1 to the perpetuity factor.

$$2,000 \times \left(1 + \frac{1}{0.09}\right) = 2,000 \times 12.11 = \$24,222$$



Additional question – Advanced annuities and perpetuities

Find the present value of the following cash flows:

- (1) A fifteen-year annuity of \$300 starting at once. Interest rates are 6%.
- (2) A seven-year annuity of \$450 starting immediately. Interest rates are 11%
- (3) A perpetuity of \$33,000 commencing immediately. Interest rates are 22%
- (4) A perpetuity of \$14,000 starting at once. Interest rates are 12.5%

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

Delayed annuities and perpetuities

Some regular cash flows may start later than T_1 .



These are dealt with by:

- (1) applying the appropriate factor to the cash flow as normal
- (2) discounting your answer back to T_0 .



Delayed annuities – An illustration

For delayed cash flows, applying the standard annuity factor will find the value of the cash flows one year before they began, which in this illustration is T_2 . To find the PV, an additional calculation is required – the value must be discounted back to T_0 .

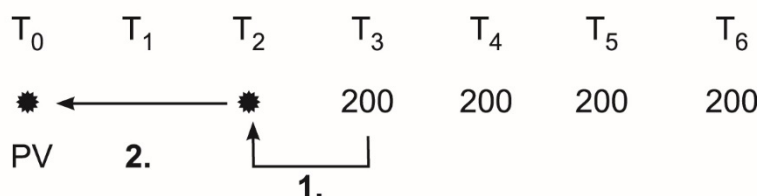
Illustration – Delayed annuities and perpetuities

What is the PV of \$200 incurred each year for four years, starting in three years' time, if the discount rate is 5%?

Solution

Method:

A four-year annuity starting at T_3
(1-4)



Step 1 Discount the annuity as usual

$$200 \times 4 \text{ yr } 5\% \text{ AF} = 200 \times 3.546 = 709.2$$

Note that this gives the value of the annuity at T_2

Step 2 Discount the answer back to T_0

$$709.2 \times 2 \text{ yr } 5\% \text{ DF} = 709.2 \times 0.907 = \$643$$

Care must be taken to discount back the appropriate number of years. The figure here was discounted back two years because the first step gave the value at T_2 . It can help to draw a timeline as above and mark on the effect of the first step (as shown with a 1. here) to help you remember.



Additional question – Delayed annuities and perpetuities

Find the present value of the following cash flows:

- (1) A fifteen-year annuity of \$300 starting at T_3 . Interest rates are 6%.
- (2) A seven-year annuity of \$450 starting in five years' time. Interest rates are 11%
- (3) A perpetuity of \$33,000 commencing at T_2 . Interest rates are 22%
- (4) A perpetuity of \$14,000 starting in ten years' time. Interest rates are 12.5%
- (5) A perpetuity of \$2,000 starting in six years' time, growing at 3% per year. Interest rates are 10%

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

Computer based exam (CBE) time saving techniques

In the constructed response spreadsheets workspace, use can be made of spreadsheet functions to speed up the construction of answers.

For instance, clicking on the bottom right corner of a cell and dragging the cursor across other cells will copy the contents of the original cell into the other cells. This is particularly useful for copying formulae, such as totals.

Formulae such as SUM, POWER and SQRT are available within the spreadsheet workspace. Make sure you are comfortable with the use of these. For guidance on the formulae available, see:

http://www.accaglobal.com/content/dam/ACCA_Global/Students/exam/Guide%20to%20CBES_FINAL.PDF

8 The Internal Rate of Return (IRR)



The IRR is another project appraisal method using DCF techniques.

The IRR represents the discount rate at which the NPV of an investment is zero. As such, it represents a breakeven cost of capital.

Decision rule:

- projects should be accepted if their IRR is greater than the cost of capital.



IRR

Using the NPV method, PVs are calculated by discounting cash flows at a given cost of capital, and the difference between the PV of costs and the PV of benefits is the NPV. In contrast, the IRR method of DCF analysis is to calculate the exact DCF rate of return that the project is expected to achieve.

If an investment has a positive NPV, it means it is earning more than the cost of capital. If the NPV is negative, it is earning less than the cost of capital. This means that if the NPV is zero, it will be earning exactly the cost of capital.

Conversely, the percentage return on the investment must be the rate of discount or cost of capital at which the NPV equals zero. This rate of return is called the IRR or the DCF yield and if it is higher than the target rate of return then the project is financially worth undertaking.

Calculating the IRR using linear interpolation

The steps in linear interpolation are:

- (1) Calculate two NPVs for the project at two different costs of capital
- (2) Use the following formula to find the IRR:

$$\text{IRR} = L + \left[\frac{N_L}{N_L - N_H} \right] \times (H - L)$$

where:

L = Lower cost of capital

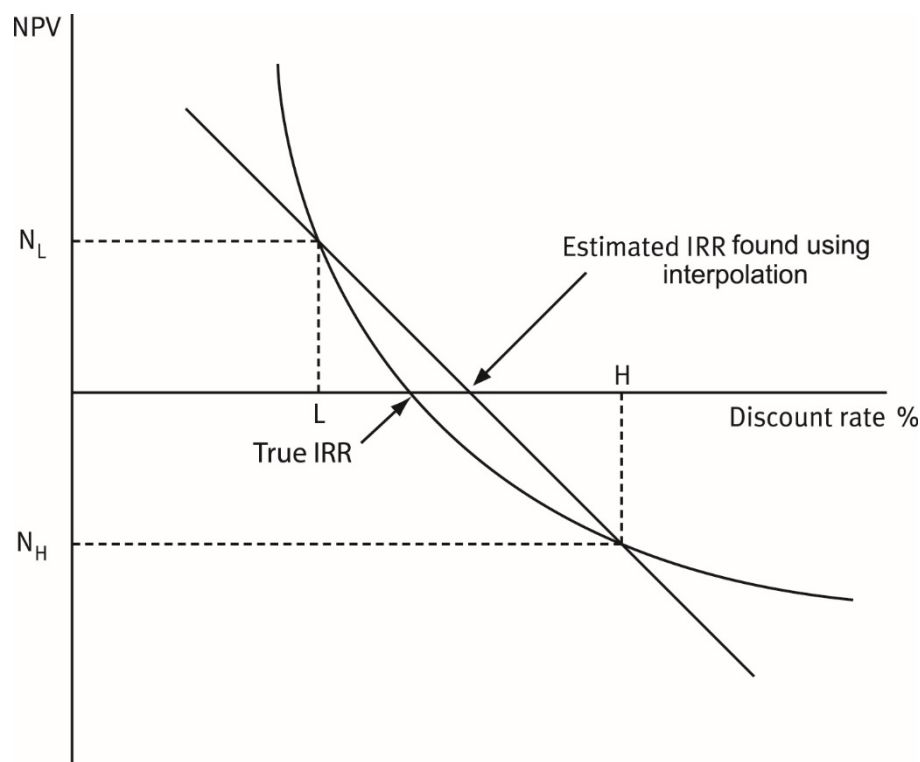
H = Higher cost of capital

N_L = NPV at lower cost of capital

N_H = NPV at higher cost of capital.

(Note that if N_H is negative, its value will end up being added to that of N_L on the bottom of the equation).

The diagram below shows the IRR as estimated by the formula.



Calculating the IRR

Linear interpolation method

The IRR may be calculated by a linear interpolation, i.e. by assuming a linear relationship between the NPV and the discount rate. Plotting a graph would give an approximate IRR, but the same point can also be found using a formula.

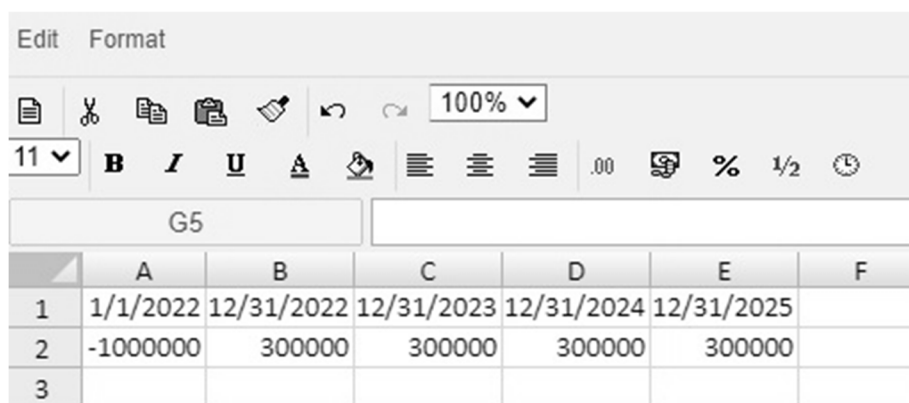
Step 1 Calculate two NPVs for the project at two different costs of capital. You can choose any costs of capital and get a fair result. However, it helps to find two costs of capital for which the NPV is close to 0, because the IRR will be a value close to them. Ideally, you should use one cost of capital where the NPV is positive and the other cost of capital where the NPV is negative, although this is not essential. You should not waste time in the exam trying to achieve this.

Step 2 Once the two NPVs have been calculated, they and their associated costs of capital can be used to calculate the IRR. In other words, we can estimate the IRR by finding the point where a line joining these points would cross the x-axis (the point where the NPV is zero) in a graph plotting the project NPV against various discount rates.

Spreadsheet function method

In the section C workspace, you may find it quicker and easier to use the IRR function.

Consider a project that has an initial cash outflow of \$1m, followed by four years of annual cash inflows of \$0.3m at equal annual intervals. The data may be entered into the constructed response workspace as follows:



The screenshot shows a spreadsheet with a toolbar at the top containing icons for editing and formatting. Below the toolbar is a table with the following data:

	A	B	C	D	E	F
1	1/1/2022	12/31/2022	12/31/2023	12/31/2024	12/31/2025	
2	-1000000	300000	300000	300000	300000	
3						

The IRR formula available for use in the exam is:

=IRR(values)

To calculate the IRR for the data above, the formula would be as follows:

=IRR(A2:E2)

which would calculate as 0.077138 or 7.7%

Note that the values used in the formula when dealing with a more complex set of cash flows, such as for a large investment project, should be the total net cash flows for each time period, i.e. those that would be multiplied by the discount factors when calculating an NPV.

If there is more than one cash flow for a time period, put the total of the net cash flows for that time period into one cell. The formula does not allow for multiple cells to be used for the same time period.

Also note that when dealing with an annuity, to use the formula each time period's cash flow must be shown individually – there is no short-cut method.



Test your understanding 9 – IRR

A potential project's predicted cash flows give an NPV of \$50,000 at a discount rate of 10% and -\$10,000 at a rate of 15%.

Calculate the IRR using linear interpolation.



For the examination, the choice of rates used to estimate the IRR using linear interpolation is less important than your ability to perform the calculation to estimate it.



Test your understanding 10 – IRR

A business undertakes high-risk investments and requires a minimum expected rate of return of 17% per year on its investments. A proposed capital investment has the following expected cash flows:

Year	\$
0	(50,000)
1	18,000
2	25,000
3	20,000
4	10,000

Required:

- (1) Calculate the NPV of the project if the cost of capital is 15%.
- (2) Calculate the NPV of the project if the cost of capital is 20%.
- (3) Use the NPVs you have calculated to estimate the IRR of the project.
- (4) Recommend, on financial grounds alone, whether this project should go ahead.
- (5) Using a spreadsheet, recalculate the IRR using the =IRR formula and compare the answer to your answer in part (3)



Calculating the IRR with annuity cash flows

There is a simpler technique available, using annuity tables, if the project cash flows are annuities.

- (1) Find the cumulative DF as Initial investment $\div$ Annual inflow
- (2) Find the life of the project, n .
- (3) Look along the n year row of the cumulative DF until the closest value is found.
- (4) The column in which this figure is found is the IRR.

Illustration – Calculating IRR of a project with even cash flows

Find the IRR of a project with an initial investment of \$1.5 million and three years of inflows of \$700,000 starting in one year.

Solution

NPV calculation:

		Cash flow	DF (c) %	PV
		\$000		\$000
Time				
0	Investment	(1,500)	1	(1,500)
1–3	Inflow	700	(b)	(a)
NPV				Nil

- The aim is to find the discount rate (c) that produces an NPV of nil.
- Therefore, the PV of inflows (a) must equal the PV of outflows, \$1,500,000.
- If the PV of inflows (a) is to be \$1,500,000 and the size of each inflow is \$700,000, the DF required (b) must be $1,500,000 \div 700,000 = 2.143$.
- The discount rate (c) for which this is the 3-year factor can be found by looking along the 3-year row of the cumulative DFS shown in the annuity table.
- The figure of 2.140 appears under the 19% column suggesting an IRR of 19% is the closest.

**Calculating the IRR with perpetuities**

$$\text{IRR of a perpetuity} = \frac{\text{Annual inflow}}{\text{Initial investment}} \times 100$$

Illustration – Calculating IRR where cash flows are perpetuities

Find the IRR of an investment that costs \$20,000 and generates \$1,600 for an indefinitely long period.

Solution

$$\text{IRR} = \frac{\text{Annual inflow}}{\text{Initial investment}} \times 100 = \frac{\$1,600}{\$20,000} \times 100 = 8\%$$



Test your understanding 11 – IRR with even cash flows

Find the IRR of an investment of \$50,000 if the inflows are:

- (a) \$5,000 in perpetuity
- (b) \$8,060 for eight years.

9 Advantages and disadvantages of IRR

Advantages

The IRR has a number of benefits, e.g. it:

- considers the time value of money
- is a percentage and therefore easily understood
- uses cash flows not profits
- considers the whole life of the project
- means a firm selecting projects where the IRR exceeds the cost of capital should increase shareholders' wealth

Disadvantages

- It is not a measure of absolute profitability.
- Interpolation only provides an estimate. An accurate estimate requires the use of a spreadsheet programme.
- It is fairly complicated to calculate, unless using a spreadsheet.
- Non-conventional cash flows may give rise to multiple IRRs, which means the interpolation method can't be used.
- Contains an inherent assumption that cash returned from the project will be re-invested at the project's IRR, which may be unrealistic.



Advantages and disadvantages of IRR

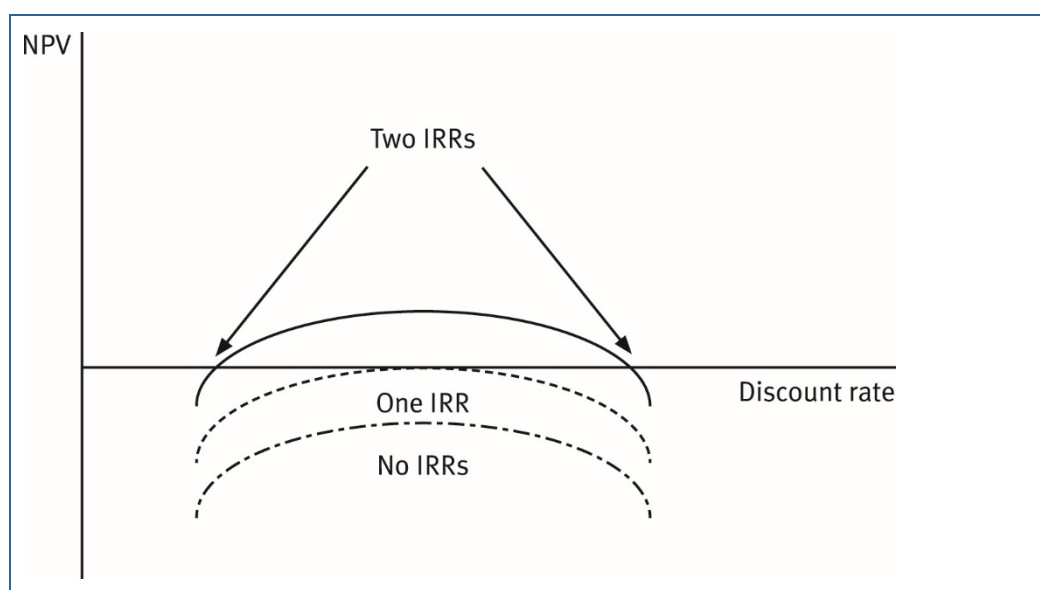
Advantages:

- IRR considers the time value of money. The current value earned from an investment project is therefore more accurately measured. As previously discussed, this is a significant improvement over the basic methods.
- IRR is a percentage and therefore easily understood. Although managers may not completely understand the detail of the IRR, the concept of a return earned is familiar and the IRR can be simply compared with the required return of the organisation.
- IRR uses cash flows, not profits. These are less subjective, as discussed above.

- IRR considers the whole life of the project rather than ignoring later flows (which would occur with payback).
- A firm selecting projects where the IRR exceeds the cost of capital should increase shareholders' wealth. This holds true provided the project cash flows follow the standard pattern of an outflow followed by a series of inflows, as in the investment examples above.

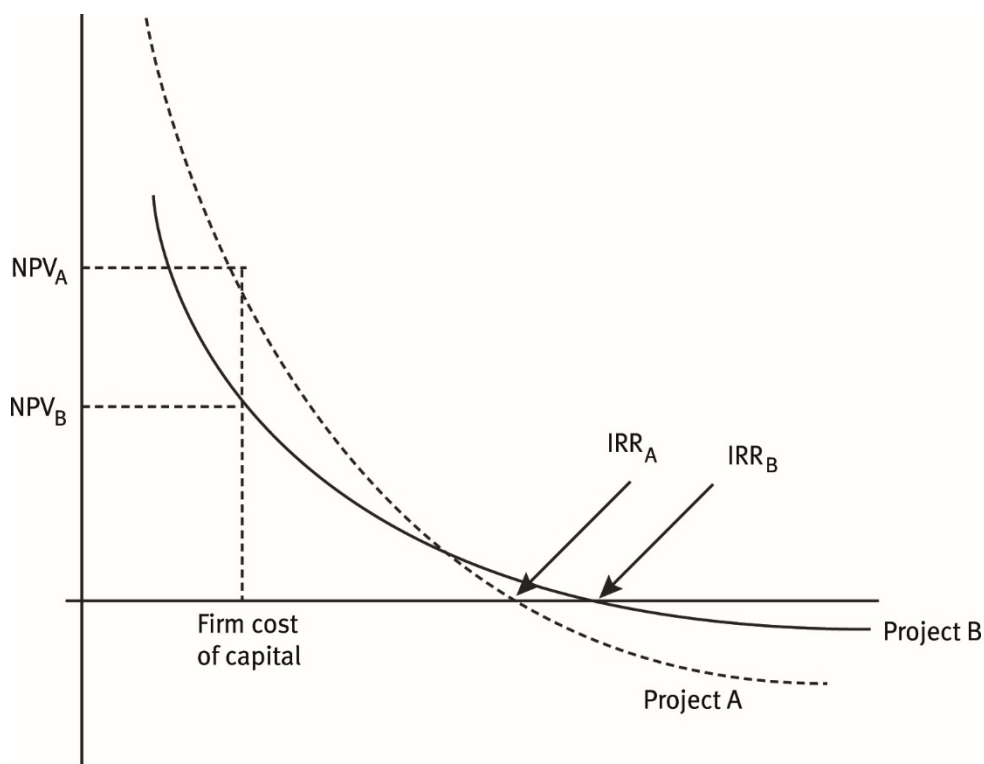
However, there are a number of difficulties with the IRR approach:

- It is not a measure of absolute profitability. A project of \$1,000 invested now and paying back \$1,100 in a year's time has an IRR of 10%. If a company's required return is 6%, then the project is viable according to the IRR rule, but most businesses would consider the absolute return of \$100 too small to be worth the investment.
- Interpolation only provides an estimate (and an accurate estimate requires the use of a spreadsheet programme). The cost of capital calculation itself is also only an estimate and if the margin between required return and the IRR is small, this lack of accuracy could actually mean the wrong decision is taken.
- For example, if the cost of capital is found to be 8% (but is actually 8.7%) and the project IRR is calculated as 9.2% (but is actually 8.5%) the project would be wrongly accepted. Note that where such a small margin exists, the project's success would be considered to be sensitive to the discount rate (see later chapter on risk). Further analysis may then be considered before the project is accepted/rejected.
- Non-conventional cash flows may give rise to no IRR or multiple IRRs. For example, a project with an outflow at T_0 and T_2 but income at T_1 could, depending on the size of the cash flows, have a number of different profiles on a graph (see below). Even where the project does have one IRR, it can be seen from the graph that the decision rule would lead to the wrong result as the project does not earn a positive NPV at any cost of capital.
- Use of the IRR contains an inherent assumption that once cash starts flowing in from the investment it will be reinvested at the same level as the project's IRR. This may be an unrealistic assumption. For instance, a business may have a cost of capital of 10% and has undertaken a project with an IRR of 20%. The assumption in the IRR calculation is that the cash inflows generated by the project will be reinvested at an equally beneficial level of return as this project, whereas the cash is more likely to be invested in projects that return at or above 10%.



10 NPV versus IRR

Both NPV and IRR are investment appraisal techniques that discount cash flows and are superior to the basic techniques discussed in the previous chapter. However only NPV can be used to distinguish between two mutually-exclusive projects, as the diagram below demonstrates:



The profile of project A is such that it has a lower IRR and applying the IRR rule would prefer project B. However, in absolute terms, A has the higher NPV at the company's cost of capital and should therefore be preferred.

NPV is therefore the better technique for choosing between projects.



The advantage of NPV is that it tells us the absolute increase in shareholder wealth as a result of accepting the project, at the current cost of capital. The IRR simply tells us how far the cost of capital could increase before the project would not be worth accepting.



Test your understanding 12 – Appraisal techniques

Paprika Co is contemplating three available investment opportunities, the cash flows of which are given below.

Project	Initial investment	Cash flow				
		Y1	Y2	Y3	Y4	Y5
	\$000	\$000	\$000	\$000	\$000	\$000
E	(125)	50	50	50	50	
F	(120)	15	15	15	15	200
G	(170)	120	80			

In each case, the initial investment represents the purchase of plant and equipment whose realisable value will be 20% of initial cost, receivable in addition to the above flow at the end of the life of each project.

Required:

For each of the three projects:

- Calculate the accounting rate of return (based on the average investment method)
- Calculate the payback period
- Calculate the net present value using a discount rate of 10%
- Calculate the internal rate of return



Test your understanding 13

In a review of a project with a large initial cash outflow followed by a few years of cash inflows that has a positive NPV, the cash inflows are shifted to 1 year later than originally predicted and the cost of capital used for discounting the project is reduced by 1%.

What are the effects of these changes on the project NPV?

- A cash flows – lower NPV, cost of capital – lower NPV
- B cash flows – lower NPV, cost of capital – higher NPV
- C cash flows – higher NPV, cost of capital – lower NPV
- D cash flows – higher NPV, cost of capital – higher NPV



Test your understanding 14

Which of the following statements about net present value (NPV) and internal rate of return (IRR) methods are correct?

- (i) An investment with a positive NPV is financially viable
 - (ii) NPV is a superior method of investment appraisal to IRR
 - (iii) The graph of NPV against discount rate has a negative slope for most projects
 - (iv) NPV is the present value of expected future net cash receipts less the cost of the investment
- A (i) and (ii) only
 - B (ii) and (iv) only
 - C (i), (ii) and (iii) only
 - D (i), (ii), (iii) and (iv)



Test your understanding 15

Which of the following is not an advantage of the IRR?

- A It considers the whole life of the project
- B It uses cash flows not profits
- C It is a measure of absolute return
- D It considers the time value of money



Test your understanding 16

Arbury Co has made an investment with a net present value (NPV) of \$42,000 at 10% and an NPV of (\$22,000) at 20%.

What is the internal rate of return of the project?

- A 31.0%
- B 16.6%
- C 15.0%
- D 13.4%



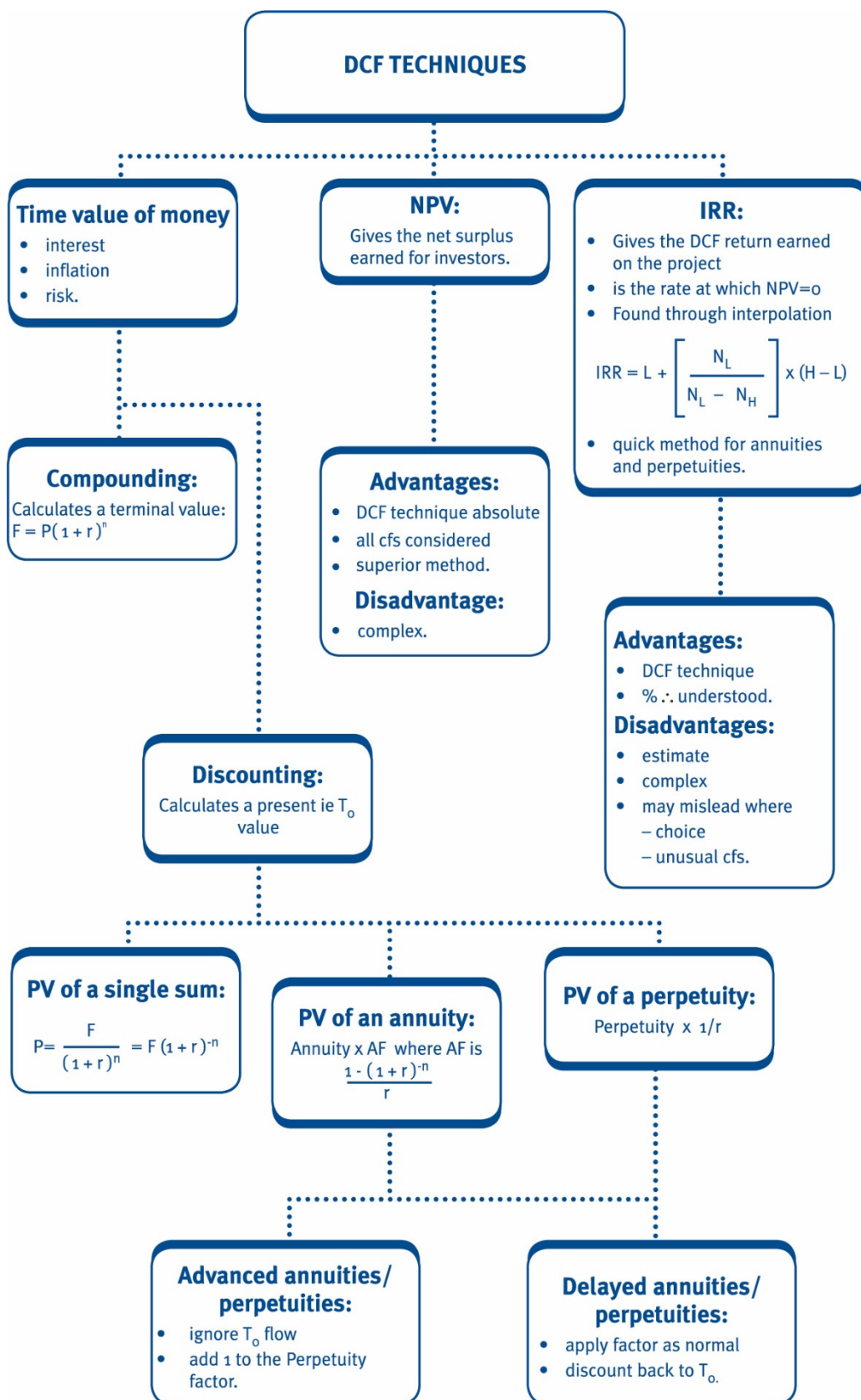
Test your understanding 17

A company is considering a two-year project, which has two annual internal rates of return, namely 10% and 25%. The sum of the undiscounted cash flows is positive.

The project will necessarily have a positive net present value, when the annual cost of capital is

- A More than 25%
- B More than 10%
- C Between 10% and 25%
- D Less than 25%

11 Chapter summary





Answer to additional question – Advanced annuities & perpetuities

- (1) This is a standard 14-year annuity with one additional payment at T_0 .
Step 1: Look up the 14-year AF : $AF = 9.295$
Step 2: Add 1 : $9.295 + 1 = 10.295$
Step 3: Calculate the PV : $300 \times 10.295 = \$3,088.50$
- (2) This is a standard 6-year annuity with one additional payment at T_0 .
Step 1: Look up the 6-year AF : $AF = 4.231$
Step 2: Add 1 : $4.231 + 1 = 5.231$
Step 3: Calculate the PV : $450 \times 5.231 = \$2,353.95$
- (3) This is simply a standard perpetuity with one additional payment at T_0 .
Step 1: Calculate the perpetuity factor : $1/0.22 = 4.545$
Step 2: Add 1 : $4.545 + 1 = 5.545$
Step 3: Calculate the PV : $33,000 \times 5.545 = \$182,985$
- (4) This is simply a standard perpetuity with one additional payment at T_0 .
Step 1: Calculate the perpetuity factor : $1/0.125 = 8$
Step 2: Add 1 : $8 + 1 = 9$
Step 3: Calculate the PV : $14,000 \times 9 = \$126,000$



Answer to additional question – Delayed annuities & perpetuities

- (1) A fifteen-year annuity of \$300 starting at T_3 .
Step 1: Look up the 15-year AF : $AF = 9.712$
Step 2: Discount the annuity as usual : $300 \times 9.712 = \$2,913.60$
 Note that this gives the value of the annuity at T_2
Step 3: Discount the answer back to T_0 : $\$2,913.60 \times 0.890 = \$2,593.10$
- (2) A seven-year annuity of \$450 starting in five years' time.
Step 1: Look up the 7-year AF : $AF = 4.712$
Step 2: Discount the annuity as usual : $450 \times 4.712 = \$2,120.40$
 Note that this gives the value of the annuity at T_4
Step 3: Discount the answer back to T_0 : $2,120.40 \times 0.659 = \$1,397.34$

- (3) A perpetuity of \$33,000 commencing at T_2 .

Step 1: Calculate the perpetuity factor : $1/0.22 = 4.545$

Step 2: Discount the perpetuity as usual : $33,000 \times 4.545 = \$149,985$

Note that this gives the value of the perpetuity at T_1

Step 3: Calculate the PV : $149,985 \times 1.22^{-1} = \$122,939$

Note: can't use figures from the present value table, as 22% is higher than the table values.

- (4) A perpetuity of \$14,000 starting in ten years' time.

Step 1: Calculate the perpetuity factor: $1/0.125 = 8$

Step 2: Discount the perpetuity as usual : $14,000 \times 8 = \$112,000$

Note that this gives the value of the perpetuity at T_9

Step 3: Calculate the PV : $112,000 \times 0.346 = \$38,752$

- (5) A perpetuity of \$2,000 starting in six years' time, growing at 3% per year.

Step 1: Calculate the perpetuity factor, with growth : $1/(0.10 - 0.03) = 14.286$

Step 2: Discount the perpetuity as usual : $2,000 \times 14.286 = \$28,572$

Note that this gives the value of the perpetuity at T_5

Step 3: Calculate the PV : $28,572 \times 0.621 = \$17,743$

Test your understanding answers

**Test your understanding 1**

$$\begin{aligned}
 F &= \$5,000 \times (1 + 0.05)^6 \\
 &= \$5,000 \times 1.3401 \\
 &= \$6,700
 \end{aligned}$$

**Test your understanding 2**

$$PV = F \times (1 + r)^{-n} = \$115,000 \times (1 + 0.06)^{-9} = \$68,068$$

**Test your understanding 3**

Year	Cash flow \$	DF at 6%	PV \$
0	(25,000)	1.000	(25,000)
1	6,000	0.943	5,658
2	10,000	0.890	8,900
3	8,000	0.840	6,720
4	7,000	0.792	5,544
			1,822

The NPV of the project is positive at \$1,822. The project should therefore be accepted.

**Test your understanding 4**

Year	Cash flow \$	DF at 9%	PV \$
0	(240,000)	1.000	(240,000)
1	80,000	0.917	73,360
2	120,000	0.842	101,040
3	70,000	0.772	54,040
4	40,000	0.708	28,320
5	20,000	0.650	13,000
NPV			29,760

The PV of cash inflows exceeds the PV of cash outflows by \$29,760, which means that the project will earn a DCF return in excess of 9%, i.e. it will earn a surplus of \$29,760 after paying the cost of financing. It should therefore be undertaken.

**Test your understanding 5 – Discounting using tables**

The amount to be invested is the PV of the future sum.

$$P = \$10,000 \times (1.08)^{-4} = \$7,350$$

$$P = \$10,000 \times 0.735 = \$7,350 \text{ (using tables).}$$

**Test your understanding 6 – Discounting annuities**

Using the formula:

$$\frac{1 - (1 + r)^{-n}}{r} = \frac{(1 - 1.08)^{-7}}{0.08} = 5.206$$

$$\$3,600 \times 5.206 = \$18,741.60$$

**Test your understanding 7 – Discounting annuities using tables**

Using the formula:

$$\frac{1 - (1 + r)^{-n}}{r} = \frac{1 - (1.05)^{-13}}{0.05} = 9.394$$

$$\$11,400 \times 9.394 = \$107,091.60$$

From the annuity factor tables, the AF = 9.394.

**Test your understanding 8 – Discounting perpetuities**

(1)

$$PV = \frac{\$3,000}{0.1} = \$30,000$$

(2)

$$PV = \frac{\$3,000}{0.1 - 0.02} = \$37,500$$

**Test your understanding 9 – IRR**

$$IRR = 10\% + \frac{50,000}{50,000 - (-10,000)} \times (15\% - 10\%) = 14.17\%$$

**Test your understanding 10 – IRR**

Year	Cash flow	DF at 15%	PV at 15%	DF at 20%	PV at 20%
\$		\$		\$	
0	(50,000)	1.000	(50,000)	1.000	(50,000)
1	18,000	0.870	15,660	0.833	14,994
2	25,000	0.756	18,900	0.694	17,350
3	20,000	0.658	13,160	0.579	11,580
4	10,000	0.572	5,720	0.482	4,820
NPV			3,440		(1,256)

The IRR is above 15% but below 20%.

Using the interpolation method:

(1) The NPV is + 3,440 at 15%.

(2) The NPV is – 1,256 at 20%.

(3) The estimated IRR is therefore:

$$\begin{aligned}
 IRR &= 15\% + \frac{3,440}{(3,440 - (-1,256))} \times (20 - 15)\% \\
 &= 15\% + 3.7\% \\
 &= 18.7\%
 \end{aligned}$$

(4) The project is expected to earn a DCF return in excess of the target rate of 17%, so on financial grounds (ignoring risk) it is a worthwhile investment.

- (5) Using a spreadsheet, the IRR would calculate to 18.6%. This is a more accurate figure than calculated using linear interpolation. Either method is appropriate to use in the exam.



Test your understanding 11 – IRR with even cash flows

(a)

$$\text{IRR} = \frac{\text{Annual inflow}}{\text{Initial investment}} \times 100 = \frac{\$5,000}{\$50,000} \times 100 = 10\%$$

(b) NPV calculation

		Cash flow	DF(c) %	PV
Time		\$		\$
0	Investment	(50,000)	1	(50,000)
1–8	Inflow	8,060	(b)	(a)
NPV				Nil

- The aim is to find the discount rate (c) that produces an NPV of nil.
- Therefore, the present value of inflows (a) must equal the present value of outflows, \$50,000.
- If the present value of inflows (a) is to be \$50,000 and the size of each inflow is \$8,060, the discount factor required must be $50,000 \div 8,060 = 6.203$.
- The discount rate (c) for which this is the 8-year factor can be found by looking along the 8-year row of the cumulative discount factors shown in the annuity table.
- The figure of 6.210 appears under the 6% column suggesting an IRR of 6% is the closest.



Test your understanding 12 – Appraisal techniques

(a) Accounting rate of return

For each project the ARR has been calculated as:

$$\text{ARR} = \frac{(\text{Total cash flows} - \text{depreciation}) \div \text{project life}}{(\text{Initial investment} + \text{scrap value}) \div 2} \times 100\%$$

and depreciation has been calculated as initial investment – scrap value.

Project E

$$ARR = \frac{(\$200k - \$100k) \div 4}{(\$125k + \$25k) \div 2} \times 100\% = 33.3\%$$

Project F

$$ARR = \frac{(\$260k - \$96k) \div 5}{(\$120k + \$24k) \div 2} \times 100\% = 45.6\%$$

Project G

$$ARR = \frac{(\$200k - \$136k) \div 2}{(\$170k + \$34k) \div 2} \times 100\% = 31.4\%$$

(b) **Payback**

Don't forget to include the realisable value cash flow if payback occurs in the last year.

Project E

$$\text{Payback} = \frac{\$125k}{\$50k} = 2.5 \text{ years}$$

Project F

$$\$120k - (4 \times \$15k) = \$60k$$

$$\$60k \div (\$200k + \$24k) = 0.268$$

Payback therefore happens after 4.268 years or approx. 4 years and 4 months

Project G

$$\$170k - \$120k = \$50k$$

$$\$50k \div (\$80k + \$34k) = 0.439$$

Payback therefore happens after 1.439 years, or approx. 1 years and 6 months

(c) **NPV @ 10%**

Project E

$$\text{NPV } (\$000s) = (125) + 50 \times 0.909 + 50 \times 0.826 + 50 \times 0.751 + (50+25) \times 0.683 = 50.53$$

Project F

$$\text{NPV } (\$000s) = (120) + 15 \times 0.909 + 15 \times 0.826 + 15 \times 0.751 + 15 \times 0.683 + (200+24) \times 0.621 = 66.64$$

Project G

$$\text{NPV } (\$000s) = (170) + 120 \times 0.909 + (80+34) \times 0.826 = 33.24$$

(d) Internal rate of return

NPV @ 20% (linear interpolation method)

Project E

$$\text{NPV (\$000s)} = (125) + 50 \times 0.833 + 50 \times 0.694 + 50 \times 0.579 + (50+25) \times 0.482 = 16.45$$

Project F

$$\text{NPV (\$000s)} = (120) + 15 \times 0.833 + 15 \times 0.694 + 15 \times 0.579 + 15 \times 0.482 + (200+24) \times 0.402 = 8.87$$

Project G

$$\text{NPV (\$000s)} = (170) + 120 \times 0.833 + (80+34) \times 0.694 = 9.08$$

IRRs =

Project E

$$10 + [50.53 / (50.53 - 16.45)] \times (20 - 10) = 24.8\%$$

Project F

$$10 + [66.64 / (66.64 - 8.87)] \times (20 - 10) = 21.5\%$$

Project G

$$10 + [33.24 / (33.24 - 9.08)] \times (20 - 10) = 23.8\%$$

Note that using annuity factors where relevant or using the NPV or IRR spreadsheet functions will give a slightly different answer to the above due to rounding differences. All of these methods are valid for the exam.

**Test your understanding 13****Answer B**

Cash flows occurring at a later date will be more heavily discounted, giving a lower present value. As the cash flows are inflows, this will lower the overall NPV.

A lower cost of capital will mean that all cash flows after T_0 will be less heavily discounted. This will have the effect of raising the NPV.

**Test your understanding 14****Answer D**

All four statements are correct. The NPV method is extensively used in practice to ascertain whether a project is a viable investment or not.



Test your understanding 15

Answer C

IRR is based on discounted cash flow principles. It therefore considers all of the cash flows in a project (A), does not include notional accounting costs such as depreciation (B) and it considers the time value of money (D).

It is not an absolute measure of return, however, as the IRR is expressed as a percentage. Two projects can have the same IRR but very different cash flows. C is therefore an incorrect statement.



Test your understanding 16

Answer B

The IRR can be calculated using the following formula.

If:

a = 10%

b = 20%

A = \$42,000

B = (\$22,000)

Then:

$$\text{IRR} = L + (N_L / (N_L - N_H)) \times (H - L)\% = 10 + 0.66 \times 10 = 16.6\%$$

If you selected 31.0%, you did not account correctly for the fact that the NPV was negative at a discount rate of 20%. You treated the \$22,000 NPV as a positive figure in the calculation. If you selected 13.4% you swapped A and B round in the formula, using A = \$22,000 and B = \$42,000. If you selected 15.0%, you have simply calculated the mid-point of the two rates instead of using the IRR formula.



Test your understanding 17

Answer A

The graph would be U-shaped with a negative NPV between 10% and 25% and positive NPVs at less than 10% or more than 25%.

Investment appraisal – Further aspects of discounted cash flows

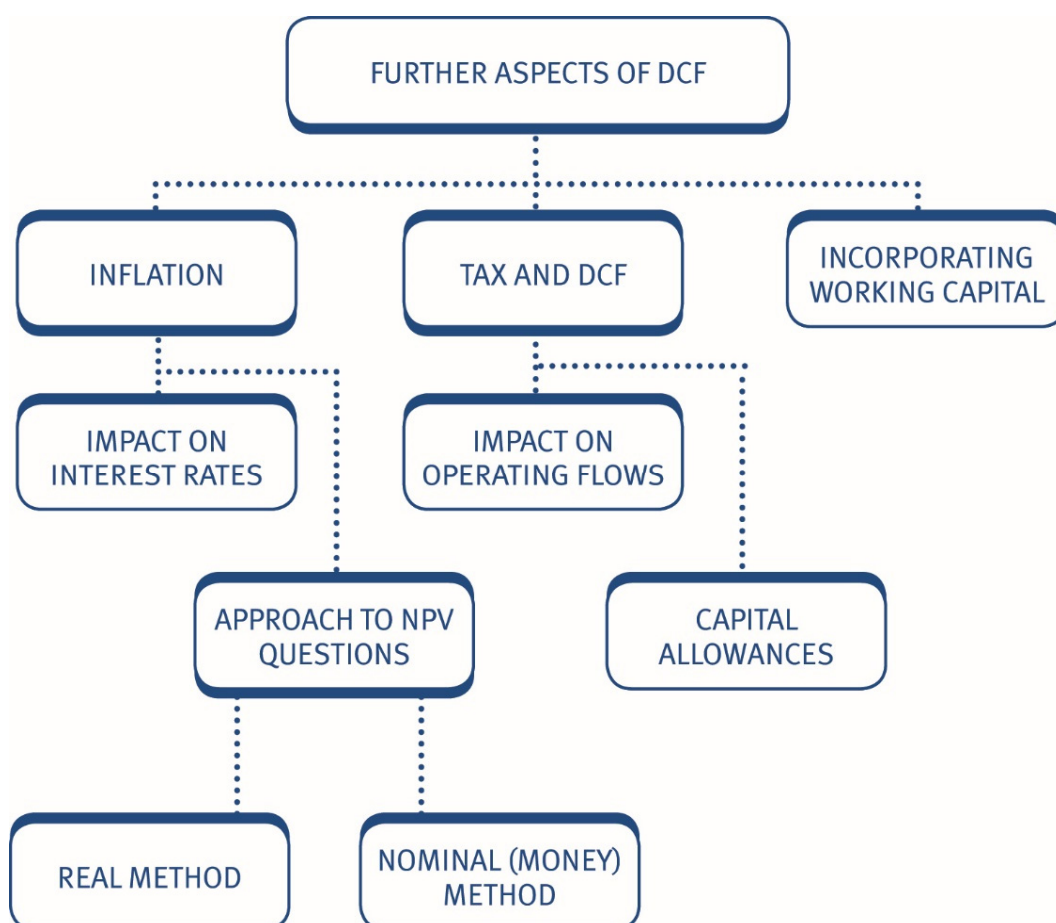
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- explain the impact of inflation on interest rates and define and distinguish between real and nominal (money) interest rates
- explain the difference between the real terms and nominal terms approaches to investment appraisal
- use the nominal (money) terms approach to appraise an investment
- use the real terms approach to appraise an investment
- explain the impact of tax on DCF appraisals
- calculate the tax cash flows associated with tax-allowable depreciation and incorporate them into net present values (NPV) calculations
- calculate the tax cash flows associated with taxable profits and incorporate them into NPV calculations
- explain the impact of working capital on an NPV calculation and incorporate working capital flows into NPV calculations.



One of the PER performance objectives (PO9) is to evaluate investment and financing decisions. You evaluate investment opportunities and their consequences and advise on their costs and benefits to the organisation. Working through this chapter should help you understand how to demonstrate that objective.



1 The impact of inflation on interest rates



Inflation is a general increase in prices leading to a general decline in the real value of money.

In times of inflation, the fund providers will require a return made up of two elements:

- real return for the use of their funds (i.e. the return they would want if there were no inflation in the economy)
- additional return to compensate for inflation.

The overall required return is called the money or nominal rate of return.



The real and money (nominal) returns are linked by the Fisher formula:

$$(1 + i) = (1 + r)(1 + h)$$

where

i = money rate

r = real rate

h = inflation



Illustration of relationship between inflation and interest rates

An investor is prepared to invest \$100 for one year and requires a real return of 10% per year.

In addition, compensation is required for the loss of purchasing power resulting from inflation, which is currently running at 5% per year.

What money rate of return will be required?

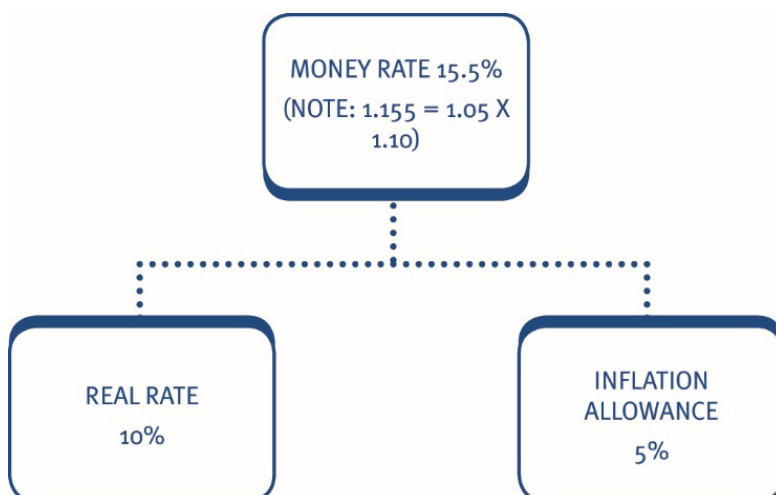
Solution

Just to compensate for inflation, the investment needs to increase by 5%, to $\$100 \times 1.05 = \105 .

To give a real return on top of this, it must further increase by 10%, to $\$105 \times 1.1 = \115.50 .

Thus, the money must increase overall by $1.05 \times 1.1 = 1.155$, i.e. by 15.5%.

Therefore, the investor's actual required return is 15.5%.



Test your understanding 1 – Money and real returns

\$1,000 is invested in an account that pays 10% interest per year. Inflation is currently 7% per year.

Find the real return on the investment.



Test your understanding 2 – Money and real returns

If the real rate of interest is 8% and the rate of inflation is 5%, what should the money rate of interest be?

2 The impact of inflation on cash flows



Where cash flows have not been increased for expected inflation they are described as being in **current prices**, or **today's prices**.



Where cash flows have been increased to take account of expected inflation they are known as **money cash flows**, or **nominal cash flows**.

Remember, if they do take inflation into account, they represent expected flows of money, hence the term 'money cash flows'.

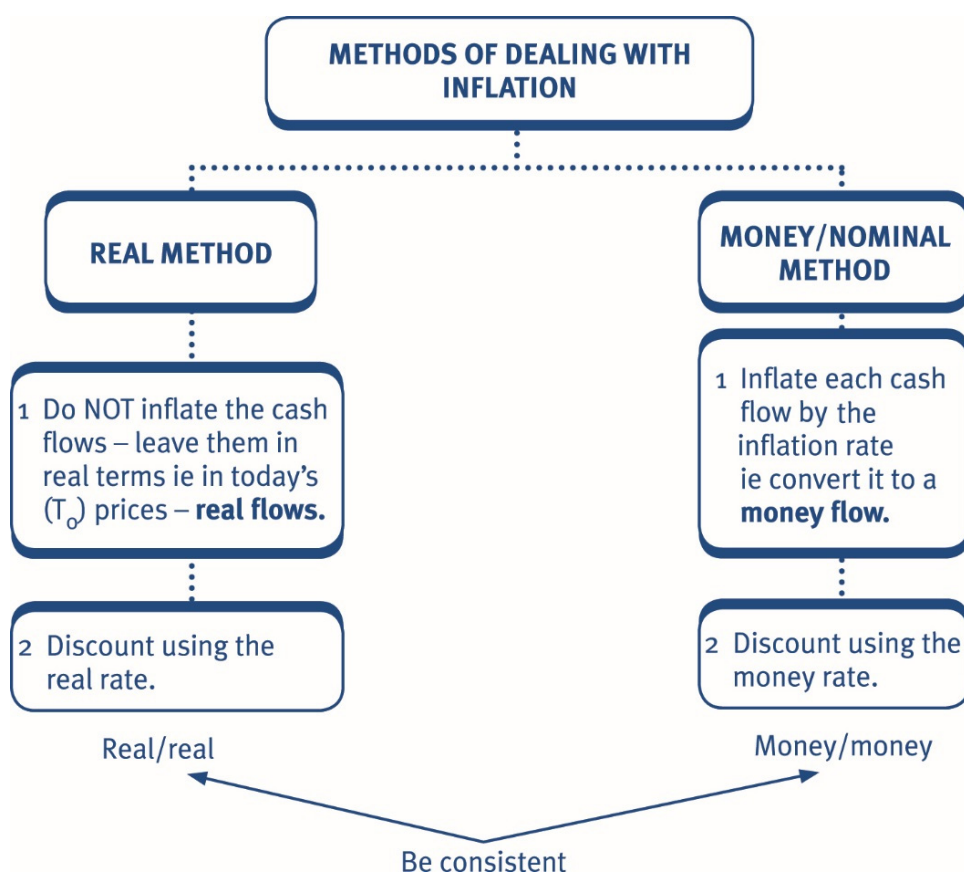
You can assume that cash flows you are given in the exam are the money cash flows unless told otherwise.

If the exam question specifies that the cash flows are in current price terms you will generally need to put these in money terms before you can discount them (although see "other methods of dealing with inflation" below).

Make sure you read the question carefully. Sometimes you will be given the cash flows in Year 1 terms with subsequent inflation.

- For example, if the question tells you that sales for the next 3 years are \$100 in current terms but are expected to inflate by 10%, then what is actually meant is that the sales will be:
 Year 1: \$110
 Year 2: \$121 } i.e. these are the cash flows in money terms
 Year 3: \$133.10
- For example, if the question says "Sales will be \$100 in the first year, but are they going to inflate by 10% for the next two years", then the sales will be:
 Year 1: \$100
 Year 2: \$110 } compare these to the previous example – make sure
 Year 3: \$121 you understand why they are different!

The impact of inflation can be dealt with in two different ways – both methods give the same NPV.



Note that the real method only applies if the rate of inflation of the specific cash flows involved is the same as the general rate of inflation.

Real cash flows are those that have not had the general rate of inflation incorporated into their value. When there is only one rate of inflation, real cash flows and current cash flows will be the same (see information on current, money and real cash flows in section 3 below).



Test your understanding 3 – Money and real methods

Storm Co is evaluating Project X, which requires an initial investment of \$50,000. Expected net cash flows are \$20,000 per year for four years at today's prices. However, these are expected to rise by 5.5% per year because of inflation. The firm's money cost of capital is 15%.

Find the NPV by:

- discounting money cash flows
- discounting real cash flows.



Test your understanding 4 – Money and real methods

A project has the following cash flows before allowing for inflation, i.e. they are stated at their T_0 values.

The company's money discount rate is 15.5%. The general rate of inflation is expected to remain constant at 5%.

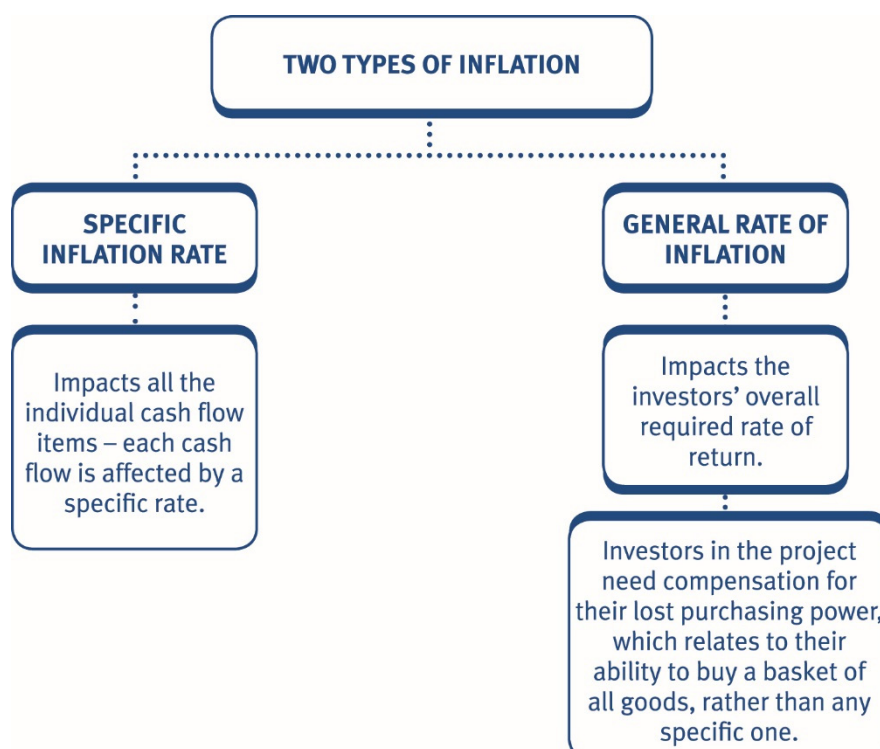
Timing	Cash flow
	\$
0	(750)
1	330
2	242
3	532

Evaluate the project in terms of:

- (a) real cash flows and real discount rates
- (b) money cash flows and money discount rates.

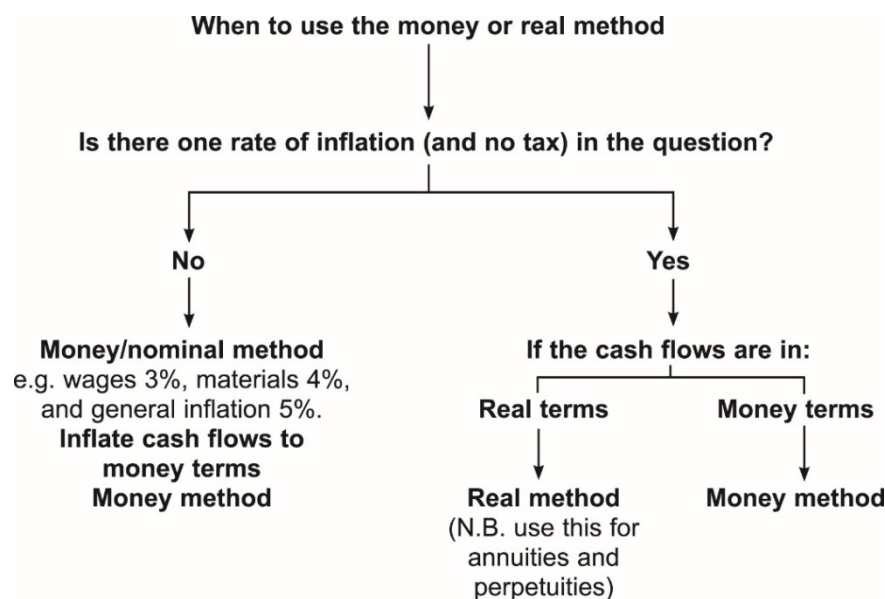
3 Specific and general inflation rates

The TYUs given above had all cash flows inflating at the general rate of inflation. In practice, inflation does not affect all cash flows to the same extent. In some investment appraisal questions, you may be given information on more than one inflation rate. In these situations, you will have information on both specific inflation rates and general inflation rates.



In situations where you are given a number of specific inflation rates, the real method outlined above cannot directly be used. This is because the cash flows would first have to be inflated at their specific inflation rates to get the money cash flows and then deflated using the general inflation rate to become 'real cash flows'.

The following gives a useful summary of how to approach examination questions.



If a question contains both tax and inflation, it is advisable to use the money method.



Specific and general inflation rates

If there is one rate of inflation in the question both the real and money method will give the same answer. However, it is easier to adjust one discount rate, rather than all the cash flows over a number of years. This is particularly true where the cash flows are annuities. The real method is the only possible method where they are perpetuities.

Although it is theoretically possible to use the real method in questions incorporating tax, it is extremely complex and would involve a lot of calculations. It is therefore much safer (and easier) to use the money/nominal method in all questions where tax is taken into account and so this is recommended.

This involves:

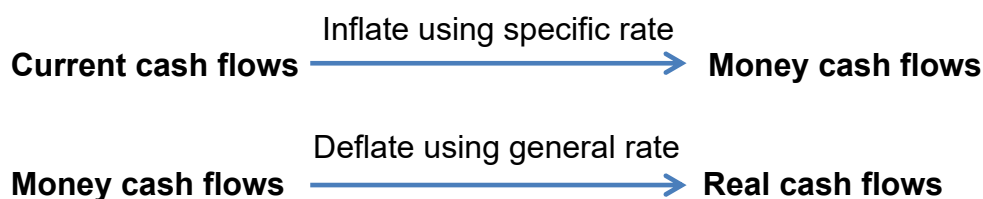
- inflating the cash flows at their specific inflation rates
- discounting using the money rate

Very often the money rate will not be given in the question but will need to be calculated. This should be done using the Fisher formula to combine the real rate and the general inflation rate.

In a couple of recent exam questions, the real method was specifically asked for. However, the money method had already been calculated in the previous part of the question. The examining team's approach is to use the cash flows from the money method calculation and deflate them using the general rate of inflation to calculate their 'real' values, before then discounting them at the real cost of capital.



To convert money (nominal) cash flows into real cash flows, they need to be deflated using the general inflation rate.



Current and real cash flows will only be the same if the specific inflation rate used for the cash flow is the same as the general inflation rate.



Illustration of current, money and real cash flows

A cash flow occurring at time period 2 has a current value of \$5,000. The specific rate of inflation that applies to the cash flow is 6% and the general rate of inflation is 4%.

Calculate the money cash flow and real cash flow values as at time period 2.

Solution

Current cash flow at t_2 = \$5,000. As this is a current value, it includes no inflation.

Apply two years' worth of **specific** inflation to calculate the **money** cash flow at time period 2:

$$\$5,000 \times 1.06 \times 1.06 = \$5,618$$

Strip out two years' worth of **general** inflation to calculate the **real** cash flow at time period 2:

$$\$5,618 \div 1.04 \div 1.04 = \$5,194$$

The real cash flow could then be discounted at the real cost of capital to find its present value.



In the examination, for a short life project, with cash flows inflating at different rates, it is recommended to set the NPV calculation out with the cash flows down the side and the time periods across the top.



Test your understanding 5 – General and specific inflation rates

A company is considering a cost-saving project. This involves purchasing a machine costing \$7,000, which will result in annual savings (in real terms) on wage costs of \$1,000 and on material costs of \$400.

The following forecasts are made of the rates of inflation each year for the next five years:

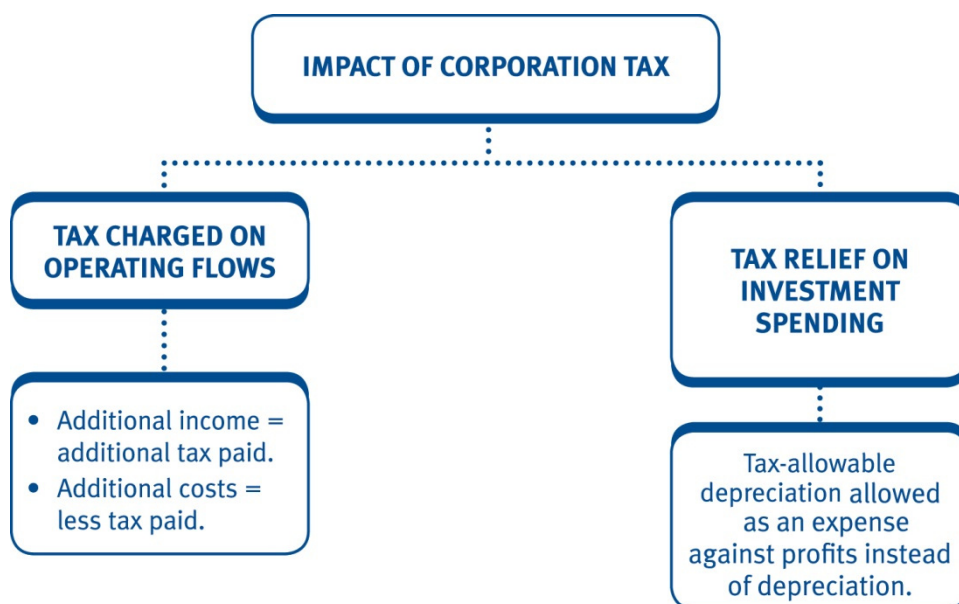
Wage costs	10%
Material costs	5%
General prices	6%

The cost of capital of the company, in real terms, is 8.5%.

Evaluate the project, assuming that the machine has a life of five years and no scrap value.

4 Dealing with tax in NPV calculations

Since most companies pay tax, the impact of corporation tax must be considered in any investment appraisal.



The impact of taxation on cash flows

Corporation tax charged on a company's profits is a relevant cash flow for NPV purposes. It is assumed, unless otherwise stated in the question, that:

- operating cash inflows will be taxed at the corporation tax rate
- operating cash outflows will be tax deductible and attract tax relief at the corporation tax rate
- investment spending attracts tax-allowable depreciation
- the company is earning net taxable profits overall (this avoids any issues of carrying losses forwards to reduce future taxation).
- tax is paid one year after the related operating cash flow is earned (unless told otherwise)



The impact of taxation on cash flows

Taxation has the following effects on an investment appraisal problem:

- Project cash flows will give rise to taxation, which itself has an impact on the project appraisal. Normally we assume that tax paid on operating cash flows is due one year after the related cash flow. However, it is possible for alternative assumptions to be made and so you should read any examination question carefully to ascertain precisely what assumptions should be used for the scenario.
- Organisations benefit from being able to claim tax-allowable depreciation. The effect of this is to reduce the amount of tax that organisations are required to pay. Again, it is important to read any examination question carefully in order to identify what treatment is expected by the examiner.

Note that the **tax-allowable depreciation is not a cash flow** and to calculate the tax impact we have to multiply each year's tax-allowable depreciation by the corporation tax rate. The effect of tax-allowable depreciation is on the amount of tax payable, the change in which is the relevant cash flow for the NPV.

In dealing with these tax effects, it is always assumed that where a tax loss arises from the project, there are sufficient taxable profits elsewhere in the organisation to allow the loss to reduce any relevant (subsequent) tax payment (and it may therefore be treated as a cash inflow) and that the company has sufficient taxable profits to obtain full benefit from tax-allowable depreciation.

Directors may choose to evaluate an investment over a fixed time period. In the investment appraisal, the tax effects of any decisions made during the fixed time period need to be included. This means, for example, that if the investment appraisal period is set at four years the appraisal itself would cover five years if tax is paid (and saved) one year in arrears.

In practice, the effects of taxation are complex, and are influenced by a number of factors including the following:

- the taxable profits and tax rate
- the company's accounting period and tax payment dates
- tax-allowable depreciation
- losses available for set-off

but many of these issues are ignored or simplified for the purposes of NPV investment appraisal in the FM exam.

Tax-allowable depreciation

For tax purposes, a business may not deduct the cost of an asset from its profits as depreciation (in the way it does for financial accounting purposes).

Instead, the cost must be deducted from taxable profits in the tax computation, in the form of tax-allowable depreciation. The basic rules are as follows:

- tax-allowable depreciation (also known as a writing down allowances or WDA) is calculated based on the written down value of the assets (this will either be on a reducing balance or straight line basis – read the question carefully)
- the total amount of tax-allowable depreciation given over the life of an asset will equate to its fall in value over the period (i.e. the cost less any scrap proceeds)
- tax-allowable depreciation is claimed as early as possible
- tax-allowable depreciation is given for every year of ownership except the year of disposal
- in the year of sale or scrap a balancing allowance (BA) or balancing charge arises (BC).

	\$
Original cost of asset	X
Cumulative tax-allowable depreciation claimed	(X)
	—
Written down value of the asset	X
Disposal value of the asset	(X)
	—
Balancing allowance or balancing charge	X
	—



Tax-allowable depreciation

If a business buys a capital asset in one year and sells it several years later, the total tax relief it will receive is the tax on the cost of the asset less its eventual disposal value.

For example, if a business buys equipment for \$100,000 in Year 0 and disposes of it in Year 5 for \$20,000, it will receive tax relief on the net cost of \$80,000. If the rate of corporation tax is 30%, the reduction in tax payments over the five years would be $30\% \times \$80,000 = \$24,000$.

Balancing allowances are given as a final deduction to ensure the full fall in value has been allowed. Balancing charges occur where the total tax-allowable depreciation claimed at this point exceeds the fall in value of the asset. The excess claimed is treated as a taxable amount in the year of disposal.



Test your understanding 6 – Balancing allowance or charge

An asset is bought for \$10,000 and will be used on a project for four years after which it will be disposed of. Tax is payable at 30%, one year in arrears, and tax-allowable depreciation is available at 25% reducing balance.

Required:

- (a) Calculate the tax-allowable depreciation and hence the tax savings for each year if the proceeds on disposal of the asset are \$2,500.
- (b) How would your answer change if the asset were sold for \$5,000?
- (c) If net trading income from the project is \$8,000 per year, based on your answer to part (a) and a cost of capital of 10%, calculate the NPV of the project.



For tax purposes, care must be taken to identify the exact time of asset purchase relative to the accounting period end. However, unless you are told otherwise in the exam you should assume that an asset is purchased on the first day of an accounting period (T_0) and that the first amount of tax-allowable depreciation is given one year later at T_1 .

Note that the treatment of tax in this exam is very much simplified compared to what you will see in the tax-specific exams such as Taxation or Advanced Taxation.

5 Incorporating working capital

Investment in a new project often requires an additional investment in working capital, i.e. the difference between short-term assets and liabilities.



The treatment of working capital is as follows:

- initial investment is a cash outflow at the start of the project
- if the investment is increased during the project, the **increase** is a relevant cash outflow
- if the investment is decreased during the project, the **decrease** is a relevant cash inflow
- at the end of the project all the working capital is 'released' and treated as a cash inflow.

To calculate the working capital cash flows, you should:

Step 1: Calculate the absolute amounts of working capital needed in each period

Step 2: Work out the incremental cash flows required each year



Test your understanding 7 – Working capital

A company expects sales for a new project to be \$225,000 in the first year growing at 5% per year. The project is expected to last for 4 years. Working capital equal to 10% of annual sales is required and needs to be in place at the start of each year.

Calculate the working capital flows for incorporation into the NPV calculation.

If you have a question including both working capital and inflation, you should always adopt the money method (inflating the cash flows). Calculate the actual money amount of the factor on which working capital is dependent (often sales or contribution) before calculating the working capital requirements.



Test your understanding 8 – Working capital

A company anticipates sales for the latest venture to be 100,000 units per year. The selling price is expected to be \$3 per unit in the first year, inflating by 8% per year over the three-year life of the project. Working capital equal to 10% of annual sales is required and needs to be in place at the start of each year.

Calculate the working capital cash flows for incorporation into the NPV calculation.

6 Laying out long NPV questions

For the majority of investment appraisal questions, the following pro-forma is recommended:

YEAR	0	1	2	3
				
	Add an extra year (if tax is delayed)			
(1) Net trading cash flows	Inflated where necessary			
The inflows and outflows from trading (e.g. sales minus operating cash flows)				
(2) Tax payable	Calculated as normal on the money flows			
The net trading cash flows $\times$ tax rate (normally delayed by one year)				
(3) Investment	Shouldn't need inflating!			
(4) Residual/scrap value	May need inflating but usually given in money terms			
(5) Tax relief on tax-allowable depreciation	Calculated as normal			
(6) Working capital cash flows	Calculate using money figures			

7 Dealing with questions with both tax and inflation

Combining tax and inflation in the same question does not make it any more difficult than keeping them separate.

Questions with both tax and inflation are best tackled using the money method.

- Inflate costs and revenues, where necessary, before determining their tax implications.
- Ensure that the cost and disposal values have been inflated (if necessary) before calculating tax-allowable depreciation.
- Always calculate working capital on these inflated figures, unless given.
- Use a post-tax money discount rate.



Test your understanding 9 – NPV with tax and inflation

Ackbono Co is considering a potential project with the following forecasts:

	Now	T ₁	T ₂	T ₃
Initial investment (\$million)	(1,000)			
Disposal proceeds (\$million)				200
Demand (millions of units)		5	10	6

The selling price per unit is expected to be \$100 and the variable cost \$30 per unit. Both of these figures are given in today's terms.

Tax is paid at 30%, one year after the accounting period concerned.

Working capital will be required equal to 10% of annual sales. This will need to be in place at the start of each year.

Tax-allowable depreciation is available at 25% reducing balance.

The company has a real required rate of return of 6.8%.

General inflation is predicted to be 3% per year but the selling price is expected to inflate at 4% and variable costs by 5% per year

Determine the NPV of the project.

N.B. work in \$ millions and round all numbers to the nearest whole million.



Test your understanding 10

A project has an annual net cash inflow (in current terms) of \$3 million, occurring at the end of each year of the project's two-year life. An investment of \$3.5 million is made at the outset. All cash inflows are subject to corporation tax of 30%, payable when the cash is received. There is no tax-allowable depreciation on the initial investment. An average inflation rate of 5% per year is expected to affect the inflows of the project.

The cost of capital in money terms is 15.5%

What is the expected net present value of the project (to the nearest \$100,000)?

- A \$93,000
- B \$147,000
- C \$287,000
- D \$367,000



Test your understanding 11

A company has a 'money' cost of capital of 16.55% per year. The 'real' cost of capital is 11% per year.

What is the inflation rate?

- A 5%
- B 5.55%
- C 11%
- D 16.55



Test your understanding 12

Data of relevance to the evaluation of a particular project is given below.

Cost of capital in real terms 10% per year

Expected inflation 5% per year

Expected increase in the project's annual cash inflow 6% per year

Expected increase in the project's annual cash outflow 4% per year

Which one of the following sets of adjustments will lead to the correct NPV being calculated?

	Cash inflow	Cash outflow	Discount percentage
A	Unadjusted	Unadjusted	10.0%
B	5% per year increase	5% per year increase	15.5%
C	6% per year increase	4% per year increase	15.0%
D	6% per year increase	4% per year increase	15.5%

**Test your understanding 13**

An asset costing \$40,000 is expected to last for three years, after which it can be sold for \$16,000. The corporation tax rate is 30%, tax-allowable depreciation at 25% is available, and the cost of capital is 10%. Tax is payable at the end of each financial year.

What is the cash flow in respect of tax-allowable depreciation that will be used at time 2 of the net present value calculation?

- A \$1,688
- B \$2,250
- C \$5,624
- D \$7,500

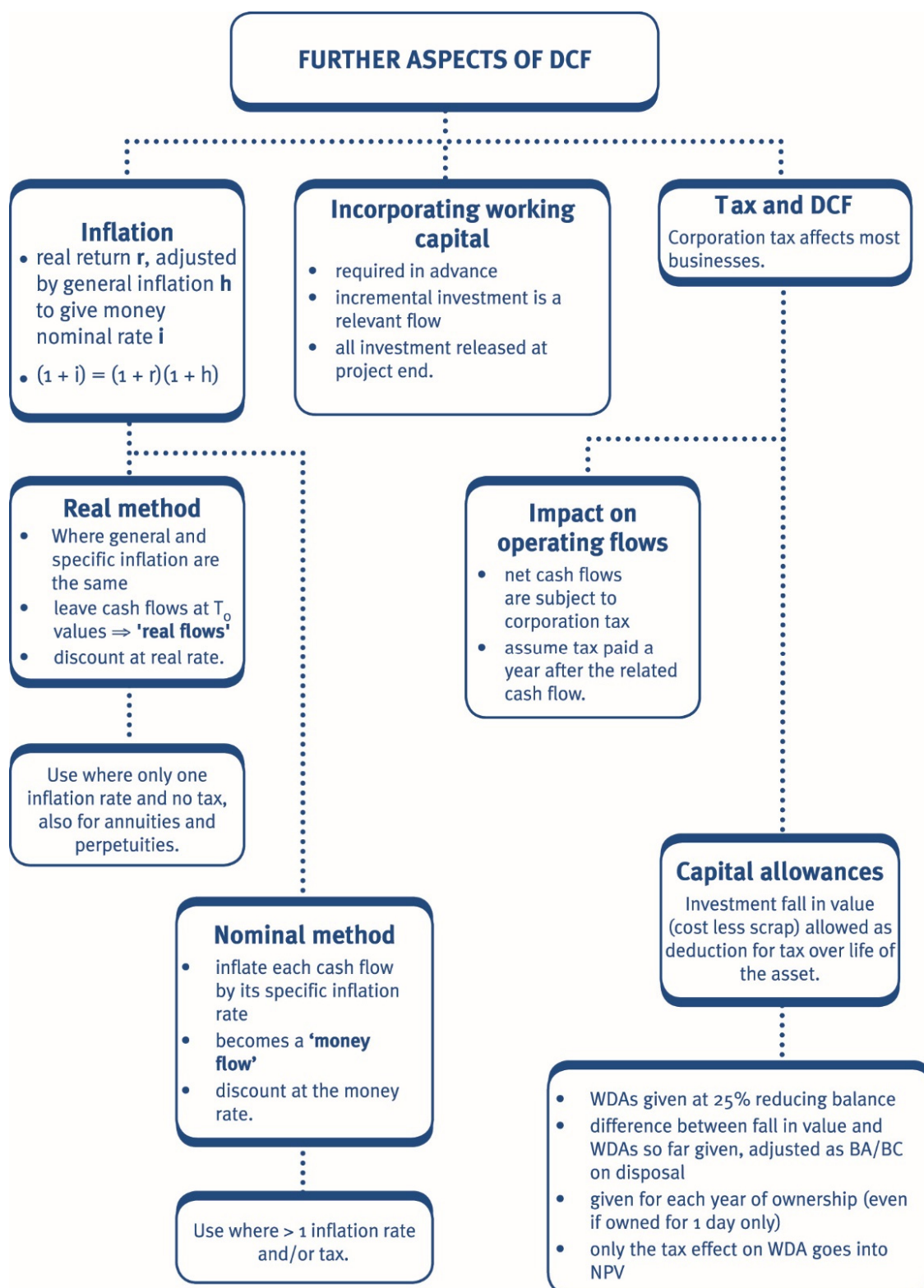
**Test your understanding 14**

A new project is expected to generate sales of 55,000 units per year. The selling price is expected to be \$3.50 per unit in the first year, growing at 6% per year. The project is expected to last for three years. Working capital equal to 12% of annual sales is required and needs to be in place at the start of each year.

What is the cash flow in respect of working capital that will be used at time 2 of the net present value calculation?

- A \$(25,955)
- B \$(24,486)
- C \$(1,386)
- D \$(1,469)

8 Chapter summary



Test your understanding answers

**Test your understanding 1 – Money and real returns**

$$(1 + r) = (1 + i) / (1 + h) = 1.10/1.07 = 1.028.$$

$$r = 0.028 \text{ or } 2.8\%.$$

**Test your understanding 2 – Money and real returns**

$$(1 + i) = (1 + r)(1 + h) = 1.08 \times 1.05 = 1.134$$

$$\text{Money rate (i)} = 13.4\%.$$

**Test your understanding 3 – Money and real returns****Discounting the money cash flows at the money rate – the money method**

Discounting money cash flow at the money rate: The cash flows at today's prices are inflated by 5.5% for every year to convert them into money flows. They are then discounted using the money cost of capital.

The money cash flows are therefore:

Time	Working	Money cash flow
		\$
0	(50,000)	= (50,000)
1	$20,000 \times 1.055$	= 21,100
2	$20,000 \times 1.055^2$	= 22,261
3	$20,000 \times 1.055^3$	= 23,485
4	$20,000 \times 1.055^4$	= 24,776

Note: The question simply refers to the 'firm's cost of capital'. You can assume this is the money rate – if you are given a real rate the examiner will always specify.

Time	Working	Discount rate	PV
		15%	\$
0	(50,000)	1	(50,000)
1	21,100	0.870	18,357
2	22,261	0.756	16,829
3	23,485	0.658	15,453
4	24,776	0.572	14,172
		NPV =	14,811

Discounting the real cash flows at the real rate – the real method.

Calculate the real rate by removing the general inflation from the money cost of capital.

$$(1 + r) = \frac{(1 + i)}{(1 + h)}$$

$$(1 + r) = \frac{(1 + 0.15)}{(1 + 0.055)}$$

$$(1 + r) = 1.09$$

Therefore, $r = 0.09$ i.e. 9%

The real rate can now be applied to the real flows without any further adjustments.

Time	Real cash flow	Discount rate	PV
	\$	9%	\$
0	(50,000)	1	(50,000)
1–4	20,000	3.240	64,800
		NPV =	14,800

Note: Differences due to rounding.


Test your understanding 4 – Money and real returns
(a) Real cash flows and real discount rates

Discount rate as per the question of 15.5% includes investor's/lender's inflation expectation of 5%. Hence 'real' discount rate, r , is given by:

$$1 + r = \frac{1 + i}{1 + h}$$

$$1 + r = \frac{1 + 0.155}{1 + 0.05} = 1.10$$

Therefore, r is 0.10 or 10%

Time	Working	Discount factor @10%	PV \$
0	(750)	1.000	(750)
1	330	0.909	300
2	242	0.826	200
3	532	0.751	400
NPV =			150

(b) Money cash flows and money discount rates

The discount rate as per the question of 15.5% is the money discount rate.

To convert real cash flows into money flows they will need to be increased by 5% each year from year 0, to allow for inflation.

Timing	Real/cash flow (a) \$	Inflation factor (b)	Money cash flow (a) × (b) \$	DF @ 15.5%	PV \$
0	(750)	1	(750)	1.000	(750)
1	330	$1 + 0.05$	346.5	0.866*	300
2	242	$(1 + 0.05)^2$	266.8	0.750	200
3	532	$(1 + 0.05)^3$	615.9	0.649	400
NPV					150

$$*1.155^{-1} = 0.866$$

$$1.155^{-2} = 0.750$$

$$1.155^{-3} = 0.649$$

Note: Either approach yields identical conclusions (allowing for rounding).



Test your understanding 5 – General and specific inflation rates

Since the question contains both specific and general inflation rates, the money method should be used

Step 1

The money method needs to be calculated using the information provided on the real rate of return and the general rate of inflation

$$(1 + i) = (1 + r)(1 + h)$$

$$(1 + i) = (1.085)(1.06) = 1.1501$$

$$i = 15\%$$

Step 2

Inflate the cash flows using the specific inflation rates and discount using the money rate calculated above.

	T ₀	T ₁	T ₂	T ₃	T ₄	T ₅
	\$	\$	\$	\$	\$	\$
Investment	(7,000)					
Wages savings (inflating @ 10%)		1,100	1,210	1,331	1,464	1,610
Materials savings (inflating @ 5%)		420	441	463	486	510
Net cash flows	(7,000)	1,520	1,651	1,794	1,950	2,120
DF @ 15%	1.000	0.870	0.756	0.658	0.572	0.497
PV of cash flow	(7,000)	1,322	1,248	1,180	1,115	1,054

Therefore NPV = \$(1,081) which suggests the project is not worthwhile.


Test your understanding 6 – Balancing allowance or charge

(a)

Time		\$	Tax saving @ 30%	Timing of tax relief
T ₀	Initial investment	10,000		
T ₁	Tax-allowable depreciation @ 25%	(2,500)	750	T ₂
	Written down value	7,500		
T ₂	Tax-allowable depreciation @ 25%	(1,875)	563	T ₃
	Written down value	5,625		
T ₃	Tax-allowable depreciation @ 25%	(1,406)	422	T ₄
	Written down value	4,219		
T ₄	Sale proceeds	(2,500)		
T ₄	Balancing Allowance	1,719	516	T ₅

Note:

- total tax-allowable depreciation = 2,500 + 1,875 + 1,406 + 1,719 = 7,500 = fall in value of the asset
- total tax relief = 750 + 563 + 422 + 516 = 2,251 $\approx$ 7,500 $\times$ 30%

(tax-allowable depreciation $\times$ tax rate)

- (b) If the scrap proceeds had been \$5,000, there would be a balancing charge in T₄ of \$5,000 – \$4,219 = \$781 and additional tax of \$234 (\$781 $\times$ 30%) would be payable in T₅.

Note:

- total tax-allowable depreciation = 2,500 + 1,875 + 1,406 – 781 = 5,000 = fall in value of the asset
- total tax relief = 750 + 563 + 422 – 234 = 1,501 $\approx$ 5,000 $\times$ 30% (tax-allowable depreciation $\times$ tax rate)

(c)

Time	T ₀	T ₁	T ₂	T ₃	T ₄	T ₅
	\$	\$	\$	\$	\$	\$
Net trading inflows		8,000	8,000	8,000	8,000	
Tax payable (30%)			(2,400)	(2,400)	(2,400)	(2,400)
Initial investment	(10,000)					
Scrap proceeds					2,500	
Tax relief on tax-allowable depreciation			750	563	422	516
Net cash flows	(10,000)	8,000	6,350	6,163	8,522	(1,884)
DF @ 10%	1.000	0.909	0.826	0.751	0.683	0.621
PV	(10,000)	7,272	5,245	4,628	5,821	(1,170)
					NPV	\$11,796



Test your understanding 7 – Working capital

Step 1: Calculate the absolute amounts of working capital needed over the project:

	T ₀	T ₁	T ₂	T ₃	T ₄
	\$	\$	\$	\$	\$
Sales		225,000	236,250	248,063	260,466
Working capital required (10% sales)	22,500	23,625	24,806	26,047	

Step 2: Work out the **incremental** investment required each year (remember that the full investment is released at the end of the project):

	T ₀	T ₁	T ₂	T ₃	T ₄
	\$	\$	\$	\$	\$
Working	22,500 – 0	23,625 – 22,500	24,806 – 23,625	26,047 – 24,806	26,047 – 0
Working capital investment	(22,500)	(1,125)	(1,181)	(1,241)	26,047



Test your understanding 8 – Working capital

Step 1: Calculate the absolute amounts of working capital needed over the project:

	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Selling price (inflating at 8% p.a.)		3.00	3.24	3.4992
Sales value (based on 100,000 units)		300,000	324,000	349,920
Working capital required	30,000	32,400	34,992	

Step 2: Work out the incremental investment required each year, remembering to release all the working capital at the end of the project

	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Working		32,400	34,992	
		– 30,000	– 32,400	
Working capital investment	(30,000)	(2,400)	(2,592)	34,992



Test your understanding 9 – NPV with tax and inflation

\$ millions	T ₀	T ₁	T ₂	T ₃	T ₄
Sales (W1)		520	1,082	675	
Variable costs (W1)		(158)	(331)	(208)	
Net trading inflows		362	751	467	
Tax payable (30%)			(109)	(225)	(140)
Initial investment	(1,000)				
Scrap proceeds				200	
Tax relief on tax-allowable depreciation (W2)			75	56	109
Working capital (W3)	(52)	(56)	40	68	
Net cash flows	(1,052)	306	757	566	(31)
DF @ 10% (W4)	1	0.909	0.826	0.751	0.683
PV	(1,052)	278	625	425	(21)
				NPV	255

Note:

When performing the calculations on a spreadsheet, instead of completing the rows for discount factor, PV and NPV, you could use the spreadsheet function as follows:

If the numbers in the 'Net cash flows' row are in the spreadsheet cells B12 to F12, use the formula:

=NPV(10%,C12:F12)

to calculate the PV of the year 1 to year 4 cash flows, which should come to around 1,308 depending on the level of roundings in your figures. Then deduct the cash outflow of (1,052) in time period 0 to complete the NPV calculation.

Use of the formula will lead to a different level of rounding in the final NPV, so don't worry if your answer is slightly different. Either method is acceptable in the FM exam.

(W1) Revenue and costs

Revenue and costs need to be expressed in money terms.

e.g. revenue at T₂ = \$10m × 100 × (1.04)² = \$1,081.6m.

(W2) Tax-allowable depreciation

Time		\$m	Tax saving \$m	Timing of tax relief
T ₀	Initial investment	1,000		
T ₁	Tax-allowable depreciation @ 25%	(250)	75	T ₂
	Written down value	750		
T ₂	Tax-allowable depreciation @ 25%	(188)	56	T ₃
	Written down value	562		
	Sale proceeds	(200)		
T ₃	Balancing Allowance	362	109	T ₄

(W3) Working capital requirements

Step 1: Calculate the absolute amounts of working capital needed over the project:

\$ millions	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Sales		520	1,082	675
Working capital required	52	108	68	

Step 2: Work out the incremental investment required each year, remembering to release all the working capital at the end of the project

\$ millions	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Working		52–108	108–68	
Capital investment	(52)	(56)	40	68

(W4) Discount rate

$(1+i) = (1+r) \times (1+h) = 1.068 \times 1.03 = 1.10$, giving a money rate (i) = 10%.



Test your understanding 10

Answer B

Time	0	1	2
	\$000	\$000	\$000
Investment	(3,500)		
	_____	_____	_____
Revenue		3,150	3,308
Tax		(945)	(992)
		_____	_____
Net cash flows	(3,500)	2,205	2,316
	_____	_____	_____
Discount factor @ 15.5%	1	0.866	0.750
PV @ 15.5%	(3,500)	1,910	1,737
NPV = \$147,000			



Test your understanding 11

Answer A

$$(1 + \text{Money rate}) = (1 + \text{Real rate}) \times (1 + \text{Inflation rate})$$

$$1.1655 = (1.11) \times (1 + \text{Inflation rate})$$

$$1 + \text{inflation rate} = 1.1655 / 1.11 = 1.05$$

$$\text{Inflation rate} = 0.05 = 5\%$$



Test your understanding 12

Answer D

As there are different rates of inflation the 'money approach' must be used, i.e. the cash flows must be inflated at their specific rates and discounted at the money cost of capital.

$$(1 + \text{Money rate}) = (1 + \text{Real rate}) \times (1 + \text{Inflation rate}) = 1.1 \times 1.05 = 1.155$$

**Test your understanding 13****Answer B**

No delay in tax so the cash flow for the tax saving occurs in the same period as the tax-allowable depreciation is earned.

Time		\$m	Tax saving \$m	Timing of tax relief
T ₀	Initial investment	40,000		
T ₁	Tax-allowable depreciation @ 25%	(10,000)	3,000	T ₁
	Written down value	30,000		
T ₂	Tax-allowable depreciation @ 25%	(7,500)	2.250	T ₂

**Test your understanding 14****Answer D**

Step 1: Calculate the absolute amounts of working capital needed over the project:

	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Selling price (inflating at 6% p.a.)		3.50	3.71	3.9326
Sales value (based on 55,000 units)		192,500	204,050	216,293
Working capital required (12%)	23,100	24,486	25,955	

Step 2: Work out the incremental investment required each year, remembering to release all the working capital at the end of the project

	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Working		24,486 – 23,100	25,955 – 24,486	
Capital investment	(23,100)	(1,386)	(1,469)	25,955

Asset investment decisions and capital rationing

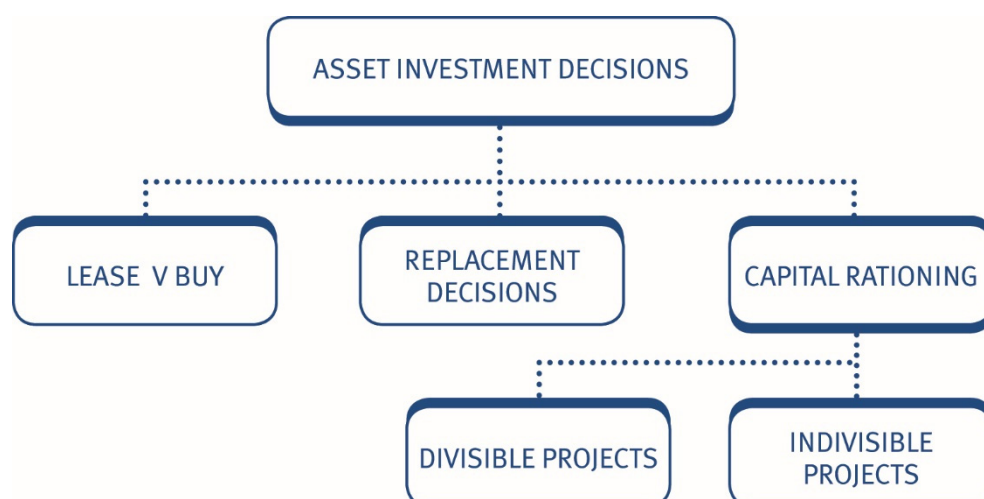
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- evaluate the choice between leasing an asset and borrowing to buy using the before- and after-tax costs of debt
- define and calculate an equivalent annual cost (EAC)
- evaluate asset replacement decisions using EACs
- explain the reasons why capital rationing might be required
- define and distinguish between divisible and indivisible projects
- calculate profitability indexes for divisible investment projects and use them to evaluate investment decisions
- calculate the net present value (NPV) of combinations of non-divisible investment projects and use the results to evaluate investment decisions.



One of the PER performance objectives (PO9) is to evaluate investment and financing decisions. You evaluate investment opportunities and their consequences and advise on their costs and benefits to the organisation. Working through this chapter should help you understand how to demonstrate that objective.



1 Lease versus buy

Once the decision has been made to acquire an asset for an investment project, a decision still needs to be made as to how to finance it. The choices that we will consider are:

- lease
- buy.

The NPVs of the financing cash flows for both options are found and compared and the lowest cost option selected.



The finance decision is considered separately from the investment decision. The operating costs and revenues generated from the investment are assumed to be common in each case.

Only the relevant cash flows arising as a result of the type of finance are included in the NPV calculation.

Leasing

The asset is never 'owned' by the user company from the perspective of the taxman.

Implications

- The finance company (lessor) receives the tax-allowable depreciation as the owner of the asset.
- The user (lessee) receives no tax-allowable depreciation but is able to offset the full rental payment against tax.

The relevant cash flows for the user would thus be:

- the lease payments
- tax relief on the lease payments.

Buying

The assumption is that buying requires the use of a bank loan (for the sake of comparability). The user is the owner of the asset.

Implication

- The user will receive tax-allowable depreciation on the asset and tax relief for the interest payable on the loan.

The relevant cash flows would be:

- the purchase cost
- any residual value
- any associated tax implications due to tax-allowable depreciation.



Do not include the interest payments or the tax relief arising on them in the NPV calculation, as this is dealt with via the cost of capital (see below).

Incremental costs

If there are other costs provided in the question scenario that are incurred in one option but not the other, they are incremental to the decision and therefore relevant to include. For example, the question may state that maintenance costs are incurred under the purchase option but not the lease option.

Cost of capital

As the interest payments attract tax relief we must use the **post-tax cost of borrowing** as our discount rate. The same rate is used for both leasing and buying, reflecting the risk level to the lender of providing finance to the company.



Post-tax cost of borrowing = Cost of borrowing $\times$ (1 – Tax rate).

(**Note:** In some questions, you may find that a company is not paying tax and so the unadjusted pre-tax borrowing rate would be appropriate.)



Lease versus buy

Where the use of an asset is required for a new project, there are effectively two decisions to be made:

- Is the project worthwhile?
- If so, should the asset be leased or bought with a loan?

If the project has been approved in principle, then you need to calculate the NPVs of the two sets of financing cash flows for buying and leasing.

To be consistent with the earlier calculations, you could use the company's existing cost of capital to discount these cash flows.

However, the decision of whether to lease or borrow to buy is actually a financing decision as opposed to an investment decision. Therefore, the appropriate discount rate is the borrowing cost (cost of debt) of the company, whether on a before or after-tax basis.

The borrowing cost will be the same for both leasing and buying as the main determinant of the cost is the risk level taken on by the lender of money, which will be the same for either finance provider – the bank via a separate loan or the lease provider and the interest charge inherent in the lease payments.



Test your understanding 1 – Leasing calculations

Walshey Co has already decided to accept a project and is now considering how to finance it.

The asset could be leased over four years at a rental of \$36,000 per year, payable at the start of each year. Tax is payable at 30%, one year in arrears. The post-tax cost of borrowing is 10%.

Required:

Calculate the net present value of the leasing option.



Test your understanding 2 – Lease versus buy

A firm has decided to acquire a new machine to neutralise the toxic waste produced by its refining plant. The machine would cost \$6.4 million and would have an economic life of five years.

Tax-allowable depreciation of 25% per year on a reducing balance basis is available for the investment.

Taxation of 30% is payable on operating cash flows, one year in arrears.

The firm intends to finance the new plant by means of a five-year fixed interest loan at a pre-tax cost of 11.4% per year, with the principal repayable in five years' time.

As an alternative, a leasing company has proposed a lease over five years at \$1.42 million per year payable in advance.

Scrap value of the machine under each financing alternative will be zero.

Evaluate the two options for acquiring the machine and advise the company on the best alternative.



Other considerations

There may be other issues to consider before a final decision is made to lease or buy, for example:

- Who receives the residual value in the lease agreement?
- Any restrictions associated with the taking on of leased equipment, e.g. leases may restrict a firm's borrowing capacity.
- Any additional benefits associated with the lease agreement, e.g. maintenance or other support services.

2 Replacement decisions using equivalent annual cost and equivalent annual benefit

Once the decision has been made to acquire an asset for a long-term project, it is quite likely that the asset will need to be replaced periodically throughout the life of the project.

Where there are competing replacements for a particular asset, we must compare the possible replacement strategies available.

A problem arises where

- equivalent assets available are likely to last for different lengths of time or
- an asset, once bought, must be replaced at regular intervals.

The decision we are concerned with here is – how often should the asset be replaced?

3 Equivalent annual costs (EACs)



In order to deal with the different timescales, the NPV of each option is converted into an annuity or an EAC.

The EAC is the equal annual cash flow (annuity) to which a series of uneven cash flows is equivalent in PV terms.

The formula used is:

$$\text{EAC} = \frac{\text{PV of costs}}{\text{Annuity factor}}$$



The **optimum replacement period (cycle)** will be the period that has the lowest EAC, although in practice other factors may influence the final decision.

Note that the above formula is just a rearrangement of the annuity present value formula, where the EAC represents the annual annuity cash flow:

$$\text{PV} = \text{annual cash flow} \times \text{annuity factor}$$



The method can be summarised as:

- (1) calculate the NPV of each strategy or replacement cycle
- (2) calculate the EAC for each one
- (3) choose the one with the lowest EAC.

Key assumptions

- Cash inflows from trading are ignored since they will be similar regardless of the replacement decision. In practice, using an older asset may result in lower quality, which in turn could affect sales.
- The operating efficiency of machines will be similar with differing machines or with machines of differing ages.
- The assets will be replaced in perpetuity or at least into the foreseeable future.
- In most questions tax and inflation are ignored.
- As with all NPV calculations, non-financial aspects such as pollution and safety are ignored. An older machine may have a higher chance of employee accidents and may produce more pollution.
- Note that if required to compare different sets of cash **inflows** with differing lifespans, an equivalent annual **benefit** calculation can be done using the same principles.



Replacement decisions

The factors to be considered when making replacement decisions are as follows:

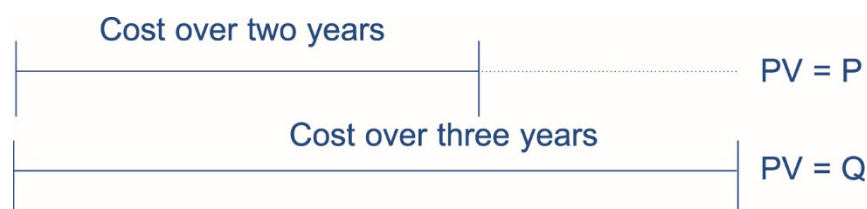
- Capital cost of new equipment – the higher cost of equipment will have to be balanced against known or possible technical improvements.
- Operating costs – operating costs will be expected to increase as the machinery deteriorates over time. This is referred to as operating inferiority, and is the result of:
 - increased repair and maintenance costs
 - loss of production due to ‘down time’ resulting from increased repair and maintenance time
 - lower quality and quantity of output.
- Resale value – the extent to which old equipment can be traded in for new. This value generally decreases with the age of the asset.
- Taxation and investment incentives.
- Inflation – both the general price level change and relative movements in the prices of inputs and outputs.

Determining the optimum replacement period (cycle) will largely be influenced by:

- the capital cost/resale value of the asset – the longer the period, the less frequently these will occur
- the annual operating costs of running the asset – the longer the period, the higher these will become.

The timescale problems

A special feature of replacement problems is that it involves comparisons of alternatives with different timescales. If the choice is between replacing an item of machinery every two years or every three years, it would be meaningless simply to compare the NPV of the two costs.



Almost certainly $P < Q$. However, this does not take account of the cost of providing an asset for the third year. One way of comparing asset replacement options is to convert the PV of cost over one replacement cycle to an equivalent annual PV of cost.

In other words, we compute the PV of costs over one cycle and then turn it into an EAC using the annuity factor for the number of years in the replacement cycle.

Thus, the costs associated with any particular cycle can be considered as equivalent to having to pay this EAC every year throughout the cycle and throughout subsequent cycles.

The alternative would be to evaluate the replacements over a period that covers a full number of cycles for each choice. For instance, evaluation over a period of 6 years would represent 3 2-year cycles and 2 3-year cycles and would give a meaningful comparison.



Test your understanding 3 – Equivalent annual costs

A machine costs \$20,000.

The following information is also available:

Running costs (payable at the end of the year):

Year 1	\$5,000
Year 2	\$5,500

Trade-in allowance

Disposal after 1 year: \$16,000

Disposal after 2 years: \$13,000

Calculate the optimal replacement cycle if the cost of capital is 10%.



Test your understanding 4 – Equivalent annual costs

A decision has to be made on replacement policy for vans. A van costs \$12,000 and the following additional information applies:

Asset sold at end of year	Trade-in allowance	Asset kept for	Maintenance cost at end of year
	\$		\$
1	9,000	1 year	0
2	7,500	2 years	1,500 in 1st year
3	7,000	3 years	2,700 in 2nd year

Calculate the optimal replacement policy at a cost of capital of 15%.

Note that the asset is only maintained at the end of the year if it is to be kept for a further year, i.e. there are no maintenance costs in the year of replacement.

Ignore taxation and inflation.

Limitations of replacement analysis

The model assumes that when an asset is replaced, the replacement is in all practical respects identical to the last one and that this process will continue for the foreseeable future. However, in practice this will not hold true owing to:

- changing technology
- inflation
- changes in production plans.



Limitations of replacement analysis

The replacement analysis model assumes that the firm replaces like with like each time it needs to replace an existing asset.

However, this assumption ignores:

- Changing technology – machines fast become obsolete and can only be replaced with a more up-to-date model, which will be more efficient and perhaps perform different functions.
- Inflation – the increase in prices over time alters the cost structure of the different assets, meaning that the optimal replacement cycle can vary over time.
- Changes in production plans – firms cannot predict with accuracy the market environment they will be facing in the future and whether they will even need to make use of the asset at that time.



Equivalent annual benefits

The equivalent annual benefit (EAB) is the annual annuity with the same value as the net present value of an investment project.

It can be calculated using a similar formula to that used for the equivalent annual cost:

$$\text{EAB} = \frac{\text{NPV of project}}{\text{Annuity factor}}$$

Calculating the EAB is particularly useful when trying to compare projects with unequal lives. The project with the highest EAB would be preferred.

For instance, consider two projects, X and Y, only one of which can be undertaken:

Project	X	Y
NPV at 12% cost of capital	\$40,000	\$30,000
Duration	3 years	2 years

On the face of it, project X is a better project as it has a higher NPV. If the projects are one-off projects that can't be repeated, project X would be preferred.

However, if, once each project is complete, it can be repeated and this can happen over and over again into the foreseeable future, the choice is not as obvious. Project X earns the higher NPV, but can only be repeated every three years, whereas Y earns a lower NPV, but that NPV can be earned every two years.

To make the project benefits more comparable, we compare their equivalent annual benefits (EABs):

Project	X	Y
NPV at 12% cost of capital	\$40,000	\$30,000
Annuity factor at 12%	2.402	1.690
EAB	\$16,653	\$17,751

Now we can see that the ability to earn the lower NPV every two years outweighs the benefit of the higher NPV earned less frequently. Project Y should be chosen.

4 Capital rationing

An introduction

Shareholder wealth is maximised if a company undertakes all possible positive NPV projects. Capital rationing is where there are insufficient funds to do so. This implies that where investment capital is rationed, shareholder wealth is not being maximised.



Ensure you are able to discuss the difference between hard (external) and soft (internal) capital rationing.

Hard capital rationing: An absolute limit on the amount of finance available is imposed by investors.

Soft capital rationing: A company may impose its own rationing on capital. This is contrary to the rational view of shareholder wealth maximisation.



Reasons for hard or soft capital rationing

Hard capital rationing

Reasons for hard capital rationing:

- Industry-wide factors limiting funds.
- Company-specific factors, such as:
 - lack of or poor track record
 - lack of asset security
 - poor management team.

Soft capital rationing

Reasons for soft capital rationing:

- Limited management skills available.
- Desire to maximise return of a limited range of investments.
- Limited exposure to external finance.
- Encourages acceptance of only substantially profitable business.
- Improving investment conditions – reluctance to commit to long-term projects.

Single and multi-period capital rationing

Single-period capital rationing: Shortage of funds for this period only.

Multi-period capital rationing: Shortage of funds in more than one period (outside syllabus). You will learn techniques to deal with this if you choose to take the Advanced Financial Management exam.

The method for dealing with single-period capital rationing is similar to the limiting factor analysis used elsewhere in decision-making.

The profitability index (PI) and divisible projects



If a project is divisible, any fraction of the project may be undertaken and the returns from the project are expected to be generated in exact proportion to the amount of investment undertaken. Projects cannot however be undertaken more than once.



The aim when managing capital rationing is to maximise the NPV earned per \$1 invested in projects.

Where the projects:

- are divisible (i.e. can be done in part)
- earn corresponding returns to scale

it is achieved by:

- (1) calculating a PI for each project (see below)
- (2) ranking the projects according to their PI
- (3) allocating funds according to the projects' rankings until they are used up.

The formula is:

$$PI = \frac{NPV}{\text{Investment}}$$



Test your understanding 5 – Divisible projects

A company has \$100,000 available for investment and has identified the following 5 investments in which to invest. All investments must be started now (Yr 0).

Project	Initial investment (Yr 0)	NPV
	\$000	\$000
C	40	20
D	100	35
E	50	24
F	60	18
G	50	(10)

Required:

Determine which projects should be chosen to maximise the return to the business.

Indivisible projects – trial and error



If a project is indivisible, it must be done in its entirety or not at all.

Where projects cannot be done in part, the optimal combination can only be found by trial and error.



Test your understanding 6 – Indivisible projects

The company from TYU5 has the same problem as before but this time the projects are indivisible. The information is reproduced below:

A company has \$100,000 available for investment and has identified the following 5 investments in which to invest. All investments must be started now (Yr 0).

Project	Initial investment (Yr 0)	NPV
	\$000	\$000
C	40	20
D	100	35
E	50	24
F	60	18
G	50	(10)

Required:

Determine the optimal project selection.



The key in the examination is to ascertain whether or not the projects are divisible.

Divisible projects can be ranked using the PI. Combinations of indivisible projects must be considered on a trial and error basis.

Mutually exclusive projects

Sometimes the taking on of projects will preclude the taking on of another, e.g. they may both require use of the same asset.

In these circumstances, each combination of investments is tried to identify which earns the higher level of returns.



Test your understanding 7 – Mutually exclusive projects

Using the same company information for A Co (divisible projects) the additional factor to be considered is that projects C and E are mutually exclusive.

The information is reproduced below.

A company has \$100,000 available for investment and has identified the following 5 investments in which to invest. All investments must be started now (Yr 0).

Project	Initial investment (Yr 0) \$000	NPV \$000	PI (NPV/Investment)
C	40	20	0.5
D	100	35	0.35
E	50	24	0.48
F	60	18	0.3
G	50	(10)	Not worth while

Required:

Determine the optimal project selection.



Test your understanding 8

Sarratt Co has calculated the following NPVs relating to two mutually exclusive projects, X and Y.

	X	Y
NPV at 10%	100	35
NPV/\$ invested	50	24

The company's cost of capital is 10%.

If the objective of Sarratt plc is the maximisation of shareholder wealth, which project(s), if any, should it accept?

- A Neither project
- B Project Y only
- C Project X only
- D Both projects



Test your understanding 9

Which of the following is not a reason for hard capital rationing?

- A Lending institutions may consider the company to be too risky
- B There are restrictions on lending due to government control
- C The costs associated with making small issues of capital may be too great
- D Desire to maximise return of a limited range of investments



Test your understanding 10

A company has a number of projects available to it but has a limit of \$20,000 on its capital investment funds. Each project has an initial outlay followed by a constant annual cash inflow in perpetuity, commencing in one year's time. The projects are as follows.

Project	Initial outlay	Inflow per year
	\$	\$
E	6,000	900
F	8,000	1,000
G	10,000	3,500
H	12,000	3,600
I	20,000	4,600

The company's cost of capital is 10% per year and all projects are independent and indivisible.

What is the maximum net present value that can be generated?

- A \$26,000
- B \$27,000
- C \$28,000
- D \$45,000



Test your understanding 11

A machine costing \$150,000 has a useful life of eight years, after which time its estimated resale value will be \$30,000. Annual running costs will be \$6,000 for the first three years of use and \$8,000 for each of the next five years. All running costs are payable on the last days of the year to which they relate.

Using a discount rate of 20% per year, what is the equivalent annual cost (to the nearest \$100) of using the machine if it were bought and replaced every eight years in perpetuity?

- A \$21,100
- B \$34,000
- C \$44,200
- D \$46,600

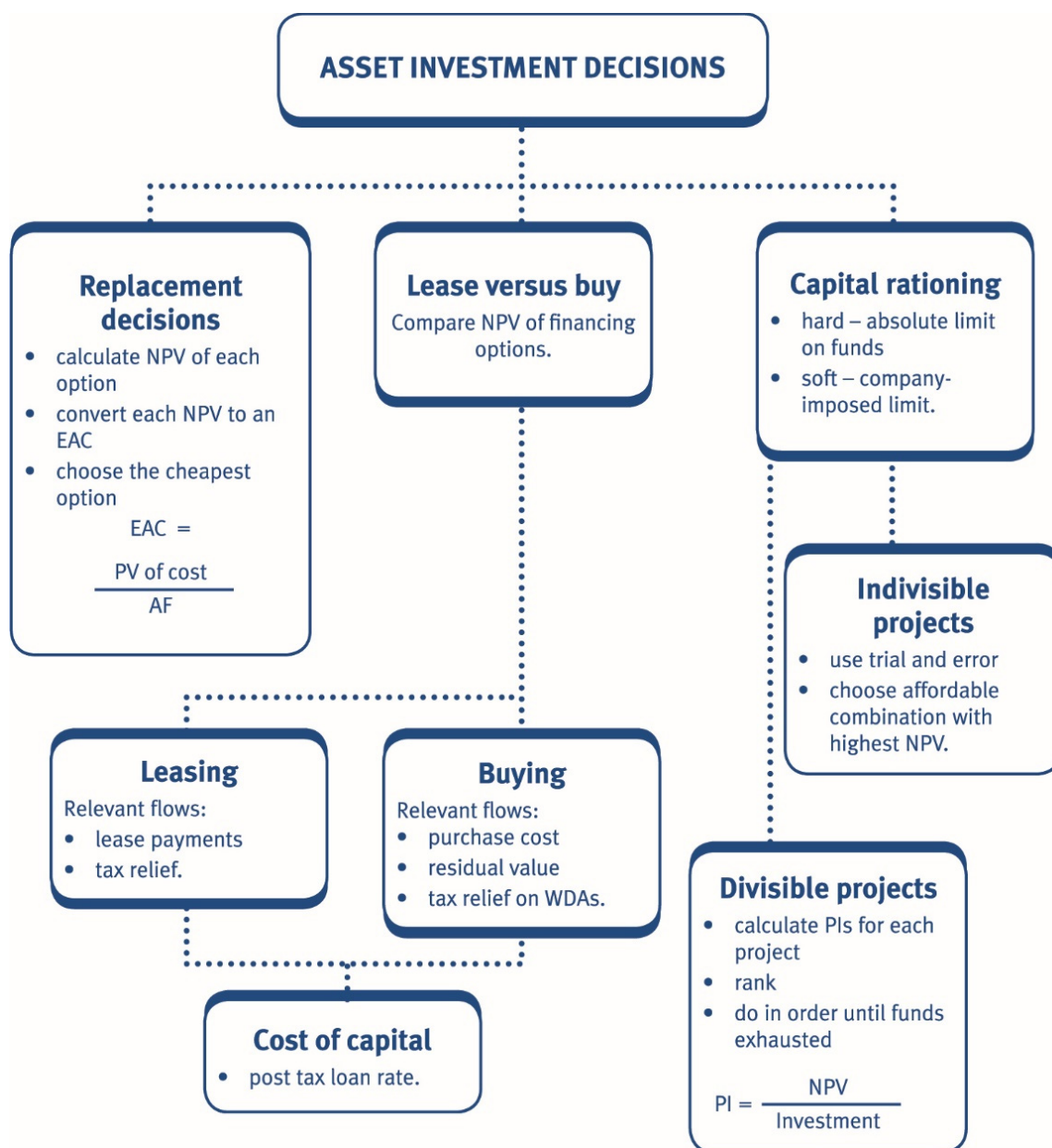


Test your understanding 12

Which of the following statements is incorrect?

- A The optimum replacement period (cycle) will be the period that has the highest equivalent annual cost, although in practice, other factors may influence the final decision
- B The replacement analysis model assumes that the firm replaces like with like each time it needs to replace an existing asset
- C The replacement analysis model ignores inflation, meaning that the optimal replacement cycle can vary over time
- D The replacement analysis model ignores the fact that firms cannot predict whether they will even need to make use of the asset in the future

5 Chapter summary



Test your understanding answers



Test your understanding 1 – Leasing calculations

Cost of leasing:

Year		Cash flows \$	DF @10%	PV \$
0–3	Rentals	(36,000)	1.000 + 2.487	(125,532)
2–5	Tax relief	10,800	3.170 × 0.909 = 2.882*	31,126
				NPV (94,406)

*Or $3.791 - 0.909 = 2.882$



Test your understanding 2 – Lease versus buy

(W1) Calculation of the tax relief on tax-allowable depreciation if asset bought:

Year	Narrative	Written down value	Tax saved/(extra tax paid) at 30%	Timing of tax flow
		\$m	\$m	
0	Cost	6.400		
1	Tax-allowable depreciation	1.600	0.480	2
		4.800		
2	Tax-allowable depreciation	1.200	0.360	3
		3.600		
3	Tax-allowable depreciation	0.900	0.270	4
		2.700		
4	Tax-allowable depreciation	0.675	0.203	5
		2.025		
5	Disposal proceeds	0.000		
		2.025		
	Balancing allowance	2.025	0.608	6

Note: The asset is bought at time $t=0$ as usual with the first amount of tax-allowable depreciation in the year ended time one. Given the one-year time lag on the tax, the first tax effect is at time $t=2$.

(W2) Calculation of the post-tax cost of borrowing.

The pre-tax cost of borrowing is 11.4%.

The post-tax cost of borrowing can be approximated by multiplying this by $(1 - \text{tax rate})$, i.e. $11.4\% \times (1 - 0.3) = 7.98\%$, say 8% (strictly this ignores the impact of a one-year time delay on tax relief, but this is acceptable).

Cost of borrowing to buy:

Time	0	1	2	3	4	5	6
	\$m	\$m	\$m	\$m	\$m	\$m	\$m
Asset	(6.400)						
Tax relief on t.a.d. (W1)			0.480	0.360	0.270	0.203	0.608
	(6.400)	0.000	0.480	0.360	0.270	0.203	0.608
PV factor @ 8% (W2)	1.000	0.926	0.857	0.794	0.735	0.681	0.630
PV	(6.400)	0	0.411	0.286	0.198	0.138	0.383
NPV = \$(4.984)m							

Cost of leasing:

Timing	Narrative	Cash flow \$m	DF @ 8%	PV \$m
0–4	Lease payments	(1.420)	1 + 3.312	(6.123)
2–6	Tax savings	0.426	3.993×0.926	1.575
	NPV			(4.548)

The cost of leasing is lower than the cost of buying and the asset should therefore be acquired under a lease.



Test your understanding 3 – Equivalent annual costs

Note: In contrast to the maintenance costs in the above illustration, running costs (e.g. petrol) are incurred in every year of ownership.

One-year replacement cycle:

	0	1
	\$	\$
Buy asset	(20,000)	
Running costs		(5,000)
Trade-in		16,000
Net cash flow	(20,000)	11,000
DF@10%	1	0.909
PV	(20,000)	9,999
NPV	(10,001)	

$$\text{Equivalent annual cost} = \frac{\$(10,001)}{1 \text{ yr AF}} = \frac{\$(10,001)}{0.909} = \$(11,002)$$

Two-year replacement cycle:

	0	1	2
	\$	\$	\$
Buy asset	(20,000)		
Running costs		(5,000)	(5,500)
Trade-in			13,000
Net cash flow	(20,000)	(5,000)	7,500
DF@10%	1	0.909	0.826
PV	(20,000)	(4,545)	6,195
NPV	(18,350)		

$$\text{Equivalent annual cost} = \frac{\$(18,350)}{2 \text{ yr AF}} = \frac{\$(18,350)}{1.736} = \$(10,570)$$

The machine should therefore be replaced after two years.



Test your understanding 4 – Equivalent annual costs

The costs incurred over a single cycle are computed and the EAC is found as follows.

(1) Replace every year

	0	1	2
	\$	\$	\$
Buy asset	(12,000)		
Maintenance costs	0		
Trade-in		9,000	
Net cash flow	(12,000)	9,000	
DF@15%	1	0.870	
PV	(12,000)	7,830	
NPV	(4,170)		
	PV of costs	(4,170)	
EAC =			
	1 year AF	0.870	

(2) Replace every two years

	0	1	2
Buy asset	(12,000)		
Maintenance costs		(1,500)	
Trade-in			7,500
Net cash flow	(12,000)	(1,500)	7,500
DF@15%	1	0.870	0.756
PV	(12,000)	(1,305)	5,670
NPV	(7,635)		
	PV of costs	(7,635)	
EAC =			
	2 year AF	1.626	

(3) Replace every three years

	0	1	2	3
Buy asset	(12,000)			
Maintenance costs		(1,500)	(2,700)	
Trade-in				7,000
	_____	_____	_____	_____
Net cash flow	(12,000)	(1,500)	(2,700)	7,000
DF@10%	1	0.870	0.756	0.658
	_____	_____	_____	_____
PV	(12,000)	(1,305)	(2,041)	4,606
	_____	_____	_____	_____
NPV	(10,740)			
		PV of costs	(10,740)	
EAC =	_____	=	_____	= \$(4,704)
	3 year AF		2.283	

Here, the optimal replacement period is every two years.

Note that the EAC is that sum that could be paid annually in arrears to finance the three replacement cycles. It is equivalent to the budget accounts that various public services encourage customers to open to spread the cost of those services more evenly. The PV of annual sums equal to the EAC is the same as the PV of the various receipts and payments needed to buy and maintain a van.



Test your understanding 5 – Divisible projects

Project	Working	PI	Ranking
C	20/40	= 0.5	1
D	35/100	= 0.35	3
E	24/50	= 0.48	2
F	18/60	= 0.3	4
G	Not worth while		

Funds available	Projects undertaken	NPV earned
\$		\$
100,000		
(40,000)	Project C	20,000
60,000		
(50,000)	Project E	24,000
10,000		
(10,000)	10/100 = 10% project	3,500
Nil	Total NPV	47,500



Test your understanding 6 – Indivisible projects

Alternatives Mix	Investment	NPV earned
	\$	\$
C,F	100,000	38,000
D	100,000	35,000
C,E	90,000	44,000*

*C and E is the best mix. There is a problem however relating to the unused funds. The assumption is that the un-utilised funds will earn a return equivalent to the cost of capital and hence will generate an NPV of 0. This may or may not be the case.


Test your understanding 7 – Mutually exclusive projects

Mix	Investment	NPV
	\$	\$
Project C mix C, 60% D	\$100,000	\$41,000 (20 + 0.6 × 35)
Project E mix E, 50% D	\$100,000	\$41,500* (24 + 0.5 × 35)

* The best mix

Note that F is not considered as it ranks below project D according to the PI.


Test your understanding 8
Answer C

The correct answer (based on NPVs) is project X.


Test your understanding 9
Answer D

This is an example of soft capital rationing.

Soft capital rationing is brought about by internal factors. Hard capital rationing is brought about by external factors.


Test your understanding 10
Answer C

Project	NPV
	\$
E	$-6,000 + 900/0.1$ 3,000
F	$-8,000 + 1,000/0.1$ 2,000
G	$-10,000 + 3,500/0.1$ 25,000
H	$-12,000 + 3,600/0.1$ 24,000
I	$-20,000 + 4,600/0.1$ 26,000

	NPV
	\$
E + F	5,000
E + G	28,000
E + H	27,000
F + G	27,000
F + H	26,000
I	26,000

The best combination is E + G, where NPV = \$28,000



Test your understanding 11

Answer C

PV of cash flows:

$$T_0 \text{ } \$ (150,000) \times 1 = \$ (150,000)$$

$$T_{1-3} \text{ } \$ (6,000) \times 2.106 = \$ (12,636)$$

$$T_{4-8} \text{ } \$ (8,000) \times 2.991 \times 0.579 = \$ (13,854) \text{ (NB: delayed annuity)}$$

$$T_8 \text{ } \$ 30,000 \times 0.233 = \$ 6,990$$

$$\text{Total PV of cash flows} = \$ (169,500)$$

Equivalent annual cost calculation:

$$\text{NPV} = \frac{\$ (169,500)}{3.837} = \$ (44,175)$$



Test your understanding 12

Answer A

The optimum replacement period (cycle) will be the period that has the lowest (not the highest) equivalent annual cost, although in practice other factors may influence the final decision.

Investment appraisal under uncertainty

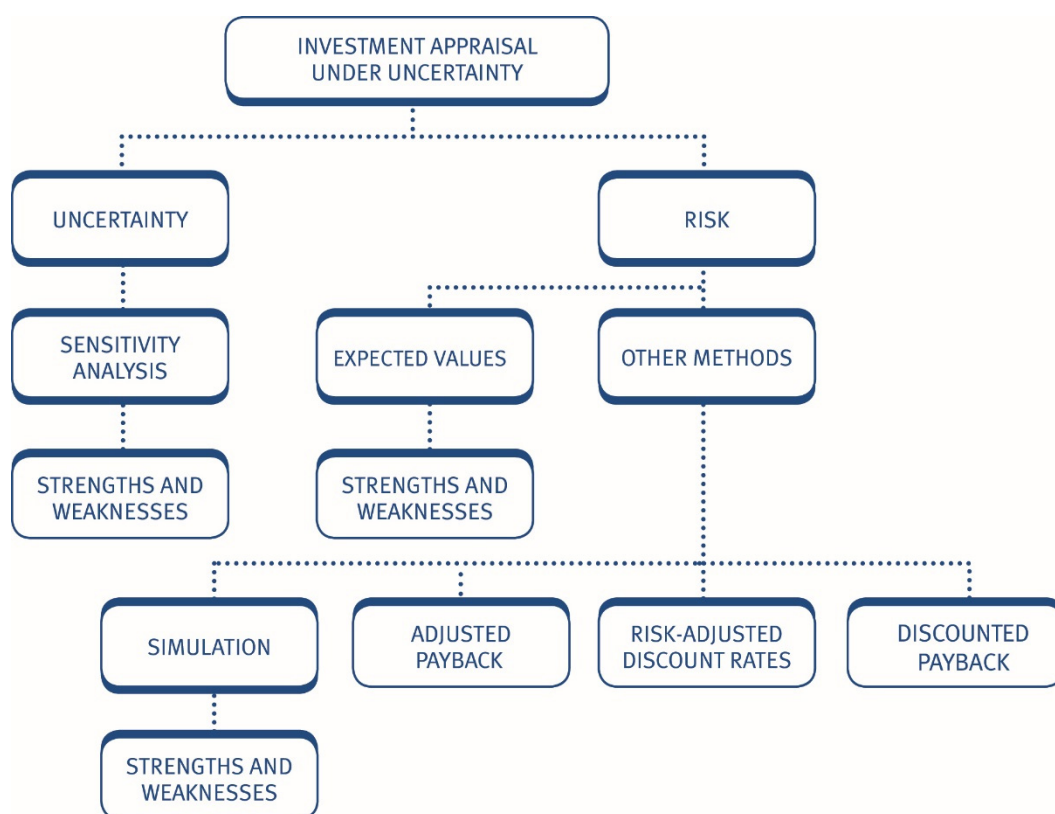
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- distinguish between risk and uncertainty in investment appraisal
- define sensitivity analysis and discuss its usefulness in assisting investment decisions
- apply sensitivity analysis to investment projects and explain the meaning of the findings
- define an expected value (EV) and discuss the usefulness of probability analysis in assisting investment decisions
- apply probability analysis to investment projects and explain the meaning of the findings
- discuss the use of simulation to take account of risk and uncertainty in investment appraisal
- discuss the use of adjusted payback in investment appraisal
- calculate the discounted payback and discuss its usefulness as an investment appraisal method
- explain the principle of adjusting discount rates to take account of risk.



One of the PER performance objectives (PO9) is to evaluate investment and financing decisions. You evaluate investment opportunities and their consequences and advise on their costs and benefits to the organisation. Working through this chapter should help you understand how to demonstrate that objective.



1 Risk and uncertainty

The difference between risk and uncertainty

Investment appraisal faces the following problems:

- all decisions are based on forecasts
- all forecasts are subject to uncertainty
- this uncertainty needs to be reflected in the financial evaluation.



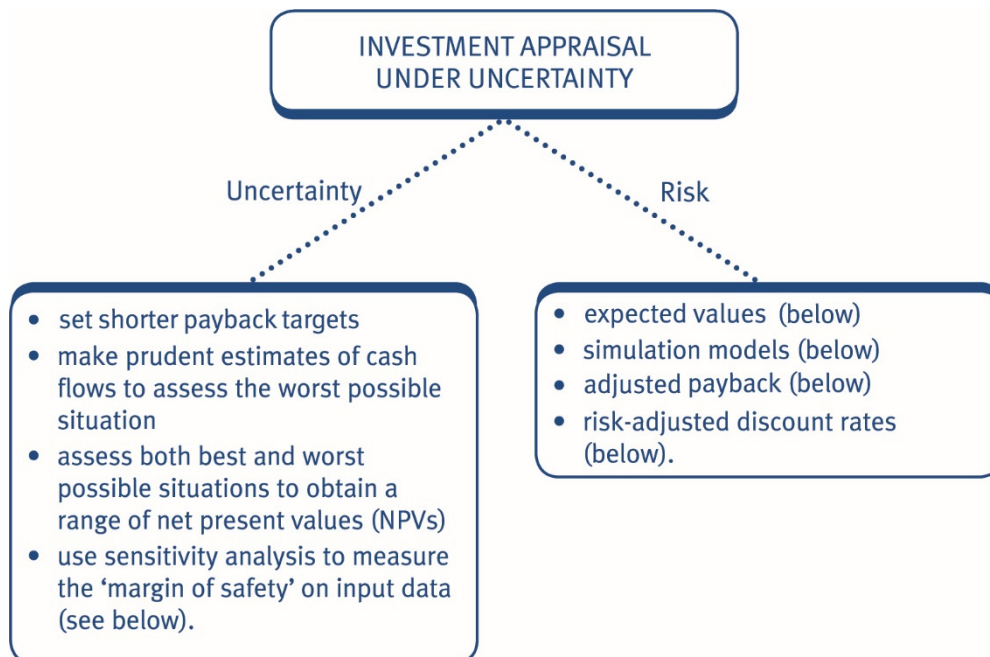
Risk is represented by the variability of potential returns. Within investment appraisals, the decision maker must also distinguish between:

- **risk** – quantifiable – possible outcomes have associated probabilities, thus allowing the use of mathematical techniques
- **uncertainty** – unquantifiable – outcomes cannot be mathematically modelled.

In investment appraisal, the areas of concern are therefore the accuracy of the estimates concerning:

- project life
- predicted cash flows and associated probabilities
- discount rate used.

Incorporating risk and uncertainty



2 Sensitivity analysis

- Sensitivity analysis typically involves posing 'what if?' questions.
- For example, what if demand fell by 10% compared to our original forecasts? Would the project still be viable?
- Ideally, we want to know how much demand could fall before the project should be rejected or, equivalently, the breakeven demand that gives an NPV of zero. We could then assess the likelihood of forecast demand being that low.

Calculating sensitivity

This maximum possible change in a variable that would bring the NPV to zero can be expressed as a percentage:

$$\text{Sensitivity margin} = \frac{\text{NPV}}{\text{Present value (PV) of cash flow under consideration}} \times 100\%$$

This would be calculated for each variable under consideration individually.

The value of the 'PV of cash flow under consideration' should include all effects of a changing variable. For instance, the cash flows affected by a change in sales volume would include both sales revenue and any variable costs, so the relevant figure for the bottom of the calculation would be the PV of contribution.

If taxation is included in the question, then it must also be considered. For instance, if the sensitivity to the sales price is to be calculated in a situation where tax is charged on profits at a rate of 30% and paid one year in arrears, then both revenue and taxation would be affected by a change in the sales price. The calculation of the PV of the cash flow under consideration would be:

[PV of revenue] **less** [PV of revenue $\times$ 30% $\times$ 1 year discount factor]

If there is no delay in the payment of tax, the multiplication by the discount factor is not needed.



The lower the sensitivity margin, the more sensitive the decision to the particular parameter being considered, i.e. small changes in the estimate could change the project decision from accept to reject.

Calculating sensitivity to the discount rate

The sensitivity value represents the percentage change in a variable that brings the project NPV to zero.

The discount rate at which the project NPV is zero is the internal rate of return (IRR).

Therefore, the sensitivity to the discount rate is calculated as the percentage change from the company's cost of capital to the IRR.



Test your understanding 1 – Sensitivity analysis

An investment of \$40,000 today is expected to give rise to annual contribution of \$25,000. This is based on selling one product, with a sales volume of 10,000 units, selling price of \$12.50 and variable costs per unit of \$10. Incremental annual fixed costs of \$10,000 will be incurred for the next four years; the discount rate is 10%.

Required:

- (a) Calculate the NPV of this investment.
- (b) Calculate the sensitivity of your calculation to the following:
 - (i) initial investment
 - (ii) selling price per unit
 - (iii) variable cost per unit
 - (iv) sales volume
 - (v) fixed costs
 - (vi) discount rate



Test your understanding 2 – Sensitivity analysis

Bacher Co is considering investing \$500,000 in equipment to produce a new type of ball. Sales of the product are expected to continue for three years, at the end of which the equipment will have a scrap value of \$80,000. Sales revenue of \$600,000 per year will be generated at a variable cost of \$350,000. Annual fixed costs will increase by \$40,000

- (a) **Determine whether, on the basis of the estimates given, the project should be undertaken, assuming that all cash flows occur at annual intervals and that Bacher Co has a cost of capital of 15%.**
- (b) **Find the percentage changes required in the following estimates for the investment decision to change:**
- (i) initial investment
 - (ii) scrap value
 - (iii) selling price
 - (iv) unit variable cost
 - (v) annual fixed cost
 - (vi) sales volume
 - (vii) cost of capital.



Test your understanding 3 – Sensitivity analysis with taxation

Elem Co is appraising a new project. The project has an NPV of \$72,000 when calculated using the company's cost of capital of 9%. There is some uncertainty about the sales price used.

A current sales price of \$65 has been predicted for the project. This price has been inflated at 5% per year over the life of the three year project. Sales volumes are due to be 5,000 units per year. The taxation rate is 30% and tax is paid a year in arrears.

Calculate the sensitivity of the project's NPV to changes in the sales price.

Advantages and disadvantages of sensitivity analysis

Advantages:

- simple
- provides more information to allow management to make subjective judgements
- identifies critical estimates.

Disadvantages:

- assumes variables change independently of each other
- does not assess the likelihood of a variable changing
- does not directly identify a correct decision.

**Advantages and disadvantages of sensitivity analysis****Strengths of sensitivity analysis**

- No complicated theory to understand.
- Information will be presented to management in a form that facilitates subjective judgement to decide the likelihood of the various possible outcomes considered.
- Identifies areas that are crucial to the success of the project. If the project is chosen, those areas can be carefully monitored.
- Indicates just how critical are some of the forecasts which are considered to be uncertain.

Weaknesses of sensitivity analysis

- It assumes that changes to variables can be made independently, e.g. material prices will change independently of other variables. This is unlikely. If material prices went up the firm would probably increase selling prices at the same time and there would be little effect on NPV. A technique called simulation (see later) allows us to change more than one variable at a time.
- It only identifies how far a variable needs to change. It does not look at the probability of such a change. It may be that sales volume appears to be the most crucial variable, but if the firm were facing volatile raw material markets a large change in raw material prices may be far more likely than a small change in sales volume.
- It is not an optimising technique. It provides information on the basis of which decisions can be made. It does not point directly to the correct decision.

3 Probability analysis

When there are several possible outcomes for a decision and probabilities can be assigned to each, a probability distribution of expected cash flows can often be estimated, recognising there are several possible outcomes, not just one. This could then be used to:

- (1) Calculate an expected value (EV);

(2) Measure risk by:

- (a) calculating the worst possible outcome and its probability;
- (b) calculating the probability that the project will fail (for example, that a negative NPV will result);
- (c) assessing the standard deviation of the outcomes.



The EV is the weighted average of all the possible outcomes, with the weightings based on the probability estimates.



The standard deviation is a statistical measure of the variability of a distribution around its mean (i.e. it measures the dispersion of possible outcomes about the EV). The tighter the distribution, the lower this measure will be. It is a measure of risk – the wider the dispersion the riskier the situation.

Calculating an EV

The formula for calculating an EV is:

$$EV = \sum px$$

where

p = the probability of an outcome

x = the value of an outcome.

The EV is not the most likely result. It may not even be a possible result, but instead it finds the long-run average outcome.



Expected values

The EV is the weighted average of the outcomes, with the weightings based on the probability estimates.

The EV does not necessarily represent what the outcome will be, nor does it represent the most likely result. What it really represents is the average pay-off per occasion if the project were repeated many times (i.e. a 'long run' average).

There are two main problems with using EV to make decisions in this way:

- The project may only be carried out once. It could result in a sizeable loss and there may be no second chance to get our money back.
- The probabilities used are simply subjective estimates of our belief, on a scale from 0 to 1. There is probably little data on which to base these estimates.



Test your understanding 4 – Expected values

A firm has to choose between three mutually exclusive projects, the outcomes of which depend on the state of the economy. The following estimates have been made:

State of the economy	Recession	Stable	Growing
Probability	0.5	0.4	0.1
	NPV	NPV	NPV
	(\$000)	(\$000)	(\$000)
Project A	100	200	1,400
Project B	0	500	600
Project C	180	190	200

Determine which project should be selected on the basis of expected market values.

Using EVs in larger NPV calculations

The EV technique can be used to simplify the available data in a larger investment appraisal question.



Test your understanding 5 – Expected values in larger NPV

Dralin Co is considering an investment of \$460,000 in a non-current asset expected to generate substantial cash inflows over the next five years. Unfortunately, the annual cash flows from this investment are uncertain, but the following probability distribution has been established:

Annual cash flow (\$)	Probability
50,000	0.3
100,000	0.5
150,000	0.2

At the end of its five-year life, the asset is expected to sell for \$40,000. The cost of capital is 5%.

Should the investment be undertaken?

Note that the investment decision based upon an expected NPV is not as simple as identifying whether the result is positive or negative. For example, where a positive expected NPV is calculated, the decision-maker should also take into account the likelihood of the result being worse and how that would affect the investment decision.

Using EVs with decision matrices

EVs are also used to deal with situations where the same conditions are faced many times. The problems involve construction of a 'decision matrix'.



Using EVs with decision matrices

Illustration – Using expected values with decision matrices

A newsagent sells a weekly magazine, which advertises local second-hand goods. The owner can buy the magazines for 15c each and sell them at the retail price of 25c. At the end of each week, unsold magazines are obsolete and have no value.

The owner estimates a probability distribution for weekly demand which looks like this:

Weekly demand in units	Probability
10	0.20
15	0.55
20	0.25
	1.00

- What is the EV of demand?
- If the owner is to order a fixed quantity of magazines per week how many should that be? Assume no seasonal variations in demand.

Solution

- EV of demand = $(10 \times 0.20) + (15 \times 0.55) + (20 \times 0.25) = 15.25$ units per week.
- Step 1:** Set up a decision matrix of possible strategies (numbers bought) and possible demand, as follows:

	(Number demanded)		
Strategy (Number bought)	10	15	20
10			
15			
20			

Step 2: The 'payoff' from each combination of action and outcome is then computed:

No sale → loss of 15c per magazine.

Sale → profit of $25c - 15c = 10c$ per magazine.

- Payoffs are shown for each combination of strategy and outcome:

Workings

- (i) If 10 magazines are bought, then 10 are sold no matter how many are demanded and the payoff is always $10 \times 10c = 100c$.
- (ii) If 15 magazines are bought and 10 are demanded, then 10 are sold at a profit of $10 \times 10c = 100c$, and 5 are scrapped at a loss of $5 \times 15c = 75c$, making a net profit of $25c$.
- (iii) The other contributions are similarly calculated.

Step 3: Probabilities are then applied to compute the expected value resulting from each possible course of action.

Alternatively, in the same matrix, probability $\times$ payoff can be inserted in each cell and totalled to give the expected payoff.

		(Number demanded)			EV
		p = 0.2 10	p = 0.55 15	p = 0.25 20	
Strategy (Number bought)	10	100	100	100	100
	15	25	150	150	125
	20	(50)	75	200	81.25

From this matrix, we can see that the best alternative is to buy 15 magazines each week.

What does this EV mean?

It means that if the strategy is followed for many weeks, then on average the profit will be $125c$ per week.

What actually happens is that eight weeks out of ten the payoff is likely to be $150c$ and two weeks out of ten it drops to $25c$.

This strategy produces the highest long-run profit for the firm.

Strengths and weaknesses of EVs**Strengths**

- Deals with multiple outcomes.
- Quantifies probabilities.
- Relatively simple calculation.
- Assists decision-making.

Weaknesses

- Subjective probabilities.
- Answer is only a long-run average.
- Ignores variability of payoffs.
- Risk neutral decision, i.e. ignores investor's attitude to risk.

Using EVs as a basis for decision-making is appropriate if three conditions are met or nearly met:

- there is a reasonable basis for making the forecasts and estimating the probability of different outcomes
- the decision is relatively small in relation to the business, so risk is small in magnitude
- the decision is for a category of decisions that are often made.



The EV technique is best suited to a problem that is repetitive and involves relatively small investments.



Advantages and limitations of EVs

Advantages of EVs

- The technique recognises that there are several possible outcomes and is, therefore, more sophisticated than single value forecasts.
- Enables the probability of the different outcomes to be quantified.
- Calculations are relatively simple.
- Leads directly to a simple optimising decision rule by reducing a range of possible outcomes into one number. However, care should be taken (see below).

Limitations of EVs

- By asking for a series of forecasts, the whole forecasting procedure is complicated. Inaccurate forecasting is already a major weakness in project evaluation. The probabilities used are also usually very subjective.
- The EV is merely a weighted average of the probability distribution, indicating the average payoff if the project is repeated many times.
- The EV gives no indication of the dispersion of possible outcomes about the EV. The more widely spread out the possible results are, the more risky the investment is usually seen to be. The EV ignores this aspect of the probability distribution.
- In ignoring risk, the EV technique also ignores the investor's attitude to risk. Some investors are more likely to take risks than others.

Joint probabilities

Questions in the exam may sometimes best be answered with the use of joint probability calculations. In the exam context, a joint probability may be described as the probability of one thing AND another thing happening.

Joint probability of A and B occurring = probability of A × probability of B

For instance, if the probability of a business's bank account having a positive cash flow in month 1 is 20% and the probability of it having a positive cash flow in month 2 is 40%, then the joint probability of it having a positive cash flow in both months (month 1 AND month 2) can be found by multiplying the individual probabilities together:

$$20\% \times 40\% = 8\% \text{ (or } 0.2 \times 0.4 = 0.08\text{)}$$

Also potentially useful is the ability to calculate the probability of one thing OR another thing happening.

If the events are mutually exclusive, this is found by adding the individual probabilities together.

Probability of A or B occurring (mutually exclusive) = probability of A + probability of B

For instance, if it is forecast that out of nine possible scenarios for a project, only two give a positive NPV, the probability of a positive NPV overall can be found by adding the individual probabilities of the two relevant scenarios .

If the two events are not mutually exclusive, you add the individual probabilities together and then deduct the joint probability of them both occurring.

Probability of A or B occurring (not mutually exclusive) = probability of A + probability of B – joint probability of A and B occurring

For instance, if the probability of receiving \$2,000 next month is 65% and the probability of receiving \$4,000 the month after is 75%, the probability of receiving one amount OR the other would be calculated as $65\% + 75\% - (65\% \times 75\%) = 91.25\%$.

As it would be possible for receipt of both amounts to occur, the events are not mutually exclusive and therefore their joint probability must be deducted. Also note that if this deduction is not made, the calculation here would produce a probability of greater than 100%, which is not possible.

Joint probability table

In exam questions that involve many possible outcomes along with probabilities attached to each, it may be best to use a joint probability table. This is simply a table that presents each potential outcome, along with their probabilities and joint probabilities.



Illustration 1 – Joint probability table

A project has the following possible sales volumes:

Year 1: 40% chance of selling 10 units and 60% chance of selling 16 units

Year 2: 25% chance of selling 13 units and 75% chance of selling 18 units.

Required:

Prepare a joint probability table of the potential total sales volumes over the 2-year period.

Solution

Y1 sales	P	Y2 sales	P	Total sales	Joint P
10	40%	13	25%	23	10%
10	40%	18	75%	28	30%
16	60%	13	25%	29	15%
16	60%	18	75%	34	45%
					100%

As there are 2 possible outcomes in year 1, each of which could lead to a further 2 outcomes in year 2, there are 4 possible outcomes. Each of these total outcomes has a row in the table and its joint probability is calculated.

Note that the total of the joint probabilities in a joint probability table that includes all possible outcomes will add up to 100%.



Test your understanding 6

A company has estimated cash flows for a project, incorporating the probability of each present cash flow value into its estimates as follows:

Year 1		Year 2	
Present cash flow	Probability	Present cash flow	Probability
\$100,000	20%	\$200,000	30%
\$200,000	80%	\$300,000	70%

The proposed investment will cost \$400,000, payable in full at the start of the project.

Calculate the following figures:

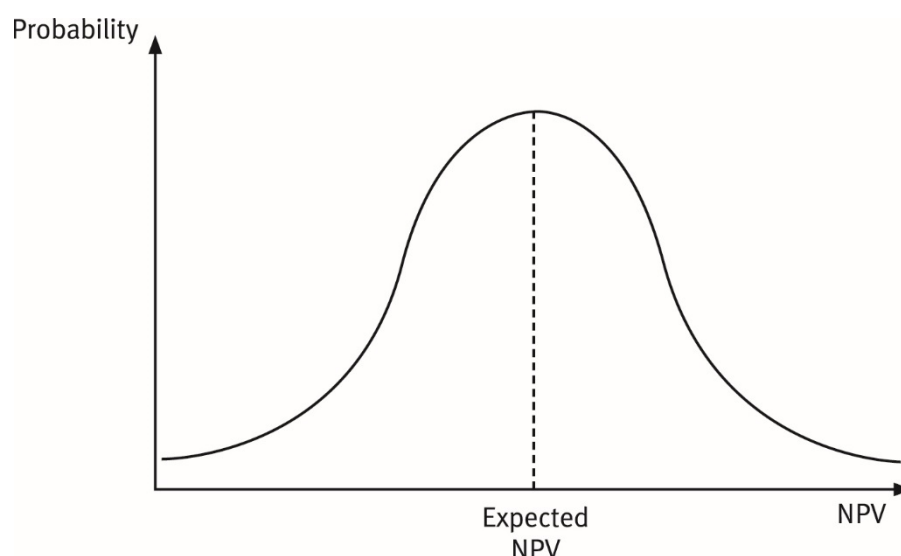
- The mean (expected) present cash flow in year 1
- The mean (expected) present cash flow in year 2
- The mean (expected) NPV of the investment
- The probability of the investment having a negative NPV
- The probability of the investment having a \$0 NPV

4 Further techniques for adjusting for risk and uncertainty

Simulation

Sensitivity analysis considers the effect of changing one variable at a time. Simulation improves on this by looking at the impact of many variables changing at the same time.

Using mathematical models, it produces a distribution of the possible outcomes from the project. The probability of different outcomes can then be calculated.



You will never be expected to carry out a simulation exercise in the exam although you may be asked to interpret the output.



Simulation

The four stages of a simulation exercise

There are four stages involved in carrying out a simulation exercise:

- (1) Specify major variables, e.g.:
 - Market details
 - Investment costs
 - Operating costs
- (2) Specify the relationships between variables to calculate an NPV, e.g.:

Sales revenue = market size × market share × selling price. Net cash flow = sales revenue – (variable costs + fixed costs + taxation), etc.
- (3) Simulate the environment:
 - select different values of each variable within the parameters set, along with their probabilities of occurring, and compute an NPV
 - repeat the process many times to create a probability distribution of returns.
- (4) The results of a simulation exercise will be a probability distribution of NPVs.
 - Instead of choosing between expected values, decision makers can now take the dispersion of outcomes and the expected return into account.

Advantages of simulation

The major advantages of simulation are as follows:

- it includes all possible outcomes in the decision-making process
- it is a relatively easily understood technique
- it has a wide variety of applications (inventory control, component replacement, corporate models, etc.).

Drawbacks of simulation

However, it does have some significant drawbacks:

- models can become extremely complex and the time and costs involved in their construction can be more than is gained from the improved decisions
- probability distributions may be difficult to formulate.

Adjusted payback

We looked at the payback form of investment appraisal in the chapter on basic investment appraisal.

One way of dealing with risk is to shorten the payback period required. This places more emphasis on the earlier cash flows, which are considered to be less risky.

However, given the disadvantages of payback as a method of investment appraisal, adjusted payback is rarely recommended as a method of adjusting for risk.

Discounted payback

It is also possible to reflect risk by using discounted cash flows in the calculation of a discounted payback period. The cash flows are first discounted using an appropriate discount rate that reflects the risk profile of the project. The cumulative discounted cash flow can then be calculated in the same manner as the cumulative cash flow is for the standard payback calculation.

The discounted payback method has the same advantages and disadvantages as for the traditional payback method except that the shortcoming of failing to account for the time value of money has been overcome.



Illustration 2 – Discounted payback period

A project with the following cash flows is under consideration:

T_0	T_1	T_2	T_3	T_4
(20,000)	8,000	12,000	4,000	2,000

Cost of capital 8%

Required:

Calculate the Discounted Payback Period.

Solution

Year	Working	Discounted cash flow	Cumulative discounted cash flow
0		(20,000)	(20,000)
1	$8,000 \times (1.08)^{-1} =$	7,407	(12,593)
2	$12,000 \times (1.08)^{-2} =$	10,288	(2,305)
3	$4,000 \times (1.08)^{-3} =$	3,175	870

Hence discounted payback period = 2 years + $(2,305/3,175) = 2.73$ years.

Risk-adjusted discount rates

The discount rate we have assumed so far is the rate that reflects either:

- the cost of borrowing funds in the form of a loan rate or
- the underlying required return of the business (i.e. the return required by the shareholder),
- or a mix of both.

For example, if an individual investment or project is perceived to be more risky than current investments, the increased risk could be used as a reason to increase the discount rate.



This is a key concept in investment appraisal. Applying the current discount rate or cost of capital to an investment assumes that the current business and gearing risk of the company will remain unchanged. If the project is significant in size and likely to result in additional risks then a project specific or risk-adjusted discount rate should be used.

The application of an increased discount rate is often successful in eliminating marginal projects. The addition to the usual discount rate is called the risk premium. The method used is examined further in the cost of capital chapter.

You will see again these techniques for dealing with risk and uncertainty and new ones if you choose to take the Advanced Financial Management exam.

Conservative values of variables

By using the most conservative (pessimistic) values of variables in an investment appraisal (e.g. lowest potential sales prices and volumes, highest potential cost levels, inflation rates, tax rates), a minimum potential NPV can be calculated.



Test your understanding 7

A company is currently evaluating a project that requires investments of \$12,000 now, and \$4,800 at the end of year 1. The cash inflow from the project will be \$16,800 at the end of year 2 and \$14,400 at the end of year 3. The cost of capital is 15%.

What is the discounted payback period (DPP) and the net present value (NPV)?

	DPP	NPV
A	2.0 years	\$6,000
B	2.36 years	\$4,400
C	2.0 years	\$4,400
D	2.36 years	\$6,000



Test your understanding 8

Petra is contemplating purchasing a machine for \$275,000, which she will use to produce 50,000 units of a product per year for five years. These products will be sold for \$10 each and unit variable costs are expected to be \$6. Incremental fixed costs will be \$70,000 per year for production costs and \$25,000 per year for selling and distribution costs. Petra has a required rate of return of 10% per year.

By how many units must the estimate of production and sale volume fall for the project to be regarded as not worthwhile?

- A 2,875
- B 6,465
- C 8,115
- D 12,315



Test your understanding 9

When using the expected value criterion, it is assumed that the individual wants to

- A Minimise risk for a given level of return
- B Maximise return for a given level of risk
- C Minimise risk irrespective of the level of return
- D Maximise return irrespective of the level of risk



Test your understanding 10

Which of the following statements is incorrect?

- A Sensitivity analysis assesses how responsive the project's NPV is to changes in the variables used to calculate that NPV
- B When calculating the sensitivity of each variable, the lower the percentage, the less sensitive is the NPV to that project variable
- C Management should review critical variables to assess whether or not there is a strong possibility of events occurring which will lead to a negative NPV
- D Sensitivity analysis does not provide a decision rule. Parameters defining acceptability must be laid down by managers



Test your understanding 11

A distinction should be made between the terms risk and uncertainty.

Which of the following statements is true?

Statement 1: Risk can be applied to a situation where there are several possible outcomes and, on the basis of past relevant experience, probabilities can be assigned to the various outcomes that could prevail.

Statement 2: Uncertainty can be applied to a situation where there are several possible outcomes but there is little past relevant experience to enable the probability of the possible outcomes to be predicted.

	Statement 1:	Statement 2:
A	True	True
B	True	False
C	False	True
D	False	False

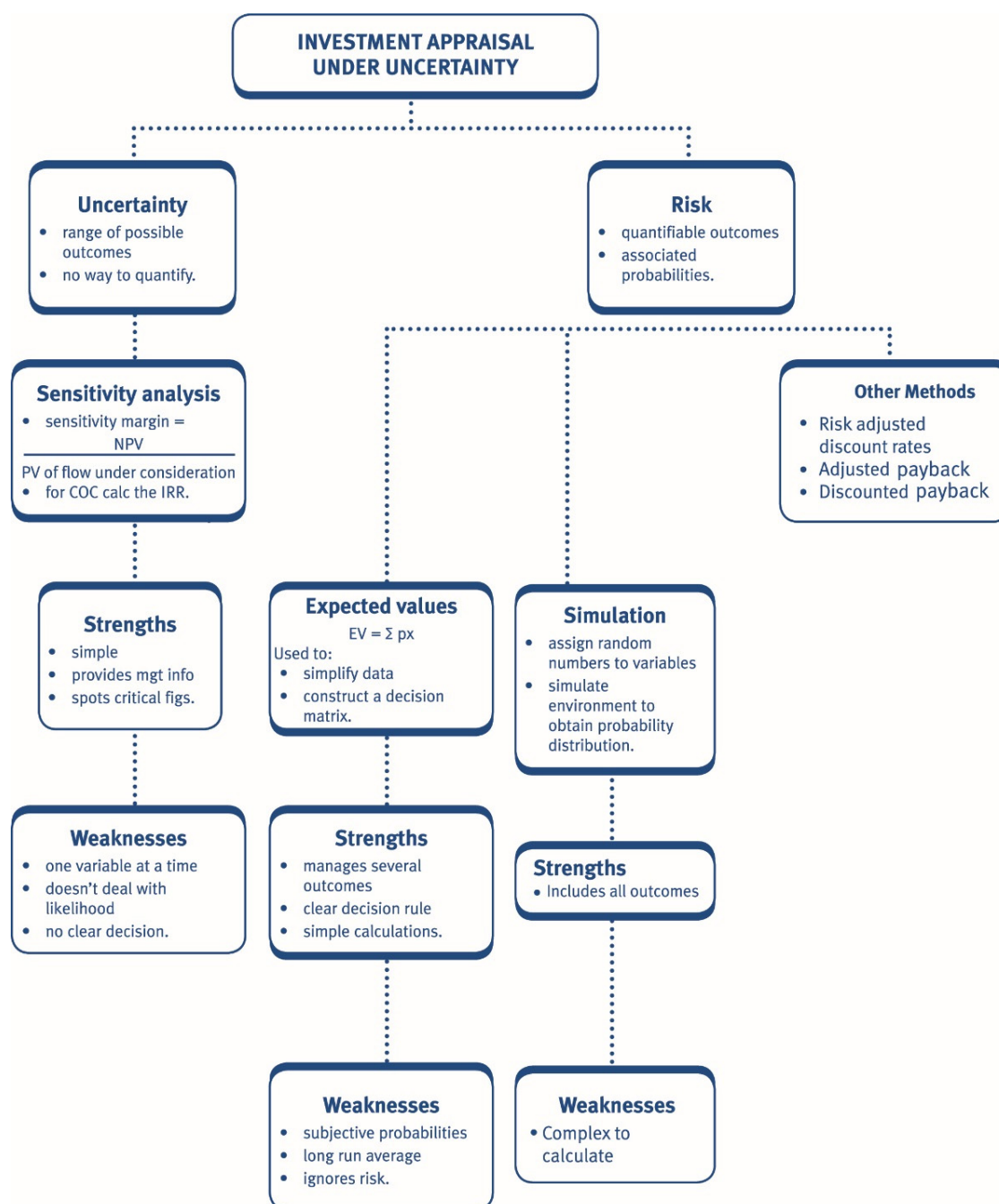


Test your understanding 12

The higher risk of a project can be recognised by decreasing

- A the cost of the initial investment of the project
- B the estimates of future cash inflows from the project
- C the internal rate of return of the project
- D the required rate of return of the project

5 Chapter summary



Test your understanding answers



Test your understanding 1 – Sensitivity analysis

(a)

Time	Narrative	Cash flow \$	DF 10%	PV \$
0	Investment	(40,000)	1.000	(40,000)
1–4	Contribution	25,000	3.170	79,250
1–4	Fixed costs	(10,000)	3.170	(31,700)
				NPV = 7,550

Therefore, the decision should be to accept the investment.

(b) (i) Sensitivity to initial investment = $\frac{7,550}{40,000} \times 100 = 18.9\%$

i.e. an 18.9% increase in the cost of the initial investment would cause the NPV to fall to zero.

(ii) Sensitivity to selling price per unit

If selling price changes, the revenue will change, so the relevant cash flow will be revenue.

PV of revenue = $\$12.50 \times 10,000 \times 3.170 = 396,250$

Sensitivity margin = $\frac{7,550}{396,250} \times 100 = 1.9\%$

i.e. if price per unit falls by more than 1.9%, the project will make a loss.

(iii) Sensitivity to variable cost per unit

If the variable cost per unit changes, the total variable cost will change, so the relevant cash flow will be total variable cost.

PV of total variable cost = $\$10 \times 10,000 \times 3.170 = 317,000$

Sensitivity margin = $\frac{7,550}{317,000} \times 100 = 2.4\%$

i.e. if cost per unit increases by more than 2.4%, the project will make a loss.

(iv) Sensitivity to sales volume

As sales volume affects both sales revenue and variable costs, being asked to find the sensitivity to sales volume is the same as sensitivity to contribution.

$$\text{PV of contribution} = (\$12.50 - \$10.00) \times 10,000 \times 3.170 = 79,250$$

$$\text{Sensitivity margin} = \frac{7,550}{79,250} \times 100 = 9.5\%$$

i.e. a 9.5% decrease in the level of contribution would cause the NPV to fall to zero.

(v) Sensitivity to fixed cost

$$\text{Sensitivity margin} = \frac{7,550}{31,700} \times 100 = 23.8\%$$

(vi) Sensitivity to the discount rate

To calculate the sensitivity to the discount rate, it is necessary to find the rate at which the project NPV is zero, i.e. the internal rate of return (IRR) of the project.

Time	Cash flow	DF %	PV \$
0	(40,000)	1,000	(40,000)
1–4	15,000	2,667 (W1)	40,000
			0

(W1)

If the project NPV is set to zero, the PV of the net cash inflows would need to be \$40,000. Hence, the discount factor needed to make the NPV go down to 0 is $40,000 \div 15,000 = 2.667$.

From tables, at four years, the closest annuity rate to 2.667 occurs at approximately 18%. This is therefore the breakeven discount rate, i.e. the IRR.

This quick method of finding the IRR will only work if all the inflows are in the form of annuities. Otherwise, the usual linear interpolation route must be followed.

The sensitivity is therefore $(18 - 10)/10 \times 100 = 80\%$, i.e. the cost of capital would have to rise by 80% before the NPV falls to zero.



Test your understanding 2 – Sensitivity analysis

Although part (a) could be completed most efficiently by finding the PV of net annual inflows (\$600,000 – \$350,000 – \$40,000), i.e. of \$210,000, part (b) would be most effectively negotiated if the separate PVs were found.

NPV calculation

Time		Cash flow	15% DF	PV
		\$000		\$000
0	Equipment	(500)	1	(500)
1–3	Revenue	600	2.283	1,370
1–3	Variable costs	(350)	2.283	(799)
1–3	Fixed costs	(40)	2.283	(91)
3	Scrap value	80	0.658	53
				—
NPV				33
				—

The project should, on the basis of these estimates, be accepted.

(a) Sensitivity analysis

(i) Initial investment

For the decision to change, the NPV must fall by \$33,000. For this to occur, the cost of the equipment must rise by \$33,000.

This is a rise of: $\frac{33}{500} \times 100 = 6.6\%$

(ii) Scrap value

If the NPV is to fall by \$33,000, the PV of scrap proceeds must fall by \$33,000. The PV of scrap proceeds is currently \$53,000. It must fall by: $33 \div 53 \times 100 = 62.26\%$, say 62%

(iii) Selling price

If sales price varies, sales revenue will vary (assuming no effect on demand). If the NPV of the project is to fall by \$33,000, the selling price must fall by:

$\frac{33}{1,370} \times 100 = 2.4\%$

(iv) Unit variable cost

The project's NPV must fall by \$33,000 therefore the PV of the variable costs must rise by \$33,000. Since the PV of variable costs is \$799,000, a rise of \$33,000 is an increase of:

$$\frac{33}{799} \times 100 = 4.1\%$$

(v) Annual fixed costs

Since the PV of fixed costs is \$91,000, a rise of \$33,000 is an increase of:

$$\frac{33}{91} \times 100 = 36\%$$

(vi) Sales volume

If sales volume falls, revenue and variable costs fall (contribution falls). If the NPV is to fall by \$33,000, volume must fall by:

$$\frac{33}{1,370 - 799} \times 100 = 5.8\%$$

(vii) Cost of capital

If NPV is to fall, cost of capital must rise. The figure which the cost of capital must rise to, that gives an NPV of zero, is the project's IRR.

To find the IRR, which is probably not much above 15%, the NPV at 20% can be found using the summarised cash flows.

$$\text{NPV (\$000)} = -500 + [210 \times 2.106] + [80 \times 0.579] = -11.$$

The IRR is therefore a little closer to 20% than 15%, but the formula can be used.

$$\begin{aligned} \text{IRR} &= 15 + \frac{33}{33 + 11} \times (20 - 15) \\ &= 18.75\% \end{aligned}$$

The cost of capital would have to increase from 15% to 18.75% before the investment decision changes.

This represents an increase of $(18.75 - 15)/15 \times 100 = 25\%$.



Test your understanding 3 – Sensitivity analysis with taxation

Given that taxation is included in the NPV and that tax cash flows are paid a year in arrears, the calculation for the sensitivity to the sales price is:

NPV / PV of sales revenue net of tax $\times 100$

The PV of revenue net of tax can be found using:

[PV of revenue] **less** [PV of revenue $\times 30\% \times 1$ year discount factor]

PV of revenue:

Time period	Sales price (\$) ($\times 1.05$ per year)	Sales volume	Revenue (\$)	Discount factor 9%	PV (\$)
1	68.25	5,000	341,250	0.917	312,926
2	71.66	5,000	358,300	0.842	301,689
3	75.24	5,000	376,200	0.772	290,426
					905,041

PV of revenue net of tax = \$905,041 – (\$905,041 $\times 30\% \times 0.917$)

PV of revenue net of tax = \$905,041 – \$248,977 = \$656,064

Sensitivity to sales price = \$72,000 / \$656,064 $\times 100 = 11.0\%$

Alternatively, making use of a spreadsheet and the =NPV formula:

	A	B	C	D	E	F
1	NPV	72000				
2						
3		t0	t1	t2	t3	t4
4	Sales price	65	68.25	71.66	75.24	
5	Sales volume		5,000	5,000	5,000	
6						
7	Sales revenue		341,250	358,300	376,200	
8	Tax at 30%			-102,375	-107,490	-112,860
9						
10	Net		341,250	255,925	268,710	-112,860
11						
12	PV	656,021				
13						
14	Sensitivity	11.0%				

Formula used in cell B12: =NPV(9%,C10:F10)

Formula used in cell B14: =B1/B12



Test your understanding 4 – Expected values

Project A

State of the economy	Probability (p)	Project NPV (x)	px
	\$000	\$000	\$000
Recession	0.5	100	50
Stable	0.4	200	80
Growing	0.1	1,400	140
Expected value			270

Project B

State of the economy	Probability (p)	Project NPV (x)	px
	\$000	\$000	\$000
Recession	0.5	0	0
Stable	0.4	500	200
Growing	0.1	600	60
Expected value			260

Project C

State of the economy	Probability (p)	Project NPV (x)	px
	\$000	\$000	\$000
Recession	0.5	180	90
Stable	0.4	190	76
Growing	0.1	200	20
Expected value			186

Based on expected values, Project A should be selected.

However, it should be noted that Project A is also the most risky option as it has the widest range of potential outcomes.



Test your understanding 5 – Expected values in larger NPV

Expected annual cash flows are:

Annual cash flow (x)	Project NPV (p)	px
50,000	0.3	15,000
100,000	0.5	50,000
150,000	0.2	30,000
Expected value		95,000

NPV calculation:

Time	Cash flow \$	DF 5%	PV \$
0	(460,000)	1,000	(460,000)
1–5	95,000	4,329	411,255
5	40,000	0.784	31,360
NPV =			(17,385)

As the expected NPV is negative, the project should not be undertaken.

An alternative approach would be to calculate three separate NPVs and then combine them, giving the following figures:

Annual cash flow \$	NPV working	NPV \$
50,000	$(\$460k) + \$50k \times 4.329 + \$40k \times 0.784 =$	(212,190)
100,000	$(\$460k) + \$100k \times 4.329 + \$40k \times 0.784 =$	4,260
150,000	$(\$460k) + \$150k \times 4.329 + \$40k \times 0.784 =$	220,710

Expected NPV = $0.3 \times (-\$212,190) + 0.5 \times \$4,260 + 0.2 \times \$220,710 = (\$17,385)$

Even though the expected NPV is negative, these figures show that there is a 70% chance of the project giving a positive NPV. Some investors may consider the project acceptable on this basis.



Test your understanding 6

- (a) EV of cash flows in year 1 = $\$100,000 \times 0.2 + \$200,000 \times 0.8 = \$180,000$
- (b) EV of cash flows in year 2 = $\$200,000 \times 0.3 + \$300,000 \times 0.7 = \$270,000$
- (c) EV of investment = $\$(400,000) + \$180,000 + \$270,000 = \$50,000$

Alternatively, calculating joint probabilities in a table:

Year 1 (\$)	Prob	Year 2 (\$)	Prob	Total PV (\$)	Joint prob	PV × jp	NPV (\$)
100,000	0.2	200,000	0.3	300,000	0.06	18,000	(100,000)
100,000	0.2	300,000	0.7	400,000	0.14	56,000	0
200,000	0.8	200,000	0.3	400,000	0.24	96,000	0
200,000	0.8	300,000	0.7	500,000	0.56	280,000	100,000
Sum of PV						450,000	
Invest-ment						(400,000)	
Expected NPV						50,000	

- (d) Having done the table, this is now an easy calculation. The only scenario that gives a negative NPV is if the cash flow in year 1 is \$100,000 AND in year 2 is \$200,000 (row 1 on the table). The joint probability of this is found by multiplying together the individual probabilities of the cash flows occurring (0.2×0.3). From the table, this has been calculated as 0.06 or 6%.
- (e) Two of the scenarios in the table give a zero NPV. Therefore the probability of a zero NPV is found by calculating the probability of the scenarios in the second row OR the third row of the table happening. The probability of the second row happening is 0.14 (14%) derived from the probability of earning \$100,000 in year 1 AND \$300,000 in year 2. The probability of the third row is 0.24 or 24%, calculated in the same way.

As the two results are mutually exclusive (can't happen together), the total probability of a zero NPV is therefore found by adding the probabilities of the scenarios in the second row and the third row together:

$$0.14 + 0.24 = 0.38 \text{ or } 38\%$$



Test your understanding 7

Answer D

DPP is 2.36 years and NPV is \$6,000

Year	Present value (PV)	Discount factor	PV
	\$	@ 15%	\$
0	(12,000)	1.000	(12,000)
1	(4,800)	0.870	(4,176)
2	16,800	0.756	12,701
3	14,400	0.658	9,475

Net present value (NPV) = \$6,000

If you selected an NPV of \$4,400 you treated the \$12,000 cash flow as occurring in year 1 and discounted it. Cash flows occurring 'now' should not be discounted.

Year	Present value (PV)	Cumulative PV
	\$	\$
0	(12,000)	(12,000)
1	(4,176)	(16,176)
2	12,701	(3,475)
3	9,475	6,000

DPP = 2 years + $(3,475/9,475 \times 1 \text{ year}) = 2.36 \text{ years}$

If you selected 2.0 years you calculated the non-discounted payback period.



Test your understanding 8

Answer C

	Cash flow	Discount factor	PV
	\$	@ 10%	\$
Time 0 machine	(275,000)	1.000	(275,000)
Time 1–5 contribution	200,000	3.791	758,200
Time 1–5 fixed costs	(95,000)	3.791	(360,145)
NPV			123,055

PV of contribution must fall by \$123,055

Sales volume must fall by $\$123,055 / 758,200 = 16.23\%$

Fall in sales volume = $0.1623 \times 50,000 = 8,115$



Test your understanding 9

Answer D

The expected value criterion is irrespective of risk.



Test your understanding 10

Answer B

When calculating the sensitivity of each variable, the lower the percentage, the more (not less) sensitive is the NPV to that project variable.



Test your understanding 11

Answer A

A risky situation is one where we can say that there is a 70% probability that returns from a project will be in excess of \$100,000 but a 30% probability that returns will be less than \$100,000. If, however, no information can be provided on the returns from the project, we are faced with an uncertain situation.



Test your understanding 12

Answer B

The internal rate of return (C) is specific to the cash flows of the project so cannot be altered independently. Decreasing the initial investment cost (A) would reflect lower risk rather than higher risk. Decreasing the required rate of return (D), or the cost of capital, would reflect lower rather than higher risk.

To reflect the higher risk, adjustments can be made to make it more difficult for the project to present a positive NPV. Decreasing the estimates of future cash inflows (B) would do this.

Working capital management

Chapter learning objectives

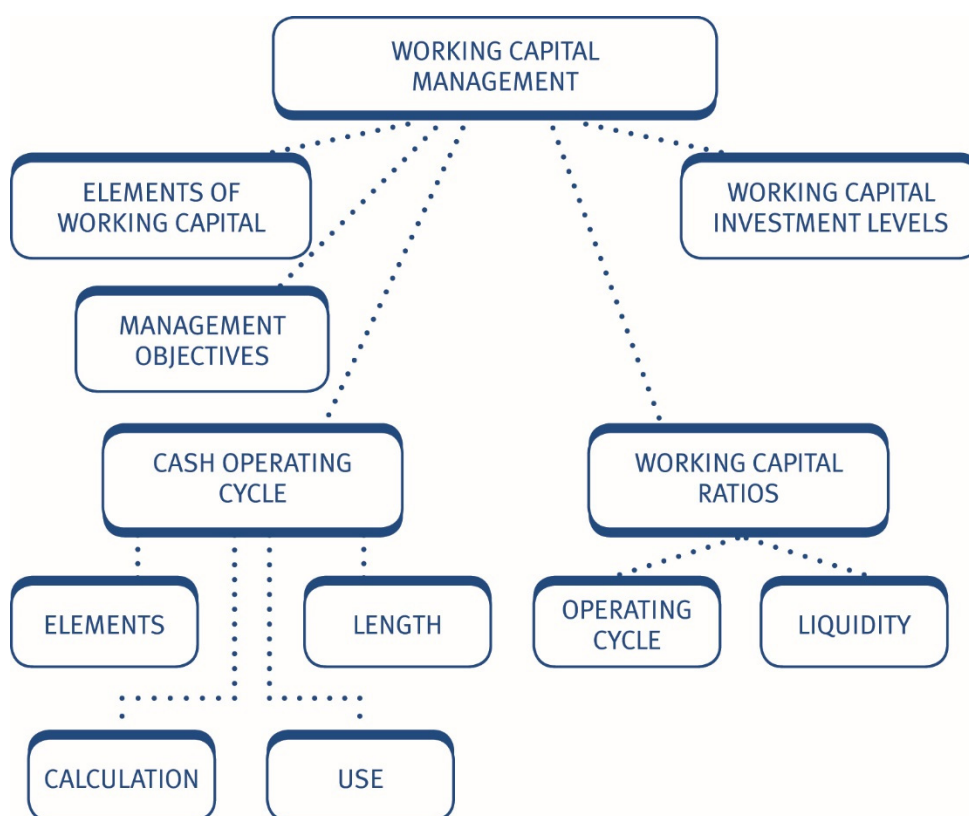
Upon completion of this chapter, you will be able to:

- define working capital and identify its elements
- explain the objectives of working capital management in terms of liquidity and profitability, and discuss the conflict between them
- explain the importance of working capital management to good financial management
- describe the principle and components of the cash operating cycle including the impact on it of accounts payable and receivable
- calculate the length of the cash operating cycle from supplied data
- calculate the current ratio and explain its relevance
- calculate the quick ratio and explain its relevance
- calculate the inventory turnover ratio and the inventory holding period and explain their relevance
- calculate the average collection period for receivables and explain its relevance
- calculate the average payable period for payables and explain its relevance
- calculate the length of a company's cash operating cycle by selecting relevant data from the company's accounts and discuss the implications for the company

- calculate the sales revenue/net working capital ratio and explain its relevance
- calculate the level of working capital investment in current assets from supplied data
- discuss the effect of a business' terms of trade on the length of the working capital cycle
- explain the policies a company may adopt on the level of investment in current assets
- discuss the effect of the industry in which the organisation operates on the length of the working capital cycle
- calculate the level of working capital investment in current assets by selecting relevant data from the company's accounts and discuss the implications for the company.



One of the PER performance objectives (PO10) is to manage and control working capital. You manage cash and working capital effectively, planning for any shortfall or surplus including receivables, payables and inventories. Working through this chapter should help you understand how to demonstrate that objective.

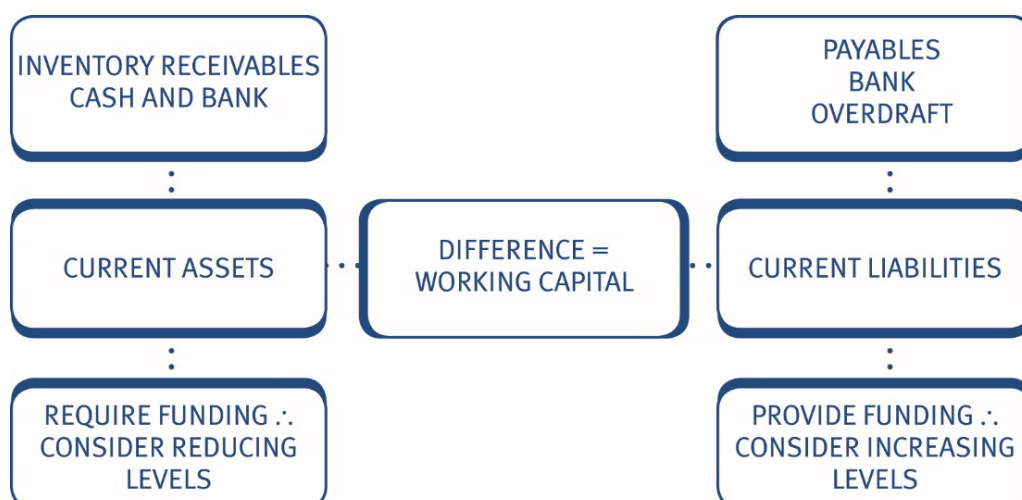


1 The elements of working capital



Working capital is the capital available for conducting the day-to-day operations of an organisation; normally the excess of current assets over current liabilities.

Working capital management is the management of all aspects of both current assets and current liabilities, to minimise the risk of insolvency while maximising the return on assets.



Investing in working capital has a cost, which can be expressed either as:

- the cost of funding it, or
- the opportunity cost of lost investment opportunities because cash is tied up and unavailable for other uses.

Working capital investment

Working capital investment refers to the levels of receivables, inventory and payables that an organisation has, which need to be supported by funding. The total investment is calculated using the balances for each:

receivables + inventory – payables.

Working capital funding

The investment in current assets, such as inventory and receivables, needs to be funded. Current liabilities, such as accounts payable, provide a free source of funding. Therefore, the net working capital investment level will need to be funded using other funding sources, such as overdrafts and loans. These will come with a cost.

Let's say that a company has an inventory balance of \$0.3 million and a receivables balance of \$0.5 million, meaning a total investment of \$0.8 million that needs to be funded.

The company's payables balance acts as a free source of finance. Let's say that the payables balance sits at \$0.2 million. This leaves (\$0.8 million – \$0.2 million) \$0.6 million of current assets that needs funding.

Such funding can be provided by short-term sources of finance, such as an overdraft, or long-term sources, such as a bank loan, or a mix of both types. Strategies for such funding are covered in a later chapter.



Illustration 1 – Working capital funding cost

Let's determine the cost of funding a receivables balance by considering the difference between making cash sales and offering credit.

A business currently has an overdraft balance of \$10,000, with interest charged at 7% per year. If it then makes a cash sale of \$1,000, the cash is received immediately into the bank and the overdraft is reduced to \$9,000.

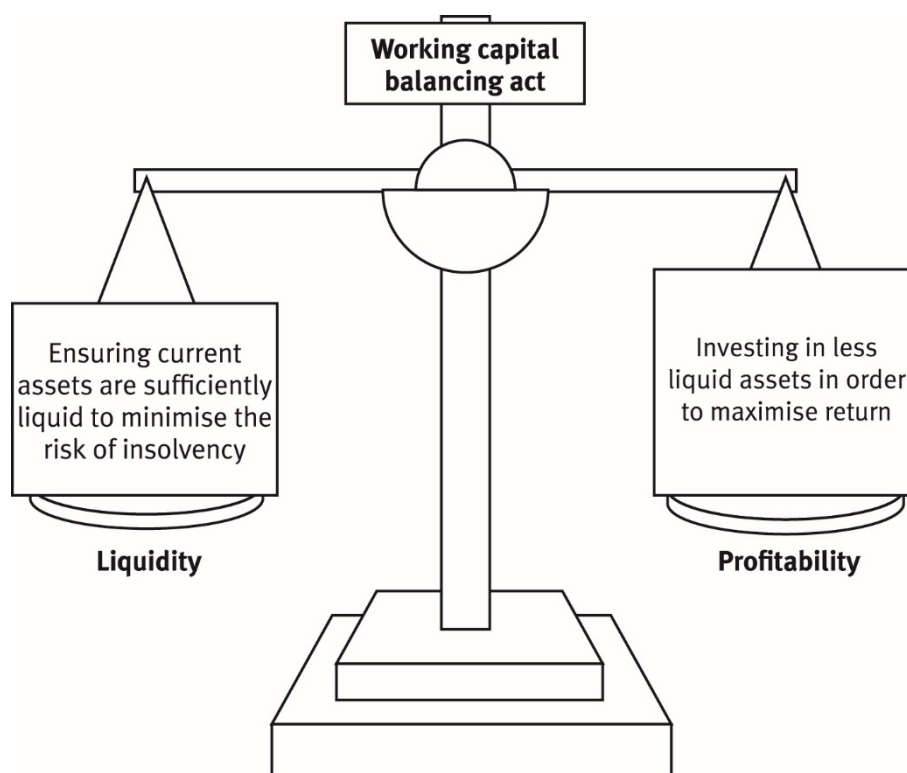
If, instead, the sale is made on 30-day credit terms, a receivables balance is generated, and the cash is not immediately received into the bank account. The overdraft remains at \$10,000 for a 30-day period until the customer pays.

Therefore, an extra \$1,000 of the overdraft is used when offering credit compared to making the sale on cash terms. This means that offering credit (and generating a receivables balance) will lead to an increased use of the overdraft for 30 days, costing an extra \$5.75 ($\$1,000 \times 7\% \times 30 / 365$) in this scenario.

The funding cost can be therefore calculated by taking the receivables balance multiplied by the working capital funding rate for the time period over which the receivables balance is in place.

2 The objectives of working capital management

The main objective of working capital management is to get the balance of current assets and current liabilities right.



For example, cash in a current account is liquid but does not earn profit. The cash could be transferred to a deposit account to earn money, but the cash may be less accessible there, hence less liquid.

The balancing act can also be seen as the trade-off between cash flow versus profits.

Current assets are a major financial position statement item and especially significant to smaller firms. Mismanagement of working capital is therefore a common cause of business failure, e.g.:

- inability to meet bills as they fall due
- demands on cash during periods of growth being too great (overtrading)
- overstocking.



Working capital management is a key factor in an organisation's long-term success. The trade-off between liquidity and profitability and its role in determining a business's overall investment in working capital is fundamental to your understanding of working capital management for the examination.



Profitability v liquidity

The decision regarding the level of overall investment in working capital is a cost/benefit trade-off – **liquidity versus profitability**, or **cash flow versus profits**.

Cash flow versus profit

It is worthwhile stressing the difference between cash flow and profits. Cash flow is as important as profit. Unprofitable companies can survive if they have liquidity. Profitable companies can fail if they run out of cash to pay their liabilities (wages, amounts due to suppliers, overdraft interest, etc.).

Some examples of transactions that have this 'trade-off' effect on cash flows and on profits are as follows:

- (a) Purchase of non-current assets for cash. The cash will be paid in full to the supplier when the asset is delivered; however, the cost will be charged to profits gradually over the life of the asset in the form of depreciation.
- (b) Sale of goods on credit. Profits will be credited in full once the sale has been confirmed; however, the cash may not be received for some considerable period afterwards.
- (c) With some payments such as tax, there may be a significant timing difference between the impact on reported profit and the cash flow.

Clearly, cash balances and cash flows need to be monitored just as closely as trading profits. The need for adequate cash flow information is vital to enable management to fulfil this responsibility.

Profitability versus liquidity

Liquidity in the context of working capital management means having enough cash or ready access to cash to meet all payment obligations when these fall due. The main sources of liquidity are usually:

- cash in the bank
- short-term investments that can be cashed in easily and quickly
- cash inflows from normal trading operations (cash sales and payments by receivables for credit sales)
- an overdraft facility or other ready source of extra borrowing.

The basis of the trade-off is where a company is able to improve its profitability but at the expense of tying up cash. For example:

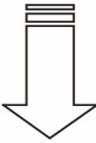
- Receiving a bulk purchase discount (improved profitability) for buying more inventory than is currently required (reduced liquidity)
- Offering credit to customers (attracts more customers so improves profitability but reduces liquidity)

Sometimes, the opposite situation can be seen where a company can improve its liquidity position but at the expense of profitability. For example, offering an early settlement discount to customers.



Additional question – Profitability v liquidity

Fill in the blanks in the table to identify the advantages of having more or less working capital.

Advantages of keeping it high 		 Advantages of keeping it low
	INVENTORY	
	+	
	RECEIVABLES	
	+	
	CASH	
	=	
	CURRENT ASSETS	
	–	
	TRADE PAYABLES	
	=	
	WORKING CAPITAL	

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

3 Policies regarding working capital management

Aggressive versus conservative approach

A firm choosing to have a lower level of working capital (including cash) than rivals is said to have an '**aggressive**' approach, whereas a firm with a higher level of working capital (including high cash balances) has a '**conservative**' approach.

An **aggressive** approach will result in lower working capital funding costs and **higher risk**, while a **conservative** approach will result in higher working capital funding costs and **lower risk**.

Why is the aggressive approach high risk?

A low working capital balance (aggressive approach) comes from having low receivables and inventory and/or high payables. Offering shorter credit terms to customers will reduce receivables levels but this can be risky as it may leave the company in a less competitive position, with customers looking elsewhere for suppliers who will offer them more generous credit. Keeping inventory levels low carries the risk of stockouts, where a company is not able to fulfil demand as it has run out of available stock. Delaying payments to suppliers will raise payables balances and reduce working capital, but runs the risk of upsetting suppliers, which could lead to them restricting supply.

Over-capitalisation and working capital

If there are excessive inventories, accounts receivable **and cash**, and very few accounts payable, there will be an over-investment by the company in current assets. Working capital will be excessive and the company will be over-capitalised.

Overtrading

Cash flow is the lifeblood of the thriving business. Effective and efficient management of the working capital investment is essential to maintaining control of business cash flow. Management must have full awareness of the profitability versus liquidity trade-off.

For example, healthy trading growth typically produces:

- increased profitability
- the need to increase investment in non-current assets and working capital.

In contrast to over-capitalisation, if the business does not have access to sufficient capital to fund the increase, it is said to be 'overtrading'. This can cause serious trouble for the business, as it is unable to pay its business creditors.

In the exam, you might be expected to diagnose overtrading from information given about a company. You should look out for the following:

- A rapid increase in revenue
- An increase in the values of the working capital period calculations, particularly receivables collection and payables payment periods (see later section)
- Most of the increase in assets being financed by credit
- A dramatic drop in the liquidity ratios (see next section) and cash balances

4 Working capital ratios – liquidity



Two key measures, the current ratio and the quick ratio, are used to assess short-term liquidity. Generally, a higher ratio indicates better liquidity.

Current ratio



Measures how much of the total current assets are financed by current liabilities.

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

A measure of 2:1 means that current liabilities can be paid twice over out of existing current assets.

Quick (acid test) ratio



The quick or acid test ratio:

- measures how well current liabilities are covered by liquid assets
- is particularly useful where inventory holding periods are long and therefore distort the current ratio.

$$\text{Quick ratio (acid test)} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$$

A measure of 1:1 means that the company is able to meet existing liabilities if they all fall due at once.



Liquidity ratios

These liquidity ratios are a guide to the risk of cash flow problems and insolvency. If a company suddenly finds that it is unable to renew its short-term liabilities (for instance if the bank suspends its overdraft facilities) there will be a danger of insolvency unless the company is able to turn enough of its current assets into cash quickly.

In general, high current and quick ratios are considered 'good' in that they mean that an organisation has the resources to meet its commitments as they fall due. However, it may indicate that working capital is not being used efficiently, for example that there is too much idle cash that should be invested to earn a return.

Conventional wisdom has it that an ideal current ratio is 2 and an ideal quick ratio is 1. It is very tempting to draw definite conclusions from limited information or to say that the current ratio should be 2, or that the quick ratio should be 1.

However, this is not very meaningful without taking into account the type of ratio expected in a similar business or within a business sector. Any assessment of working capital ratios must take into account the nature of the business involved.

For example, a supermarket business operating a just in time system will have relatively little inventory and since most of sales are for cash they will have few receivables. In addition, the ability to negotiate long credit periods with suppliers can result in a relatively large payables figure. This can result in net current liabilities and a current ratio below 1 – but does not mean the business has a liquidity problem.

Some companies use an overdraft as part of their long-term finance, in which case the current and quick ratios may appear worryingly low. In such questions, you could suggest that the firm replace the overdraft with a loan. Not only would this be cheaper, but it would also improve liquidity ratios.

5 The cash operating cycle

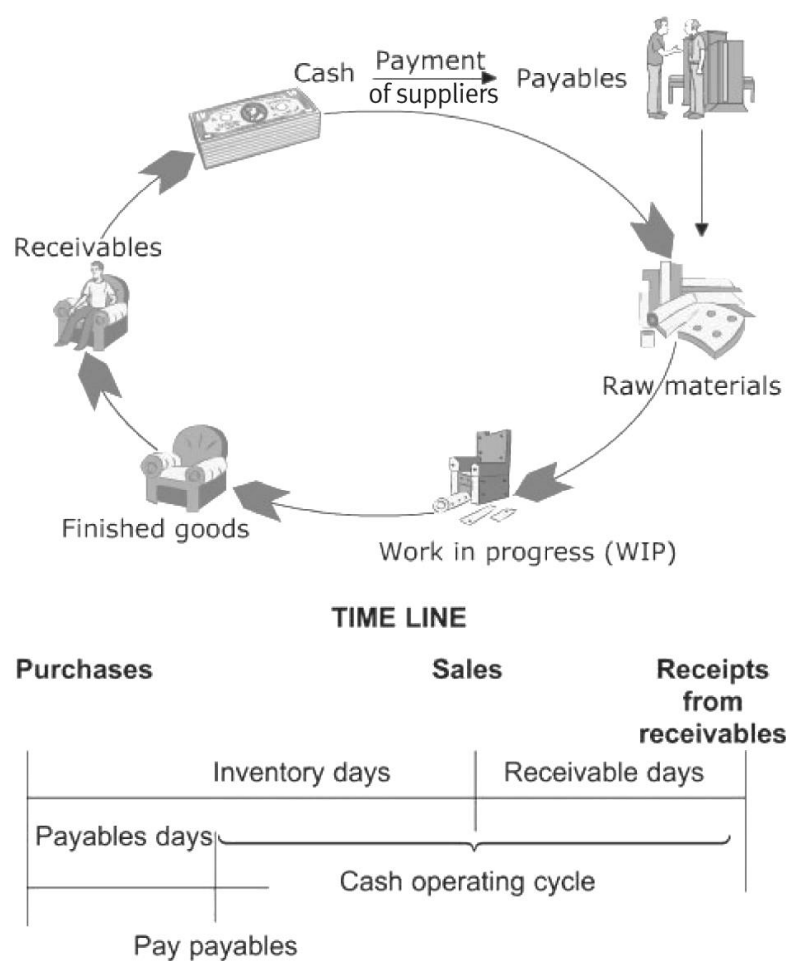
The elements of the operating cycle



The cash operating cycle is the length of time between the company's outlay on raw materials, wages and other expenditures and the inflow of cash from the sale of goods.



The faster a firm can 'push' items around the cycle the lower its investment in working capital will be.



The cash operating cycle

The cash operating cycle reflects a firm's investment in working capital as it moves through the production process towards sales. The investment in working capital gradually increases, first being only in raw materials, but then in labour and overheads as production progresses. This investment must be maintained throughout the production process, the holding period for finished goods and up to the final collection of cash from trade receivables.

(Note: The net investment can be reduced by taking trade credit from suppliers.)

Calculation of the cash operating cycle

For a manufacturing business, the cash operating cycle is calculated as

Raw materials holding period	X
Less: accounts payable payment period	(X)
WIP holding period	X
Finished goods holding period	X
Accounts receivable collection period	X
	—
	X
	—

For a wholesale or retail business, there will be no raw materials or WIP holding periods, and the cycle simplifies to:

Inventory holding period	X
Less: accounts payable payment period	(X)
Accounts receivable collection period	X
	—
	X
	—

The cycle may be measured in days, weeks or months and it is advisable, when answering an exam question, to use the measure used in the question.



Test your understanding 1 – Cash operating cycle

A company has provided the following information:

Accounts receivable collection period	56 days
Raw material inventory holding period	21 days
Production period (WIP)	14 days
Suppliers' payment period	42 days
Finished goods holding period	28 days

Calculate the length of the operating cycle

Factors affecting the length of the operating cycle

Length of the cycle depends on:

- liquidity versus profitability decisions
- terms of trade
- management efficiency
- industry norms, e.g. retail versus construction.

The optimum level of working capital is the amount that results in no idle cash or unused inventory, but that does not put a strain on liquid resources.



The length of the cycle

The length of the cycle depends on how the balancing act between liquidity and profitability is resolved, the efficiency of management and the nature of the industry.

The optimum level is the amount that results in no idle cash or unused inventory, but that does not put a strain on liquid resources. Trying to shorten the cash cycle may have detrimental effects elsewhere, with the organisation lacking the cash to meet its commitments and losing sales since customers will generally prefer to buy from suppliers who are prepared to extend trade credit, and who have items available when required.

Additionally, any assessment of the acceptability or otherwise of the length of the cycle must take into account the nature of the business involved.

A supermarket chain will tend to have a very low or negative cycle – they have very few, if any, credit customers, they have a high inventory turnover (and therefore a low inventory holding period) and they can negotiate quite long credit periods with their suppliers.

A construction company will have a long cycle – their projects tend to be long-term, often extending over more than a year, and whilst progress payments may be made by the customer (if there is one), the bulk of the cash will be received towards the end of the project.

The amount of cash required to fund the operating cycle (and therefore the funding cost for working capital) will increase as either:

- the cycle gets longer
- the level of activity/sales increases.

This can be summed up as follows:

Activity/sales	Length of cycle	Funds needed increase proportionate to:
Stays constant =	Increases ↑	Days in cycle
Increase ↑	Stays constant =	Sales

Where level of activity (sales) is constant and the number of days of the operating cycle increase, the amount of funds required for working capital will increase in approximate proportion to the number of days.

Where the cycle remains constant but activity (sales) increase, the funds required for working capital will increase in approximate proportion to sales.

By monitoring the operating cycle, the manager gains a macro view of the relative efficiency of the working capital utilisation. Further, it may be a key target to reduce the operating cycle to improve the efficiency of the business.

6 Working capital ratios – operating cycle

The periods used to determine the cash operating cycle are calculated by using a series of working capital ratios.

The ratios for the individual components (inventory, receivables and payables) are normally expressed as the number of days/weeks/months of the relevant statement of profit or loss figure they represent.

These ratios are also relevant for the Performance Management exam and are useful when analysing financial statements.



Illustration 2 – Working capital ratios – Operating cycle

X Co has the following figures from its most recent accounts:

	\$m
Receivables	4
Trade payables	2
Inventory	4.3
Sales (80% on credit)	30
Materials purchases (all on credit)	18
Cost of sales	25

Required:

Calculate the relevant working capital ratios, to the nearest day.

Inventory holding period



The length of time inventory is held between purchase and sale.

Calculated as:

$$= \frac{\text{Inventory}}{\text{cost of sales}} \times 365$$



Solution 1 to Illustration 2

$$\frac{\$4.3\text{m}}{\$25\text{m}} \times 365 = 63 \text{ days}$$

In some questions, a more detailed breakdown of inventory may be provided, with values given for raw materials, work-in-progress and finished goods stock. Where this is the case, more detailed calculations can be performed as follows:

Raw material inventory holding period



The length of time raw materials are held between purchase and being used in production.

Calculated as:

$$= \frac{\text{Raw material inventory}}{\text{Material usage}} \times 365$$

NB. Where usage cannot be calculated, purchases gives a good approximation.



Solution 2 to Illustration 2

Suppose you are told that raw materials inventory was \$1m of the total inventory held and that materials usage was \$20m. The raw material inventory holding period could be calculated as:

$$\frac{\$1\text{m}}{\$20\text{m}} \times 365 = 18 \text{ days}$$



WIP holding period

The length of time goods spend in production.

Calculated as:

$$= \frac{\text{Work-in-progress inventory held}}{\text{Production cost}} \times 365$$

NB. Where production cost cannot be calculated, cost of goods sold gives a good approximation.



Solution 3 to Illustration 2

If you were told this company had a WIP balance of \$1.3m and a total production cost of \$23m, the WIP holding period could be calculated as:

$$\frac{\$1.3\text{m}}{\$23\text{m}} \times 365 = 21 \text{ days}$$



Finished goods inventory holding period

The length of time finished goods are held between completion or purchase and sale.

Calculated as:

$$= \frac{\text{Finished goods inventory held}}{\text{Cost of goods sold}} \times 365$$

For all inventory holding period ratios, a low ratio is usually seen as a sign of good working capital management. It is very expensive to hold inventory and thus minimum inventory holding usually points to good practice.



Solution 4 to Illustration 2

If you were told that the remaining inventory of \$2m was finished goods inventory, you could calculate the finished goods holding period as:

$$\frac{\$2\text{m}}{\$25\text{m}} \times 365 = 29 \text{ days}$$

Note: The total of the three individual holding periods calculated is slightly more than the one calculated as an overall inventory holding period. This is because the more detailed approach allows us to take account of the costs within each element by using materials usage and production cost within the calculation rather than just cost of sales. The more detailed approach is more accurate but won't always be possible due to the information available.



Inventory turnover

For each ratio, the corresponding turnover ratio can be calculated as:

$$\text{Inventory turnover (no of times)} = \frac{\text{Cost}}{\text{Average inventory held}}$$

Generally, this is less useful in the examination.

Using finished goods information from Illustration 1:

$$\text{Inventory turnover} = \frac{\$25\text{m}}{\$2\text{m}} = 12.5 \text{ times}$$

Thus, finished goods inventory turns round/is turned into sales 12.5 times in the year.



Accounts receivable collection period

The length of time credit is extended to customers.

Calculated as:

$$= \frac{\text{Receivables}}{\text{Credit sales}} \times 365$$

Generally, shorter credit periods are seen as financially sensible but the length will also depend upon the nature of the business.



Solution 5 to Illustration 2

$$\frac{\$4\text{m}}{\$30\text{m} \times 80\%} \times 365 = 61 \text{ days}$$



Interpreting trade accounts receivable collection periods

Businesses which sell goods on credit terms specify a credit period. Failure to send out invoices on time or to follow up late payers will have an adverse effect on the cash flow of the business. The accounts receivable collection period measures the average period of credit allowed to customers.

In general, the shorter the collection period the better because receivables are effectively 'borrowing' from the company. Remember, however, that the level of receivables reflects not only the ability of the credit controllers but also the sales and marketing strategy adopted, and the nature of the business. Any change in the level of receivables must therefore be assessed in the light of the level of sales.



Accounts payable payment period

The average period of credit extended by suppliers.

Calculated as:

$$= \frac{\text{Trade payables}}{\text{Credit purchases}} \times 365$$



Solution 6 to Illustration 2

$$\frac{\$2\text{m}}{\$18\text{m}} \times 365 = 41 \text{ days}$$

Generally, increasing the accounts payable payment period suggests advantage is being taken of available credit but there are risks:

- losing supplier goodwill
- losing prompt payment discounts
- suppliers increasing the price to compensate.

The operating cycle

The ratios can then be brought together to produce the cash operating cycle.

	Days
Inventory holding period	63
Accounts payable payment period	(41)
Accounts receivable collection period	61
	—
Length of cash operating cycle	83

The cash operating cycle tells us it takes X Co 83 days between paying out for material inventories and eventually receiving cash back from customers.

As always, this must then be compared with prior periods or industry average for meaningful analysis.

Additional points for calculating ratios

The ratios may be needed to provide analysis of a company's performance or simply to calculate the length of the operating cycle.



There are a few simple points to remember which will be of great use in the examination:

- Where the period is required in days, the multiple in the ratios is 365, for months the multiple is 12, or 52 for weeks, unless the question specifies otherwise.
- To avoid getting a result that is unduly influenced by unusual balances at a period in time, your calculations should ideally use average balances for inventory, receivables and payables. If not given, these could be calculated by adding the opening and closing financial position statement figures and dividing by two. If you are required to compare ratios between two financial position statements, it is acceptable to base each holding period on financial position statement figures, rather than an average, in order to see whether the ratio has increased or decreased.
- For each ratio calculated above, the corresponding turnover ratio can be calculated by inverting the ratio given and removing the multiple.
- When using the ratios to appraise performance, it is essential to compare the figure with others in the industry or identify the trend over a number of periods.
- Ratios have their limitations and care must be taken because:
 - the financial position statement values at a particular time may not be typical
 - balances used for a seasonal business may not represent average levels, e.g. a fireworks manufacturer
 - ratios can be subject to window dressing/manipulation
 - ratios concern the past (historic) rather than the future
 - figures may be distorted by inflation and/or rapid growth.
- If the proper figure for the calculation is not given in the exam, use the closest available figure. For instance, the payables payment period is calculated using a credit purchases figure. If this is not, given, you may use total purchases or cost of sales figures instead.

Working capital turnover

One final ratio that relates to working capital is the working capital turnover ratio and is calculated as:

$$\frac{\text{Sales revenue}}{\text{Net working capital}}$$

This measures how efficiently management is utilising its investment in working capital to generate sales and can be useful when assessing whether a company is overtrading.

It must be interpreted in the light of the other ratios used.



Test your understanding 2 – Calculation of cash operating cycle

Marlboro Co estimates the following figures for the coming year.

Sales – all on credit	\$3,600,000
Receivables	\$306,000
Gross profit margin*	25% on sales
Finished goods	\$200,000
Work in progress	\$350,000
Raw materials (balance held)	\$150,000
Trade payables	\$130,000

Inventory levels are constant.

*Raw materials are 80% of cost of sales – all on credit.

Required:

Calculate the cash operating cycle.



From an examination point of view, calculating the ratios is only the start – interpretation is the key to a good answer. Try to build a cumulative picture, e.g. the current ratio looks good until we find out from calculating the inventory holding period that there are high levels of illiquid inventory.

Do not be afraid to point out further information that may be required to provide a better interpretation of your calculations, e.g. the company credit policy, industry benchmarks, etc.


Test your understanding 3 – Appraisal of working capital

You have been given the following information for a company:

Summarised statements of financial position at 30 June

	20X7		20X6	
	\$000	\$000	\$000	\$000
Non-current assets (carrying value)		130		139
Current assets:				
Inventory	42		37	
Receivables	29		23	
Bank	3		5	
	—		—	
		74		65
		—		—
Total assets		204		204
		—		—
Equity and liabilities				
Ordinary share capital (\$0.50 shares)		35		35
Share premium account		17		17
Revaluation reserve		10		—
Profit and loss account		31		22
		—		—
		93		74
Non-current liabilities				
5% secured loan notes		40		40
8% Preference shares (\$1 shares)		25		25
Current liabilities:				
Trade payables	36		55	
Taxation	10		10	
	—		—	
		46		65
		—		—
Total equity and liabilities		204		204
		—		—

Summarised statements of profit or loss for the year ended 30 June				
	20X7		20X6	
	\$000	\$000	\$000	\$000
Revenue		209		196
Opening inventory	37		29	
Purchases	162		159	
	—		—	
	199		188	
Closing inventory	42		37	
	—		—	
		157		151
		—		—
Gross profit		52		45
Interest expense	2		2	
Depreciation	9		9	
Sundry expenses	14		11	
	—		—	
		25		22
		—		—
Operating profit		27		23
Taxation		10		10
		—		—
Profit for the year		17		13
Dividends				
Ordinary shares	6		5	
Preference shares	2		2	
	—		—	
		8		7
		—		—
Retained profit		9		6
		—		—
Required:				
(a) Calculate the liquidity ratios in 20X6 and 20X7				
(b) Calculate the length of the cash operating cycle in 20X6 and 20X7				
(c) Comment on your results and appraise how effectively the working capital is being managed				

7 Working capital investment levels

The working capital ratios can be used to predict the future levels of investment (the financial position statement figure) required. This is done by re-arranging the formulas. For example:

$$\text{Trade receivables balance} = \frac{\text{Accounts receivable collection period}}{365} \times \text{Credit sales}$$



Working capital investment levels

The level of working capital required is affected by the following factors:

- (1) The nature of the business, e.g. manufacturing companies need more inventory than service companies.
- (2) Uncertainty in supplier deliveries. Uncertainty would mean that extra inventory needs to be carried in order to cover fluctuations.
- (3) The overall level of activity of the business. As output increases, receivables, inventory, etc. all tend to increase.
- (4) The company's credit policy. The tighter the company's policy the lower the level of receivables.
- (5) The length of the operating cycle. The longer it takes to convert material into finished goods into cash the greater the investment in working capital
- (6) The credit policy of suppliers. The less credit the company is allowed to take, the lower the level of payables and the higher the net investment in working capital.



Test your understanding 4 – Working capital investment levels

X Co has the following expectations for the forthcoming period.

	\$m
Sales	10
Materials	(6)
Other costs	(2)
Profit	2

The following working capital ratios are expected to apply.

Inventory holding period	30 days
Accounts receivable collection period	60 days
Accounts payable payment period	40 days

Required:

Compute the working capital requirement.



Additional question – Working capital investment levels (1)

A company's annual sales are \$8 million with a mark-up on cost of 60%. It normally settles payables two months after purchases are made, holding one month's worth of demand in inventory. It allows receivables 1½ months' credit and its cash balance currently stands at \$1,250,000.

What are its current and quick ratios?

The answer to this question can be found after the chapter summary diagram at the end of this chapter.



Additional question – Working capital investment levels (2)

The following data relate to Mugwump Co, a manufacturing company.

Sales revenue for year:	\$1,500,000
Costs as percentage of sales:	
Direct materials	30%
Direct labour	25%
Variable overheads	10%
Fixed overheads	15%
Selling and distribution	5%

Average statistics relating to working capital are as follows:

- receivables take 2 ½ months to pay
- raw materials are in inventory for three months
- WIP represents two months' half-produced goods
- finished goods represent one month's production
- credit is taken

– Materials	2 months
– Direct labour	1 week
– Variable overheads	1 month
– Fixed overheads	1 month
– Selling and distribution	½ month

WIP and finished goods are valued at the cost of material, labour and variable expenses.

Compute the working capital requirement of Mugwump Co assuming that the labour force is paid for 50 working weeks in each year.

The answer to this question can be found after the chapter summary diagram at the end of this chapter.



Test your understanding 5

A company has a liquidity ratio equal to 0.5. The directors believe that the company has to reduce its bank overdraft and have agreed to alter the company's credit terms to customers from two months to one month.

What would be the effects on the company's cash operating cycle and liquidity ratio if this change were to be achieved?

	Cash operating cycle	Liquidity ratio
A	Decrease	Decrease
B	Decrease	No change
C	Decrease	Increase
D	Increase	Increase



Test your understanding 6

The key trade-off that lies at the heart of working capital management is that between

- A Business stability and solvency
- B Debtors and creditors
- C Current assets and current liabilities
- D Liquidity and profitability



Test your understanding 7

Which of the following is not a typical symptom of overtrading?

- A A rapid increase in sales revenue
- B A bank overdraft which may reach the limit of the facilities agreed by the bank
- C A decrease in the current ratio and the quick ratio
- D A decrease in the trade accounts payable payment period



Test your understanding 8

Archie Co manufactures plastic cutlery. The company buys raw materials from suppliers that allow the company 2.5 months credit. The raw materials remain in inventory for 2 months and it takes Archie plc 2 months to produce the goods, which are sold immediately production is completed. Customers take an average of 1.5 months to pay.

What is Archie plc's cash operating cycle?

- A 2 months
- B 2.5 months
- C 3 months
- D 7 months

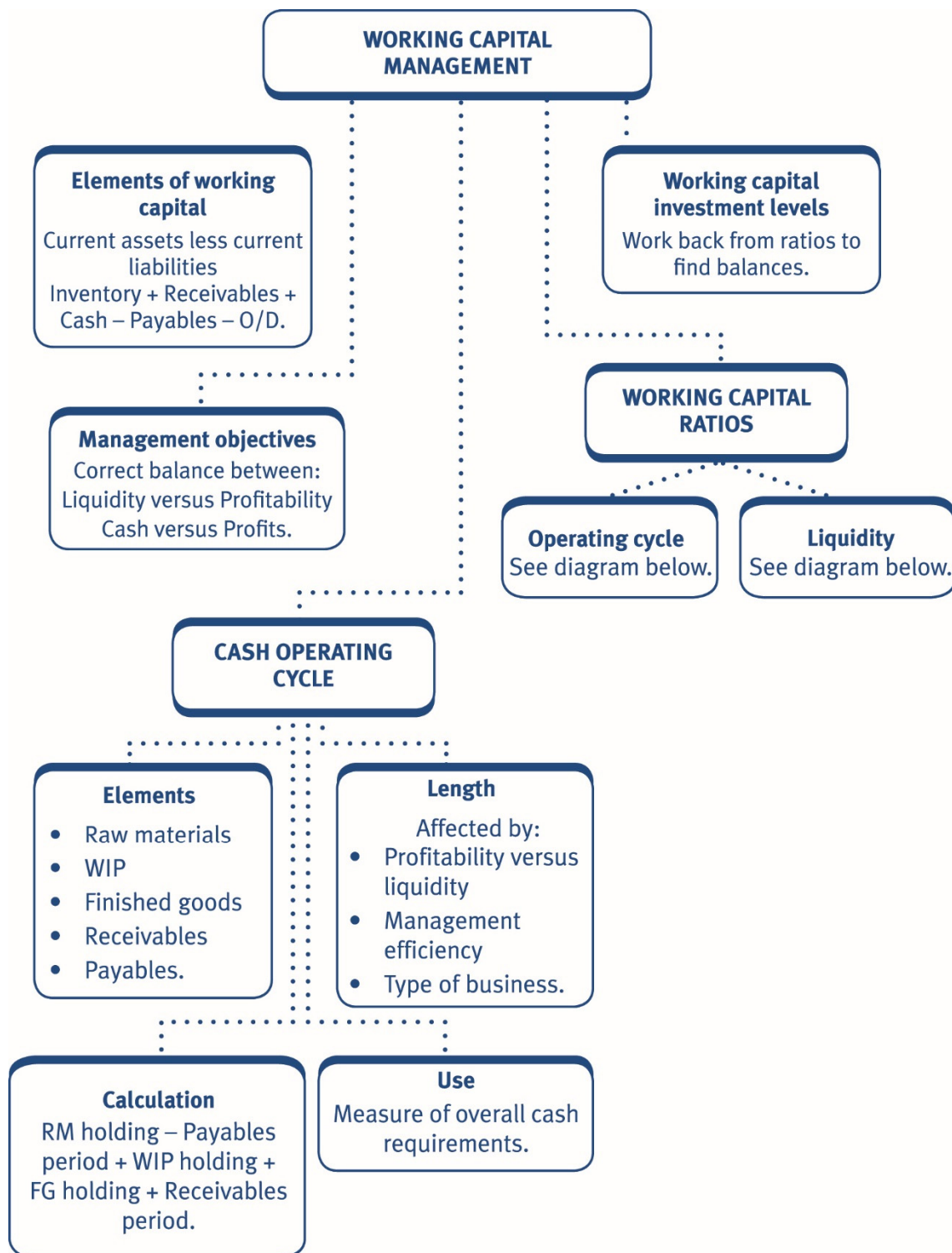


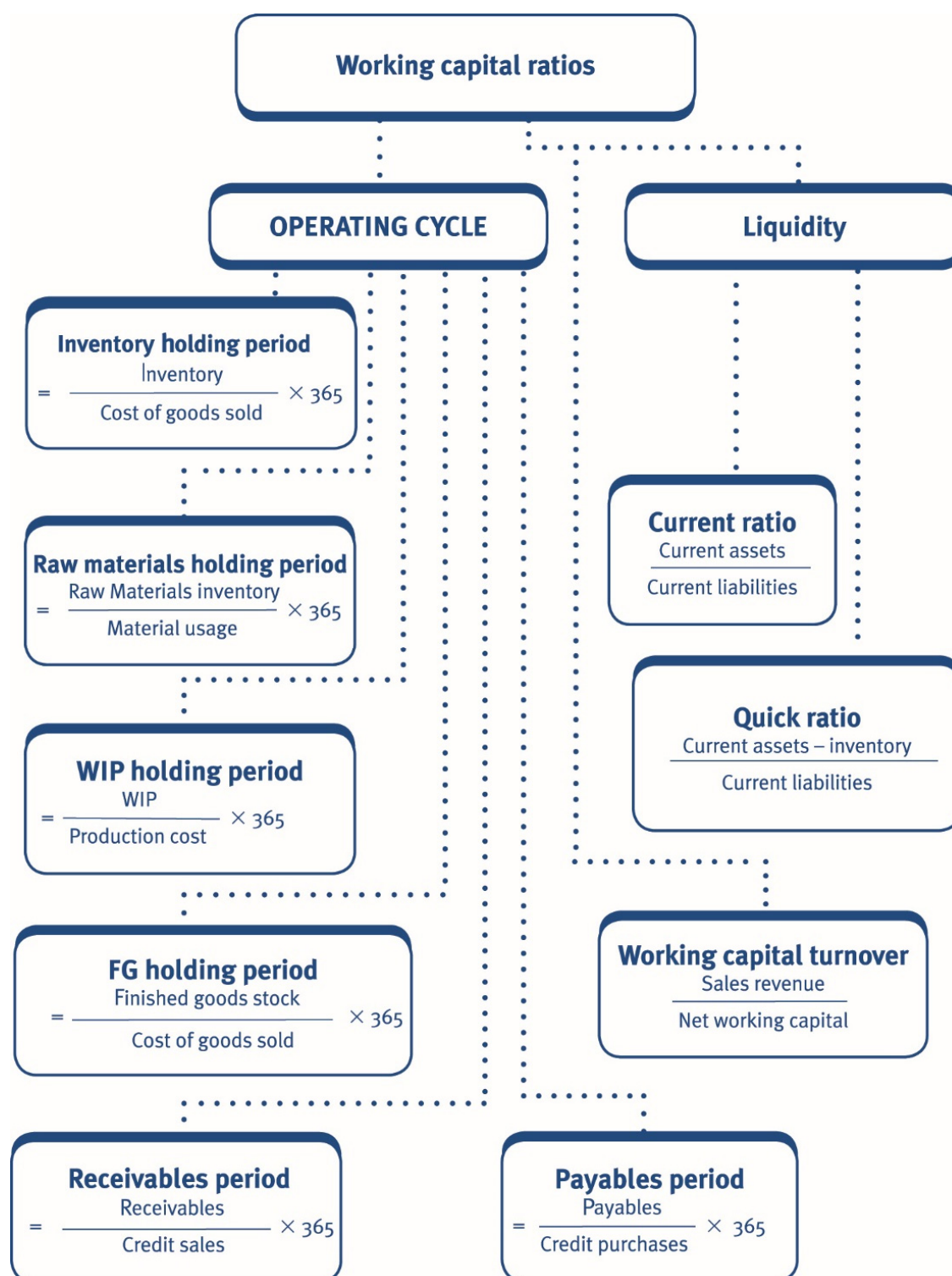
Test your understanding 9

Which of the following will shorten the cash operating cycle?

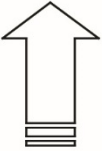
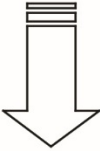
- A An increase in the raw materials inventory holding period
- B An increase in the trade accounts payable payment period
- C An increase in the trade accounts receivable collection period
- D An increase in the production period

8 Chapter summary






Answer to additional question – Profitability v liquidity

Advantages of keeping it high 		 Advantages of keeping it low
– Few stockouts – Bulk purchase discounts – Reduced ordering costs	INVENTORY	– Less cash tied up in inventory – Lower storage costs – Flexibility
	+	
– Customers like credit – ∴ profitable as attracts more sales	RECEIVABLES	– Less cash tied up – Less chance of irrecoverable debts – Reduced costs of credit control
	+	
– Able to pay bills on time – Take advantage of unexpected opportunities – Avoid high borrowing costs	CASH	– Can invest surplus to earn higher returns – Less vulnerable to takeover
	=	
	CURRENT ASSETS	
– Preserves own cash – cheap source of finance	– TRADE PAYABLES	– Can take advantage of prompt payment discounts – Retain good credit status – Move favourable supplier treatment
	=	
	WORKING CAPITAL	


Answer to additional question – Working capital investment levels (1)
Step 1 Calculate annual cost of sales, using the cost structure.

	%	\$m
Sales	160	8
Cost of sales (COS)	100	5
Gross profit	60	3

Step 2 Calculate payables, receivables and inventory.

$$\text{Payables} = \frac{2}{12} \times \text{annual COS} = \frac{2}{12} \times \$5\text{m} = \$0.833\text{m}$$

$$\text{Receivables} = \frac{1.5}{12} \times \text{annual sales} = \frac{1.5}{12} \times \$8\text{m} = \$1\text{m}$$

$$\text{Inventory} = \frac{1}{12} \times \text{annual COS} = \frac{1}{12} \times \$5\text{m} = \$0.417\text{m}$$

Step 3 Calculate the ratios.

$$\text{Current ratio} = \frac{\text{Inventory} + \text{receivables} + \text{cash}}{\text{Payables}} = \frac{0.417 + 1 + 1.25}{0.833} = 3.2$$

$$\text{Quick ratio} = \frac{\text{Receivables} + \text{cash}}{\text{Payables}} = \frac{1 + 1.25}{0.833} = 2.7$$


Answer to additional question – Working capital investment levels (2)

(1) Costs incurred		
		\$
Direct materials 30% of \$1,500,000		450,000
Direct labour 25% of \$1,500,000		375,000
Variable overheads 10% of \$1,500,000		150,000
Fixed overheads 15% of \$1,500,000		225,000
Selling and distribution 5% of \$1,500,000		75,000
(2) Average value of current assets		
	\$	\$
Finished goods $1/12 \times \$975,000$		81,250
Raw materials $3/12 \times \$450,000$		112,500
WIP:		
(2 months @ half produced – 1 month equivalent cost)		
Materials		
$1/12 \times \$450,000$	37,500	
Labour $1/12 \times \$375,000$	31,250	
Variable overheads $1/12 \times \$150,000$	12,500	
		81,250
Receivables $(\$1,500,000 \times 2.5/12)$		312,500
		587,500
(3) Average value of current liabilities		
Materials $2/12 \times \$450,000$	75,000	
Labour $1/50 \times \$375,000$	7,500	
Variable overheads $1/12 \times \$150,000$	12,500	
Fixed overheads $1/12 \times \$225,000$	18,750	
Selling and distribution $1/24 \times \$75,000$	3,125	
		(116,875)
(4) Working capital required		
		470,625

Test your understanding answers

**Test your understanding 1 – Cash operating cycle**

	Days
Raw materials inventory holding period	21
Less: suppliers' payment period	(42)
WIP holding period	14
Finished goods holding period	28
Accounts receivable collection period	56
Operating cycle (days)	77

**Test your understanding 2 – Calculation of cash operating cycle****Statement of profit or loss**

	\$	\$
Revenue		3,600,000
Cost of sales		
Materials – 80% (given)	2,160,000	
Other (balancing figure)	540,000	
		2,700,000
Gross profit – 25% (given)		900,000

Cash operating cycle

Raw materials holding period

$$\frac{\$150,000}{\$2,160,000} \times 365 = 25 \text{ days}$$

Trade accounts payable payment period

$$\frac{\$130,000}{\$2,160,000} \times 365 = (22) \text{ days}$$

WIP holding period

$$\frac{\$350,000}{\$2,700,000} \times 365 = 47 \text{ days}$$

Finished goods holding period

$$\frac{\$200,000}{\$2,700,000} \times 365 = 27 \text{ days}$$

Accounts receivable collection period

$$\frac{\$306,000}{\$3,600,000} \times 365 = 31 \text{ days}$$

108 days



Test your understanding 3 – Appraisal of working capital

The current ratio:

20X7

$$\frac{74}{46} = 1.6$$

20X6

$$\frac{65}{65} = 1.0$$

The quick (or acid test) ratio:

20X7

$$\frac{32}{46} = 0.7$$

20X6

$$\frac{28}{65} = 0.4$$

Both of these ratios show an improvement. The extent of the change between the two years seems surprising and would require further investigation. It would also be useful to know how these ratios compare with those of a similar business, since typical liquidity ratios for supermarkets, say, are quite different from those for heavy engineering firms.

In 20X7, current liabilities were well covered by current assets. Liabilities payable in the near future are 70% covered by cash and receivables (a liquid asset, close to cash).

To better understand the liquidity ratios we can then look at each individual component of working capital.

The inventory holding period:

20X7	20X6
$\frac{42,000}{157,000} \times 365 = 98 \text{ days}$	$\frac{37,000}{151,000} \times 365 = 89 \text{ days}$

The inventory holding period has lengthened. In general, the shorter the stock holding period the better. It is very expensive to hold stock and thus minimum stock holding usually points to good management.

The current ratio calculation now seems less optimistic, considering the holding period for inventory of 98 days. Inventory that takes nearly four months to sell is not very liquid! It would be better to focus attention on the acid test ratio.

Accounts receivable collection period:

	20X7	20X6
Average daily sales	$\frac{\$209,000}{365} = \573	$\frac{\$196,000}{365} = \537
Closing trade receivables	\$29,000	\$23,000

Accounts receivable collection period:

$$\frac{\$29,000}{\$573} = 51 \text{ days} \quad \frac{\$23,000}{\$537} = 43 \text{ days}$$

Or more quickly:

20X7	20X6
$\frac{29,000}{209,000} \times 365 \text{ days} = 50.6 \text{ days}$	$\frac{23,000}{196,000} \times 365 = 42.8 \text{ days}$

Compared with 20X6, the accounts receivable collection period has worsened in 20X7. It would be important to establish the company policy on credit allowed. If the average credit allowed to customers was, say, 30 days, then something is clearly wrong. Further investigation might reveal delays in sending out invoices or failure to 'screen' new customers.

This situation suggests yet a further review of the liquidity ratios. The acid test ratio ignores inventory but still assumes receivables are liquid. If debt collection is a problem then receivables too are illiquid and the company could struggle to pay its current liabilities were they all to fall due in a short space of time.

Accounts payable payment period:

	20X7	20X6
Average daily purchases	$\frac{\$162,000}{365} = \444	$\frac{\$159,000}{365} = \436
Closing trade payables	\$36,000	\$55,000
Payables' payment period	81 days	126 days

Or more quickly:

20X7	20X6
$\frac{36,000}{162,000} \times 365 \text{ days} = 81.1 \text{ days}$	$\frac{55,000}{159,000} \times 365 = 126.3 \text{ days}$

The accounts payable payment period has reduced substantially from last year. It is, however, in absolute terms still a high figure. Often, suppliers request payment within 30 days. The company is taking nearly three months. Trade creditors are thus financing much of the working capital requirements of the business, which is beneficial to the company.

A high level of creditor days may be good in that it means that all available credit is being taken, but there are three potential disadvantages of taking extended credit:

- Future supplies may be endangered.
- Availability of cash discounts is lost.
- Suppliers may quote a higher price for the goods knowing the company takes extended credit.

Additionally, when viewed alongside the previous ratios calculated, this might suggest a cash flow problem causing suppliers to be left unpaid.

Length of the operating cycle:

	20X7 days	20X6 days
Inventory holding period	98	89
+		
Accounts receivable collection period	51	43
–		
Accounts payable payment period	(81)	(126)
=		
Cash operating cycle	68 days	6 days

Our example shows that, in 20X7, there is approximately a 68-day gap between paying cash to suppliers for goods, and receiving the cash back from customers. However, in 20X6, this was only 6 days. An investigation should be made as to why this has happened, as it will be costing the business money to support cash being out of the business for longer.



Test your understanding 4 – Working capital investment levels

We need to use the ratios to calculate financial position statement values in order to construct the projected working capital position.

			\$m
Inventory	$= 30 \div 365$	$\times \$6m$	$= 0.49$
Receivables	$= 60 \div 365$	$\times \$10m$	1.64
Trade payables	$= 40 \div 365$	$\times \$6m$	$= (0.66)$
Working capital required			1.47



Test your understanding 5

Answer A

Cash operating cycle decreases, liquidity ratio decreases.

The reduction in the amount of time taken for customers to pay their bills would reduce the cash operating cycle.

The reduction in receivables and in the overdraft mean that the numerator and denominator of the liquidity ratio would both reduce by the same amount. Therefore, the ratio would decrease.



Test your understanding 6

Answer D

All businesses face a trade-off between being profitable (providing a return) and being liquid (staying in business).



Test your understanding 7

Answer D

Overtrading happens when a business tries to do too much too quickly with too little long-term capital. The business tries to support too large a volume of trade with the capital resources at its disposal. As such, the payment period to accounts payable is likely to lengthen (not shorten).



Test your understanding 8

Answer C

	Months
The average time the raw materials are in inventory	2.0
Less: The time taken to pay suppliers	(2.5)
Time taken to produce the goods	2.0
Time taken by customers to pay for the goods	1.5
Total	3.0



Test your understanding 9

Answer B

The calculation of the trade accounts payable payment period gives the average period of credit extended to suppliers. As this amount increases, the cash operating cycle will shorten.

Working capital management – Inventory control

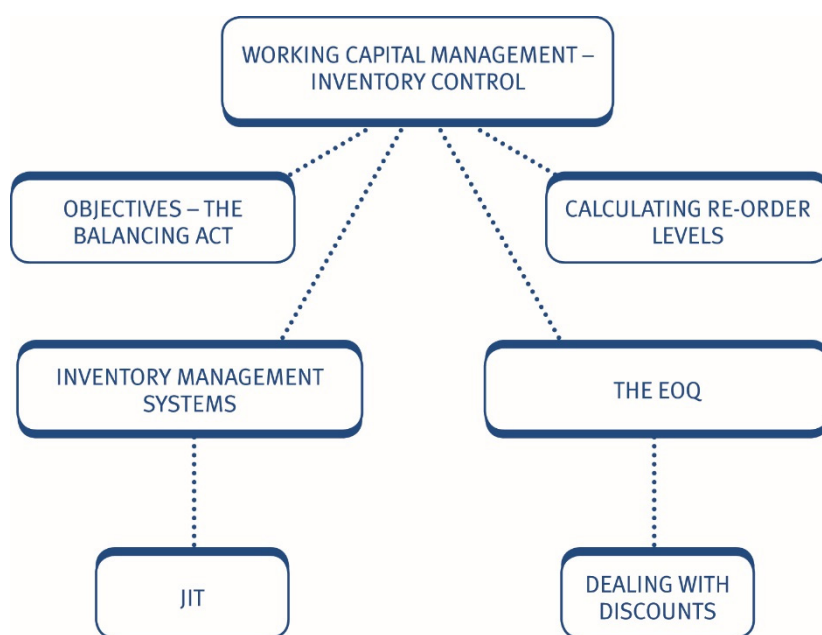
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- explain the objective of inventory management
- define and explain lead-time and buffer inventory
- explain and apply the basic economic order quantity (EOQ) formula to data provided
- calculate the EOQ taking account of quantity discounts and calculate the financial implications of discounts for bulk purchases
- define and calculate the re-order level where demand and lead-time are known
- describe and evaluate the main inventory management systems including Just-In-Time (JIT) techniques
- suggest appropriate inventory management techniques for use in a scenario.



One of the PER performance objectives (PO10) is to manage and control working capital. You manage cash and working capital effectively, planning for any shortfall or surplus including receivables, payables and inventories. Working through this chapter should help you understand how to demonstrate that objective.

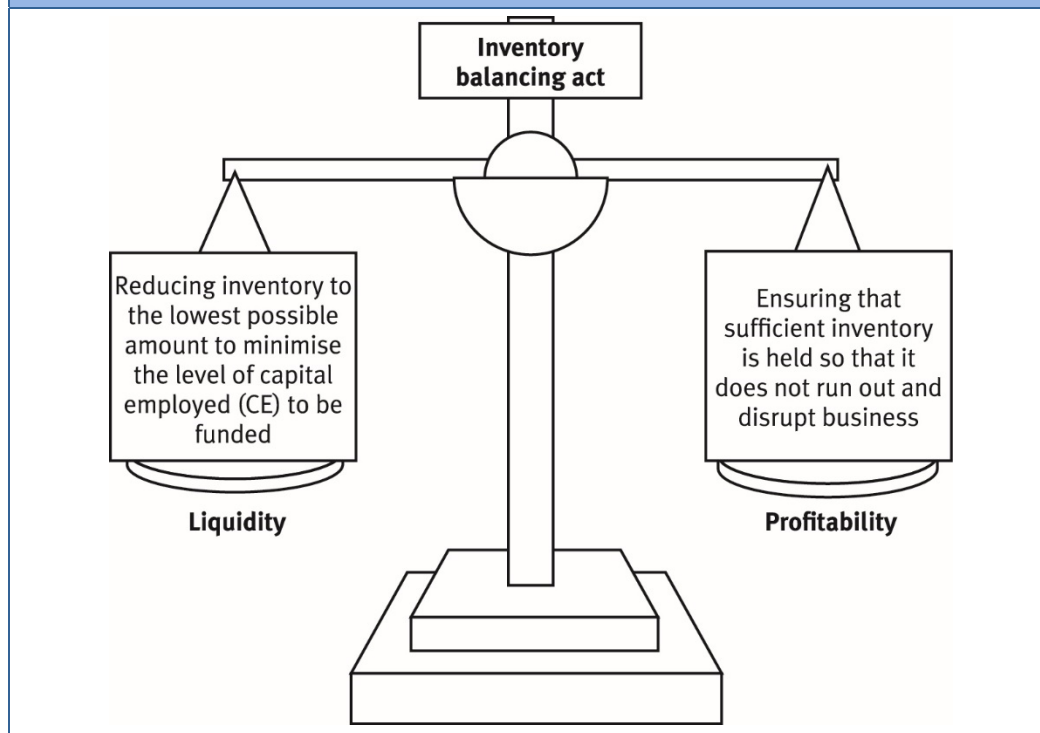


1 The objectives of inventory management

Inventory is a major investment for many companies. Manufacturing companies can easily be carrying inventory equivalent to between 50% and 100% of the revenue of the business. It may therefore be desirable to reduce the levels of inventory held to the necessary minimum.



The balancing act



Costs of high inventory levels

Keeping inventory levels high is expensive owing to:

- foregone interest from tying up capital in inventory
- holding costs:
 - storage
 - stores administration
 - risk of theft/damage/obsolescence.



Costs of high inventory levels

Carrying inventory involves a major working capital investment and therefore levels need to be very tightly controlled. The cost is not just that of purchasing the goods, but also storing, insuring, and managing them once they are in inventory.

Purchase costs: once goods are purchased, capital is tied up in them and until sold on (in their current state or converted into a finished product), the capital earns no return. This lost return is an opportunity cost of holding the inventory. Alternatively, if the business is less cash rich, a funding source such as an overdraft may be needed and this comes with an associated interest cost.

Storage and stores administration: in addition, the goods must be stored. The company must incur the expense of renting out warehouse space, or if using space they own, there is an opportunity cost associated with the alternative uses the space could be put to. There may also be additional requirements such as controlled temperature or light, which require extra funds.

Other risks: once stored, the goods will need to be insured. Specialist equipment may be needed to transport the inventory to where it is to be used. Staff will be required to manage the warehouse and protect against theft and if inventory levels are high, significant investment may be required in sophisticated inventory control systems.

The longer inventory is held, the greater the risk that it will deteriorate or become out of date. This is true of perishable goods, fashion items and high-technology products, for example.

2 Costs of low inventory levels

If inventory levels are kept too low, the business faces alternative problems:

- stockouts:
 - lost contribution
 - production stoppages
 - emergency orders
- high re-order/setup costs
- lost quantity discounts.



Costs of low inventory levels

Stockout: if a business runs out of a particular product used in manufacturing it may cause interruptions to the production process – causing idle time, stockpiling of work-in-progress (WIP) or possibly missed orders. Alternatively, running out of goods held for onward sale can result in dissatisfied customers and perhaps future lost orders if custom is switched to alternative suppliers. If a stockout looms, the business may attempt to avoid it by acquiring the goods needed at short notice. This may involve using a more expensive or poorer quality supplier.

Re-order/setup costs: each time inventory runs out, new supplies must be acquired. If the goods are bought in, the costs that arise are associated with administration – completion of a purchase requisition, authorisation of the order, placing the order with the supplier, taking and checking the delivery and final settlement of the invoice. If the goods are to be manufactured, the costs of setting up the machinery will be incurred each time a new batch is produced.

Lost quantity discounts: purchasing items in bulk will often attract a discount from the supplier. If only small amounts are bought at one time in order to keep inventory levels low, the quantity discounts will not be available.

The challenge



The objective of good inventory management is therefore to determine:

- the optimum re-order level – how many items are left in inventory when the next order is placed, and
- the optimum re-order quantity – how many items should be ordered when the order is placed

for all material inventory items.

In practice, this means striking a balance between holding costs on the one hand and stockout and re-order costs on the other.



The balancing act between liquidity (keeping inventory levels low) and profitability (maintaining high inventory levels, although this does require extra funding which impacts profitability negatively), which might also be considered to be a trade-off between holding costs and stockout/re-order costs, is key to any discussion on inventory management.



Terminology

Other key terms associated with inventory management include:

- lead time – the lag between when an order is placed and the item is delivered
- buffer inventory – the basic level of inventory kept for emergencies. A buffer is required because both demand and lead-time will fluctuate and predictions can only be based on best estimates.

Ensure you can distinguish between the various terms used: re-order level, re-order quantity, lead-time and buffer inventory.

3 Economic order quantity (EOQ)

For businesses that do not use just in time (JIT) inventory management systems (discussed in more detail below), there is an optimum order quantity for inventory items, known as the EOQ.

The challenge

The aim of the EOQ model is to minimise the total cost of holding and ordering inventory.



EOQ explanation

To minimise the total cost of holding and ordering inventory, it is necessary to balance the relevant costs. These are:

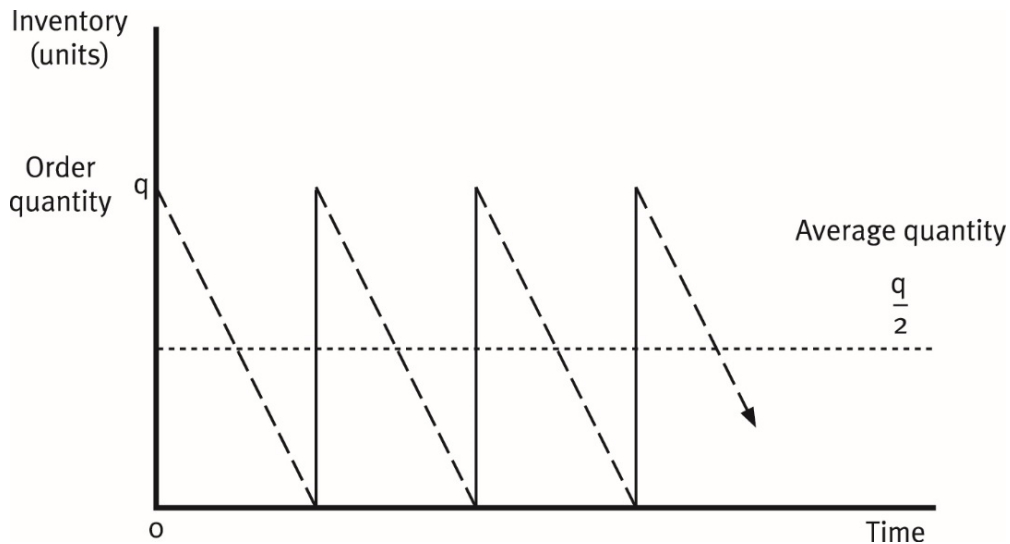
- the variable costs of holding the inventory
- the fixed costs of placing the order

Holding costs

The model assumes that it costs a certain amount to hold a unit of inventory for a year (referred to as C_H in the formula). Therefore, as the average level of inventory increases, so too will the total annual holding costs incurred.

Because of the assumption that demand per period is known and is constant (see below), conclusions can be drawn over the average inventory level in relationship to the order quantity.

When new batches or items of inventory are purchased or made at periodic intervals, the inventory levels are assumed to exhibit the following pattern over time.

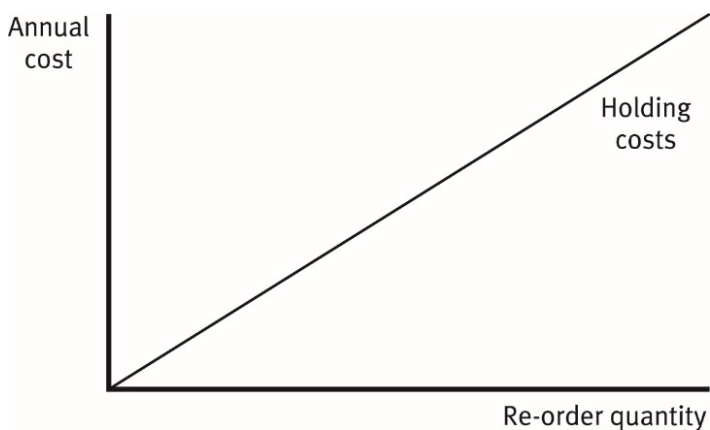


If q is the quantity ordered, the annual holding cost would be calculated as:

Holding cost per unit $\times$ Average inventory:

$$C_H \times \frac{q}{2}$$

We therefore see an upward sloping, linear relationship between the reorder quantity and total annual holding costs.



Note that if buffer inventory is held, the annual holding cost becomes: $C_H \times (q/2 + \text{buffer})$.

Ordering costs

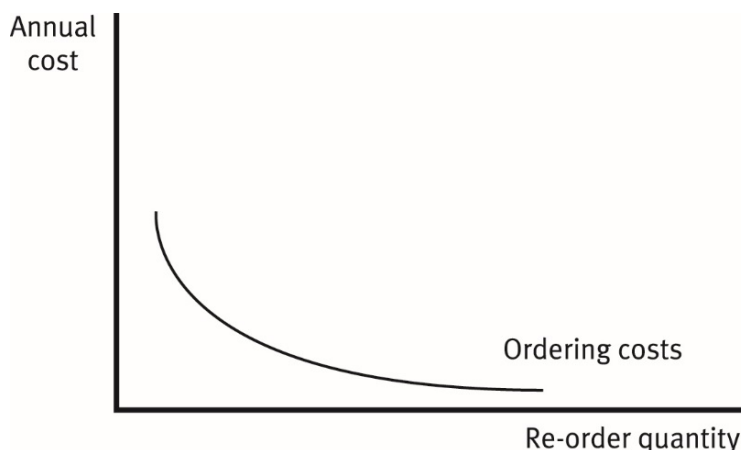
The model assumes that a fixed cost is incurred every time an order is placed (referred to as C_O in the formula). Therefore, as the order quantity increases, there is a fall in the number of orders required, which reduces the total ordering cost.

If D is the annual expected sales demand, the annual order cost is calculated as:

Order cost per order $\times$ no. of orders per year.

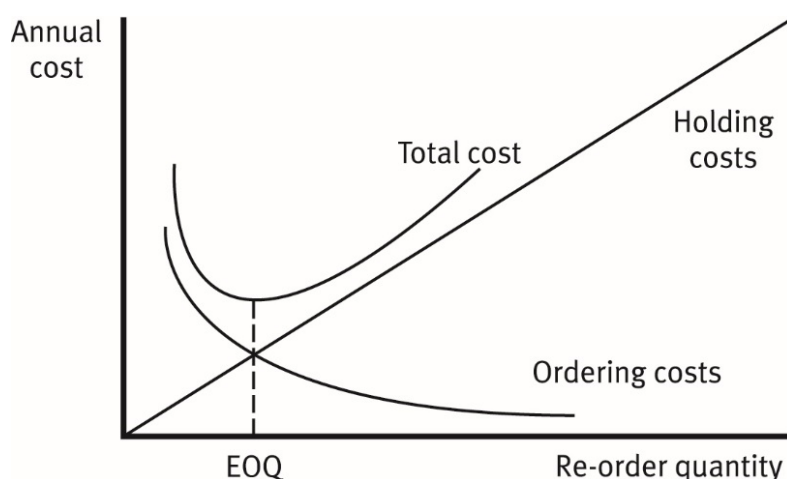
$$C_o \times \frac{D}{q}$$

However, the fixed nature of the cost results in a downward sloping, curved relationship.



Because you are trying to balance these two costs (one which increases as re-order quantity increases and one which falls), total costs will always be minimised at the point where the total holding costs equals the total ordering costs. This point will be the economic order quantity.

When the re-order quantity chosen minimises the total cost of holding and ordering, it is known as the EOQ.



Assumptions

The following assumptions are made:

- demand and lead-time are constant and known
- purchase price is constant
- no buffer inventory held (not needed).



These assumptions are critical and should be discussed when considering the validity of the model and its conclusions, e.g. in practice, demand and/or lead-time may vary.

The calculation

The EOQ can be more quickly found using a formula (given in the examination):

$$EOQ = \sqrt{\frac{2C_o D}{C_H}}$$

where:

C_o = cost per order

D = annual demand

C_H = cost of holding one unit for one year.



Test your understanding 1 – EOQ I

A company requires 1,000 units of material X per month. The cost per order is \$30 regardless of the size of the order. The holding costs are \$2.88 per unit per year.

Required:

Investigate the total cost of buying the material in quantities of 400, 500, or 600 units at one time. What is the cheapest option?

Use the EOQ formula to prove your answer is correct.



Test your understanding 2 – EOQ II

Monthly demand for a product is 10,000 units. The purchase price is \$10/unit and the company's cost of finance is 15% per year. Warehouse storage costs per unit per year are \$2/unit. The supplier charges \$200 per order for delivery.

Calculate the EOQ.

Dealing with quantity discounts

Discounts may be offered for ordering in large quantities. If the EOQ is smaller than the order size needed for a discount, should the order size be increased above the EOQ?



To work out the answer you should carry out the following steps:

Step 1: Calculate EOQ, ignoring discounts.

Step 2: If the EOQ is below the quantity qualifying for a discount, calculate the total annual inventory cost arising from using the EOQ:

Total annual inventory cost = purchase costs ($D \times P$ where P is purchase price) + ordering costs ($C_o \times D/Q$) + holding costs ($C_H \times Q/2$)

Step 3: Recalculate total annual inventory costs using the order size required to just obtain each discount. Take the available discount into account within the purchase costs.

Step 4: Compare the totals from steps 2 and 3 and select the lowest cost option.

Step 5: Repeat for all discount levels.



Test your understanding 3 – EOQ with discounts I

W Co is a retailer of barrels. The company has an annual demand of 30,000 barrels. The barrels cost \$2 each. Fresh supplies can be obtained immediately, with ordering and transport costs amounting to \$200 per order. The annual cost of holding one barrel in stock is estimated to be \$1.20.

A 2% discount is available on orders of at least 5,000 barrels and a 2.5% discount is available if the order quantity is 7,500 barrels or above.

Required:

Calculate the EOQ ignoring the discount and determine if it would change once the discount is taken into account.



Test your understanding 4 – EOQ with discounts II

D Co uses component V22 in its construction process. The company has a demand of 45,000 components per year. They cost \$4.50 each. There is no lead-time between order and delivery, and ordering costs amount to \$100 per order. The annual cost of holding one component in inventory is estimated to be \$0.65.

A 0.5% discount is available on orders of at least 3,000 components and a 0.75% discount is available if the order quantity is 6,000 components or above.

Calculate the optimal order quantity.

4 Calculating the re-order level (ROL)

Known demand and lead time

Having decided how much inventory to re-order, the next problem is when to re-order. The firm needs to identify a level of inventory which can be reached before an order needs to be placed.



The **ROL** is the quantity of inventory on hand when an order is placed.

When demand and lead-time are known with certainty the ROL may be calculated exactly:

$$\text{ROL} = \text{demand in the lead-time.}$$

If a buffer inventory is held, this should be adjusted to:

$$\text{ROL} = \text{demand in the lead time} + \text{buffer}$$



Test your understanding 5 – Calculating the re-order level I

Using the data for W Co, assume that the company adopts the EOQ as its order quantity and that it now takes two weeks for an order to be delivered.

How frequently will the company place an order? How much inventory will it have on hand when the order is placed?



Test your understanding 6 – Calculating the re-order level II

Using the data relating to D Co, and ignoring discounts, assume that the company adopts the EOQ as its order quantity and that it now takes three weeks for an order to be delivered.

- (a) How frequently will the company place an order?
- (b) How much inventory will it have on hand when the order is placed?

ROL with variable demand or variable lead-time

When lead-time and demand are known with certainty, $\text{ROL} = \text{demand during lead-time}$. Where there is uncertainty, an optimum level of buffer inventory must be found. This depends on:

- variability of demand
- cost of holding inventory
- cost of stockouts.

You will not be required to perform this calculation in the examination.



Re-order levels

If there were certainty, then the last unit of inventory would be sold as the next delivery is made. In the real world, this ideal cannot be achieved. Demand will vary from period to period, and ROL must allow some buffer (or safety) inventory, the size of which is a function of maintaining the buffer (which increases as the levels increase), running out of inventory (which decreases as the buffer increases) and the probability of the varying demand levels.

5 Inventory management systems

A number of systems have been developed to simplify the inventory management process:

- periodic review
- Just in time (JIT)

Periodic review system (constant order cycle system)

Inventory levels are reviewed at fixed intervals, e.g. every four weeks. The inventory in hand is then made up to a predetermined level, which takes account of:

- likely demand before the next review
- likely demand during the lead-time.

Thus a four-weekly review in a system where the lead time was two weeks would demand that inventory be made up to the likely maximum demand for the next six weeks.

Under this system orders are evenly spread, so it is popular with suppliers.



Additional question – Periodic review systems

A company has estimated that for the coming season weekly demand for components will be 80 units. Suppliers take three weeks on average to deliver goods once they have been ordered and a buffer inventory of 35 units is held.

If the inventory levels are reviewed every six weeks, how many units will be ordered at a review where the count shows 250 units in inventory?

The answer to this question can be found after the chapter summary diagram at the end of this chapter.



Slow moving inventory

Certain items may have a high individual value, but be subject to infrequent demand.

In most organisations, about 20% of items held make up 80% of total usage (the 80/20 rule).

Slow-moving inventory

Management need to review inventory usage to identify slow-moving inventory. An aged inventory analysis should be produced and reviewed regularly so that action can be taken. Actions could include:

- elimination of obsolete items
- slow-moving inventory items only ordered when actually needed (unless a minimum order quantity is imposed by the supplier)
- review of demand level estimates on which re-order decisions are based.

A regular report of slow-moving items is useful in that management is made aware of changes in demand and of possible obsolescence. Arrangements may then be made to reduce or eliminate inventory levels or, on confirmation of obsolescence, for disposal.

Just in Time (JIT) systems



JIT is a series of manufacturing and supply chain techniques that aim to minimise inventory levels and improve customer service by manufacturing not only at the exact time customers require, but also in the exact quantities they need and at competitive prices.

In JIT systems, the balancing act is dispensed with. Inventory is reduced to an absolute minimum or eliminated altogether.

Aims of JIT are:

- a smooth flow of work through the manufacturing plant
- a flexible production process, which is responsive to the customer's requirements
- reduction in capital tied up in inventory.

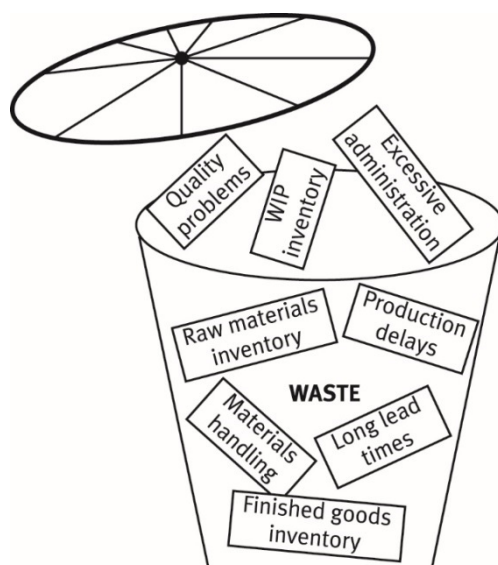
This involves the elimination of all activities performed that do not add value = waste.



Just in time systems

JIT extends much further than a concentration on inventory levels. It centres on the elimination of waste. Waste is defined as any activity performed within a manufacturing company that does not add value to the product. Examples of waste are:

- raw material inventory
- WIP inventory
- finished goods inventory
- materials handling
- quality problems (rejects and reworks, etc.)
- queues and delays on the shop floor
- long raw material lead times
- long customer lead times
- unnecessary clerical and accounting procedures.



JIT attempts to eliminate waste at every stage of the manufacturing process, notably by the elimination of:

- WIP, by reducing batch sizes (often to one)
- raw materials inventory, by the suppliers delivering direct to the shop floor JIT for use
- scrap and rework, by an emphasis on total quality control of the design, of the process, and of the materials
- finished goods inventory, by reducing lead times so that all products are made to order
- material handling costs, by re-design of the shop floor so that goods move directly between adjacent work centres.

The combination of these concepts in JIT results in:

- a smooth flow of work through the manufacturing plant
- a flexible production process which is responsive to the customer's requirements
- reduction in capital tied up in inventory.

A JIT manufacturer looks for a single supplier who can provide **high quality**, frequent and **reliable** deliveries, rather than the lowest price. In return, the supplier can expect more business under long-term **purchase orders**, thus providing **greater certainty** in forecasting activity levels. Very often, the suppliers will be located close to the company.

Long-term contracts and single sourcing strengthen buyer-supplier relationships and tend to result in a higher quality product. Inventory problems are shifted back onto suppliers, with deliveries being made as required.

The spread of JIT in the production process inevitably affects those in delivery and transportation. Smaller, more frequent loads are required at shorter notice. The haulier is regarded as almost a partner to the manufacturer, but tighter schedules are required of hauliers, with penalties for non-delivery.

Reduction in inventory levels reduces the time taken to count inventory and the clerical cost. However, with JIT, although inventory holding costs are close to zero, inventory ordering costs are high.



You need to be able to outline the key features of each stock control system and the impact they may have on ordering and holding costs and/or order dates and quantities. Ensure you can discuss the implications of JIT for production processes and supplier relationships as well as inventory levels.

Note that the presentation of inventory in the financial statements is not part of this syllabus and was covered in the ACCA Financial Accounting exam or in your other previous studies.



Test your understanding 7

Gogo Co is a retailer of large storage boxes. The company has an annual demand of 120,000 units. The costs incurred each time an order is placed are \$200. The carrying cost per unit of the item each month is estimated at \$3. The purchase price of each unit is \$4. The economic order quantity formula is:

$$EOQ = \sqrt{\frac{2C_oD}{C_H}}$$

When using this formula to find the optimal quantity to be ordered, which of the following amounts are not included in the calculation?

- A Cost per order (\$200)
- B Carrying cost per unit (\$3)
- C Purchase price per unit (\$4)
- D Estimated usage of the inventory item over a particular period (120,000 units per year)



Test your understanding 8

The Economic Order Quantity (EOQ):

- A is a formula that calculates a realistic purchase price for an item
- B determines the lowest order quantity by balancing the cost of ordering against the cost of holding inventory
- C is used to calculate how much safety inventory should be carried
- D should be calculated once a year



Test your understanding 9

Which of the following statements is true?

Statement 1: The re-order level is the measure of inventory at which a replenishment order should be made.

Statement 2: Use of a re-order level builds in a measure of safety inventory and minimises the risk of the organisation running out of inventory.

Statement 1 *Statement 2*

- | | | |
|---|-------|-------|
| A | True | True |
| B | True | False |
| C | False | True |
| D | False | False |

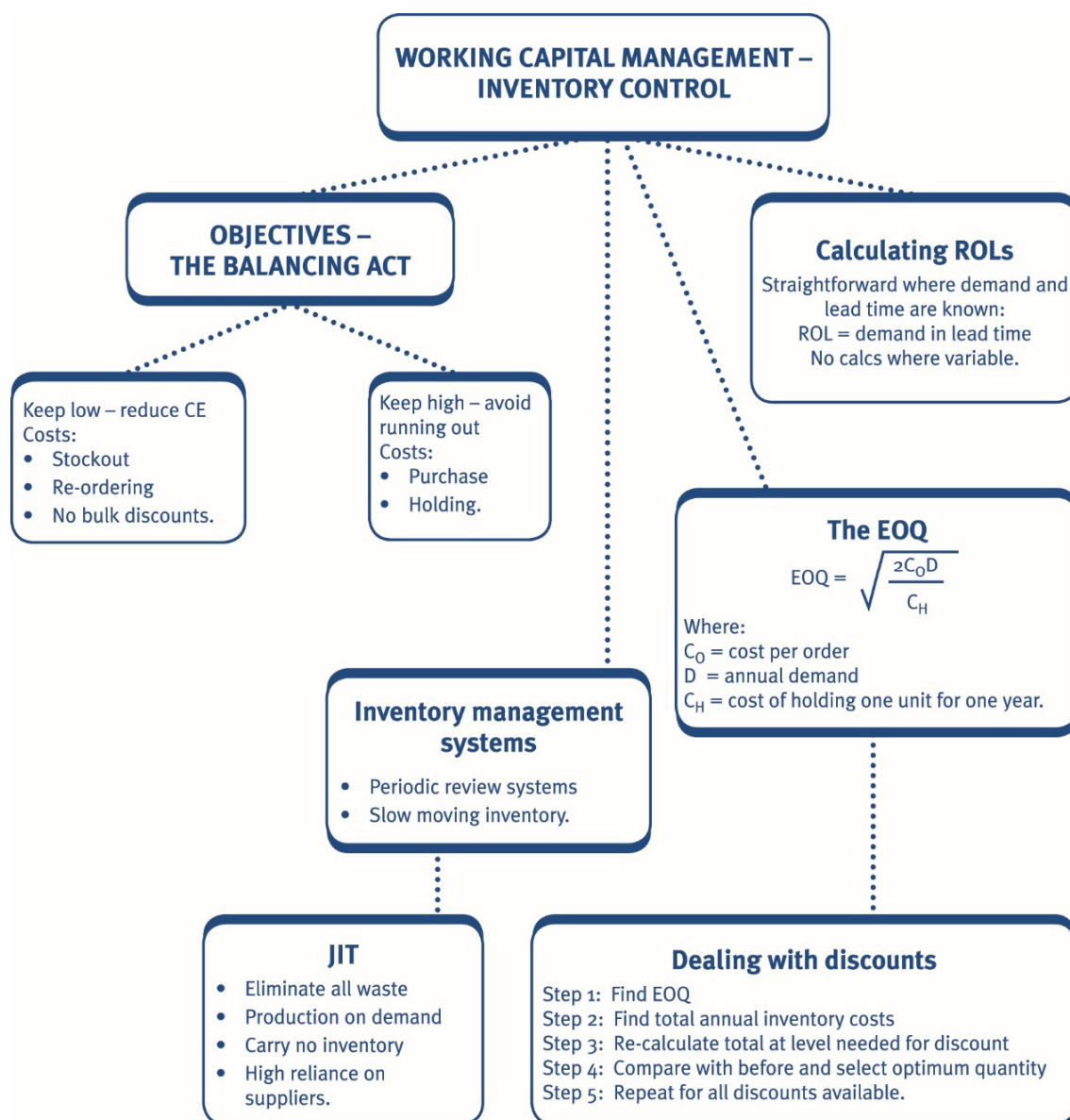


Test your understanding 10

Periodic review means

- A Ordering inventory at a fixed and regular time interval
- B Ordering inventory when it falls below the designated safety inventory level
- C Ordering inventory in consultation with suppliers relative to their available capacity
- D Ordering inventory at a pre-determined re-order level

6 Chapter summary





Answer to additional question – Periodic review systems

Demand per week is 80 units. The next review will be in six weeks by which time $80 \times 6 = 480$ units will have been used.

An order would then be placed and during the lead-time – three weeks – another $80 \times 3 = 240$ units will be used.

The business therefore needs to have $480 + 240 = 720$ units in inventory to ensure a stockout is avoided. Since buffer inventory of 35 units is required, the total number needed is 755 units.

Since the current inventory level is 250 units, an order must be placed for $755 - 250 = 505$ units.

Test your understanding answers



Test your understanding 1 – EOQ I

	Order quantity		
	400 units	500 units	600 units
Average inventory	200	250	300
No. of orders per year	30	24	20
Holding cost – average units × \$2.88	\$576	\$720	\$864
Ordering cost – no. of orders × \$30	\$900	\$720	\$600
Total cost	\$1,476	\$1,440	\$1,464

Therefore, the best option is to order 500 units each time.

Note: That this is the point at which total cost is minimised and the holding costs and order costs are equal.

Solution using the formula:

$$EOQ = \sqrt{\frac{2C_oD}{C_H}}$$

$$C_o = 30$$

$$D = 1,000 \times 12 = 12,000$$

$$C_H = 2.88$$

$$EOQ = \sqrt{\frac{2 \times 30 \times 12,000}{2.88}} = 500 \text{ units per order}$$



Test your understanding 2 – EOQ II

$$EOQ = \sqrt{\frac{2C_oD}{C_H}}$$

$$C_o = \$200$$

$$D = 10,000 \text{ units} \times 12 = 120,000 \text{ units per year}$$

$$C_H = (\$10 \times 0.15) + \$2 = \$3.5 \text{ per unit per year}$$

Note that the holding cost includes the 'opportunity cost' of holding inventory, i.e. the cost of funding the inventory held. This is calculated as the purchase price of a unit of inventory multiplied by the cost of capital.

It represents cash that could otherwise have been used to grow the business if it wasn't tied up in inventory.

$$EOQ = \sqrt{\frac{2 \times 200 \times 120,000}{3.5}} = 3,703 \text{ units per order}$$



Test your understanding 3 – EOQ with discounts I

Step 1 Calculate EOQ, ignoring discounts.

$$EOQ = \sqrt{\frac{2C_oD}{C_H}}$$

$$C_o = \$200$$

$$D = 30,000 \text{ barrels per year}$$

$$C_H = \$1.20 \text{ per barrel per year}$$

$$EOQ = \sqrt{\frac{2 \times 200 \times 30,000}{1.2}} = 3,162 \text{ barrels per order}$$

Step 2 As this is below the level for discounts, calculate total annual inventory costs.

Total annual costs for the company will comprise holding costs plus reordering costs.

$$= (\text{Average inventory} \times C_H) + (\text{Number of re-orders per year} \times C_o)$$

$$= \frac{3,162}{2} \times \$1.20 + \frac{30,000}{3,162} \times \$200$$

$$= \$1,897.20 + \$1,897.53$$

$$= \$3,794.73$$

$$= \$3,795 \text{ per year}$$

Step 3 Recalculate total annual inventory costs using the order size required to just obtain the discount.

At order quantity 5,000, total costs are as follows.

$$\begin{aligned}
 &= (\text{Average inventory} \times C_H) + (\text{Number of re-orders per year} \times C_O) \\
 &= \frac{5,000}{2} \times \$1.20 + \frac{30,000}{5,000} \times \$200 \\
 &= \$3,000 + \$1,200 \\
 &= \$4,200 \text{ per year}
 \end{aligned}$$

	\$
Extra costs of ordering in batches of 5,000 (4,200 – 3,795)	(405)
Less: Saving on discount 2% × \$2 × 30,000	1,200
Step 4 Net cost saving	795

Hence, batches of 5,000 are worthwhile.

Step 3 (again)

At order quantity 7,500, total costs are as follows:

$$\begin{aligned}
 &= (\text{Average inventory} \times C_H) + (\text{Number of re-orders per year} \times C_O) \\
 &= \frac{7,500}{2} \times \$1.20 + \frac{30,000}{7,500} \times \$200 \\
 &= \$4,500 + \$800 \\
 &= \$5,300 \text{ per year}
 \end{aligned}$$

	\$
Extra costs of ordering in batches of 7,500 (5,300 – 4,200)	(1,100)
Less: Saving on extra discount (2.5% – 2%) × \$2 × 30,000	300
Step 4 Net additional cost	(800)

So a further cost saving cannot be made on orders of 7,500 units. The company should therefore opt for buying 5,000 units in order to maximise the benefit.

Note: If Step 1 produces an EOQ at which a discount would have been available, and the holding cost would be reduced by taking the discount, i.e. where C_H is based on the purchase price × the cost of finance, the EOQ must be recalculated using the new C_H before the above steps are followed.



Test your understanding 4 – EOQ with discounts II

Step 1

$$EOQ = \sqrt{\frac{2C_oD}{C_H}}$$

C_o = \$100 per order

D = 45,000 components per year

C_H = \$0.65 per component per year

$$EOQ = \sqrt{\frac{2 \times 100 \times 45,000}{0.65}} = 3,721 \text{ components per order}$$

which would qualify
for a 0.5% discount

Step 2

Total annual costs for the company will comprise holding costs plus reordering costs.

= (Average inventory $\times C_H$) + (Number of re-orders per year $\times C_o$)

$$= \left(\frac{3,721}{2} \times \$0.65 \right) + \left(\frac{45,000}{3,721} \times \$100 \right)$$

= \$2,419 per year

Step 3

At an order quantity of 6,000 components, total costs are as follows.

$$(6,000 \times \$0.65/2) + (45,000 \times \$100 \div 6,000) = \$2,700$$

\$

Extra costs of ordering in batches of 6,000 (2,700 – 2,419) (281)

Less: Saving on extra discount

(0.75% – 0.5%) $\times$ \$4.5 $\times$ 45,000 506.25

Step 4

225.25

Net cost saving

So a saving can be made on orders of 6,000 components.



Test your understanding 5 – Calculating the re-order level I

- Annual demand is 30,000. The original EOQ is 3,162 barrels per order. The company will therefore place an order once every $3,162 \div 30,000 \times 365 \text{ days} = 38 \text{ days}$
- The company must be sure that there is sufficient inventory on hand when it places an order to last the two weeks' lead-time. It must therefore place an order when there is two weeks' worth of demand in inventory:
i.e. Re-order level = $2 \div 52 \times 30,000 = 1,154$ barrels remaining in inventory



Test your understanding 6 – Calculating the re-order level II

- (a) Annual demand is 45,000 components. The original EOQ is 3,721 components per order.
The company will therefore place an order once every $3,721 \div 45,000 \times 365 \text{ days} = 30 \text{ days}$
- (b) The company must be sure that there is sufficient inventory on hand when it places an order to last the three weeks' lead-time. It must therefore place an order when there is three weeks' worth of demand in inventory:
 $3/52 \times 45,000 = 2,596$ components in inventory.



Test your understanding 7

Answer C

In the formula, C_o = the cost of placing one order; D = the estimated usage of an inventory item over a particular period; and C_H = the cost of holding one unit of inventory for that period.

The purchase price per unit is not a constituent part of the formula.



Test your understanding 8

Answer B

The EOQ is the optimal ordering quantity for an item of inventory which will minimise costs.



Test your understanding 9

Answer A

The re-order level is the quantity of inventory on hand when an order is placed. The level will depend on variability of demand, cost of holding inventory and cost of stockouts.



Test your understanding 10

Answer A

Inventory levels are reviewed at fixed time. The inventory in hand is then made up to a predetermined level.

Working capital management – Accounts receivable and payable

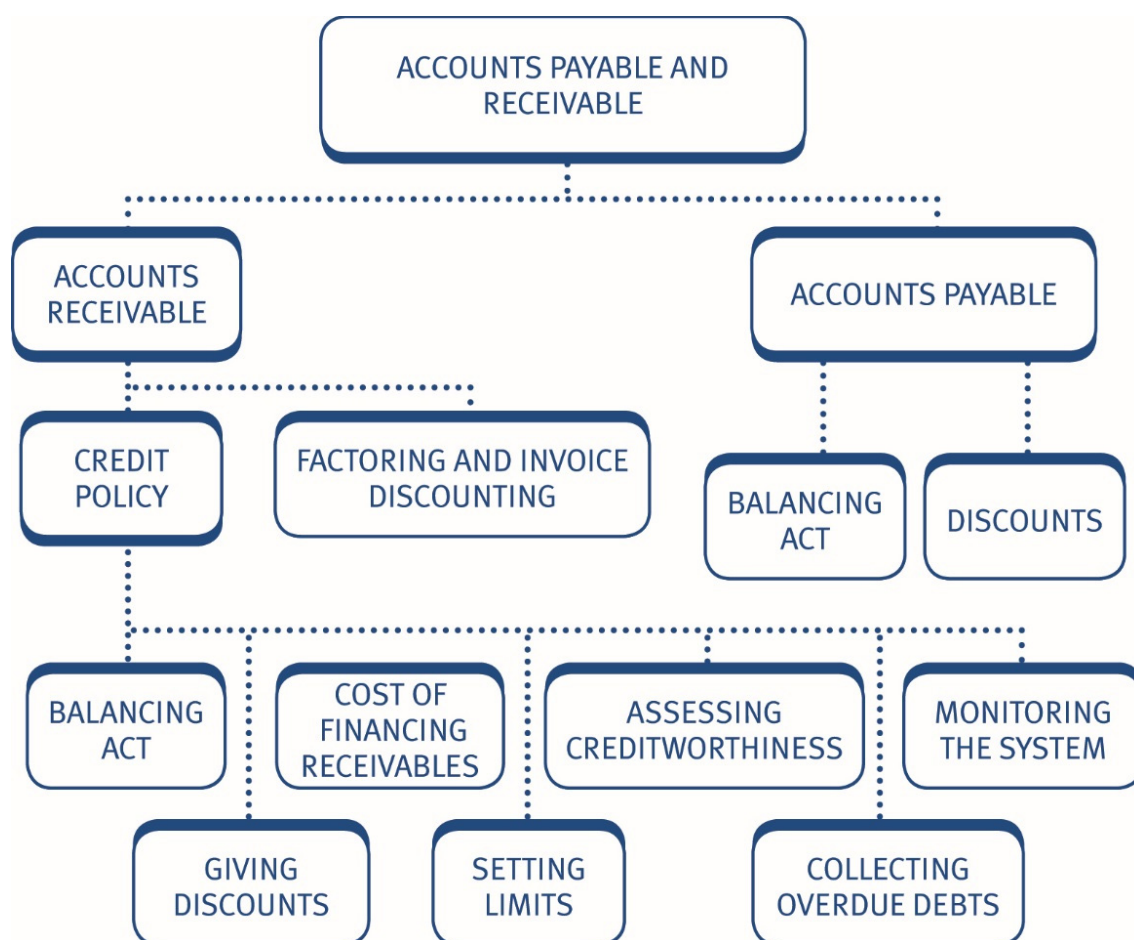
Chapter learning objectives

Upon completion of this chapter, you will be able to:

- explain how to establish and implement a credit policy for accounts receivable
- explain the administration involved in collecting amounts owing from accounts receivable
- explain the pros and cons of offering early settlement discounts to accounts receivable
- calculate the financial implications of offering discounts for early settlement
- define and explain the features of factoring
- discuss the advantages and disadvantages of factoring
- define and explain the features of invoice discounting
- suggest and evaluate suitable techniques for managing accounts receivable within a scenario question
- explain the factors involved in the effective management of trade credit
- calculate the effective cost of an early settlement discount offered on an account payable
- discuss the advantages and disadvantages of trade credit as a source of short-term finance
- explain the specific factors to be considered when managing foreign accounts receivable and payable.



One of the PER performance objectives (PO10) is to manage and control working capital. You manage cash and working capital effectively, planning for any shortfall or surplus including receivables, payables and inventories. Working through this chapter should help you understand how to demonstrate that objective.



You should have a good understanding of the nature of accounts receivable and payable from your previous studies, for instance from the ACCA Financial Accounting exam.

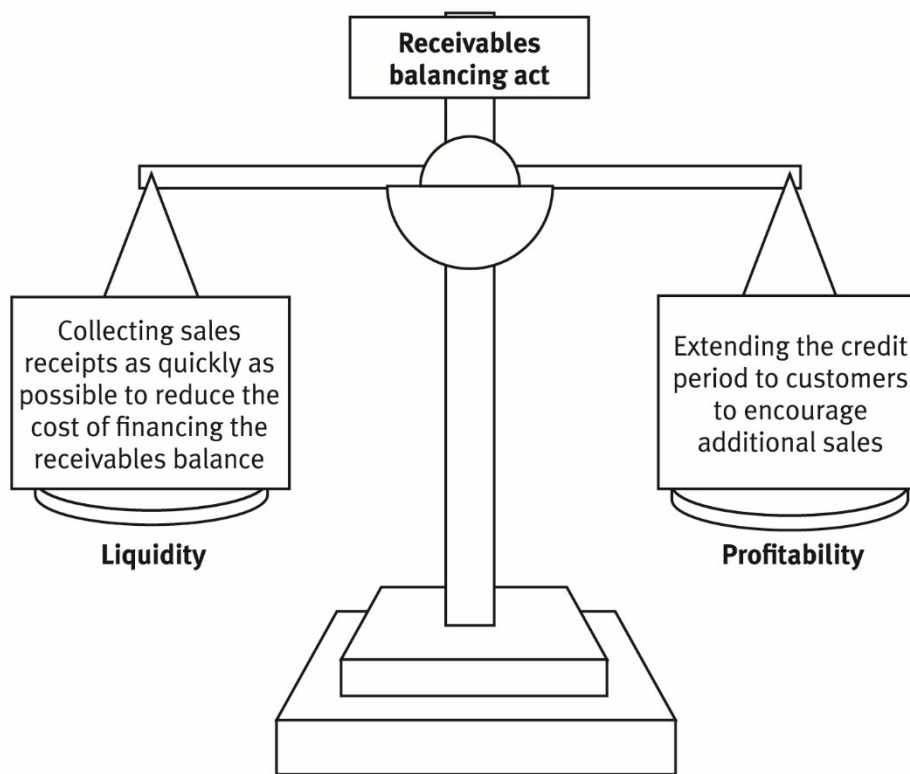
1 The objectives of accounts receivable management

The optimum level of trade credit extended represents a balance between two factors:

- profit improvement from sales obtained by allowing credit
- the cost of credit allowed.



The balancing act



Remember this trade-off is a key factor in determining the company's working capital investment.

2 Accounts receivable – establishing a credit policy

Management must establish a credit policy.



For accounts receivable, the company's policy will be influenced by:

- demand for products
- competitors' terms
- risk of irrecoverable debts
- financing costs
- costs of credit control.



Why have a credit policy?

A firm must establish a policy for credit terms given to its customers. Ideally, the firm would want to obtain cash with each order delivered, but that is uncompetitive unless substantial settlement (or cash) discounts are offered as an inducement. It must be recognised that credit terms are part of the firm's marketing policy. If the trade or industry has adopted a common practice, then it is probably wise to keep in step with it.

A lenient credit policy may well attract additional customers, but at a disproportionate increase in cost.

A credit policy has four key aspects:

- (1) Assess creditworthiness.
- (2) Credit limits.
- (3) Invoice promptly and collect overdue debts.
- (4) Monitor the credit system.



This is a useful structure to adopt for examination questions that ask about the management of receivables.

Assessing creditworthiness

A firm should assess the creditworthiness of:

- all new customers immediately
- existing customers periodically.

Information may come from:

- bank references
- trade references
- competitors
- published information
- credit reference agencies
- company sales records
- credit scoring.



Assessing creditworthiness

To minimise the risk of irrecoverable debts occurring, a company should investigate the creditworthiness of all new customers (credit risk), and should review that of existing customers from time to time, especially if they request that their credit limit should be raised. Information about a customer's credit rating can be obtained from a variety of sources.

These include:

- Bank references – A customer's permission must be sought. These tend to be fairly standardised in the UK, and so are not perhaps as helpful as they could be.
- Trade references – Suppliers already giving credit to the customer can give useful information about how good the customer is at paying bills on time. There is a danger that the customer will only nominate those suppliers that are being paid on time.
- Competitors – in some industries such as insurance, competitors share information on customers, including creditworthiness.
- Published information – The customer's own annual accounts and reports will give some idea of the general financial position of the company and its liquidity.
- Credit reference agencies – Agencies such as Dun & Bradstreet publish general financial details of many companies, together with a credit rating. They will also produce a special report on a company if requested. The information is provided for a fee.
- Company's own sales records – For an existing customer, the sales ledgers will show how prompt a payer the company is, although they cannot show the ability of the customer to pay.
- Credit scoring – Indicators such as family circumstances, home ownership, occupation and age can be used to predict likely creditworthiness. This is useful when extending credit to the public where little other information is available. A variety of software packages is available which can assist with credit scoring.

Credit limits

Credit limits should be set to reflect both the:

- amount of credit available
- length of time allowed before payment is due.

The ledger account should be monitored to take account of orders in the pipeline as well as invoiced sales, before further credit is given, to ensure that limits are not breached.

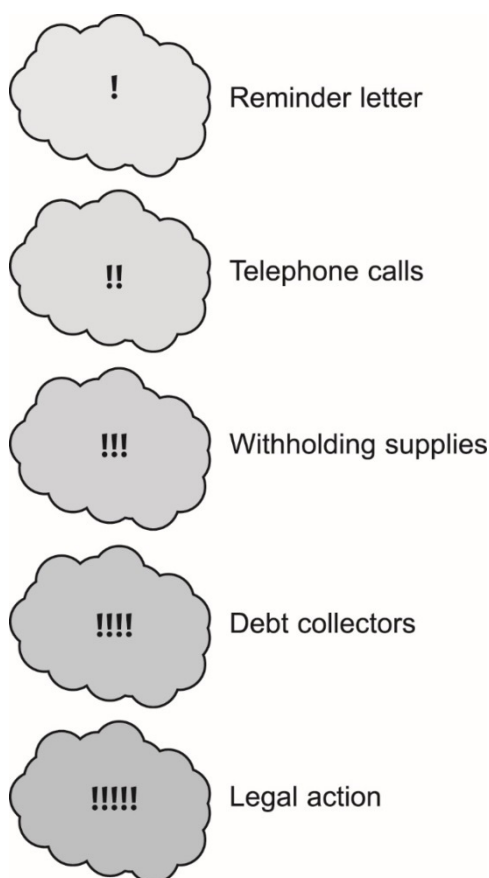
Invoicing and collecting overdue debts

A credit period only begins once an invoice is received so prompt invoicing is essential. If debts go overdue, the risk of default increases, therefore a system of follow-up procedures is required.



Invoicing and collecting overdue debts

The longer a debt is allowed to run, the higher the probability of eventual default. A system of follow-up procedures is required, bearing in mind the risk of offending a valued customer to such an extent that their business is lost.



Techniques for 'chasing' overdue debts include the following:

- **Reminder letters:** these are often regarded as being a relatively poor way of obtaining payment, as many customers simply ignore them. Sending reminders by email is usually more productive than using the post.
- **Telephone calls:** these are more expensive than reminder letters (which can be automatically generated by most accounting systems) but where large sums are involved, they can be an efficient way of speeding up payment.

- Withholding supplies: putting customers on the 'stop list' for further orders or spare parts can encourage rapid settlement of debts.
- Debt collection agencies and trade associations: these offer debt collection services on a fixed fee basis or on 'no collection no charge' terms. The quality of service provided varies considerably and care should be taken in selecting an agent.
- Legal action: this is often seen as a last resort. A solicitor's letter often prompts payment and many cases do not go to court. Court action is usually not cost effective but it can discourage other customers from delaying payment.

Monitoring the system

The position of receivables should be regularly reviewed as part of managing overall working capital and corrective action taken when needed.

Methods include:

- age analysis
- ratios
- statistical data.



Monitoring the system

Management will require regular information to take corrective action and to measure the impact of giving credit on working capital investment. Typical management reports on the credit system will include the following points.

- Age analysis of outstanding debts.
- Ratios, compared with the previous period or target, to indicate trends in credit levels and the incidence of overdue and irrecoverable debts.
- Statistical data to identify causes of default and the incidence of irrecoverable debts among different classes of customer and types of trade

3 Accounts receivable – calculations

Costs of financing receivables



Key workings:

Finance cost = Receivable balance × Interest (overdraft) rate

$$\text{Receivable balance} = \text{Credit sales} \times \frac{\text{Accounts receivable collection period}}{365}$$



Test your understanding 1 – Cost of financing receivables

Paisley Co has sales of \$20 million for the previous year, receivables at the year-end were \$4 million, and the cost of financing receivables is covered by an overdraft at the interest rate of 12% per year.

Required:

- calculate the accounts receivable collection period for Paisley
- calculate the annual cost of financing receivables.

Early settlement discounts

Cash discounts are given to encourage early payment by customers. The cost of the discount is balanced against the savings the company receives from having less capital tied up due to a lower receivables balance and a shorter average collection period. Discounts may also reduce the number of irrecoverable debts.

The calculation of the annual cost can be expressed as a formula:

$$\text{Annual cost of discount} = \left(1 + \frac{\text{discount}}{\text{amount let to pay}} \right)^{\text{no. of periods}} - 1$$



where no. of periods = $\frac{365/52/12}{\text{no. of days/weeks/months earlier the money is received}}$

Note: if the discount offered is 1%, the amount left to pay will be 99%. A discount of 1.5% would have an amount left to pay of 98.5%. The two numbers will always add to 100.

If the cost of offering the discount exceeds the rate of overdraft interest, then the discount should not be offered.



Test your understanding 2 – Early settlement discounts I

A company is offering a cash discount of 2.5% to receivables if they agree to pay debts within one month. The usual credit period taken is three months.

What is the effective annualised cost of offering the discount and should it be offered, if the bank would loan the company at 18% per year?

The above calculation gives a quick way to evaluate the annual cost and reach a decision on whether to offer an early settlement discount. This method should be used in situations where all customers are expected to take advantage of the discount, or when the question does not specify the expected take up (as in test your understanding 2).

However, in many scenarios, you may be provided with more in-depth information, in particular information relating to the proportion of customers that are expected to take up the offer. In these situations, it is necessary to perform a longer calculation.

The calculation involves comparing the annual monetary cost of offering the discount (the cash that will no longer be received) with the annual benefit that will be generated (the lower financing costs resulting from the reduced receivables balance).

The **monetary cost of the discount** can be simply calculated as:

credit sales $\times$ proportion of customers that take the discount $\times$ discount %

The **saving resulting from the reduction in the finance cost** is more difficult to calculate. There are multiple methods of calculating it. Here, we will focus on one approach, using the following steps:

- 1 Use the sales proportions and the receivables balance formula (on the previous page) to calculate receivables balances separately for those who take the discount and those who do not.
- 2 Then, add those balances together to determine the new total receivables balance.
- 3 Then, use the finance cost formula (on the previous page) to determine the finance cost on the new total receivables.
- 4 This can then be compared to the finance cost before the discount was offered to work out the saving.

Once the cost and benefit have been calculated, they can be compared to decide whether to offer the discount or not.



Illustration of early settlement discount calculations

A company has credit sales of \$5 million. Customers take 90 days to pay and working capital is funded using an overdraft costing 8% per year.

The company considers offering a discount for early settlement of 1.5% if customers pay in 30 days.

It is anticipated that 30% of customers will take up this offer and the rest will continue paying in 90 days.

Assume that there are 360 days in a year.

Calculate the net cost or benefit of offering this discount.

Cost of the discount:

Cost of the discount = credit sales × proportion of customers that take the discount × discount %

Cost of the discount = \$5,000,000 × 30% × 1.5% = **\$22,500**

Benefit of the discount:

The benefit comes from the reduced funding cost, but that only applies to the receivables related to the 30% of customers that take the discount.

First, we must work out the current receivables level and finance cost:

Receivables = credit sales × receivables collection period / 360

Current receivables = \$5,000,000 × 90 / 360 = \$1,250,000

Finance cost = receivables × interest rate

Current finance cost = \$1,250,000 × 8% = \$100,000

Now we can follow the four steps from above:

- 1 Use the sales proportions and the receivables balance formula to calculate receivables balances separately for those who take the discount and those who do not:

Receivables (for those who take the discount): \$5,000,000 × 30% × 30 / 360 = \$125,000

Receivables (for those who don't take the discount): \$5,000,000 × 70% × 90/360 = \$875,000

- 2 Then, add those balances together to determine the new total receivables balance:

Total receivables = \$125,000 + \$875,000 = \$1,000,000

- 3 Then, use the finance cost formula to determine the finance cost on the new total receivables:

New finance cost = \$1,000,000 × 8% = \$80,000

- 4 This can then be compared to the finance cost before the discount was offered to work out the saving:

Finance cost saving = difference between old and new finance costs =
 $\$100,000 - \$80,000 = \$20,000$

Now that we have calculated both the cost and the benefit, we can compare them to determine if it is worth offering the discount:

Cost \$22,500

Benefit \$20,000

Overall, there would be a **net cost of \$2,500** if the company offered the discount, so it should not be offered.



Test your understanding 3 – Early settlement discounts II

Paisley Co has sales of \$20 million for the previous year, receivables at the year-end of \$4 million and the cost of financing receivables is covered by an overdraft at the interest rate of 12% per year. It is now considering offering a cash discount of 2% for payment of debts within 10 days.

Should it be introduced if 40% of customers will take up the discount?

4 Accounts receivable – invoice discounting and factoring

Invoice discounting and factoring are both ways of speeding up the receipt of funds from accounts receivable.

Invoice discounting



Invoice discounting is a method of raising finance against the security of receivables without using the sales ledger administration services of a factor.

While specialist invoice discounting firms exist, this is a service also provided by a factoring company. Selected invoices are used as security against which the company may borrow funds (like a house being used as security for a mortgage). This is a temporary source of finance, as the receivables invoices are only temporary, repayable when the debt is cleared.

The key advantage of invoice discounting is that it is a confidential service, and the customer need not know about it.

In some ways, it is similar to the financing part of the factoring service without control of credit passing to the factor.

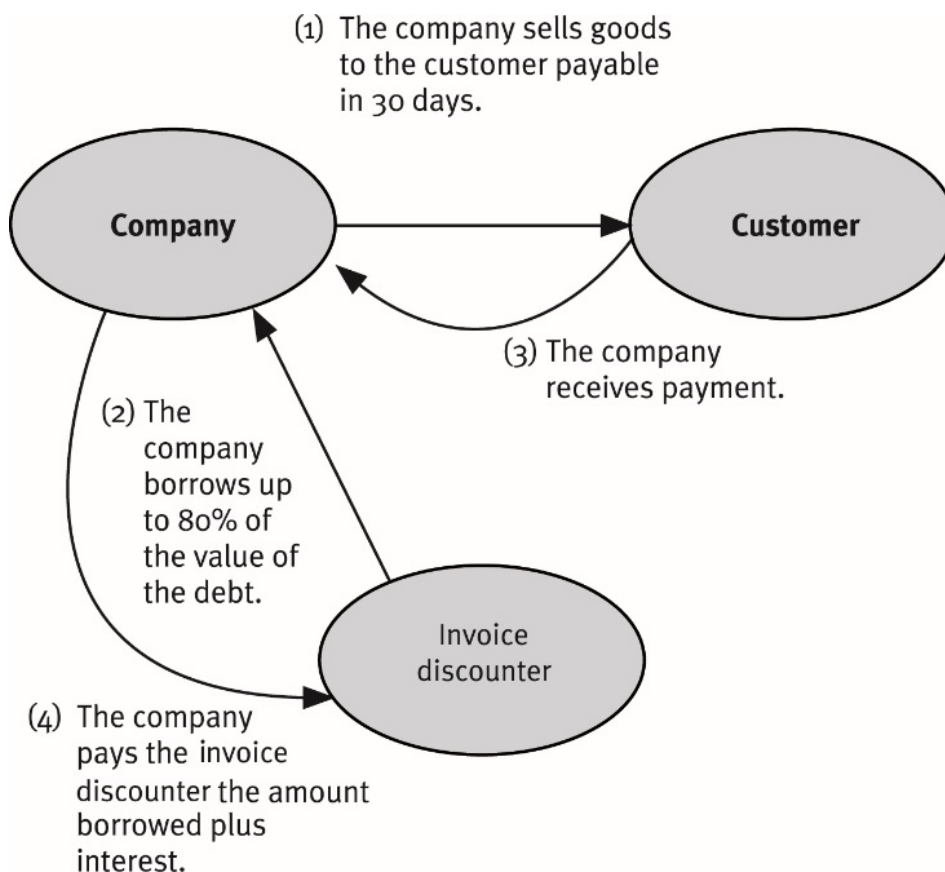


Ensure you can explain the difference between factoring and invoice discounting, and the situations where one may be more appropriate than the other



Invoice discounting

Typical arrangement



Invoice discounting is a method of raising finance against the security of receivables without using the sales ledger administration services of a factor. With invoice discounting, the business retains control over its sales ledger, and confidentiality in its dealings with customers. Firms of factors will also provide invoice discounting to clients.

The method works as follows:

- The business sends out invoices, statements and reminders in the normal way, and collects the debts. With 'confidential invoice discounting', its customers are unaware that the business is using invoice discounting.
- The invoice discounter provides cash to the business for a proportion of the value of the invoice, as soon as it receives a copy of the invoice and agrees to discount it. The discounter may exclude invoices from customers determined to be poor payers. It will then advance cash up to 80% of face value of the remaining outstanding invoices.

- When the business eventually collects the payment from its customer, the debt is removed from the loan advance amount, effectively meaning that the business pays back what was advanced from the invoice discounter.

Invoice discounting can help a business that is trying to improve its cash flows, but does not want a factor to administer its sales ledger and collect its debts. It is therefore equivalent to the financing service provided by a factor.

Administration charges for this service are around 0.5–1 % of a client's revenue. It is more risky than factoring since the client retains control over its credit policy. Consequently, such facilities are usually confined to established companies with high sales revenue, and the business must be profitable. Finance costs are usually in the range 3–4% above base rate, although larger companies and those which arrange credit insurance may receive better terms.

The invoice discounter will check the sales ledger of the client regularly, perhaps every three months, to check that its debt collection procedures are adequate.



Illustration of the invoice discounting process

At the beginning of August, Guntur Co sells goods for a total value of \$300,000 to regular customers but decides that it requires payment earlier than the agreed 30-day credit period for these invoices.

A discounter agrees to finance 80% of their face value, i.e. \$240,000, at an interest cost of 9% per year.

The invoices were due for payment in early September, but were subsequently settled in mid-September, exactly 45 days after the initial transactions. The invoice discounter's service charge is 1% of invoice value. A special account is set up with a bank, into which all payments are made.

The sequence of cash flows is:

August	Guntur receives cash advance of \$240,000.	
Mid- September	Customers pay \$300,000.	
	Invoice discounter receives the full \$300,000 paid into the special bank account.	
	Guntur receives the balance payable, less charges, i.e.	
	Service fee = $1\% \times \$300,000 =$	\$3,000
	Finance cost = $9\% \times \$240,000 \times 45/365 =$	\$2,663
	Total charges	\$5,663
	Guntur receives:	
	Balance of payment from customer	\$60,000
	Less charges	\$5,663
		\$54,337
Summary	Total receipts by Guntur: \$240,000 + \$54,337	\$294,337
	Invoice discounter's fee and interest charges	\$5,663
	Total amount invoiced	\$300,000

Factoring



Factoring is the outsourcing of the credit control department to a third party.

The debts of the company are effectively sold to a factor (normally owned by a bank). The factor takes on the responsibility of collecting the debt for a fee. The company can choose some or all of the following three services offered by the factor:

- (1) debt collection and administration – recourse (the company bears the cost of bad debts) or non-recourse (the factor bears the cost of bad debts)
- (2) financing
- (3) credit insurance.



These are of particular value to:

- smaller firms
- fast growing firms.

Make sure you can discuss the various services offered and remember that non-recourse factoring is more expensive as the factor bears the costs of any irrecoverable debts. An exam question can, for example, test students' knowledge that under non-recourse factoring, the cost of any bad debts previously borne by the company would reduce to zero with the factor in place.



Factoring

Debt collection and administration – the factor takes over the whole of the company's sales ledger, issuing invoices and collecting debts.

Financing provision – in addition to the above, the factor will advance up to 80% of the value of a debt to the company; the remainder (minus finance costs) being paid when the debts are collected. The factor becomes a source of finance. Finance costs are usually 1.5% to 3% above bank base rate and charged on a daily basis.

Credit insurance – the factor agrees to insure the irrecoverable debts of the client. The factor would then determine to whom the company was able to offer credit.

Some companies realise that, although it is necessary to extend trade credit to customers for competitive reasons, they need payment earlier than agreed in order to assist their own cash flow. Factors exist to help such companies.

Factoring is most suitable for:

- small and medium-sized firms which often cannot afford sophisticated credit and sales accounting systems, and
- firms that are expanding rapidly. These often have a substantial and growing investment in inventory and receivables, which can be turned into cash by factoring the debts. Factoring debts can be a more flexible source of financing working capital than an overdraft or bank loan.

Factoring is primarily designed to allow companies to accelerate cash flow, providing finance against outstanding trade receivables. This improves cash flow and liquidity.

Factoring can be arranged on either a 'without recourse' basis or a 'with recourse' basis.

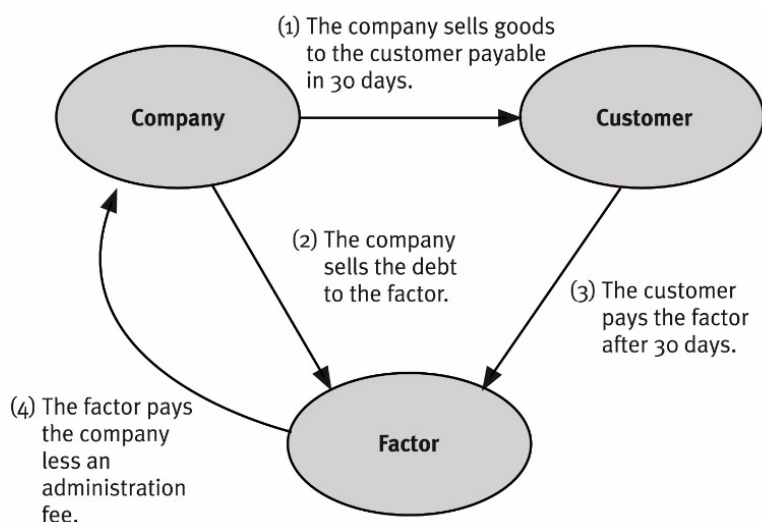
- When factoring is without recourse or 'non-recourse', the factor provides protection for the client against irrecoverable debts. The factor has no 'comeback' or recourse to the client if a customer defaults. When a customer of the client fails to pay a debt, the factor bears the loss and the client receives the money from the debt.

- When the service is with recourse ('recourse factoring'), the client must bear the loss from any irrecoverable debt, and so has to reimburse the factor for any money it has already received for the debt.

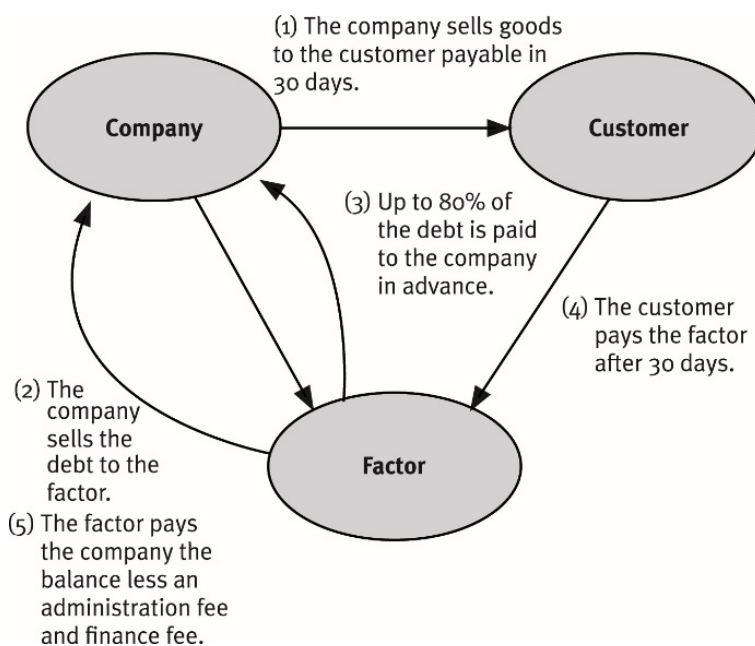
Credit protection is provided only when the service is non-recourse and this is obviously more costly.

Typical factoring arrangements

Administration and debt collection



Including financing



Advantages	Disadvantages
(1) Saving in administration costs. (2) Reduction in the need for management control. (3) Particularly useful for small and fast growing businesses where the credit control department may not be able to keep pace with volume growth.	(1) Likely to be costlier than an efficiently run internal credit control department. (2) Factoring has a bad reputation associated with failing companies; using a factor may suggest your company has money worries. (3) Customers may not wish to deal with a factor. (4) Once you start factoring it is difficult to revert easily to an internal credit control system. (5) The company may give up the opportunity to decide to whom credit may be given (non-recourse factoring).



Benefits and problems with factoring

The benefits of factoring are as follows.

- A business improves its cash flow, because the factor provides finance for up to 80% or more of debts within 24 hours of the invoices being issued. A bank providing an overdraft facility secured against a company's unpaid invoices will normally only lend up to 50% of the invoice value. (Factors will provide 80% or so because they set credit limits and are responsible for collecting the debts.)
- A factor saves the company the administration costs of keeping the sales ledger up to date and the costs of debt collection.
- A business can use the factor's credit control system to assess the creditworthiness of both new and existing customers.
- Non-recourse factoring is a convenient way of obtaining insurance against irrecoverable debts.

Problems with factoring

- Although factors provide valuable services, companies are sometimes wary about using them. A possible problem with factoring is that the intervention of the factor between the factor's client and the debtor company could endanger trading relationships and damage goodwill. Customers might prefer to deal with the business, not a factor.
- When a non-recourse factoring service is used, the client loses control over decisions about granting credit to its customers.

- For this reason, some clients prefer to retain the risk of irrecoverable debts, and opt for a 'with recourse' factoring service. With this type of service, the client and not the factor decides whether extreme action (legal action) should be taken against a non-payer.
- On top of this, when suppliers and customers of the client find out that the client is using a factor to collect debts, it may arouse fears that the company is beset by cash flow problems, raising fears about its viability. If so, its suppliers may impose more stringent payment terms, thus negating the benefits provided by the factor.
- Using a factor can create problems with customers who may resent being chased for payment by a third party, and may question the supplier's financial stability.



Test your understanding 4 – Factoring arrangements

Edden Co is a medium-sized company producing a range of engineering products, which it sells to wholesale distributors. Recently, its sales have begun to rise rapidly due to economic recovery. However, it is concerned about its liquidity position and is looking at ways of improving cash flow.

Its sales are \$16 million per year, and average receivables are \$3.3 million (representing about 75 days of sales).

One way of speeding up collection from receivables is to use a factor.

Required:

Determine the relative costs and benefits of using the factor in each of the following scenarios.

- The factor will operate on a service-only basis, administering and collecting payment from Edden Co's customers. This is expected to generate administrative savings of \$100,000 each year.

The factor has undertaken to pay outstanding debts after 45 days, regardless of whether the customers have actually paid or not. The factor will make a service charge of 1.75% of Edden Co's revenue. Edden Co can borrow at an interest rate of 8% per year.
- It is now considering a factoring arrangement with a different factor where 80% of the book value of invoices is paid immediately, with finance costs charged on the advance at 10% per year.

Suppose that this factor will charge 1% of sales as their fee for managing the sales ledger, that there will be administrative savings of \$100,000 as before, but that outstanding balances will be paid after 75 days (i.e. there is no change in the typical payment pattern by customers this time).

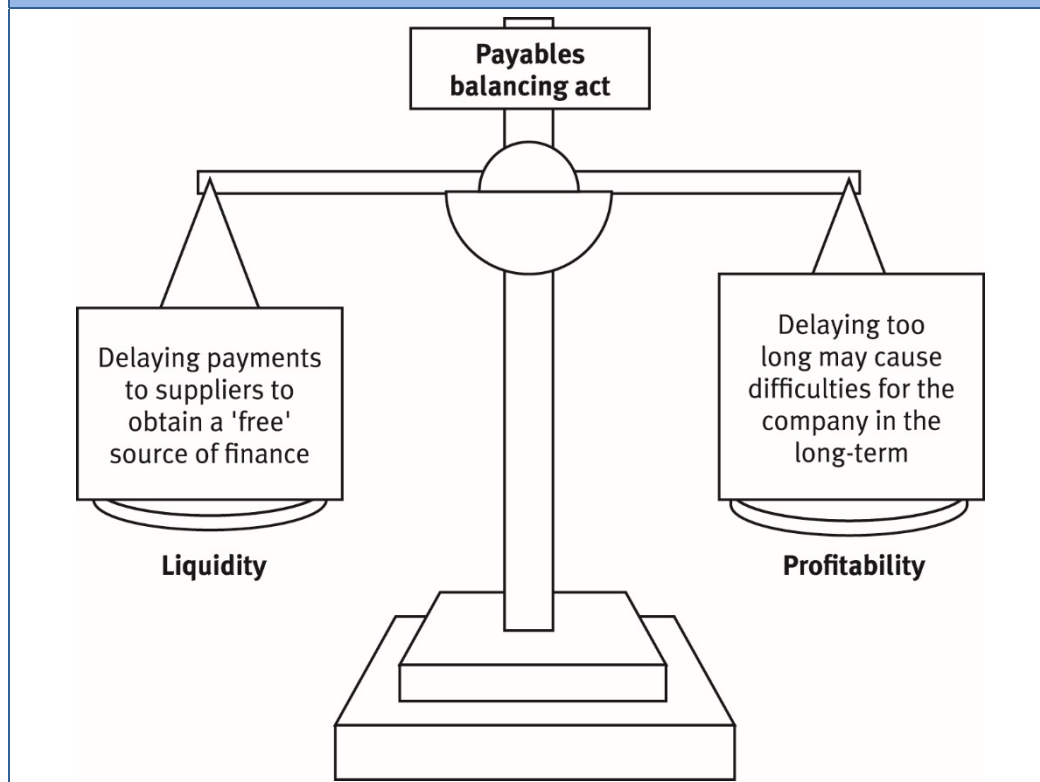
5 Accounts payable – managing trade credit

Trade credit is the simplest and most important source of short-term finance for many companies.

Again, it is a balancing act between liquidity and profitability.



The balancing act



By delaying payment to suppliers, companies face possible problems:

- supplier may refuse to supply in future
- supplier may only supply on a cash basis
- there may be loss of reputation
- supplier may increase price in future or withhold discounts.

Trade credit is normally seen as a 'free' source of finance. Whilst this is normally true, it may be that the supplier offers a discount for early payment. In this case delaying payment is no longer free, since the cost will be the lost discount.



In the examination, you need to be able to calculate the cost of this discount foregone.

It can be done using the same techniques we saw under accounts receivable, although note that the calculations are now from the customer perspective rather than the supplier one. A calculation of whether to accept an early settlement discount offered by a supplier will assess:

- the cost of the discount in terms of the increased working capital funding cost (earlier payment leads to a lower payables balance and a larger working capital balance that now requires extra funding – the annual percentage cost of which will be, for instance, an overdraft charge).
- the benefit of the discount in that less cash is paid to the supplier. The annual benefit as a percentage is calculated using the same formula as used for the annual cost of offering an early settlement discount to a customer:

$$\text{Annual benefit of discount} = \left(1 + \frac{\text{discount}}{\text{amount let to pay}}\right)^{\text{no. of periods}} - 1$$



where no. of periods = $\frac{365/52/12}{\text{no. of days/weeks/months earlier the money is received}}$



Test your understanding 5 – Discount for early payment

One supplier has offered a discount to Box Co of 2% on an invoice for \$7,500, if payment is made within one month, rather than the three months normally taken to pay. If Box's overdraft rate is 10% per year, is it financially worthwhile for them to accept the discount and pay early?



Test your understanding 6 – Equivalent annual cost of discount

Work out the equivalent annual cost of the following credit terms: 1.75% discount for payment within three weeks; alternatively, full payment must be made within eight weeks of the invoice date. Assume there are 50 weeks in a year.

Hint: Consider a \$100 invoice.



Trade payables

Under **trade credit**, a firm is able to obtain goods (or services) from a supplier without immediate payment, the supplier accepting that the firm will pay at a later date.

Trade credit periods vary from industry to industry and each industry will have what is a generally accepted norm, which would be from seven days upwards. The usual terms of credit range from four weeks to the period between the date of purchase and the end of the month following the month of purchase.

However, considerable scope for flexibility exists and longer credit periods are sometimes offered, particularly where the type of business activity requires a long period to convert materials into saleable products, e.g. farming.

A proportion of the firm's suppliers will normally offer early **settlement** discounts, which should be taken up where possible by ensuring prompt payment within the specified terms where settlement discount is allowed. However, if the firm is short of funds, it might wish to make maximum use of the credit period allowed by suppliers regardless of the settlement discounts offered.

It can be a mistake to reduce working capital by holding on to creditors' money for a longer period than is allowed as, in the long-term, this could affect the supplier's willingness to supply goods and raw materials, and cause further embarrassment to the firm.

Favourable credit terms are one of several factors that influence the choice of a supplier. Furthermore, the act of accepting settlement discounts has an opportunity cost, i.e. the cost of finance obtained from another source to replace that not obtained from creditors.

Whilst trade credit may be seen as a source of **free credit**, there will be **costs** associated with extending credit taken beyond the norm – lost discounts, loss of supplier goodwill, more stringent terms for future sales.

In order to compare the cost of different sources of finance, all costs are usually converted to a rate per year basis. The cost of extended trade credit is usually measured by loss of discount, but the calculation of its cost is complicated by such variables as the number of alternative sources of supply, and the general economic conditions.

Certain assumptions have to be made concerning (a) the maximum delay in payment that can be achieved before the supply of goods is withdrawn by the supplier, and (b) the availability of alternative sources of supply.

6 Accounts receivable and payable – managing foreign trade

Overseas accounts receivable and payable bring additional risks that need to be managed:



Export credit risk is the risk of failure or delay in collecting payments due from foreign customers.



Foreign exchange risk is the risk that the value of the currency will change between the date of contract and the date of settlement. For more details, see Chapter 13.



Risks of overseas trades

Whilst all of the basic management principles and techniques discussed so far apply equally to overseas receivables and payables, there are additional risks that will need to be managed, including:

- export credit risk; and
- foreign exchange transaction exposure.

Possible causes of loss from export credit risk, which apply to all export trade of whatever size, include the following.

- Illiquidity or insolvency of the customer. This also occurs in domestic trading. When an export customer cannot pay however, suppliers have extra problems in protecting their positions in a foreign legal and banking system.
- Bankruptcy or failure of a bank in the remittance chain.
- A poorly-specified remittance channel.
- Inconvertibility of the customer's currency, and lack of access to the currency in which payment is due. This can be caused by deliberate exchange controls or by an unplanned lack of foreign exchange in the customer's central bank.
- Political risks. Their causes can be internal (change of regime, civil war) or external (war, blockade) to the country concerned.

Exporters can protect themselves against these risks by the following means.

- Use banks in both countries to act as the collecting channel for the remittance and to control the shipping documents so that they are only released against payment or acceptance of negotiable instruments (bills of exchange or promissory notes).
- Commit the customer's bank through an international letter of credit (ILC).
- Require the ILC to be confirmed (effectively guaranteed) by a first class bank in the exporter's country. This makes the ILC a confirmed ILC (CILC).
- Obtain support from third parties, e.g.:
 - get a guarantee of payment from a local bank
 - get a letter from the local finance ministry or central bank confirming availability of foreign currency.
- Take out export credit cover.
- Use an intermediary such as a confirming, export finance, factoring or forfeiting house to handle the problems on their behalf; or possibly by giving no credit or selling only through agents who accept the credit risk (del credere agents) and are themselves financially strong.

None of these devices will enable the exporter to escape from certain hard facts of life.

- The need to avoid giving credit to uncreditworthy customers. Weak customers cannot obtain an ILC from their own bank, nor would they be cleared for credit by a credit insurer or intermediary.
- The need to negotiate secure payment terms, procedures and mechanisms which customers do not find congenial. An ILC and especially a CILC are costly to customers, and restrict their flexibility: if they are short of cash at the end of the month, they must still pay out if their bank is committed.
- Exporters can only collect under a letter of credit if they present exactly the required documents. They will not be able to do this if they have sent the goods by air and the credit requires shipping documents; or if they need to produce the customer's inspection certificates and the customer's engineer is mysteriously unavailable to inspect or sign.
- The need to insist that payment is in a convertible currency and in a form that the customer's authorities will permit to become effective as a remittance to where the exporters need to have the funds, usually in their own country. Often this means making the sale subject to clearance under exchange controls or import licensing regulations.

Introductory points about foreign exchange:

Many companies trade in foreign currencies, either buying from abroad in a foreign currency or denominating sales to export customers in a foreign currency.

If so they might need to:

- buy foreign currency to pay a supplier, or
- convert foreign currency receipts into domestic currency.

Like domestic trade, foreign trade is arranged on credit terms.

Companies usually know in advance what foreign currency they will need to pay out and what currencies they will be receiving. Foreign exchange risk arises in these situations:

- If a company has to obtain foreign currency at a future date to make a payment, there is a risk that the cost of buying the currency will rise (from what it would cost now) if the exchange rate moves and the currency strengthens in value.
- If a company will want to convert currency earnings into a domestic currency at a future date, there is a risk that the value of the earnings will fall (from its current value) if the exchange rate moves and the currency falls in value.

Exposures to these risks of adverse changes in an exchange rate are known as 'foreign exchange transaction exposures'. These exposures can be 'hedged' (reduced or offset). The methods available to hedge the risks are covered in Chapter 13.

In a question that deals with overseas receivables and payables, remember to consider export credit risk and foreign transaction exposure as well as the normal points on the management of receivables and payables.



Management of foreign accounts receivable

Foreign accounts receivable present some additional challenges to a business that are not present with domestic-based customers.

It is harder for a business to pursue any overdue amounts from a business in another country with a different legal system. One option for a business is to simply trust the foreign customer to pay within the stated credit period without demanding additional security, a method known as 'open account'. This option means the business faces a level of non-payment risk that some businesses may find unacceptable.

Reducing investment in foreign accounts receivable

A company can reduce its investment in foreign accounts receivable by asking for full or part payment in advance of supplying goods. However, this may be resisted by consumers, particularly if competitors do not ask for payment up front.

Another approach is for the seller (exporter) to arrange for a bank to give cash for foreign accounts receivable, sooner than the seller would normally receive payment.

Forfaiting

One method of doing this is forfaiting. Forfaiting involves the purchase of foreign accounts receivable from the seller by a forfaiter. The forfaiter takes on all of the credit risk from the transaction (without recourse) and therefore the forfaiter purchases the receivables from the seller at a discount. The purchased receivables become a form of debt instrument (such as bills of exchange) which can be sold on the money market.

The non-recourse side of the transaction makes this an attractive arrangement for businesses, but as a result, the cost of forfaiting is relatively high.

Forfaiting is usually available for large receivable amounts (over \$250,000) and also is only for major convertible currencies. It is usually only available for medium-term or longer transactions.

Letter of credit

This is a further way of reducing the investment in foreign accounts receivable and can give a business a risk-free method of securing payment for goods or services.

There are a number of steps in arranging a letter of credit:

- Both parties set the terms for the sale of goods or services
- The purchaser (importer) requests their bank to issue a letter of credit in favour of the seller (exporter)
- The letter of credit is issued to the seller's bank, guaranteeing payment to the seller once the conditions specified in the letter have been complied with. Typically, the conditions relate to presenting shipping documentation and dispatching the goods before a certain date
- The goods are dispatched to the customer and the shipping documentation is sent to the purchaser's bank
- The bank then issues a banker's acceptance
- The seller can either hold the banker's acceptance until maturity or sell it on the money market at a discounted value

As can be seen from the above process, letters of credit take up a significant amount of time and therefore are slow to arrange and must be in place before the sale occurs. The use of letters of credit may be considered necessary if there is a high level of non-payment risk.

Customers with a poor or no credit history may not be able to obtain a letter of credit from their own bank. Letters of credit are costly to customers and also restrict their flexibility: if they are short of cash when the payment to the bank is due, the commitment under the letter of credit means that the payment must be made.

Collection under a letter of credit depends on the conditions in the letter being fulfilled. Collection only occurs if the seller presents exactly the documents stated in the conditions. This means that letters of credit provide protection to both the purchaser and the seller. However, the seller will not be able to claim payment if, for example, goods have been sent by air but the letter of credit stated that shipping documents were required.

Countertrading

In a countertrade arrangement, goods or services are exchanged for other goods or services instead of for cash.

The benefits of countertrading include the fact that it facilitates conservation of foreign currency and can help a business enter foreign markets that it may not otherwise be able to.

The main disadvantage of countertrading is that the value of the goods or services received in exchange may be uncertain, especially if the goods being exchanged experience price volatility. Other disadvantages of countertrade include complex negotiations and logistical issues, particularly if a countertrade deal involves more than two parties.

Export credit insurance

Export credit insurance protects a business against the risk of non-payment by a foreign customer. Exporters can protect their foreign accounts receivable against a number of risks that could result in non-payment. Export credit insurance usually insures the seller against commercial risks, such as insolvency of the purchaser or slow payment, and also insures against certain political risks, for example war, riots, and revolution, which could result in non-payment. It can also protect against currency inconvertibility and changes in import or export regulations.

Export credit insurance therefore helps reduce the risk of non-payment, but its disadvantages include the relatively high cost of premiums and the fact that the insurance does not typically cover 100% of the value of the foreign sales.

Export factoring

An export factor provides the same functions in relation to foreign accounts receivable as a factor covering domestic accounts receivable and therefore can help with the cash flow of a business. However, export factoring can be more costly than export credit insurance and it may not be available for all countries, particularly developing countries.

Other considerations

The purchaser may be able to get a local bank to guarantee payment to the exporter, but this may only be suitable in an arrangement where the purchaser has no power over the exporter.

General policies for foreign accounts receivable

None of the methods detailed above would allow the selling company to escape from the basic fact that credit should only be given to customers who are creditworthy.

A seller should insist that any payment is made in a convertible currency and in a form that the customer's authorities will permit to become effective as a remittance to the seller. This may mean, for example, that the sale will be subject to clearance under exchange controls or any import regulations.

Written by a member of the FM examining team



Test your understanding 7

In order to improve operational cash flows, indicate whether a company would need to increase or decrease their receivables balance and payables balance.

	Receivables	Payables
A	Increase	Increase
B	Increase	Decrease
C	Decrease	Decrease
D	Decrease	Increase



Test your understanding 8

Akina Co usually takes 2 months to collect its debts from credit customers. It has just issued an invoice to Takeo Co for \$100 and offers a cash discount of 2% if payment is made within 1 month.

What is the effective annualised cost of the discount if Alfie Co does settle within 1 month?

- A 27.4%
- B 34.4%
- C 20.0%
- D 32.6%



Test your understanding 9

The main aspects of debt factoring include

- (1) Administration of the client's invoicing, sales accounting and debt collection service
- (2) Making payments to the client in advance of collecting the debts
- (3) Credit protection when the service is non-recourse

- A (1) only
- B (1) and (2) only
- C (2) and (3) only
- D All of the above

**Test your understanding 10**

Which of the following statements, concerning invoice discounting, are true?

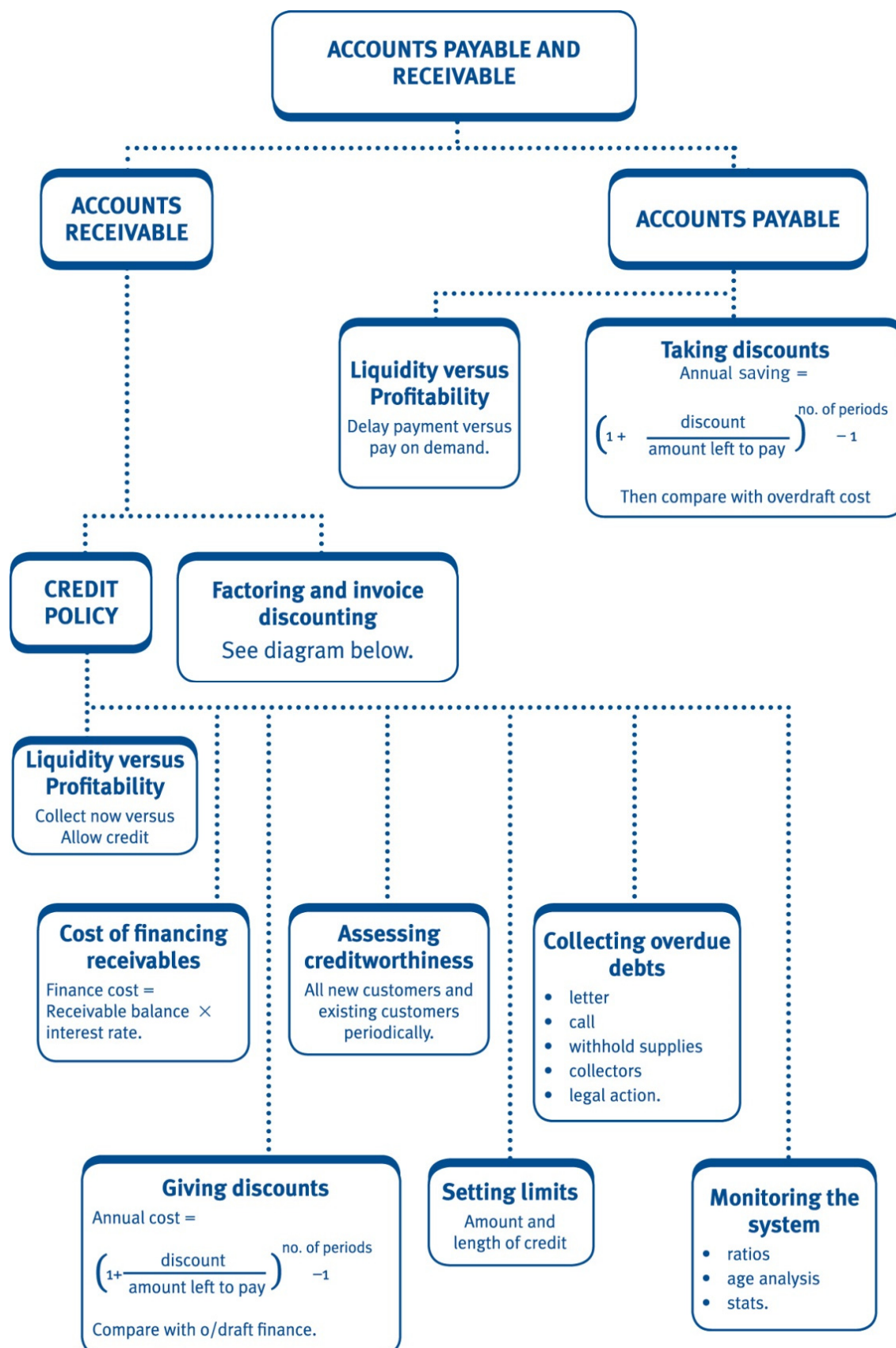
- (1) Invoice discounting is the purchase of a selection of invoices (by the provider of the discounting service) of trade debts at a discount
 - (2) With 'confidential discounting' the client's customers are unaware that the business is using invoice discounting
 - (3) The invoice discounter does not take over the administration of the client's sales ledger
- A (1) only
B (1) and (2) only
C (2) and (3) only
D All of the above

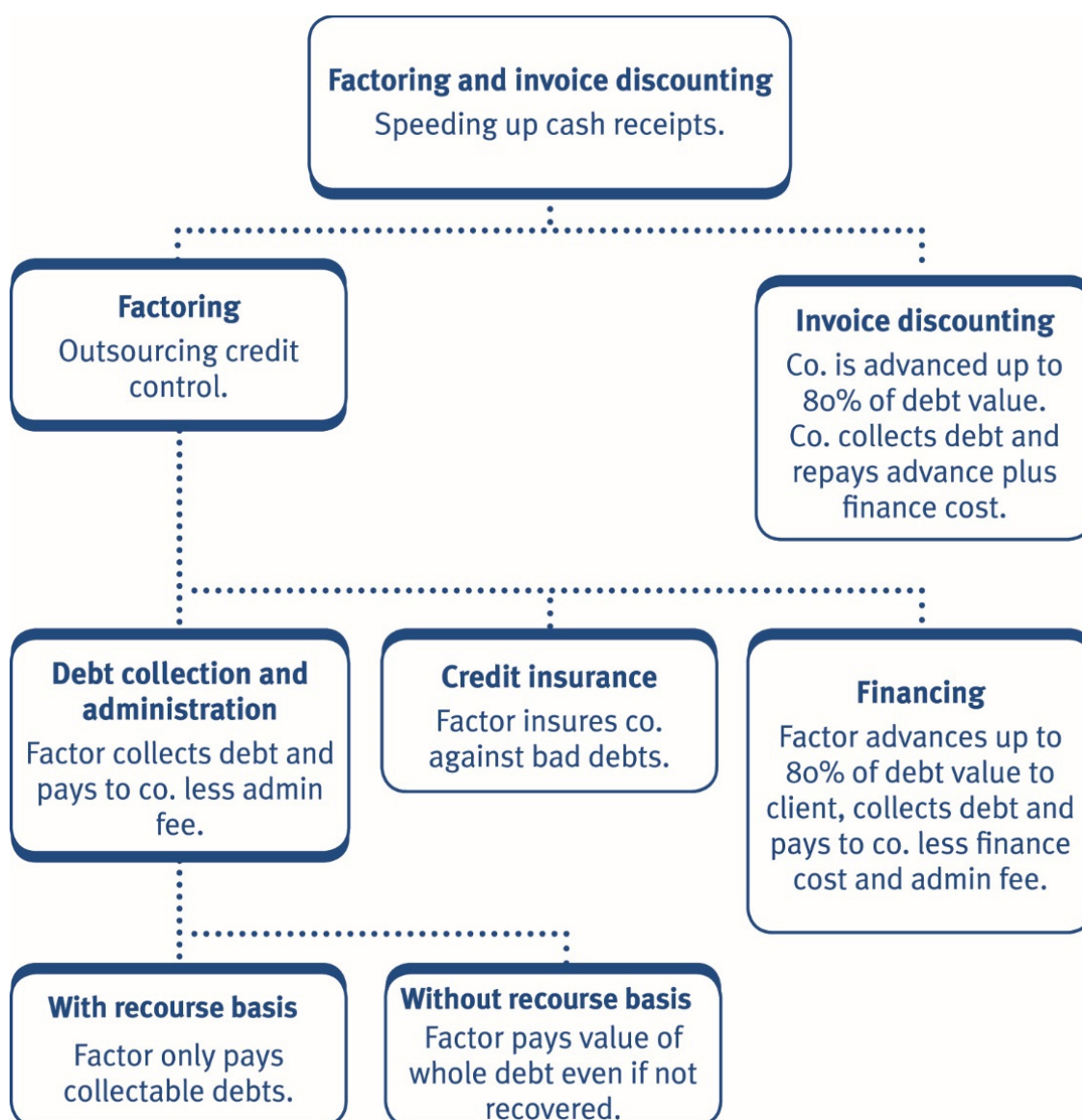
**Test your understanding 11**

Which of the following statements, concerning receivables management, is incorrect?

- A Credit limits should be reviewed periodically
- B Credit analysis depends on the provision of relevant information, for example trade references
- C Delaying payment of invoices is likely to make receivables management more effective
- D Longer-term credit may increase revenue but also increases the risk of bad debts

7 Chapter summary





Test your understanding answers



Test your understanding 1 – Cost of financing receivables

- (a) Accounts receivable collection period = $\$4\text{m} \div \$20\text{m} \times 365 = 73$ days
- (b) Cost of financing receivables = $\$4\text{m} \times 12\% = \$480,000$.



Test your understanding 2 – Early settlement discounts I

Discount as a percentage of amount paid = $2.5/97.5 = 2.56\%$

Saving is 2 months and there are $12/2 = 6$ periods in a year.

Annualised cost of discount % is

$$(1 + 0.0256)^6 - 1 = 0.1638 = 16.38\%$$

The loan rate is 18%.

It would therefore be worthwhile offering the discount.



Test your understanding 3 – Early settlement discounts II

To determine if the early settlement discount should be introduced, a comparison must be made between the cost and the benefit.

Cost

$$\$20\text{m of revenue} \times 40\% \text{ take up} \times 2\% \text{ discount} = \mathbf{\$160,000}$$

Benefit

The benefit is a reduction in the receivables balance, which will result in a higher cash balance, thereby reducing interest payable (say, on an overdraft balance).

First, we need the current financing cost, which can be found by using the current receivables balance of \$4m given in the question:

$$\text{Finance cost} = \text{receivables balance} \times \text{interest rate}$$

$$\text{Current finance cost} = \$4\text{m} \times 12\% = \$480,000 \text{ per year}$$

Next, we calculate the new financing cost using the following steps:

- 1 Use the sales proportions and the receivables balance formula to calculate receivables balances separately for those who take the discount and those who do not.
- 2 Then, add those balances together to determine the new total receivables balance.
- 3 Then, use the finance cost formula to determine the finance cost on the new total receivables.

- 4 This can then be compared to the finance cost before the discount was offered to work out the saving.

Step 1 – Use the sales proportions and the receivables balance formula to calculate receivables balances separately for those who take the discount and those who do not.

Note: as the current receivables collection period is not given to us, we will need to calculate it for use in this step, using the current receivables and credit sales figures:

$$\$4\text{m} / \$20\text{m} \times 365 = 73 \text{ days}$$

Since 40% of customers will now pay within 10 days, and the remaining 60% will continue paying as before (i.e. taking 73 days), the calculations for the new levels of receivables are:

$$\text{Those who take the discount: } \$20\text{m} \times 40\% \times 10/365 = \$219,178$$

$$\text{Those who do not take the discount: } \$20\text{m} \times 60\% \times 73/365 = \$2,400,000$$

Step 2 – Then, add those balances together to determine the new total receivables balance.

$$\text{Total new receivables: } \$219,178 + \$2,400,000 = \$2,619,178$$

Step 3 – Then, use the finance cost formula to determine the finance cost on the new total receivables.

$$\text{New financing cost} = \$2,619,178 \times 12\% \text{ interest} = \$314,301$$

Step 4 – This can then be compared to the finance cost before the discount was offered to work out the saving.

$$\text{The benefit of the lower receivables balance is therefore } \$480,000 - \$314,301 = \$165,699$$

Comparison of cost and benefit

Now we can compare the cost of offering the discount with the benefit.

$$\text{Cost} = \$160,000$$

$$\text{Benefit} = \$165,699$$

$$\text{Net benefit} = \$5,699$$

The conclusion is that this discount should be offered.

Note: You may feel that in step 1, the calculation for the receivables balance of those who take the discount should take into account the fact that 2% of the revenue for these customers will not be received:

$$10/365 \times \$20\text{m} \times 98\% \times 40\% = \$214,795, \text{ instead of } \$219,178$$

It has been confirmed that either version of the calculation is acceptable in the FM exam.



Test your understanding 4 – Factoring arrangements

(a)

Financing cost saving:

Current receivables balance = \$3,300,000

Current financing cost = $\$3,300,000 \times 8\% = \$264,000$

New receivables balance = $\$16\text{m} \times 45 / 365 = \$1,972,603$

New financing cost = $\$1,972,603 \times 8\% = \$157,808$

Financing cost saving = $\$264,000 - \$157,808 = \$106,192$

Factor service charge (1.75% of revenue):

$\$16\text{m} \times 1.75\% = \$280,000$

Admin savings:

Given as \$100,000

Overall position:

Net position = $\$106,192 - \$280,000 + \$100,000 = \$(73,808)$

Using the factor will cost Edden Co \$73,808 per year.

Edden will have to balance this cost against the security offered by improved cash flows and greater liquidity.

(b)

In this situation, collections from customers will not be speeded up, so the receivables balance doesn't change, remaining at \$3.3m. However, the factor will now finance 80% of these receivables at a cost of 10%. The overdraft will no longer be used to finance these, but it will still be used to finance the remaining 20%.

New finance cost:

Financed by factor: $\$3.3\text{m} \times 80\% \times 10\% = \$264,000$

Financed by the overdraft: $\$3.3\text{m} \times 20\% \times 8\% = \$52,800$

Total new finance cost = $\$264,000 + \$52,800 = \$316,800$

Change in finance cost = $\$264,000 - \$316,800 = \$(52,800)$ extra cost.

Factor fee (1% of revenue):

$\$16\text{m} \times 1\% = \$160,000$

Admin savings:

Given as \$100,000

Net position = $\$100,000 - \$52,800 - \$160,000 = \$(112,800)$

Using the factor will cost Edden Co \$112,800 per year.

As before, the firm will have to balance this cost against the security offered by improved cash flows and greater liquidity.



Test your understanding 5 – Discount for early payment

Discount saves 2% of \$7,500 = \$150

Financed by overdraft for extra two months in order to pay early:

Overdraft fee saved if discount not accepted:

$$\$7,500 \times 10\% \times 3/12 = \$187.50$$

Overdraft fee saved if discount accepted:

$$(\$7,500 - \$150) \times 10\% \times 1/12 = \$61.25$$

$$\text{Saving lost} = \$187.50 - \$61.25 = \$126.25$$

$$\text{Net saving} = \$150 - \$126.25 = \$23.75$$

It is worth accepting the discount.

Alternatively:

$$\text{Discount as a percentage of amount paid} = \frac{150}{7,350} = 2.04\%$$

$$\text{Saving is 2 months and there are } \frac{12}{2} = 6 \text{ periods in a year}$$

Annualised cost of not taking the discount

$$\begin{aligned} \text{(and therefore borrowing from the supplier) is } & (1 + 0.0204)^6 - 1 \\ & = 0.1288 = 12.88\% \end{aligned}$$

The overdraft rate is 10%.

It would be cheaper to borrow the money from the bank to pay early and accept the discount.



Test your understanding 6 – Equivalent annual cost of discount

Step 1

Work out the discount available and the amount due if the discount were taken.

Discount available on a \$100 invoice = $1.75\% \times \$100 = \1.75 .

Amount due after discount = $\$100 \times \$1.75 = \$98.25$

Step 2

The effective interest cost of not taking the discount is:

$1.75 \div 98.25 = 0.0178$

for an $8 - 3 =$ five-week period.

Step 3

Calculate the equivalent annual rate. There are ten five-week periods in a year.

The equivalent interest annual rate is $(1 + 0.018)^{10} - 1 = 0.195$ or 19.5%.



Test your understanding 7

Answer D

In order to improve cash flow, a business needs to decrease receivables and increase the credit period taken from suppliers.



Test your understanding 8

Answer A

Discount as a percentage of the amount paid = $2/98 = 2.04\%$

Saving is 1 month and there are $12/1 = 12$ periods in a year.

Annualised cost of discount % is

$(1 + 0.0204)^{12} - 1 = 27.4\%$



Test your understanding 9

Answer D

The main aspects of debt factoring include administration, financing and credit insurance.

**Test your understanding 10****Answer D**

Invoice discounting is related to factoring and many factors will provide an invoice discounting service. In some ways, invoice discounting is similar to the financing parts of the factoring service without the control of credit passing to the factor.

**Test your understanding 11****Answer C**

Delaying payment of invoices is an action relating to payables management, not receivables management.

Working capital management – Cash and funding strategies

Chapter learning objectives

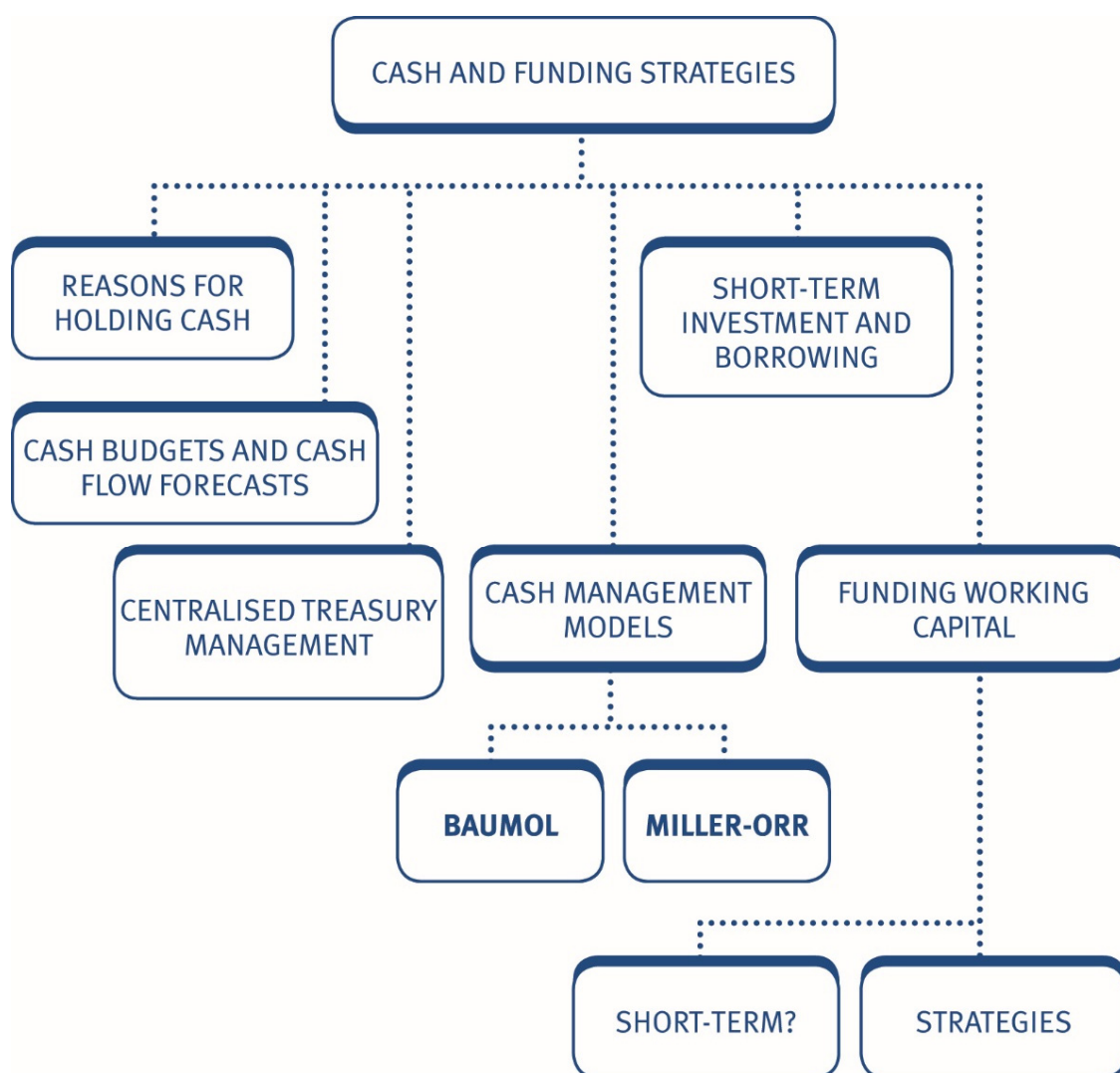
Upon completion of this chapter, you will be able to:

- explain the main reasons for a business to hold cash
- define and explain the use of cash budgets and cash flow forecasts
- prepare a cash flow forecast to determine future cash flows and cash balances
- explain the points addressed by the Baumol cash management model
- calculate the optimum cash management strategy using the Baumol cash management model
- explain the logic of the Miller-Orr cash management model
- calculate the optimum cash management strategy using the Miller-Orr cash management model
- explain the ways in which a firm can invest cash short-term
- explain the ways in which a firm can borrow cash short-term
- calculate the level of working capital investment in current assets from supplied data
- Evaluate and discuss the main strategies available for the funding of working capital
- explain the distinction between permanent and fluctuating current assets

- explain the relative costs and risks of short-term and long-term finance
- explain the logic behind matching short- and long-term assets and funding
- explain the relative costs and benefits of aggressive, conservative and matching funding policies
- explain the impact that factors such as management attitudes to risk, previous funding decisions and organisation size might have on the strategy chosen to fund working capital.
- discuss the advantages and disadvantages of an overdraft as a source of short-term finance
- discuss the advantages and disadvantages of a short-term loan as a source of short-term finance



One of the PER performance objectives (PO10) is to manage and control working capital. You manage cash and working capital effectively, planning for any shortfall or surplus including receivables, payables and inventories. Working through this chapter should help you understand how to demonstrate that objective.



1 Reasons for holding cash

Although cash needs to be invested to earn returns, businesses need to keep a certain amount readily available. The reasons include:

- transactions motive
- precautionary motive
- investment (speculative) motive.

Failure to carry sufficient cash levels can lead to:

- loss of settlement discounts
- loss of supplier goodwill
- poor industrial relations
- potential liquidation.



Reasons for holding cash

Cash is required for a number of reasons:

Transactions motive – cash required to meet day-to-day expenses, e.g. payroll, payment of suppliers, etc.

Precautionary motive – cash held to give a cushion against unplanned expenditure (the cash equivalent of buffer inventory).

Speculative motive – cash kept available to take advantage of market investment opportunities, such as the opportunity to buy a strategically useful piece of land once its current owners have taken it through the planning permission process and made it available for sale.

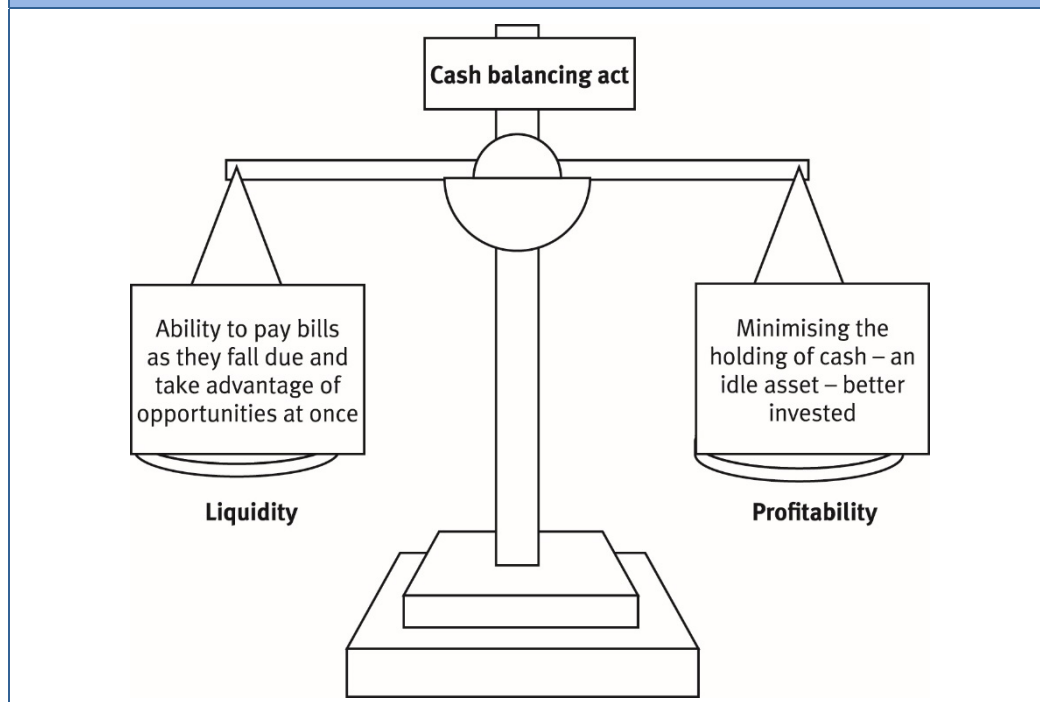
The cost of running out of cash depends on the firm's particular circumstances but may include not being able to pay debts as they fall due which can have serious operational repercussions:

- trade suppliers refuse to offer further credit, charge higher prices or downgrade the priority with which orders are processed
- if wages are not paid on time, industrial action may well result, damaging production in the short-term and relationships and motivation in the medium-term
- the court may be petitioned to wind up the company if it consistently fails to pay bills as they fall due.

Once again, therefore the firm faces a balancing act between liquidity and profitability.



The balancing act





Remember to consider the three motives for holding cash and the liquidity/profitability trade-off in a question that asks for a discussion of cash management.

2 Cash budgets and cash flow forecasts



A **cash forecast** is an estimate of cash receipts and payments for a future period under existing conditions.



A cash budget is a commitment to a plan for cash receipts and payments for a future period after taking any action necessary to bring the forecast into line with the overall business plan.

Cash budgets are used to:

- assess and integrate operating budgets
- plan for cash shortages and surpluses
- compare with actual spending.

Cash forecasts can be prepared based on:

- Receipts and payments forecast. This is a forecast of cash receipts and payments based on predictions of sales and cost of sales and the timings of the cash flows relating to these items.
- Statement of financial position forecast. This is a forecast derived from predictions of future statements of financial positions. Predictions are made of all items except cash, which is then derived as a balancing figure.
- Working capital ratios. Future cash and funding requirements can be determined from the working capital ratios seen in the chapter on working capital management.

You could be asked to do any of these in the exam.



Cash budgets and cash flow forecasts

It is important to distinguish between a budget and a forecast. A **cash forecast** is an estimate of cash receipts and payments for a future period under existing conditions before taking account of possible actions to modify cash flows, raise new capital, or invest surplus funds.

A **cash budget** is a commitment to a plan for cash receipts and payments for a future period after taking any action necessary to bring the preliminary cash forecast into conformity with the overall plan of the business.

Companies are likely to prepare a cash budget as part of the annual master budget, but then to continually prepare revised cash forecasts throughout the year, as a means of monitoring and managing cash flows.



Using planned receipts and payments

Preparing a cash flow forecast from receipts and payments

Every type of cash inflow and receipt, along with their timings, must be forecast. Note that cash receipts and payments differ from sales and cost of sales in the statement of profit or loss because:

- not all cash receipts or payments affect the statement of profit or loss, e.g. the issue of new shares or the purchase of a non-current asset
- some statement of profit or loss items are derived from accounting conventions and are not cash flows, e.g. depreciation or the profit/loss on the sale of a non-current asset
- the timings of cash receipts and payments do not coincide with the statement of profit or loss accounting period, e.g. a sale is recognised in the statement of profit or loss when the invoice is raised, yet the cash payment from the receivable may not be received until the following period or later.

The following approach should be adopted for examination questions.

Step 1 – Prepare a proforma

Month:	1	2	3	4
	\$	\$	\$	\$
Receipts	X	X	X	X
(few lines)	X	X	X	X
Sub total	X	X	X	X
Payments	X	X	X	X
(Many lines)	X	X	X	X
	X	X	X	X
Sub total	X	X	X	X
Net cash flow	X	X	X	X
Opening balance	X	X	X	X
Closing balance	X	X	X	X

Step 2 – Fill in the simple figures

Some payments need only a small amount of work to identify the correct figure and timing and can be entered straight into the proforma. These would usually include:

- wages and salaries
- fixed overhead expenses
- dividend payments
- purchase of non-current assets.

Step 3 – Work out the more complex figures

The information on sales and purchases can be more time consuming to deal with, e.g.:

- timings for both sales and purchases must be found from credit periods
- variable overheads may require information about production levels
- purchase figures may require calculations based on production schedules and inventory balances.

**Additional question – Full cash flow forecast**

In the near future, a company will purchase a manufacturing business for \$315,000, this price to include goodwill (\$150,000), equipment and fittings (\$120,000), and inventory of raw materials and finished goods (\$45,000).

A delivery van will be purchased for \$15,000 as soon as the business purchase is completed. The delivery van will be paid for in the second month of operations.

The following forecasts have been made for the business following purchase:

- (i) Sales (before discounts) of the business's single product, at a mark-up of 60% on production cost will be:

Month	1	2	3	4	5	6
(\$000)	96	96	92	96	100	104

25% of sales will be for cash; the remainder will be on credit, for settlement in the month following that of sale. A discount of 10% will be given to selected credit customers, who represent 25% of gross sales.

- (ii) Production cost will be \$5 per unit. The production cost will be made up of:
- | | |
|----------------|--------|
| Raw materials | \$2.50 |
| Direct labour | \$1.50 |
| Fixed overhead | \$1.00 |
- (iii) Production will be arranged so that closing inventory at the end of any month is sufficient to meet sales requirements in the following month. A value of \$30,000 is placed on the inventory of finished goods, which was acquired on purchase of the business. This valuation is based on the forecast of production cost per unit given in (ii) above.
- (iv) The single raw material will be purchased so that inventory at the end of a month is sufficient to meet half of the following month's production requirements. Raw material inventory acquired on purchase of the business (\$15,000) is valued at the cost per unit that is forecast as given in (ii) above. Raw materials will be purchased on one month's credit.
- (v) Costs of direct labour will be met as they are incurred in production.
- (vi) The fixed production overhead rate of \$1.00 per unit is based upon a forecast of the first year's production of 150,000 units. This rate includes depreciation of equipment and fittings on a straight-line basis over the next five years. Fixed production overhead is paid in the month incurred.
- (vii) Selling and administration overheads are all fixed, and will be \$208,000 in the first year. These overheads include depreciation of the delivery van at 30% per year on a reducing balance basis. All fixed overheads will be incurred on a regular basis, and paid in the month incurred, with the exception of rent and rates. \$25,000 is payable for the year ahead in month one for rent and rates.

Required:

- (a) Prepare a monthly cash flow forecast. You should include the business purchase and the first four months of operations following purchase.
- (b) Calculate the inventory, receivables, and payables balances at the end of the four-month period. Comment briefly upon the liquidity situation.

The answer to this question can be found after the chapter summary diagram at the end of this chapter.



Using a statement of financial position

Preparing a cash flow forecast from a statement of financial position

Used to predict the cash balance at the end of a given period, this method will typically require forecasts of:

- changes to non-current assets (acquisitions and disposals)
- future inventory levels
- future receivables levels
- future payables levels
- changes to share capital and other long-term funding (e.g. bank loans)
- changes to retained profits.



Additional question – Statement of Financial Position

Zed Co has the following statement of financial position at 30 June 20X3:

	\$	\$
Non-current assets:		
Plant and machinery		192,000
Current assets:		
Inventory	16,000	
Receivables	80,000	
Bank	2,000	
		98,000
Total assets		290,000
Equity and liabilities:		
Issued share capital		216,000
Retained profits		34,000
		250,000
Current liabilities		
Trade payables	10,000	
Dividend payable	30,000	
		40,000
Total equity and liabilities		290,000

- (a) The company expects to acquire further plant and machinery costing \$8,000 during the year to 30 June 20X4.
- (b) The levels of inventories and receivables are expected to increase by 5% and 10% respectively by 30 June 20X4, due to business growth.
- (c) Trade payables and dividend liabilities are expected to be the same at 30 June 20X4.
- (d) No share issue is planned, and retained profits for the year to 30 June 20X4 are expected to be \$42,000.
- (e) Plant and machinery is depreciated on a reducing balance basis, at the rate of 20% per year, for all assets held at the statement of financial position date.

Required:

Produce a financial position statement forecast as at 30 June 20X4, and predict what the cash balance or bank overdraft will be at that date.

The answer to this question can be found after the chapter summary diagram at the end of this chapter.



Using working capital ratios

Preparing a cash flow forecast from working capital ratios

Working capital ratios can also be used to forecast future cash requirements.

The first stage is to use the ratios to work out the working capital requirement, as we have already seen in the working capital management chapter.

This technique is used to help forecast overall cash flow. The proforma below is used.

		\$
	Operating profit	X
	Add: Depreciation	X
	Cash flow from operations	X
Add:	Cash from sale of non-current assets	X
	Long-term finance raised	X
Less:	Purchase of non-current assets	(X)
	Redemption of long-term funds	(X)
	Interest paid	(X)
	Tax paid	(X)
	Dividend paid	(X)
	Increase in working capital	(X)
	Net cash flow	X

Note that the proforma given is not strictly in accordance with International Accounting Standards, which govern the production of cash flow statements in a company's financial statements. However, here we are calculating projected future cash flows rather than the historical cash flow that is covered by the standards.



Additional question – Forecasts from working capital ratios

X Co had the following results for last year.

Statement of profit or loss	\$m
Sales	200
Cost of sales (including \$20m depreciation)	120
Operating profit	80
Interest expense	5
Profit before tax	75
Tax	22
Profit for the year	53
Dividend proposed	10
Retained earnings	43

Statement of financial position (extract)

	\$m	\$m
Non-current assets		480
Current assets:		
Inventory	25	
Receivables	33	
Cash	40	
		98
Current liabilities:		
Trade payables	20	
Dividend payable	10	
Tax payable	22	
		52
Long term loan @ 10%		50

X Co expects the following for the forthcoming year.

Sales will increase by	10%
Plant and machinery will be purchased costing	\$12m
Inventory holding period	80 days
Accounts receivable collection period	75 days
Accounts payable payment period	50 days
Depreciation will be	\$15m

Required:

Prepare a cash flow projection for the forthcoming period.

The answer to this question can be found after the chapter summary diagram at the end of this chapter.

Note that cash budgets are one part of the overall master budgets produced by businesses. These were introduced in the ACCA Management Accounting exam.

3 Cash management models

Cash management models are aimed at minimising the total costs associated with movements between:

- a current account (very liquid but not earning interest) and
- short-term investments (less liquid but earning interest).

The models are devised to answer the questions:

- at what point should funds be moved?
- how much should be moved in one go?

The Baumol cash management model

Baumol noted that cash balances are very similar to inventory levels, and developed a model based on the economic order quantity (EOQ).

Assumptions:

- cash use is steady and predictable
- cash inflows are known and regular
- day-to-day cash needs are funded from current account
- buffer cash is held in short-term investments.

The formula calculates the amount of funds to inject into the current account or to transfer into short-term investments at one time:

$$Q = \sqrt{\frac{2C_o D}{C_H}}$$

where:

C_o = transaction costs (brokerage, commission, etc.)

D = demand for cash over the period

C_H = cost of holding cash.



The model suggests that when interest rates are high, the cash balance held in non-interest-bearing current accounts should be low. However, its weakness is the unrealistic nature of the assumptions on which it is based.



Explanation of terminology

C_o = transaction costs are the costs of making a trade in securities or moving funds in and out of interest-bearing deposit accounts.

D = the demand for cash over the period relates to cash needs for day-to-day transactions.

C_H = the cost of holding cash will be the opportunity cost relating to either the return it could have earned in marketable securities or deposit accounts or the cost of borrowing in order to acquire cash.



Test your understanding 1 – Baumol

A company generates \$10,000 per month excess cash, which it intends to invest in short-term securities. The interest rate it can expect to earn on its investment is 5% per year. The transaction costs associated with each separate investment of funds is constant at \$50.

Required:

- What is the optimum amount of cash to be invested in each transaction?
- How many transactions will arise each year?
- What is the cost of making those transactions per year?
- What is the opportunity cost of holding cash per year?



Test your understanding 2 – Baumol

A company faces a constant demand for cash totalling \$200,000 per year. It replenishes its current account (which pays no interest) by selling constant amounts of treasury bills, which are held as an investment earning 6% per year. The cost per sale of treasury bills is a fixed \$15 per sale.

What is the optimum amount of treasury bills to be sold each time an injection of cash is needed in the current account, how many transfers will be needed and what will the overall transaction cost be?

The Miller-Orr cash management model

The **Miller-Orr** model controls irregular movements of cash by the setting of upper and lower control limits on cash balances.

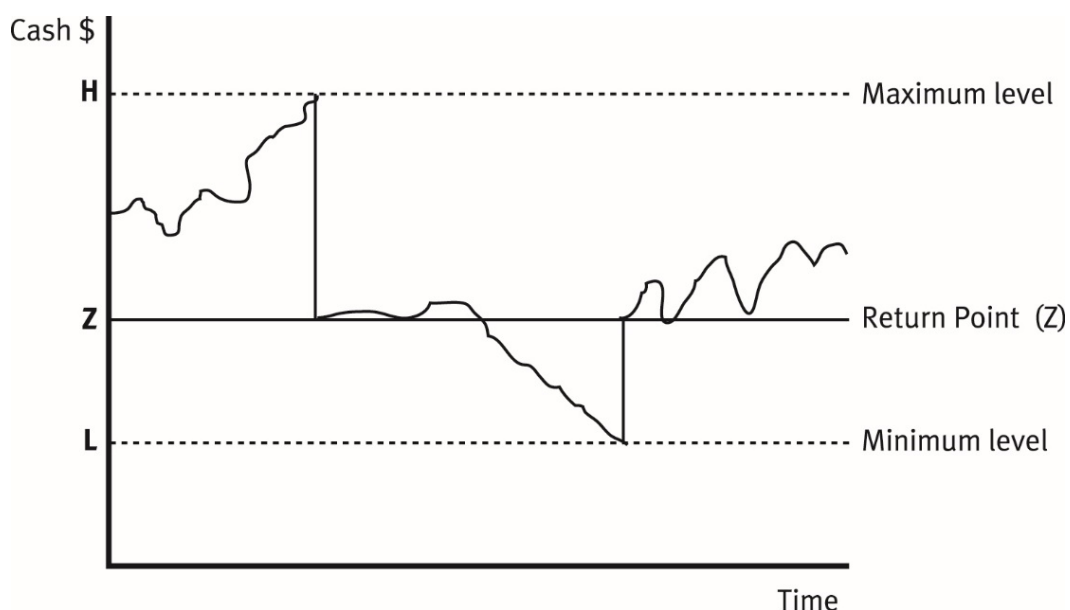


The **Miller-Orr** model is used for setting the target cash balance.

It has the advantage (over the Baumol model) of incorporating uncertainty in the cash inflows and outflows and so may be more appropriate than the Baumol model when cash flows are erratic.

The diagram below shows how the model works over time.

- The model sets higher and lower control limits, H and L, respectively, and a target cash balance, Z.
- When the cash balance reaches H, then $(H-Z)$ dollars are transferred from cash to marketable securities, i.e. the firm buys $(H-Z)$ dollars of securities.
- Similarly, when the cash balance hits L, then $(Z-L)$ dollars are transferred from marketable securities to cash.



The lower limit, L is set by management depending upon how much risk of a cash shortfall the firm is willing to accept, and this, in turn, depends both on access to borrowings and on the consequences of a cash shortfall.

The formulae (given in the examination) for the **Miller-Orr** model are:

Return point = Lower limit + $(1/3 \times \text{spread})$

Spread = $3 \times [(3/4 \times \text{Transaction cost} \times \text{Variance of cash flows}) \div \text{Interest rate}]^{1/3}$

Note: Variance and interest rates should be expressed in daily terms. If the question provides you with the standard deviation of daily cash flows, you will need to square this number to obtain the variance.



Miller Orr

The **Miller-Orr** model controls irregular movements of cash by the use of upper and lower limits on cash balances.

The lower limit has to be specified by the firm and the upper limit is calculated by the model. The cash balance of the firm is allowed to vary freely between the two limits but if the cash balance on any day goes outside these limits, action must be taken.

If the cash balance reaches the lower limit it must be replenished in some way, e.g. by the sale of marketable securities or withdrawal from a deposit account. The size of this withdrawal is the amount required to take the balance back to the return point. It is the distance between the return point (usually set in **Miller-Orr** as the lower limit plus one third of the distance up to the upper limit) and the lower limit.

If the cash balance reaches the upper limit, an amount must be invested in marketable securities or placed in a deposit account, sufficient to reduce the balance back to the return point. Again, this is calculated by the model as the distance between the upper limit and the return point.

The minimum cost upper limit is calculated by reference to brokerage costs, holding costs and the variance of cash flows. The model has some fairly restrictive assumptions, e.g. normally distributed cash flows but, in tests, **Miller and Orr** found it to be fairly robust and claim significant potential cost savings for companies.



Test your understanding 3 – Miller Orr

The minimum cash balance of \$20,000 is required at Miller-Orr Co, and transferring money to or from the bank costs \$50 per transaction. Inspection of daily cash flows over the past year suggests that the standard deviation is \$3,000 per day, and hence the variance (standard deviation squared) is \$9 million. The interest rate is 0.03% per day.

Calculate:

- (i) the spread between the upper and lower limits
- (ii) the upper limit
- (iii) the return point.



Test your understanding 4 – Miller Orr

A company sets its minimum cash balance at \$5,000 and has estimated the following:

- transaction cost = \$15 per sale or purchase of gilts
 - standard deviation of cash flows = \$1,200 per day (i.e. variance = \$1.44 million per day)
 - interest rate = 7.3% per year = 0.02% per day.
- (i) What is the spread between the upper and lower limits?
 - (ii) What is the upper limit?
 - (iii) What is the return point?

4 Short-term investment and borrowing solutions

The cash management models discussed above assumed that funds could be readily obtained when required either by liquidating short-term investments or by taking out short-term borrowing.

A company must choose from a range of options to select the most appropriate source of investment/funding.

Short-term cash investments

Short-term cash investments are used for temporary cash surpluses. To select an investment, a company has to weigh up three potentially conflicting objectives and the factors surrounding them.

The objectives can be categorised as follows:

Liquidity: the cash must be available for use when needed.

Safety: no risk of loss must be taken.

Profitability: subject to the above, the aim is to earn the highest possible after-tax returns.



Ensure you can discuss the three key objectives affecting the choice of short-term investment.



Short-term investment

Short-term investment opportunities present themselves when cash surpluses arise. The company's attitude to risk and working capital management will determine the planned cash holdings. This section deals with the practical aspects of the management of a portfolio of short-term investments.

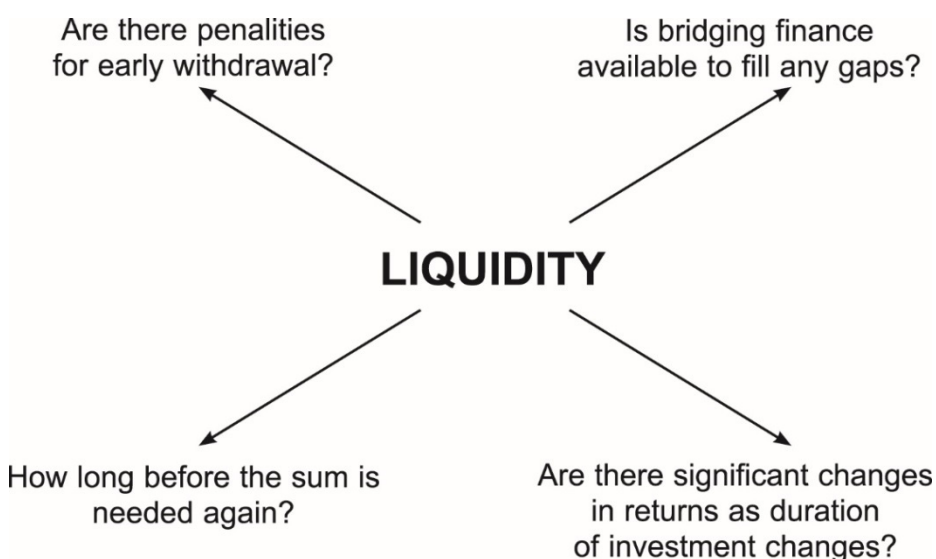
Surplus cash comprises liquid balances held by a business, which are neither needed to finance current business operations nor held permanently for short-term investment. The availability of surplus cash is temporary, awaiting employment either in existing operations or in new investment opportunities (whether already identified or not). The 'temporary' period can be of any duration, from one day to the indefinite future date at which the new investment opportunity may be identified and seized.

Where balances are held temporarily for conversion to other, more important, business uses, absolute priority must be given to the avoidance of risk over maximising returns. The usual principle of finding the optimal mix between risk and return does not apply here, because the investment is secondary and incidental to the ultimate business use of the asset, not an end in itself.

Objectives in the investment of surplus cash

Each of the three objectives raises problems.

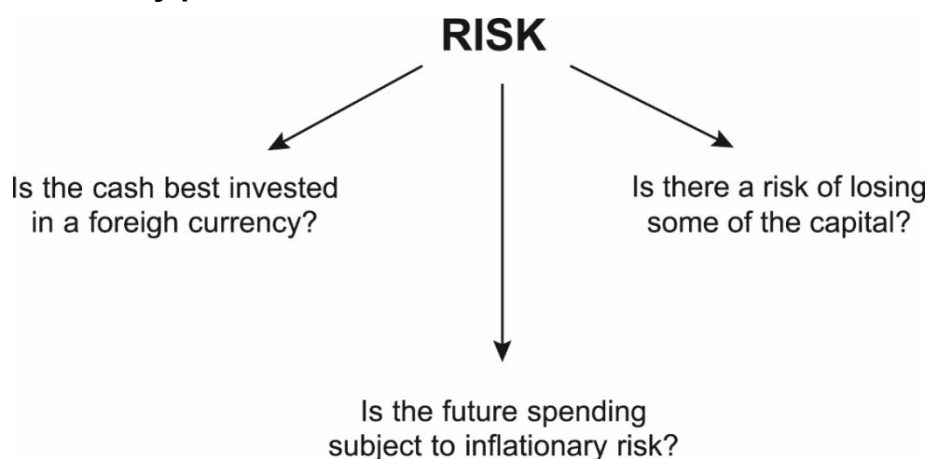
The liquidity problem



At first sight, this problem is simple enough. If a company knows that it will need the funds in three days (or weeks or months), it simply invests them for just that period at the best rate available with safety. The solution is to match the maturity of the investment with the period for which the funds are surplus. However, there are a number of factors to consider:

- The exact duration of the surplus period is not always known. It will be known if the cash is needed to meet a loan instalment, a large tax payment or a dividend. It will not be known if the need is unidentified, or depends on the build-up of inventory, the progress of construction work, or the hammering out of an acquisition deal.
- Expected future trends in interest rates affect the maturity of investments.
- Bridging finance may be available to bridge the gap between the time when the cash is needed and the subsequent date on which the investment matures.
- An investment may not need to be held to maturity, if either an earlier withdrawal is permitted by the terms of the instrument without excessive penalty, or there is a secondary market and its disposal in that market causes no excessive loss.
- A good example of such an investment is a certificate of deposit (CD), where the investor 'lends' the bank a stated amount for a stated period, usually between one and six months. As evidence of the debt and its promise to pay interest, the bank gives the investor a CD. There is an active market for CDs issued by the commercial banks and turning a CD into cash is easy and cheap.

The safety problem



Safety means there is no risk of capital loss. Superficially, this again looks simple. The concept certainly includes the absence of credit risk. For example, the firm should not deposit with a bank that might conceivably fail within the maturity period and thus not repay the amount deposited.

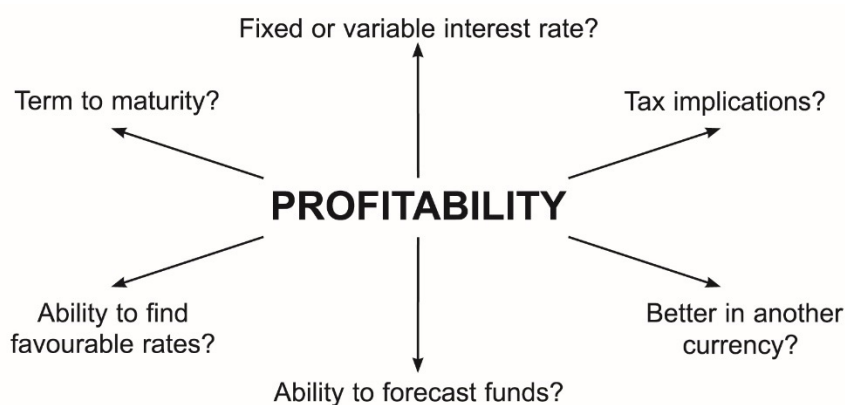
However, safety is not necessarily to be defined as certainty of getting the original investment repaid at 100% of its original home currency value. If the purpose for which the surplus cash is held is not itself fixed in the local currency, then other criteria of safety may apply.

Examples

If the cash is being held to meet a future commitment, the ultimate amount of the commitment may be subject to inflationary rises (e.g. payment to building contractors for a new factory). In this case, a safer investment instrument may be an index-linked gilt-edged bond with a maturity date close to the expected date of the payment.

The profitability problem

If the cash is being held to meet a future payment in a foreign currency, the only riskless investment would be one denominated in that currency.



Ensure you can discuss the three key factors affecting the choice of short-term investment.

The profitability objective looks deceptively simple at first: go for the highest rate of return subject to the overriding criteria of safety and liquidity. However, here there are even more complications.

Factor being considered	Rule of thumb course of action
Fixed or variable rates	Invest long (fixed interest investments with late maturity dates – subject to the liquidity rule) if there are good reasons to expect interest rates to fall.
Term to maturity	Invest short (fixed interest investments with early maturity dates) or at variable rates if there are good grounds for expecting rates to go up.

Tax effects	Aim to optimise net cash flows after tax. Tax payments are a cash outflow. There are many tax efficient investments for surplus cash, e.g. use of tax havens or government securities which may be exempt from capital gains tax (CGT).
Use of other currencies	Investing in currencies other than the company's operating currency in which it has the bulk of its assets and in which it reports to its owners is clearly incompatible with the overriding requirement of safety, except in two possible sets of circumstances: • the investment is earmarked for a payment due in another currency (as seen earlier) or • both principal and interest are sold forward or otherwise hedged against the operating currency (hedging is covered in greater detail later in Chapters 13 and 14).
Difficulty in forecasting available funds	Segregate receipts and payments into the following categories: • The steadier and more forecastable flows, such as cash takings in retail trades. There may be predictable peaks, say at the end of the week and in the pre-Christmas period. • The less predictable but not individually large items. • Controllable items such as payments to normal suppliers. • Items such as collections from major customers, which are individually so large that it pays to spend some management time on them. This segregation can even be taken to the point where separate bank accounts are used for the different categories.
Difficulty in finding the most favourable rates	Know the available instruments and their current relative benefits. Shop around for the 'best buy' among investees who offer the most appropriate instrument.

Short-term borrowing

Short-term cash requirements can also be funded by borrowing from the bank. There are two main sources of bank lending:

- bank overdraft
- bank loans.

Bank overdrafts are mainly provided by the clearing banks and are an important source of company finance.

Advantages

- Flexibility
- Only pay for what is used, so cheaper

Disadvantages

- Repayable on demand
- May require security
- Variable finance costs

Bank loans are a contractual agreement for a specific sum, loaned for a fixed period, at an agreed rate of interest. They are less flexible and more expensive than overdrafts but provide greater security.



Short-term borrowing

Finance costs on bank loans and overdrafts are normally variable, i.e. they alter in line with base rates. Fixed rate loans are available, but are less popular with firms (and providers of finance).

Bank overdrafts

A common source of short-term financing for many businesses is a bank overdraft. These are mainly provided by the clearing banks and represent permission by the bank to make payments even though the firm has insufficient funds deposited in the account to meet the payment value.

An overdraft limit will be placed on this facility, but provided the limit is not exceeded, the firm is free to make as much or as little use of the overdraft as it desires. The bank charges interest on amounts outstanding at any one time, and the bank may also require repayment of an overdraft at any time.

The advantages of overdrafts are the following.

- Flexibility – they can be used as required.
- Cheapness – interest is only payable on the finance actually used, usually at 2–5% above base rate (and all interest is a tax deductible expense).

The disadvantages of overdrafts are as follows.

- Overdrafts are legally repayable on demand. Normally, however, the bank will give customers assurances that they can rely on the facility for a certain time period, say six months.
- Security is usually required by way of fixed or floating charges on assets or sometimes, in private companies and partnerships, by personal guarantees from owners.
- Interest costs vary with bank base rates. This makes it harder to forecast and exposes the business to future increases in interest rates.

Overall, bank overdrafts are one of the most important sources of short-term finance for industry.

Bank loans

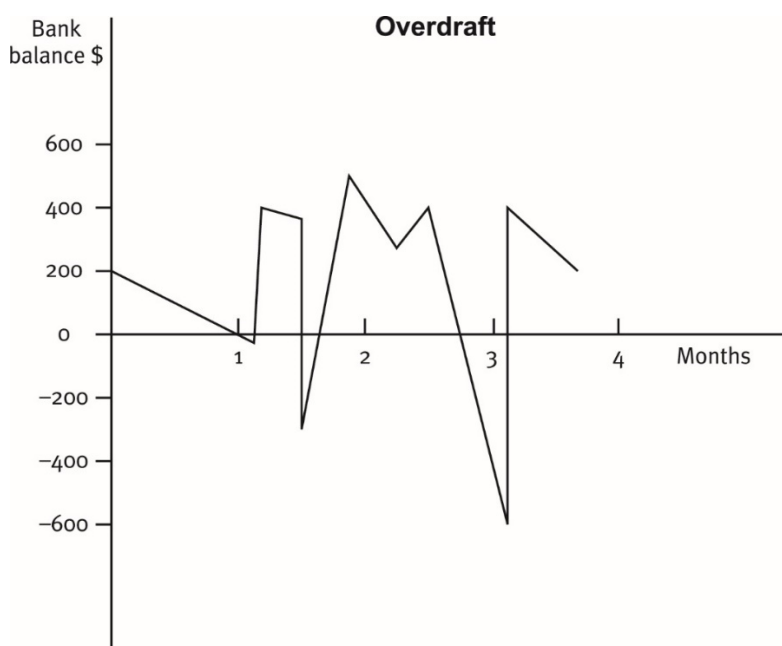
A bank loan represents a formal agreement between the bank and the borrower, that the bank will lend a specific sum for a specific period (one to seven years being the most common). Interest must be paid on the whole of this sum for the duration of the loan.

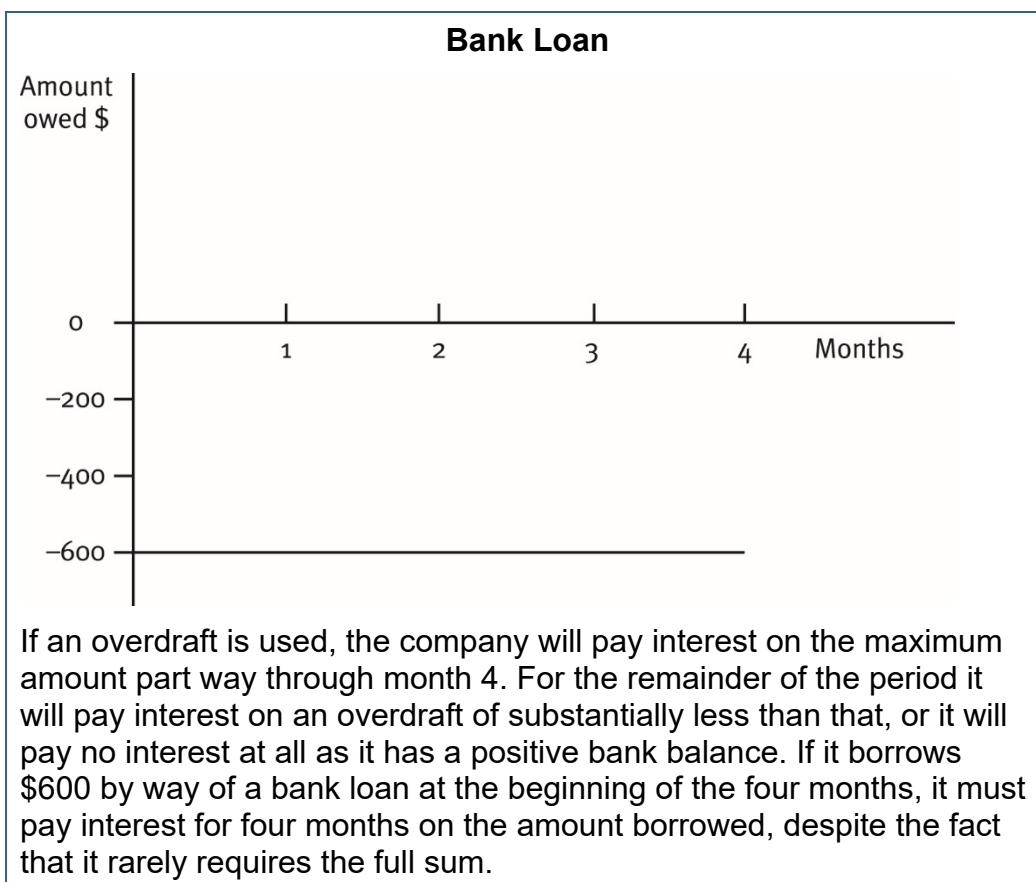
This source is, therefore, liable to be more expensive than the overdraft and is less flexible but, on the other hand, there is no danger that the source will be withdrawn before the expiry of the loan period. Interest rates and requirements for security will be similar to overdraft lending.

Comparison of bank loans and overdrafts

Consider a company that requires a maximum of \$600 over the next four months. However, it is only halfway through month four that it actually requires the full amount.

The difference can be shown as follows:





Calculation of repayment on a loan

The annual repayments on a loan can be calculated using the annuity factors seen in Chapter 3.

For example, a \$100,000 loan taken out by a business at a rate of 7%, repayable over 5 years will have annual repayments of:

$$\text{Annual payment} = \frac{\$100,000}{4.100} = \$24,390.25$$

where 4.100 is the 5 year, 7% annuity factor.

Each payment can then be split between the repayment of capital and interest on the outstanding balance.

Year	Balance b/f	Interest @ 7%	Annual payment	Balance c/f
	\$	\$	\$	\$
1	100,000	7,000	24,390	82,610
2	82,610	5,783	24,390	64,003
3	64,003	4,480	24,390	44,093
4	44,093	3,087	24,390	22,790
5	22,790	1,600*	24,390	0

* Rounding difference

5 Strategies for funding working capital

In the same way as for long-term investments, a firm must make a decision about what source of finance is best used for the funding of working capital requirements.

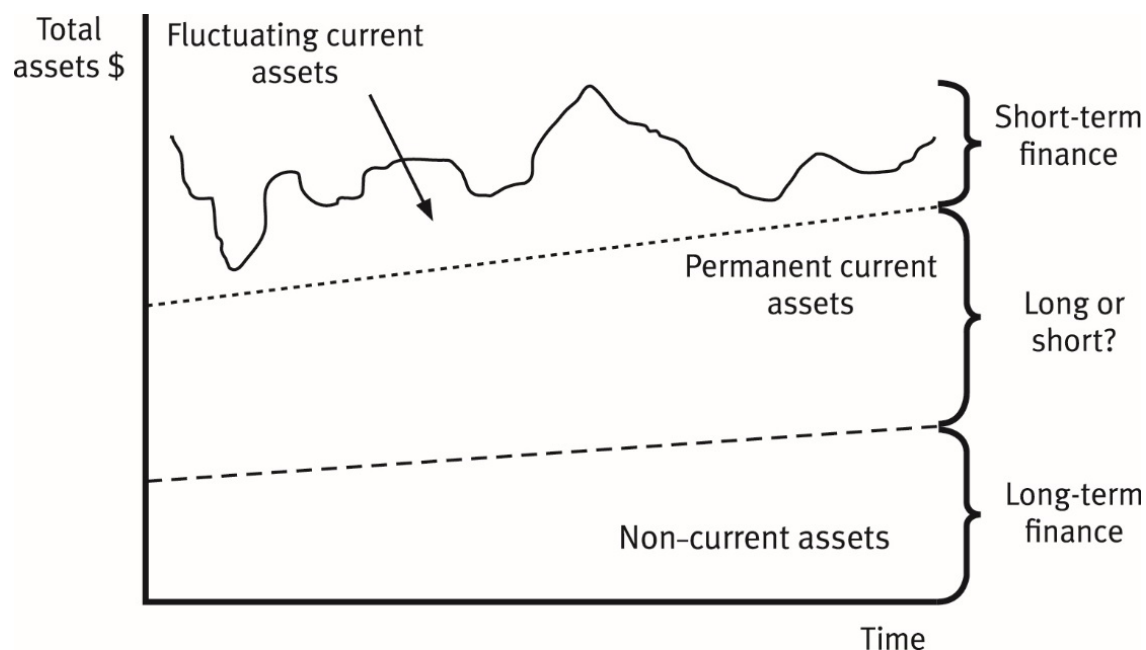
The decision about whether to choose short- or long-term options depends upon a number of factors:

- the extent to which current assets are permanent or fluctuating
- the costs and risks of short-term finance
- the attitude of management to risk

Permanent or fluctuating current assets

In most businesses a proportion of the current assets are fixed over time, i.e. 'permanent'. For example:

- buffer inventory,
- receivables during the credit period,
- minimum cash balances.



The choice of how to finance the permanent current assets is a matter for managerial judgement, but includes an analysis of the cost and risks of short-term finance.



Costs and risks of short-term finance

The cost of short-term finance

Short-term finance is usually **cheaper** than long-term finance. This is largely due to the risks taken by creditors. For example, if a bank were considering two loan applications, one for one year and the other for 20 years, all other things being equal it would demand a higher interest rate on the 20-year loan. This is because it feels more exposed to risk on long-term loans, as more could go wrong with the borrower over a period of twenty years than a period of one year (although it should be noted that occasionally this situation is reversed, with rates of return being higher on short-term finance).

Short-term finance also tends to be more flexible. For example, if funds are raised on overdraft and used to finance a fluctuating investment in current assets, they can be paid off when not required and interest saved. On the other hand, if funds were borrowed for the long term, early repayment may not be possible, or, if allowed, early repayment penalties may be experienced. The flexibility of short-term finance may, therefore, reduce its overall cost.

Short-term finance includes items such as trade payables, which are normally regarded as low cost funds, whereas long-term finance will include debt and equity. Equity finance is particularly expensive, its required returns being high, and dividends paid are non-tax deductible for the company.

The risks of short-term finance

Short-term financing has already been established as generally 'the cheaper option'. However, the price paid for reduced cost is increased risk for the borrower.

There may be:

Renewal problems – short-term finance may need to be continually renegotiated as various facilities expire and renewal may not always be guaranteed.

Unstable interest rates – if the company constantly has to renew its funding arrangements, it will be at the mercy of fluctuations in short-term interest rates.



Working capital investment levels v working capital financing

Working capital investment policy is concerned with the level of investment in current assets, with one company being compared with another. Working capital financing policy is concerned with the relative proportions of short-term and long-term finance used by a company.

NB: Make sure you know the difference between the two topics as, for example, a question on working capital financing policies will give no marks to answers that discuss the trade-off between profitability and liquidity as it relates to working capital investment levels.

While working capital investment policy is therefore assessed on an intercompany comparative basis, assessment of working capital financing policy involves analysis of financial information for one company alone.

Both working capital investment policy and working capital financing policy use the terms conservative, moderate and aggressive. In investment policy, the terms are used to indicate the comparative level of investment in current assets on an inter-company basis. One company has a more aggressive approach compared to another company if it has a lower level of investment in current assets, and vice versa for a conservative approach to working capital investment policy. In working capital financing policy, the terms are used to indicate the way in which fluctuating current assets and permanent current assets are matched to short-term and long-term finance sources.

The attitude of management to risk – Aggressive, conservative and matching funding policies

There is no ideal funding package, but three approaches may be identified.

- Aggressive – finance most current assets, including 'permanent' ones, with short-term finance. Risky but profitable (as the finance is cheaper).
- Conservative – long-term finance is used for most current assets, including a proportion of fluctuating current assets. Stable but expensive.
- Matching – the duration of the finance is matched to the duration of the investment.



The attitude of management to risk

Illustration – Aggressive, conservative and matching funding policies

The following three companies have current asset financing structures that may be considered as aggressive, matching and conservative:

Statement of financial position

	Aggressive	Matching	Conservative
	\$000	\$000	\$000
Non-current assets	50	50	50
Current assets	50	50	50
	100	100	100
Equity (50,000 \$1 shares)	50	50	50
Long-term debt (average cost 10% per year)	–	25	40
Current liabilities (average cost 3% per year)	50	25	10
	100	100	100
Current ratio	1:1	2:1	5:1

Statement of profit or loss

	\$	\$	\$
EBIT	15,000	15,000	15,000
Less: Interest	1,500	3,250	4,300
Earnings before tax	13,500	11,750	10,700
Corporation tax @ (say) 40%	5,400	4,700	4,280
Earnings available to equity	8,100	7,050	6,420
Earnings per share (EPS)	16.2c	14.1c	12.84c

The aggressive company is so termed as it is prepared to take the risk of financing more of its business investment with short-term credit (its current liabilities). The defensive company, at the other 'extreme', takes on board a high proportion of longer-term debt with, consequently, less short-term credit risk.

It can be seen that the aggressive company returns a higher profit but at the cost of greater risk. It is interesting to note that this higher risk is revealed in its relatively poor current ratio.



Essentially the final choice of working capital funding is down to the management of the individual companies, bearing in mind:

- the size of the organisation
- the willingness of creditors to lend
- the risks of their commercial sector
- previous funding decisions (past experiences will improve the managers' ability to anticipate the effect of working capital fluctuations and proactively make decision to curtail the negative impact of such fluctuations)

Smaller companies may, by necessity, finance almost all their needs from short-term finance, since long-term debt and equity may be difficult to raise without marketable shares, few assets to use as security and a good track record.

If the commercial sector in which the company operates has volatile earnings, management may consider it best to take a conservative approach to funding to avoid particular problems in problem years.

The attitude to risk of the decision maker will ultimately determine the decision taken (see Chapter 17 for more analysis of attitude to risk), and this in turn may be influenced by the attitudes and decisions of those who have gone before.



Test your understanding 5

Which of the following should not be included in a cash flow forecast?

- A Funds from the issue of share capital
- B Repayment of a bank loan
- C Receipts of dividends from outside the business
- D Revaluation of a non-current asset



Test your understanding 6

A company's projected revenue for 20X4 is \$350,000. It is forecast that 12% of sales will occur in January and remaining sales will be equally spread among the other eleven months. All sales are on credit.

Receivables accounts are settled 50% in the month of sale, 45% in the following month, and 5% are written off as bad debts after two months.

Which of the following amounts represents the budgeted cash collections for March?

- A \$24,500
- B \$26,600
- C \$28,000
- D \$32,900



Test your understanding 7

Which of the following actions would be appropriate if the cash budget identified a short-term cash deficit?

- A Issue shares
- B Pay suppliers early
- C Arrange an overdraft
- D Invest in a short-term deposit account



Test your understanding 8

In the Miller-Orr cash management model

Return point = Lower limit + × spread

- A One half
- B One third
- C One quarter
- D One fifth



Test your understanding 9

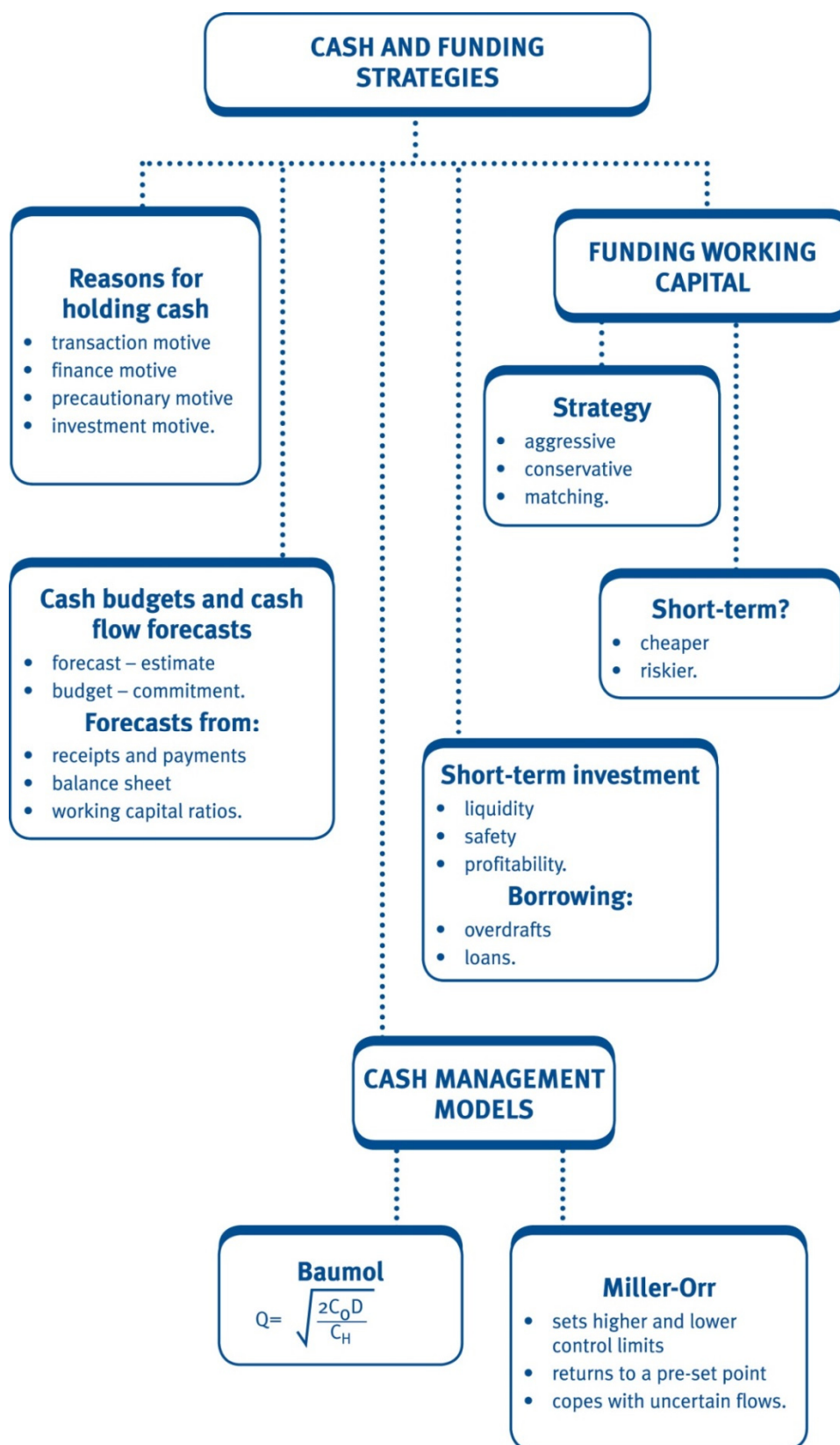
Which of the following statements is true?

Statement 1: An aggressive working capital investment policy aims to finance most of its current assets with long-term finance.

Statement 2: A conservative working capital investment policy aims to finance most of its current assets with short-term finance.

	<i>Statement 1</i>	<i>Statement 2</i>
A	True	True
B	True	False
C	False	True
D	False	False

6 Chapter summary




Answer to additional question – Full cash flow forecast
(a) Monthly cash budget

	Month 1	Month 2	Month 3	Month 4
	\$	\$	\$	\$
Cash inflows:				
Cash sales	24,000	24,000	23,000	24,000
Credit sales		72,000	72,000	69,000
Less: Discounts		(2,400)	(2,400)	(2,300)
Total inflow	24,000	93,600	92,600	90,700
Cash outflows:				
Purchases (W1)	–	44,375	29,375	30,625
Labour (W1)	27,000	17,250	18,000	18,750
Production overhead (W2)	10,500	10,500	10,500	10,500
Selling and administration overhead (W3)	39,875	14,875	14,875	14,875
Purchase of business	315,000	–	–	–
Purchase of van	–	15,000	–	–
Total outflow	392,375	102,000	72,750	74,750
Net cash flow for month	(368,375)	(8,400)	19,850	15,950
Opening balance	0	(368,375)	(376,775)	(356,925)
Closing balance	(368,375)	(376,775)	(356,925)	(340,975)

Workings

(W1)

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Sales (\$)	96,000	96,000	92,000	96,000	100,000	104,000
Sales units	12,000	12,000	11,500	12,000	12,500	13,000
+ Closing inventory	12,000	11,500	12,000	12,500	13,000	
– Opening inventory	6,000	12,000	11,500	12,000	12,500	13,000
Production (units)	18,000	11,500	12,000	12,500	13,000	
Raw material usage (Production × \$2.50)	45,000	28,750	30,000	31,250	32,500	
+ Closing inventory	14,375	15,000	15,625	16,250		
– Opening inventory	15,000	14,375	15,000	15,625		
Purchases (one month delay)	44,375	29,375	30,625	31,875		
Labour cost (production × \$1.50)	27,000	17,250	18,000	18,750		

(W2) Production overheads

	\$
Annual overheads (150,000 × \$1)	150,000
Depreciation (120,000/5)	(24,000)
	126,000
Monthly cash outflow (126,000/12)	10,500

(W3) Selling and administration overheads

Annual overheads	\$
	208,000
Depreciation (15000×0.3)	(4,500)
	203,500
Less: Rent and rates in month 1	25,000
Monthly cash outflow – months 2, 3, 4 ($178500/12$)	178,500
	14,875
Month 1: $25,000 + 14,875$	39,875

(b) Closing balances:

Inventory:	\$
Finished goods ($12,500 \times \$5$)	62,500
Raw materials	16,250
	78,750
Receivables:	
Month 4 credit sales ($96,000 \times 0.75$)	72,000
Less: Discount ($10\% \times 0.25 \times 96,000$)	(2,400)
	69,600
Payables	31,875

Apart from the purchase of the business, which will require separate long-term finance, the cash flow forecast suggests that there will be sufficient cash inflows to meet the cash outflows on an ongoing basis. The current assets and receivables provide sufficient funds to cover the payables.


Answer to additional question – Statement of financial position
Zed Co – Statement of financial position at 30 June 20X4

	\$	\$
Non-current assets:		
Plant and machinery $[(192,000 + 8,000) \times 80\%]$		160,000
Current assets:		
Inventory $(16,000 \times 105\%)$	16,800	
Receivables $(80,000 \times 110\%)$	88,000	
Bank (balancing figure)	67,200	
		172,000
Total assets		332,000
Equity and liabilities		
Issued share capital		216,000
Retained profits $(34,000 + 42,000)$		76,000
		292,000
Current liabilities:		
Trade payables	10,000	
Dividend payable	30,000	
		40,000
Total equity and liabilities		332,000

The forecast is that the bank balance will increase by \$65,200 (i.e. \$67,200 – \$2,000). This can be reconciled as follows:

	\$	\$
Retained profit		42,000
Add: Depreciation (20% of (\$192,000 + \$8,000))		40,000
		82,000
Less: Non-current asset acquired		(8,000)
		74,000
Increase in inventory	800	
Increase in receivables	8,000	
		(8,800)
Increase in cash balance		65,200



Answer to additional question – Forecasts from working capital

Here we will assume that the gross profit percentage will remain unaltered in cash terms.

	\$m
Last year	
Sales	200
Cost of sales less depreciation	100
Operating cash flow	100
Gross profit percentage	50%
	\$m
This year	
Sales 110% × \$200m	220
Cost of sales 50% × \$220m	110
Operating cash flow	110

Next we calculate the working capital requirements (to the nearest \$m).

		\$m
Inventory	$80 \div 365 \times \$110m =$	24
Receivables	$75 \div 365 \times \$220m =$	45
Trade payables	$50 \div 365 \times \$110m =$	15

Now we assemble the information in the proforma given earlier.

Note		\$m
1	Operating cash flow	110
2	Interest	(5)
3	Tax	(22)
3	Dividend	(10)
4	Purchase of plant and machinery	(12)
5	Reduction in inventory (\$24m vs \$25m)	1
5	Increase in receivables (\$45m vs \$33m)	(12)
5	Reduction in trade payables (\$15m vs \$20m)	(5)
	Net cash flow	45

Notes:

- (1) We have already calculated operating cash flow so do not need to adjust for depreciation of \$15m.
- (2) It is assumed that this is the same as last period, as the loans have not changed.
- (3) The tax and dividend payables in last year's statement of financial position will be paid in the forthcoming period.
- (4) This was given in the question.
- (5) Increases in current assets are an outflow, reductions are an inflow. The reverse is the case for trade payables.

Test your understanding answers

**Test your understanding 1**

- (a) $Q \text{ (cash investment)} = \sqrt{\frac{2 \times 50 \times 10,000 \times 12}{0.05}} = \$15,492$
- (b) $\text{Number of transactions per year} = \frac{120,000}{15,492} = 7.75$
- (c) $\text{Annual transaction cost} = 7.75 \times \$50 = \$387$
- (d) $\text{Annual opportunity cost (holding cost)} = 5\% \times \frac{15,492}{2} = \387

**Test your understanding 2 – Baumol**

The optimum amount of treasury bills sold, Q, for each cash injection into the current account will be:

$$Q = \sqrt{(2 \times 200,000 \times 15 \div 0.06)} = \$10,000$$

The total number of transactions will be

$$200,000 \div 10,000 = 20$$

and the total transaction cost will be $20 \times \$15 = \300 .

**Test your understanding 3 – Miller Orr**

- (i) $\text{Spread} = 3 \left(\frac{3}{4} \times 50 \times 9,000,000 / 0.0003 \right)^{1/3} = \$31,201$
- (ii) $\text{Upper limit} = 20,000 + 31,201 = \$51,201$
- (iii) $\text{Return point} = 20,000 + 31,201/3 = \$30,400$

**Test your understanding 4 – Miller Orr**

The spread is calculated as:

$$3 \times \left[\left(\frac{3}{4} \times 15 \times 1.44\text{m} \right) \div 0.0002 \right]^{1/3} = \$12,980$$

Therefore:

- lower limit (set by the company) = \$5,000
- upper limit = $5,000 + 12,980 = \$17,980$
- return point = $5,000 + (1/3 \times 12,980) = \$9,327$.


Test your understanding 5
Answer D

Revaluation of a non-current asset does not represent a cash flow.


Test your understanding 6
Answer B

January sales are $\$350,000 \times 12\% = \$42,000$.

Sales in each of the other months are $(\$350,000 - \$42,000)/11 = \$28,000$.

March cash collections will be:

50% of March sales = $\$28,000 \times 50\% = \$14,000$

45% of February sales = $\$28,000 \times 45\% = \$12,600$

Total = $\$14,000 + \$12,600 = \$26,600$

If you incorrectly calculated the cash collections in March as \$28,000 then you probably calculated the collections using 50% on both March and February sales.

If you incorrectly calculated the cash collection in March as \$32,900 then you probably used January sales of $\$42,000 \times 45\%$ and February sales of $\$28,000 \times 50\%$ in error.


Test your understanding 7
Answer C

Arranging short-term finance in the form of an overdraft will alleviate the short-term cash deficit problem.


Test your understanding 8
Answer B

One third.



Test your understanding 9

Answer D

A conservative working capital investment policy aims to reduce the risk of system breakdown by holding high levels of working capital.

An aggressive working capital investment policy aims to reduce financing costs and increase profitability by reducing inventories, speeding up collections from customers and delaying payments to suppliers.

The statements regarding the use of short-term or long-term finance related to working capital financing policies rather than investment policies.

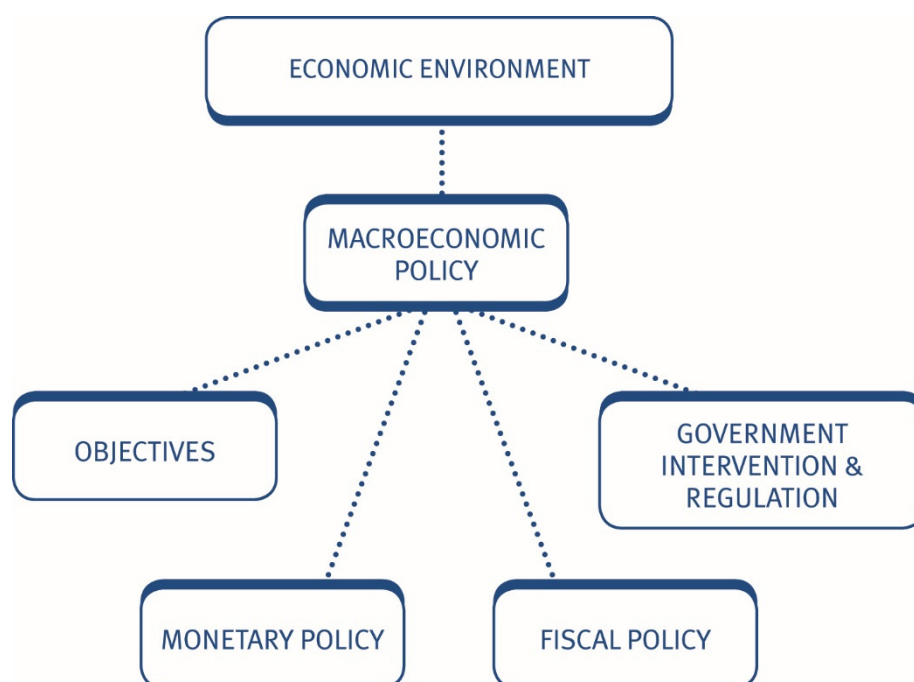
The economic environment for business

Chapter learning objectives

Upon completion of this chapter, you will be able to:

- explain the main objectives of macroeconomic policy
- explain the potential conflict between the main objectives of macroeconomic policy and its impact on policy targets
- explain the impact of general macroeconomic policy on planning and decision making in the business sector
- define monetary policy and explain the main tools used
- discuss the general role of monetary policy in the achievement of macroeconomic policy targets
- discuss use of interest rates in the achievement of macroeconomic policy targets
- define exchange rate policy and discuss its role in the achievement of macroeconomic policy targets
- explain the impact of specific economic policies on planning and decision making in the business sector
- define fiscal policy and explain the main tools used
- discuss the role of fiscal policy in the achievement of macroeconomic policy targets.
- explain the need for competition policy and its interaction with business planning and decision-making

- explain the need for government assistance for business and its interaction with business planning and decision-making
- explain the need for green policies and consideration of sustainability issues and their interaction with business planning and decision-making
- explain the need for corporate governance regulation and its interaction with business planning and decision-making



1 Macroeconomic policy

The objectives of macroeconomic policy

Macroeconomics is the branch of economics that deals with large-scale or general economic factors. It represents economics at a national and even global level.

Macroeconomic policy is the management of the economy by government in such a way as to influence the performance and behaviour of the economy as a whole.

The principal objectives of macroeconomic policy will be to achieve the following:

- full employment of resources
- price stability (low inflation)
- economic growth
- balance of payments equilibrium
- an appropriate distribution of income and wealth.



The objectives of macroeconomic policy

The full employment of resources applies in particular to the labour force. The aim is both full and stable employment.

Price stability means little or no inflation putting upward pressure on prices.

Economic growth is measured by changes in national income from one year to the next and is important for improving living standards.

The balance of payments relates to the ratio of imports to exports. A payment surplus would mean the value of exports exceeds that of imports. A payment deficit would occur where imports exceed exports.

What is considered an appropriate distribution of income and wealth will depend upon the prevailing political view at the time.

Potential for conflict

The pursuit of macroeconomic objectives may involve trade-offs – where one objective has to be sacrificed for the sake of another, e.g.:

Full employment

versus

Price stability

Economic growth

versus

Balance of payments



Potential for conflict

Both economic theory and the experience of managing the economy suggest that the simultaneous achievement of all macroeconomic objectives may be extremely difficult. Two examples of possible conflict may be cited here.

There may be conflict between full employment and price stability. It is suggested that inflation and employment are inversely related. The achievement of full employment may therefore lead to excessive inflation through an excess level of aggregate demand in the economy.

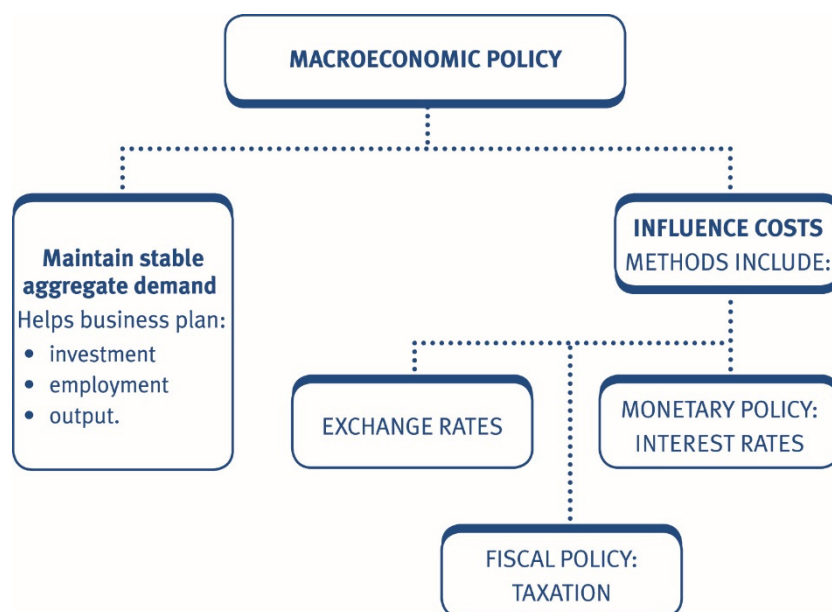
Rapid economic growth may, in the short-term at least, have damaging consequences for the balance of payments since rapidly rising incomes may lead to a rising level of imports.

Government reputation and business confidence will both be damaged if the government is seen to be pursuing policy targets that are widely regarded as incompatible.

Policy objectives may conflict and hence governments have to consider trade-offs between objectives. The identification of targets for policy should reflect this.

Making an impact – How macroeconomic policy affects the business sector

In order for macroeconomic policy to work, its instruments must have an impact on economic activity. This means that it must affect the business sector. It does so in two broad forms:



Aggregate demand

Aggregate Demand (AD) is the total demand for goods and services in the economy.

Note: National income is AD that has been satisfied by the provision of goods and services, etc.

The broad thrust of macroeconomic policy is to influence the level of AD in the economy. This is because the level of AD is central to the determination of the level of unemployment and the rate of inflation. If AD is too low, unemployment might result; if AD is too high, inflation induced by excess demand might result.

Changes in AD will affect all businesses to varying degrees.

Thus effective business planning requires that businesses can:

- predict the likely thrust of macroeconomic policy in the short- to medium-term
- predict the consequences for sales growth of the overall stance of macroeconomic policy and any likely changes in it.

The more stable government policy is, the easier it is for businesses to plan, especially in terms of investment, employment and future output capacity.

Business costs

Macroeconomic policy may influence the costs of the business sector.

Not only will the demand for goods and services be affected by macroeconomic policy, it also has important implications for the costs and revenues of businesses. Three important areas may be identified:

Exchange rates

Macroeconomic policy may involve changes in exchange rates. This may have the effect of raising the domestic price of imported goods. Most businesses use some imported goods in the production process; hence this leads to a rise in production costs.

Taxation

Fiscal policy involves the use of taxation: changes in tax rates or the structure of taxation will affect businesses, e.g. a change in the employers' national insurance contribution (NIC) will have a direct effect on labour costs for all businesses. Changes in indirect taxes (e.g. a rise in sales tax or excise duties) will either have to be absorbed or the business will have to attempt to pass on the tax to its customers.

Interest rates

Monetary policy involves changes in interest rates. These changes will directly affect firms in two ways:

- Costs of servicing debts will change, especially for highly-g geared firms.
- The viability of investment will be affected since all models of investment appraisal include the rate of interest as one, if not the main, variable.

2 Monetary policy



Monetary policy is concerned with influencing the overall monetary conditions in the economy in particular:

- directly controlling the volume of money in circulation – the money supply
- indirectly controlling the money supply through altering the price of money – interest rates.



Monetary policy

It is clear that money is crucial to the way in which a modern economy functions. **Money** is any financial asset that has liquidity and fulfils the task of a medium of exchange.

Monetary policy is concerned with influencing the overall monetary conditions in the economy.

Two particular problems with the use of monetary policy are:

- the choice of targets
- the effects of interest rate changes.

The choice of targets

A fundamental problem of monetary policy concerns the choice of variable to operate on. The ultimate objective of monetary policy is to influence some important variable in the economy – the level of demand, the rate of inflation, the exchange rate for the currency, etc. However, monetary policy has to do this by targeting some intermediate variable which, it is believed, influences, in some predictable way, the ultimate object of the policy.

The broad choice here is between targeting the **stock of money** or the **rate of interest**:

- The volume of money in circulation. The stock of money in the economy (the 'money supply') is believed to have important effects on the volume of expenditure in the economy. This in turn may influence the level of output in the economy or the level of prices.
- The **price** of money. The price of money is the rate of interest. If governments wish to influence the amount of money held in the economy or the demand for credit, they may attempt to influence the level of interest rates.

The monetary authorities may be able to control either the supply of money in the economy or the level of interest rates but cannot do both simultaneously. In practice, attempts by governments to control the economy by controlling the money supply have failed and have been abandoned. However, growth in the money supply is monitored, because excessive growth could be destabilising.

The effects of interest rates

The problem for the monetary authorities is that controlling the level of interest rates is rather easier than controlling the overall stock of money but the effects of doing so are less certain.

If governments choose to target interest rates as the principal means of conducting monetary policy, this may have a series of undesirable effects. These principally relate to the indiscriminate nature of interest rate changes and to the external consequences of monetary policy.

When interest rates are changed, it is expected that the general level of demand in the economy will be affected. Thus, a rise in interest rates will discourage expenditure, by raising the cost of credit. However, the effects will vary:

- (1) **Investment may be affected more than consumption.** The rate of interest is the main cost of investment by businesses, whether it is financed by internal funds or by debt. However, most consumption (by individuals) is not financed by credit and hence is less affected by interest rate changes. Since the level of investment in the economy is an important determinant of economic growth and international competitiveness there may be serious long-term implications arising from high interest rates.
- (2) Even where consumption is affected by rising interest rates, the **effects are uneven.** The demand for consumer durable goods and houses is most affected since these are normally credit-based purchases. Hence active interest policy may induce instability in some sectors of business.

The second problem arises from the openness of modern economies and their economic interdependence. There is now a very high degree of capital mobility between economies: large sums of short-term capital move from one financial centre to another in pursuit of higher interest rates. Changes in domestic interest rates relative to those in other financial centres will produce large inflows and outflows of short-term capital. Inflows of capital represent a demand for the domestic currency and hence push up the exchange rate. Outflows represent sales of the domestic currency and hence depress the exchange rate. This may bring about unacceptable movements in the exchange rate.

Direct control of the money supply

Governments may attempt to exert a more direct control over the money supply by engaging in the following:

- Open market operations. By buying or selling its own securities, a government can either push money into the economy or withdraw it from the economy.
- Amending reserve asset requirements for banks. The higher the level of cash that banks must hold in reserve, the less is available to lend to customers. In addition, banks can be required to hold special deposits with the central bank, which has the same effect.
- Direct control of the amounts that banks may lend.

Monetary policy in the UK

It is useful to look at the current monetary policy in the UK. Similar policies are pursued in the US and the Euro-zone countries.

In the UK, the central bank has been given responsibility by the government for controlling short-term interest rates. Short-term interest rates are controlled with a view to influencing the rate of inflation in the economy over the long-term. In broad terms, an increase in interest rates is likely to reduce demand in the economy and so lower inflationary pressures, whereas a reduction in interest rates should give a boost to spending in the economy, but could result in more inflation. The aim of economic policy is to find a suitable balance between economic growth and the risks from inflation.

Central governments can control short-term interest rates through their activities in the money markets. This is because the commercial banks need to borrow regularly from the central bank. The central bank lends to the commercial banks at a rate of its own choosing (a rate known in the UK as the repo rate). This borrowing rate for banks affects the interest rates that the banks set for their own customers. Action by a central bank to raise or lower interest rates normally results in an immediate increase or reduction in bank base rates.

In November 2017 after a decade of UK interest rates being at their lowest ever level, the Bank of England raised rates, partially to combat rising inflation at the time. The rate was raised again in 2018 as the initial rise had not brought inflation to an acceptable level. During the COVID 19 crisis in 2020, the bank reduced interest rates back to unprecedented levels with an aim to stimulate the economy.

During 2022, as a consequence of supply chain shortages arising from the COVID 19 pandemic and the war in Ukraine as well as rising demand levels once the pandemic started to ease, inflation levels rose sharply, described as a 'cost of living crisis'. To curb inflation, the Bank of England increased interest rates on a regular basis through the year.

These rises continued through until August 2023. In August 2024, once inflation had started to ease, rates were lowered slightly.

Interest rate smoothing

Interest rate smoothing is the policy of some central banks to move official interest rates in a sequence of relatively small steps in the same direction, rather than waiting until making a single larger change.

This is usually for the following reasons:

- economic (e.g. to avoid instability and the need for reversals in policy) and
- political (e.g. higher rates are broken to the electorate gently).

Impact of monetary policy on business decision making

Factors affected	Achieved by controlling supply	Achieved by increasing interest rates
Availability of finance	Credit restrictions ⇒ small businesses struggle to raise funds	
Cost of finance	Reduced supply pushes up the cost of funds ⇒ discourages expansion	Shareholders require higher returns ⇒ if not met, share price falls
Level of consumer demand	Too difficult to raise funds to spend	Saving becomes more attractive, spending less attractive
Exchange rates		High interest rates attracts foreign investment ⇒ increase in exchange rates: • exports dearer • imports cheaper.



All the above factors will also therefore influence inflation, which has a significant impact on business cash flows and profits. Inflation is the general increase in prices over time leading to a decrease in purchasing power and may be:

- demand-pull inflation – excess demand
- cost-push inflation – high production costs.

Both can have a negative impact on cash flows and profits.



Impact of monetary policy

Changes in monetary policy will influence the following factors.

The availability of finance. Credit restrictions achieved via the banking system or by direct legislation will reduce the availability of loans. This can make it difficult for small- or medium-sized new businesses to raise finance. The threat of such restrictions in the future will influence financial decisions by companies, making them more likely to seek long-term, stable finance for projects.

The cost of finance. Any restrictions on the stock of money, or restrictions on credit, will raise the cost of borrowing, making fewer investment projects worthwhile and discouraging expansion by companies. Also, any increase in the level of general interest rates will increase shareholders' required rates of return so unless companies can increase their return, share prices will fall as interest rates rise.

Thus, in times of 'tight' money and high interest rates, organisations are less likely to borrow money and will probably contract rather than expand operations.

The level of consumer demand. Periods of credit control and high interest rates reduce consumer demand. Individuals find it more difficult and more expensive to borrow to fund consumption, whilst saving becomes more attractive. The lowering of demand for goods and services is another reason for organisations to have to contract operations.

The level of exchange rates. Monetary policy which increases the level of domestic interest rates is likely to raise exchange rates as capital is attracted into the country. Very many organisations now deal with both suppliers and customers abroad and thus cannot ignore the effect of future exchange rate movements. Financial managers must consider methods of hedging exchange rate risk and the effect of changes in exchange rates on their positions as importers and exporters.

The level of inflation. Monetary policy is often used to control inflation. Rising price levels and uncertainty as to future rates of inflation make financial decisions more difficult and more important. As prices of different commodities change at different rates, the timing of purchase, sale, borrowing and repayment of debt becomes critical to the success of organisations and their projects. This is discussed further below:

Impact of inflation on business cash flows and profits

The real effects on the level of profits and the cash flow position of a business of a sustained rate of inflation depend on the form that inflation is taking and the nature of the markets in which the company is operating. One way of analysing inflation is to distinguish between demand-pull inflation and cost-push inflation.

Demand-pull inflation might occur when excess aggregate monetary demand in the economy and hence demand for particular goods and services enable companies to raise prices and expand profit margins.

Cost-push inflation will occur when there are increases in production costs independent of the state of demand, e.g. rising raw material costs or rising labour costs. The initial effect is to reduce profit margins and the extent to which these can be restored depends on the ability of companies to pass on cost increases as price increases for customers.

One would expect that the effect of cost-push inflation on company profits and cash flow would always be negative, but that with demand-pull inflation, profits and cash flow might be increased, at least in nominal terms and in the short run. In practice, however, even demand-pull inflation may have negative effects on profits and cash flow.

Demand-pull inflation may in any case work through cost. This is especially true if companies use pricing strategies in which prices are determined by cost plus some mark-up.

- Excess demand for goods leads companies to expand output.
- This leads to excess demand for factors of production, especially labour, so costs (e.g. wages) rise.
- Companies pass on the increased cost as higher prices.
- In most cases inflation will reduce profits and cash flow, especially in the long run.

3 Fiscal policy



Fiscal policy is the manipulation of the government budget in order to influence the level of aggregate demand and therefore the level of activity in the economy. It covers:

- government spending
- taxation
- government borrowing

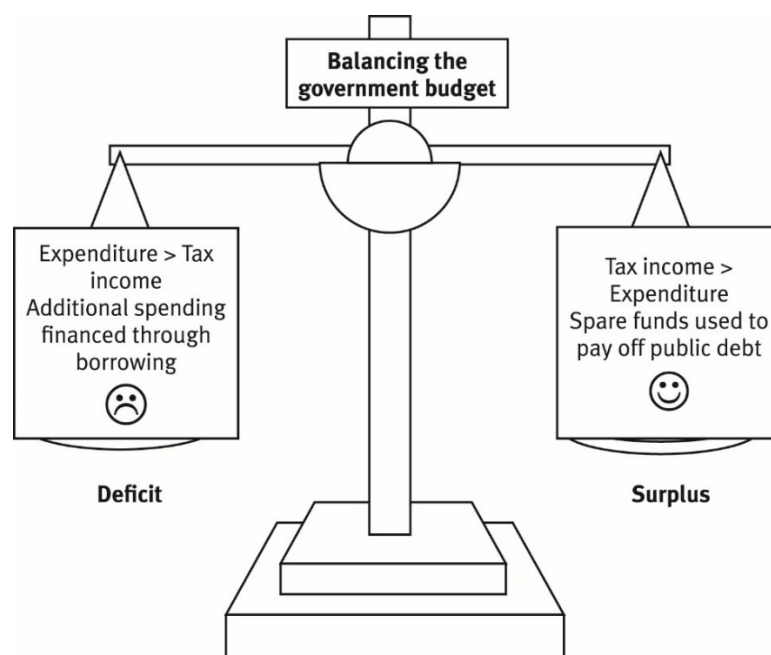
which are linked as follows:

public expenditure = taxes raised + government borrowing (+ sundry other income)

Those who view fiscal policy as crucial in the control of macroeconomic activity are known as Keynesians (after the economist John Maynard Keynes).



Fiscal policy



Balancing the budget

All governments engage in public expenditure, although levels vary somewhat from country to country. This expenditure must be financed either by taxation or by borrowing. Thus the existence of public expenditure itself raises issues of policy, notably how to tax and whom to tax. But, in addition, the process of expenditure and taxation permits the use of fiscal policy in a wider sense: the government budget can be manipulated to influence the level of AD in the economy and hence the level of economic activity.

The government budget (including central and local government) is a statement of public expenditure and income over a period of one year. Expenditure can be financed either by taxation or by borrowing. The relationship of expenditure to taxation indicates the state of the budget.

Three budget positions can be identified.

A **balanced budget**: total expenditure is matched by total taxation income.

A **deficit budget**: total expenditure exceeds total taxation income and the deficit must be financed by borrowing. In the UK the budget deficit was known as the public sector borrowing requirement (PSBR) but has been renamed to the (PSNCR) public sector net cash requirement.

A **surplus budget**: total expenditure is less than total taxation income and the surplus can be used to pay back public debt incurred as a result of previous deficits.

Taxation

The obvious means by which public expenditure can be financed is by taxation. The government receives some income from direct charges in the public sector (e.g. health prescription charges) and from trading profits of some public sector undertakings, but the bulk of its income comes from taxation.

Taxes are divided into broad groups.

Direct taxes are taxes levied directly on income receivers whether they are individuals or organisations. These include income tax, NICs, corporation tax, and inheritance tax.

Indirect taxes are levied on one set of individuals or organisations but may be partly or wholly passed on to others and are largely related to consumption rather than income. These include VAT and excise duties. By their very nature, indirect taxes tend to be regressive which means they have a relatively greater impact on individuals with lower incomes.

Impacts of excessive taxation

Taxation can raise very large flows of income for the government. However, excessive taxation may have undesirable economic consequences. Those most frequently cited are as follows:

- Personal disincentives to work and effort: this may be related mainly to the form of taxation, e.g. progressive income tax (earn more, pay more), rather than the overall level of taxation.
- Discouragement to business, especially the disincentive to invest and engage in research and development (R&D), which results from high business taxation.
- Disincentive to foreign investment: multinational firms may be dissuaded from investing in economies with high tax regimes.
- A reduction in tax revenue may occur if taxpayers are dissuaded from undertaking extra income-generating work and are encouraged to seek tax-avoidance schemes.

If the tax rate exceeds a certain level, the total tax revenue falls. It should be noted that these disincentive effects, while apparently clear in principle, are difficult to identify in the real world and hence their impact is uncertain.

Government borrowing

Broadly the government can undertake two types of borrowing:

It can borrow directly or indirectly **from the public** by issuing relatively illiquid debt. This includes National Savings certificates, premium bonds, and long-term government bonds. This is referred to as 'funding' the debt.

It can borrow **from the banking system** by issuing relatively liquid debt such as Treasury bills. This is referred to as 'unfunded' debt.

Long-term government bonds (gilts) are issued for long-term financing requirements, whereas Treasury bills are issued to fund short-term cash flow requirements.

Problems of fiscal policy

Two difficulties associated with fiscal policy have dominated debates about macroeconomic management in recent years:

- the problem of 'crowding out'
- the incentive effects of taxation.

Both have had a major impact on the way in which fiscal policy has been conducted, especially in the US and UK.

Crowding out

It is suggested that fiscal policy can lead to 'financial crowding out', whereby government borrowing leads to a fall in private investment. This occurs because increased borrowing leads to higher interest rates by creating a greater demand for money and loanable funds and hence a higher 'price'.

The private sector, which is sensitive to interest rates, will then reduce investment due to a lower rate of return. This is the investment that is crowded out. The weakening of fixed investment and other interest-sensitive expenditure counteracts the economy boosting benefits of government spending. More importantly, a fall in fixed investment by business can hurt long-term economic growth.

Doubts exist over the likely size of any crowding out effect of government borrowing on other borrowers but a very large PSNCR may well lead to a fall in private investment.

However, when the economy is depressed, and there is not much new private sector investment, government spending programmes could help to give a boost to the economy.

Incentives

It is likely that all taxes have some effect. Indeed, the structure of taxes is designed to influence particular economic activities: in particular, taxes on spending are used to alter the pattern of consumption. Here are two examples.

- High excise duties on alcohol and tobacco products reflect social and health policy priorities.
- Policies to use excise duties to raise the real price of petrol over time are designed to discourage the use of private cars because of the environmental effects.

Thus, taxes as instruments of fiscal policy can fulfil a variety of useful functions.

However, there has been a growing concern among some economists that taxes have undesirable side effects on the economy, notably on incentives. As we have seen, it is argued that high taxes, especially when they are steeply progressive, act as a disincentive to work.

Moreover, some taxes have more specific effects. For example, employer national insurance (NI) payments raise the cost of labour and probably reduce employment.

Supply side policies

Supply side policies involve government action to increase the productivity and efficiency of an economy.

For instance, privatising government-run organisations increases their exposure to the market forces of supply and demand and competitiveness and encourages improved efficiency.

Supply side policies can be fiscal in nature. For instance, offering low tax rates to small companies encourages private enterprise.

Government spending on infrastructure improvements leads to increased investment by businesses.

4 Government intervention and regulation

As well as the general measures to impact business operations discussed above, governments can also take more specific measures to regulate business. Governments are motivated to take such actions in cases of market failure, where the market performs inefficiently, which results in negative consequences for the economy as a whole.

These measures will cover aspects such as:

- Competition policy
- Privatisation
- Provision of government assistance
- Green policies and sustainability issues
- Corporate governance guidelines



Competition policy

It is not the government's place to set prices for goods and services provided by businesses in the private sector. In a competitive marketplace, prices will be set within the industry according to demand and supply. All producers have to accept the prevailing market price, unless they are able to add distinguishing features to their products (e.g. branding, superior technology) that enable them to charge higher prices.

However, some markets are characterised by various degrees of market power, in particular monopolies, whereby producers become price **makers** rather than price **takers**.

(A **monopoly** is where a firm has a sufficient share of the market to enable it to restrict output and raise prices.)

The absence of competition results in disadvantages to the economy as a whole.

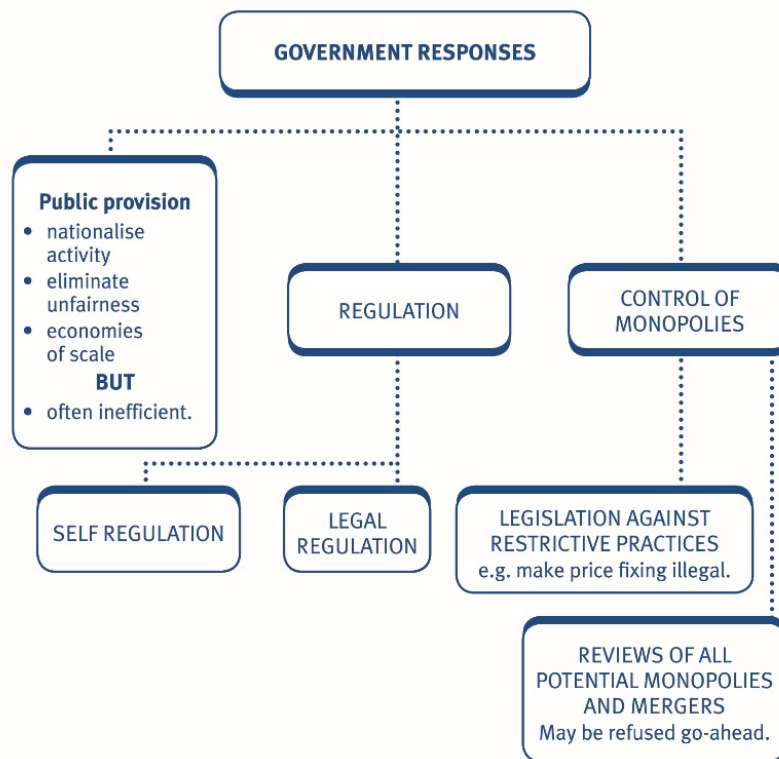
- Economic inefficiency: output is produced at a higher cost than necessary. For example, there may be no incentive to reduce costs by improving the technology used.
- Monopolies may be able to engage in price discrimination: charging different prices to different customers for the same goods or services, e.g. peak and off-peak pricing. This may act against the interests of customers.
- Disincentive to innovate: the absence of competition may reduce the incentive to develop new products or new production processes.
- Pricing practices: monopolies may adopt pricing practices to make it uneconomic for new firms to enter the industry, thus reducing competition in the long run.

These potential problems of companies with monopoly power must be considered in the light of some possible advantages that may be associated with such firms:

- Large firms may secure economies of scale: it is possible that there are significant economies of scale, reducing production costs, but that these require large firms and hence the number of firms in an industry is restricted. In this case, the benefits of economies of scale may offset the inefficiencies involved.
- The special case of natural monopolies: this is the case where the economies of scale in the provision of some basic infrastructure are so great that only one producer is feasible. This may be the case of the public utilities in energy and water.
- Research and development: it may be that monopoly profits are both the reward and the source of finance, which then enable technological and organisational innovation. Thus, some disadvantages have to be accepted in order to ensure a dynamic and innovative business sector.

Economic theory concludes that, all other things being equal, economic welfare is maximised when markets are competitive.

Government responses



Fair competition: public provision and regulation

The response to the problems associated with monopoly power can take a variety of forms. The first of these is public provision. This is where an economic activity is nationalised. The advantages of this are as follows:

- unfair pricing practices and/or excessive prices can be eliminated
- cost advantages of economies of scale can be reaped.

Traditionally, public utilities have presented the most convincing case for nationalisation because they are natural monopolies. However, nationalised industries may have disadvantages, notably a greater potential for cost-inefficiency.

The alternative response to the dangers of monopoly is regulation.

Self-regulation is common in many professions where the profession itself establishes codes of conduct and rules of behaviour (e.g. the Law Society, the British Medical Association).

The alternative is some form of public or **legal regulation**. An example in the UK is the establishment of regulatory bodies for the privatised utilities such as OFTEL and OFGAS. These were established in recognition of the monopoly power of the privatised utilities and have a degree of power over both prices and services in these industries.

Another UK regulatory body is the Financial Services Authority (FSA) that has responsibility for regulating the financial services, banking and insurance markets. The FSA does not have to control monopolies, but it is concerned with ensuring fair competition and protecting the public against unfair treatment by financial organisations.

Fair competition: the control of monopoly

Formal competition policy has typically centred upon two broad issues.

- monopolies and mergers
- restrictive practices.

The concern for monopolies is with firms which have a degree of monopoly power (defined as having more than 25% of the market) or with mergers which may produce a new company with more than 25% of market share. The underlying presumption is that monopolies are likely to be inefficient and may act against the interests of customers.

The concern over restrictive practices is with trading practices of firms that may be deemed to be uncompetitive and act against the interests of consumers.

Legislation typically prohibits:

- (a) anti-competitive agreements (such as price-fixing cartels); and
- (b) abuse of a dominant position in a market.



Privatisation

In an attempt to increase the productivity and efficiency of organisations operating in the economy, governments may sell off state-owned departments / organisations to the private sector. This also has the effect of reducing government spending (to keep state-owned organisations running) and increasing government revenue (in the form of taxation income from the private companies).

In the UK, state-owned organisations such as British Rail, British Gas and Royal Mail have been privatised.



Government assistance

The political and social objectives of a government, as well as its economic objectives, could be pursued through official aid intervention such as grants and subsidies. The government provides support to businesses both financially, in the form of grants, and through access to networks of expert advice and information.

For example, to:

- boost enterprise
- encourage innovation
- speed urban renewal
- revive flagging industries
- train labour force
- sponsor important research.

There is always strong competition for the grants and the criteria for awards are stringent. These vary but are likely to include the location, size and industry sector of the business.



Green policies and sustainability issues

When a firm appraises a project it may, rationally, only include those costs it will itself incur. However, for the good of society, external costs (such as damage to wildlife) need to be taken into account.

This has led to:

- green legislation
- punitive taxation on damaging practices

which force companies to consider the negative impacts of potential projects

Externalities are costs (benefits) which are not paid (received) by the producers or consumers of the product but by other members of society.

The need for green policies arises from the existence of external environmental costs associated with some forms of production or consumption. An example of each is given below:

- production: river pollution from various manufacturing processes
- consumption: motor vehicle emissions causing air pollution and health hazards.

If external costs (and benefits) exist in the production or consumption process and if they are large in relation to private costs and benefits, the price system may be a poor mechanism for the allocation of resources. This is because private producers and consumers ignore (and indeed may be unaware of) the external effects of their activities.

Thus, the price system, in which prices are determined by the interaction of supply and demand and which determines the allocation of resources, may lead to a misallocation of resources.

It is likely that policies to control damage to the environment will become more and more common as concern for the environment increases. Many countries now impose a charge on the use of plastic bags in order to discourage consumers from using and immediately disposing of them. There is also a continuing debate in the EU over the possible introduction of a wider 'carbon tax'.

Externalities lead to a **misallocation of resources** and imply the need for policies to correct this. In particular, **green policies** are needed to tackle externalities that affect the **environment**.

Sustainability

There is no one definition of sustainability, but the United Nations has described it as 'meeting the needs of the present without compromising the ability of future generations to meet their own needs'.

Having appropriate environmental policies in place forms a large part of meeting sustainability targets. Moves to encourage recycling of business' products and aims to become net zero carbon producers will tie into sustainability goals.

Sustainability also incorporates social and economic aspects. Governments may offer funding for business development in deprived areas to reduce economic and social inequalities. New skills training may be offered to those whose skills have become out of date (such as in communities built upon the success of mining operations that have now ceased) in order to improve standards of living. Without such assistance, communities may become more deprived or may break down as people leave them in search of work that fits their skill set.

Corporate governance



Corporate governance is defined as 'the system by which companies are directed and controlled' and covers issues such as ethics, risk management and stakeholder protection.

Regulation

Following the collapse of several large businesses due to directorial mismanagement and widespread concern about the standard of corporate governance across the business community, a new corporate governance framework was introduced.

A variety of rules have been introduced in different countries but the principles, common to all, contain regulations on:

- separation of the supervisory function and the management function
- transparency in the recruitment and remuneration of the board
- appointment of non-executive directors (NEDs)
- establishment of an audit committee and a remuneration committee

- establishment of risk control procedures to monitor strategic, business and operational activities.

Corporate governance was also discussed in the chapter on the financial management function.



Key issues in corporate governance

A further constraint on a company's activities is the system of corporate governance regulation that has been introduced to improve the standard of corporate governance in companies. The framework was introduced following a series of high profile international business collapses (research Enron and WorldCom if you want to know more). There was widespread belief that the traditional systems with the board of directors responsible for a company and the auditors appointed by the shareholders but remunerated by the directors, was insufficient to control modern companies.

Governance reports, codes and legislation vary from country to country but typically cover the following:

- Membership of the board to achieve a suitable balance of power. The chairperson and chief executive officer (CEO) should not be the same individual. There should be a sufficient number of non-executive directors (NEDs) on the board, and most of these should be independent.
- NEDs on the board should prevent the board from being dominated by the executive directors. The role of the NEDs is seen as critical in preventing a listed company from being run for the personal benefit of its senior executive directors.
- A remuneration committee to be established to decide on the remuneration of executive directors. Service contracts for directors should not normally exceed one year. There have also been efforts to give shareholders greater influence over directors' remuneration.
- The role of the audit committee of the board. This should consist of NEDs, and should work with the external auditors.
- The responsibility of the board of directors for monitoring all aspects of risk, not just the internal control system. For example, the Turnbull Committee was set up by the ICAEW to report on the risk management element of corporate governance.

The Turnbull report produced recommendations, and a risk report is now provided by listed companies in their annual report and accounts.

Audit committees

Many global listed companies now have an audit committee, or equivalent.

An audit committee:

- is a committee of the board of directors, normally three to five
- is made up of NEDs with no operating responsibility
- has a primary function to assist the board to fulfil its stewardship responsibilities by reviewing the:
 - systems of internal control
 - audit process
 - financial information which is provided to shareholders.

Note that while the topics in this chapter do not have a direct correlation to other ACCA exams, students should have an awareness of these issues when, for example, making strategic financial decisions as part of the ACCA Advanced Financial Management exam should they choose to take it.



Test your understanding 1

The policies pursued by a government may serve various objectives.

Which of the following is not one of the principle objectives of macroeconomic policy?

- A Economic growth
- B Price stability
- C Balance of payments surplus
- D Full employment



Test your understanding 2

Which of the following statements is true?

- Statement 1:* Monetary policy seeks to regulate the economy by influencing such variables as the level of interest rates and conditions for availability of credit.
- Statement 2:* Fiscal policy seeks to influence the economy by managing the amounts which the government spends and the amounts it collects through taxation.

	<i>Statement 1</i>	<i>Statement 2</i>
A	True	True
B	True	False
C	False	True
D	False	False



Test your understanding 3

Which of the following statements is correct?

- A A surplus budget occurs when total expenditure is less than total taxation income
- B Indirect taxes include income tax, corporation tax and inheritance tax
- C Direct taxes include VAT and excise duties
- D Cost-push inflation might occur when excess aggregate monetary demand enables companies to raise prices and expand profit margins



Test your understanding 4

Which of the following is not a commonly cited factor which can influence the level of exchange rates?

- A Comparative inflation rates
- B Comparative interest rates
- C Speculation
- D Government borrowing

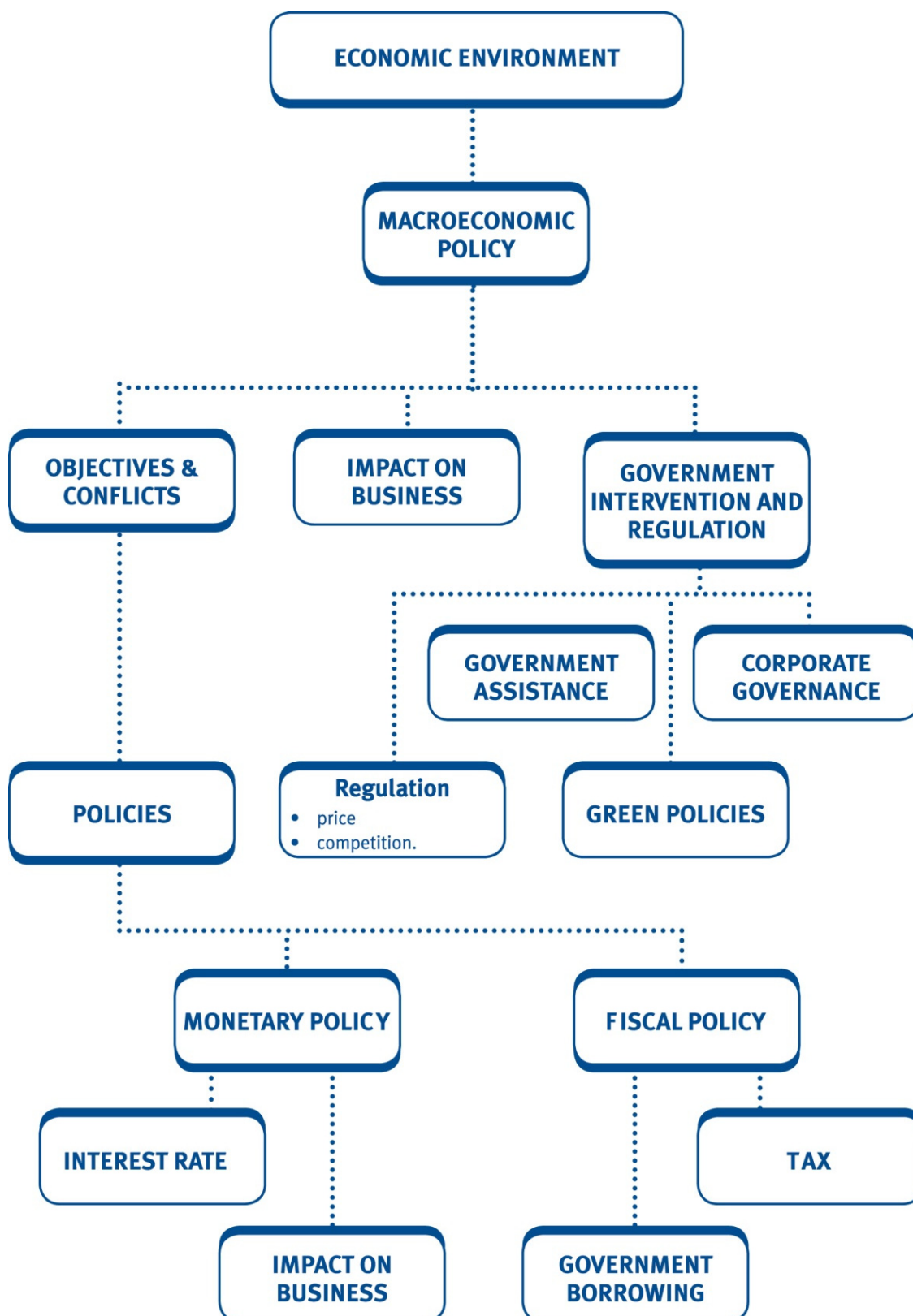


Test your understanding 5

What is the situation called where a firm has sufficient share of the market to enable it to restrict output and raise prices?

- A Duopoly
- B Oligopoly
- C Monopoly
- D Totopoly

5 Chapter summary



Test your understanding answers



Test your understanding 1

Answer C

One of the principle objectives of macroeconomic policy is to achieve a balance of payments equilibrium, not a surplus.

The wealth of a country relative to others, a country's creditworthiness as a borrower, and the goodwill between countries in international relations might all depend on the achievement of an external trade balance over time.



Test your understanding 2

Answer A

Monetary policy aims to influence monetary variables such as the rate of interest and the money supply in order to achieve the targets set.

Fiscal policy is action by the government to spend money or to collect money in taxes, with the purpose of influencing the condition of the national economy.



Test your understanding 3

Answer A

Direct taxes include income tax, corporation tax and inheritance tax (B). Indirect taxes include VAT and excise duties (C).

Demand-pull inflation might occur when excess aggregate monetary demand in the economy and hence demand for particular goods and services enables companies to raise prices and expand profit margins.



Test your understanding 4

Answer D

The exchange rate between two currencies is determined primarily by supply and demand in the foreign exchange markets. Supply and demand in turn are subject to a number of influences such as the rate of inflation, compared with the rate of inflation in other countries (A); interest rates, compared with interest rates in other countries (B); and speculation (C).



Test your understanding 5

Answer C

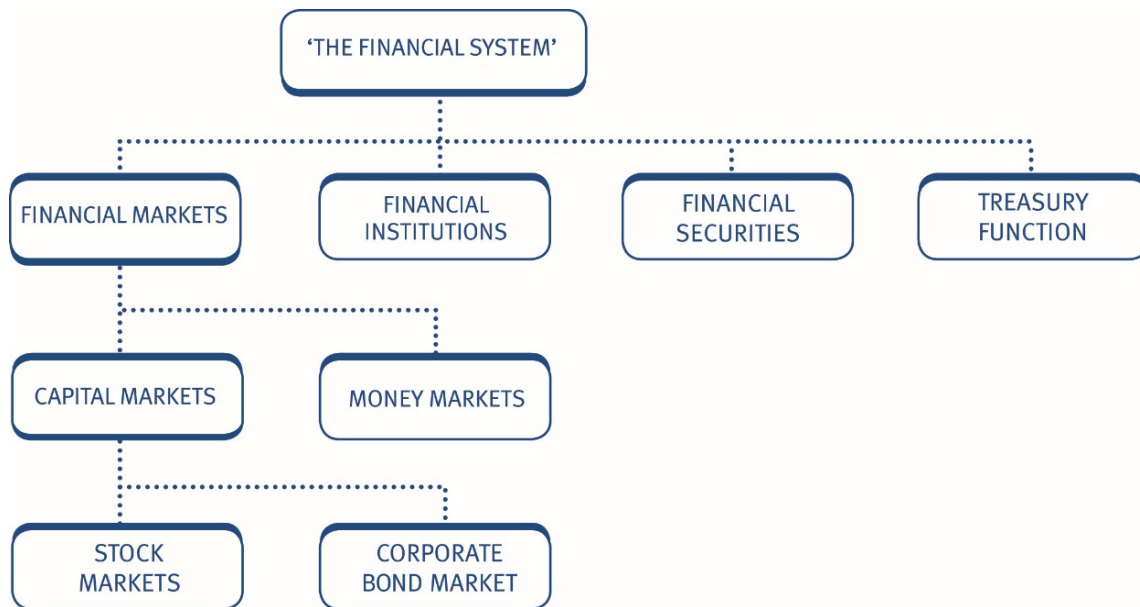
Financial markets and the treasury function

Chapter learning objectives

Upon completion of this chapter, you will be able to:

- identify the nature and role of money and capital markets, both nationally and internationally
- explain the role of financial intermediaries
- explain the main functions of a stock market and a corporate bond market
- explain the impact of Fintech in changing the nature and role of financial markets and institutions
- describe the role of the money markets in providing short-term liquidity to industry and the public sector
- describe the role of the money markets in providing short-term trade finance
- describe the role of the money markets in allowing an organisation to manage its exposure to foreign currency and interest rate risk
- explain the role of the banks and other financial institutions in the operation of the money markets
- explain and apply the characteristics and role of interest-bearing and discount instruments and their use in the money markets
- explain and apply the characteristics and role of derivatives and their use in the money markets
- discuss the advantages and disadvantages of centralised treasury management and cash control.

1 Introduction – The financial system



'The financial system' is an umbrella term covering the following:

- Financial markets – e.g. stock exchanges, money markets.
- Financial institutions – e.g. banks, building societies, insurance companies and pension funds.
- Financial securities – e.g. mortgages, bonds, bills and equity shares.

Collectively the financial system does the following:

- (1) Channels funds from lenders to borrowers.
- (2) Provides a mechanism for payments – e.g. direct debits, cheque clearing system.
- (3) Creates liquidity and money – e.g. banks create money through increasing their lending.
- (4) Provides financial services such as insurance and pensions.
- (5) Offers facilities to manage investment portfolios – e.g. to hedge risk.



The term **security** is a generic term for a medium of investment, for example, a share, a corporate bond or a money market instrument (see later for a definition of this). Many securities are traded on a market.

For every security there is a buyer (the investor who is providing the finance) and a seller (the receiver of the finance).

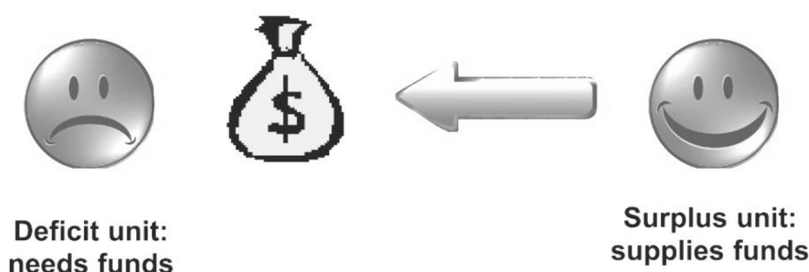
Some examples would be an individual (the buyer) investing in the shares (the security) of a listed company (the seller) or a bank (the buyer) investing in some treasury bills (the security) issued by the government (the seller).

2 The role of financial markets

Within each sector of the economy (households, firms and governmental organisations) there are times when there are cash surpluses and times when there are deficits.

- In the case of surpluses, the party concerned will seek to invest/deposit/lend funds to earn an economic return.
- In the case of deficits, the party will seek to borrow funds to manage their liquidity position.

The financial markets are mechanisms where those requiring finance (deficit units) can get in touch with those able to supply it (surplus units), i.e. allowing the buyers and sellers of finance to get together.



There are two main types of financial market: capital and money markets, and within each of these are primary and secondary markets:

Primary market – deals in new securities

Secondary market – deals in 'second-hand' securities



Details of financial markets

The financial markets can be divided into different types, depending on the products being issued/bought/sold:

- Capital markets which consist of stock-markets for shares and bond markets.
- Money markets, which provide short-term (< 1 year) debt financing and investment.
- Commodity markets, which facilitate the trading of commodities (e.g. oil, metals and agricultural produce).
- Derivatives markets, which provide instruments for the management of financial risk, such as options and futures contracts (see later for an explanation of derivatives).
- Insurance markets, which facilitate the redistribution of various risks.
- Foreign exchange markets, which facilitate the trading of foreign exchange.



Primary and secondary markets

Primary markets provide a focal point for borrowers and lenders to meet. The forces of supply and demand should ensure that funds find their way to their most productive usage. Primary markets deal in new issues of loanable funds. They raise new finance for the deficit units.

Secondary markets allow holders of financial claims (surplus units) to realise their investments before the maturity date by selling them to other investors. They therefore increase the willingness of surplus units to invest their funds. A well-developed secondary market should also reduce the price volatility of securities, as regular trading in 'second-hand' securities should ensure smoother price changes. This should further encourage investors to supply funds.

Secondary markets help investors achieve the following ends.

(a) **Diversification**

By giving investors the opportunity to invest in a wide range of enterprises, it allows them to spread their risk. This is the familiar 'don't put all your eggs in one basket' strategy.

(b) **Risk shifting**

Deficit units, particularly companies, issue various types of security on the financial markets to give investors a choice of the degree of risk they take. For example, company loan stocks secured on the assets of the business offer low risk with relatively low returns, whereas equities carry much higher risk with correspondingly higher returns.

(c) **Hedging**

Financial markets offer participants the opportunity to reduce risk through hedging, which involves taking out counterbalancing contracts to offset existing risks, e.g. if a UK exporter is awaiting payment in euros from a French customer they are subject to the risk that the euro may decline in value over the credit period. To hedge this risk, they could enter a counterbalancing contract and arrange to sell the euros forward (agree to exchange them for pounds at a fixed future date at a fixed exchange rate). In this way, they have used the foreign exchange market to insure their future sterling receipt. Similar hedging possibilities are available on interest rates (see Chapter 14).

(d) Arbitrage

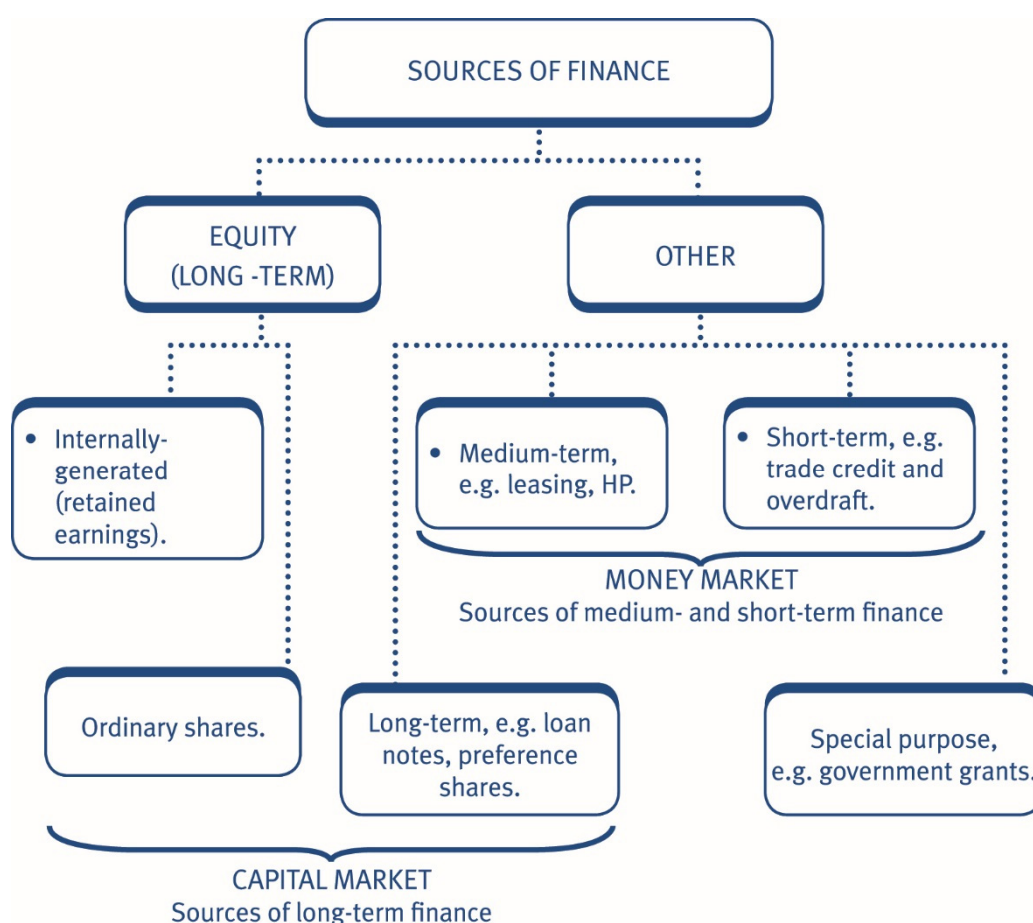
Arbitrage is the process of buying a security at a low price in one market and simultaneously selling in another market at a higher price to make a profit.

Although it is only the primary markets that raise new funds for deficit units, well-developed secondary markets are required to fulfil the above roles for lenders and borrowers. Without these opportunities more surplus units would be tempted to keep their funds 'under the bed' rather than putting them at the disposal of deficit units.

However, the emergence of disintermediation (reduction in the use of intermediaries) and securitisation (conversion into marketable securities), where companies lend and borrow funds directly between themselves, has provided a further means of dealing with cash flow surpluses and deficits.

Division of sources of finance

For the deficit units (those requiring finance), the source of finance can be divided between short- and long-term:



Durations of each are roughly:

Short-term up to one year.

Medium-term 1–7 years

Long-term 7 years or more

As you can see from the above diagram, which financial market the deficit unit will need will depend on the duration of the funds required.

Equally, for the surplus unit (the provider of finance/investor), which market they deal in will depend on how long they are willing to invest for.

The money markets

Money markets deal in short-term funds and transactions are conducted by phone or online. It is not one single market but a number of closely-connected markets.



All the markets in a money market closely inter-mesh with each other and in that way the market may be regarded as an entity. The players are the same and they pass the ball between each other.

However, since the global "credit crunch" of 2008, the liquidity in the money markets has reduced as the different players have begun to view each other with suspicion.



Illustration 1

- A large company might deposit \$500,000 with Bull's Bank, which issues it with a Certificate of Deposit (CD).
- Bull's Bank then looks at the local authority market, decides that rates there are rather low, and instead lends the money for a week on the inter-bank market to another bank that is short of funds.
- A week later local authority rates have improved and Bull's Bank lends the \$500,000 to a big city council.

- Meanwhile, the large company has decided to bring forward an investment project and wants its \$500,000 quickly to help pay for some sophisticated new electronic equipment. It sells the CD to a bank, which might either carry it to maturity or sell it to any of the banks – except Bull's Bank.

All these transactions, with the possible exception of the CD deals, will have taken place through a broker who sits at the end of a telephone switching the funds from one market to another as rates move and potential borrowers and lenders acquaint the broker with information about their requirements.

Money markets play a key role in:

- providing short-term liquidity to companies, banks and the public sector (see text below)
- providing short-term trade finance (see text below)
- allowing an organisation to manage its exposure to foreign currency risk and interest rate risk (see chapters on foreign exchange and interest rate risk)



Money markets – Providing short-term liquidity

Companies

- For many companies bank borrowing is the simplest method of short-term finance.
 - Loans are usually fixed term, may be for variable or fixed rates and are normally secured. Overdrafts are normally variable rate and unsecured.
 - Rates will depend on perceived credit risk.
- Factoring can be useful where sales are made to low risk customers.
- Companies with high credit ratings often take advantage of commercial paper markets, where they may be able to borrow at lower rates than those offered by banks.

Banks

- Banks participate in 'inter-bank markets', either lending to other banks, or borrowing from them.
- This is a very important source of funds to banks. Banks use various different measures instead of base rate to determine the interest payable on some types of company borrowing.

Examples of these rates are risk-free rates (RFRs), such as the Sterling Overnight Index Average rate (SONIA), ESTR (Euro short-term rate), or SOFR (secured overnight financing rate).

Local authorities

Local authorities obtain their borrowing requirements from:

- central government and money markets
- stock exchanges, and
- from advertising for loans from the general public.



Money markets – Providing short-term trade finance

As well as having routine cash surpluses /deficits that need managing, firms often need short-term finance for individual (usually large) trade deals. A supplier faces two problems in such cases:

- Default risk from the customer.
- The time delay between cash outflows to make the product and receipt of payment.

The most common way of managing these is by using banker's acceptances by creating a 'letter of credit'.



Letter of credit illustration

Step 1 Initial trade deal

- Company A agrees to buy goods from Company B for \$10 million (say), payable in 30 days' time.

Step 2 Create letter of credit (LOC)

- Co. A agrees payment terms with their bank (Bank A).
- Bank A issues Co. B's bank (Bank B) with a letter of credit, stating that it will pay the \$10m in 30 days.
- Bank B informs Company B of the receipt.

Step 3 Supply of goods

- Co. B then ships the goods to Co. A.
- Co. B passes shipping and other necessary documents to Bank B.

Step 4 Create banker's acceptance

- Bank B passes the letter of credit and supporting documentation back to Bank A.
- If all is in order, Bank A 'accepts' the letter of credit, accepting responsibility for payment. This creates the 'banker's acceptance', which is passed back to Bank B.

Step 5 Options for using the banker's acceptance.

Bank B could:

- wait until payment in 30 days
- request immediate payment (of the discounted value) from Bank A – effectively selling it back to Bank A
- sell the acceptance (at its discounted value) to the money markets.

Step 6 Banker's acceptance redeemed

- After 30 days Bank A redeems the acceptance, paying the current holder of the LOC.

The capital markets

Capital markets deal in longer-term finance, mainly via a stock exchange. The major types of securities dealt on capital markets are as follows:

- public sector and foreign stocks
- company securities (shares and corporate bonds)
- Eurobonds.



Eurobonds are bonds denominated in a currency other than that of the national currency of the issuing company (nothing to do with Europe or the Euro!). They are also called international bonds.

International capital markets

An international financial market exists where domestic funds are supplied to a foreign user or foreign funds are supplied to a domestic user. The currencies used need not be those of either the lender or the borrower.



International capital markets

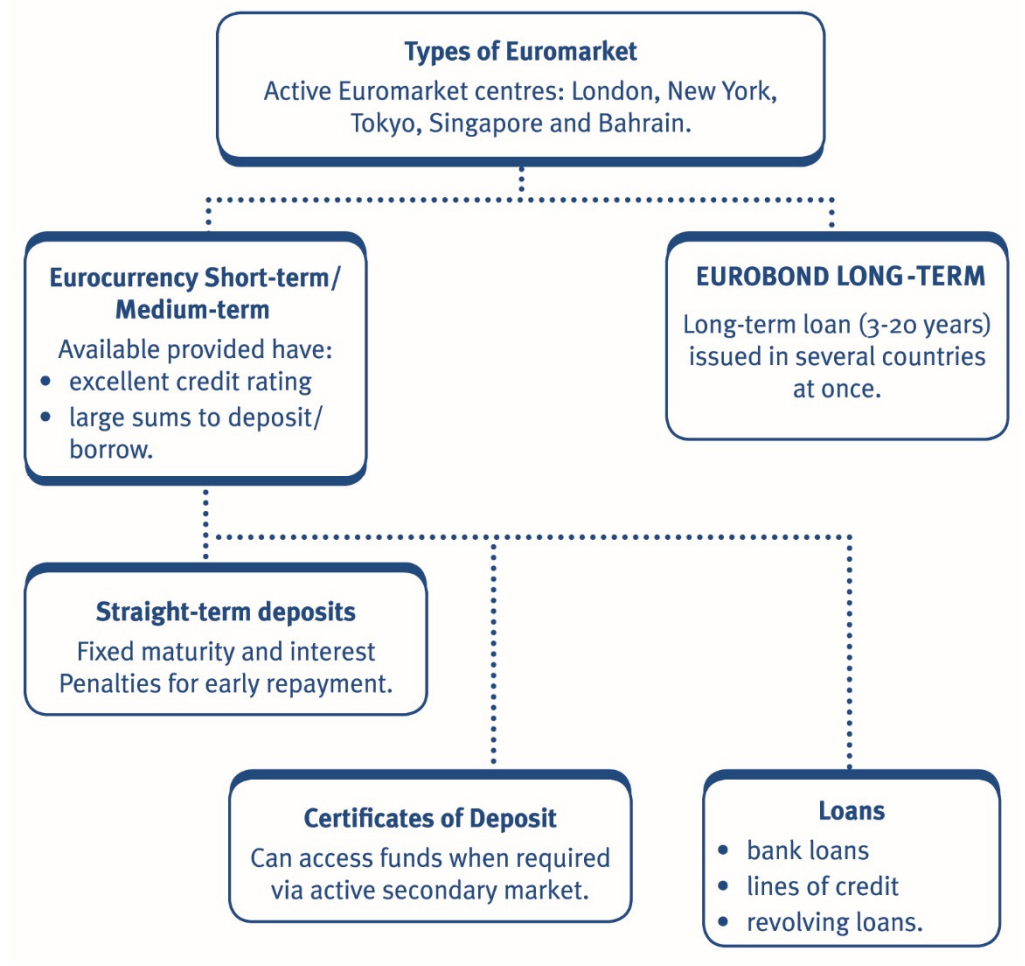
The most important international markets are:

- the Euromarkets
- the foreign bond markets.

Eurocurrency is money deposited with a bank outside its country of origin, e.g. money in a US dollar account with a bank in London is Eurodollars.

Note that these deposits need not be with European banks, although originally most of them were.

Once in receipt of these Eurodeposits, banks then lend them to other customers and a Euromarket in the currency is created.



The Eurocurrency market

This incorporates the short- to medium-term end of the Euromarket. It is a market for borrowing and lending Eurocurrencies. Various types of deposits and loans are available.

Deposits vary from overnight to five years. Deposits can be in the form of straight-term deposits, with funds placed in a bank for a fixed maturity at a fixed interest rate. However, these carry the problem of interest rate penalties if early repayment is required.

Alternatively, deposits can be made in the form of negotiable Certificates of Deposit (CDs). There is an active secondary market in CDs and investors are therefore able to have access to their funds when required. Deposits can be made in individual currencies or in the form of 'currency cocktails' to allow depositors to take a diversified currency position. One common cocktail is the Special Drawing Right, consisting of US dollars, yen, Euros and sterling.

Euromarket loans may be in the form of straight bank loans, lines of credit (similar to overdraft facilities) and revolving commitments (a series of short-term loans over a given period with regular interest rate reviews). Small loans may be arranged with individual banks, but larger ones are usually arranged through syndicates of banks.

Much of the business on the Eurocurrency market is interbank, but there are also a large number of governments, local authorities and multinational companies involved. Firms wishing to use the market must have excellent credit standing and wish to borrow (or deposit) large sums of money.

The Eurobond market

A Eurobond is a bond issued in more than one country simultaneously, usually through a syndicate of international banks, denominated in a currency other than the national currency of the issuer.

This represents the long-term end of the Euromarket.

The bonds can be privately placed through the banks or quoted on stock exchanges. They may run for periods of between three and 20 years, and can be fixed or floating rate. Convertible Eurobonds (similar to domestic convertible loan stocks) and Option Eurobonds (giving the holder the option to switch currencies for repayment and interest) are also used.

The major borrowers are large companies, international institutions like the World Bank, and the EC. The most common currencies are the US dollar, the Euro, the Swiss franc, and to a lesser extent sterling.

Stock markets and corporate bond markets



The syllabus does not require you to have a detailed knowledge of any specific country's stock market or securities exchanges. We may, however, make reference to the markets that operate in the UK for illustration purposes.

The role of the stock market is to:

- facilitate trade in stocks such as:
 - issued shares of public companies
 - corporate bonds
 - government bonds
 - local authority loans.
- allocate capital to industry
- determine a fair price for the assets traded.

Speculative trading on the market can assist by:

- smoothing price fluctuations
- ensuring shares are readily marketable.

Market-makers:

- maintain stocks of securities in a number of quoted companies
- continually quote prices for buying and prices for selling the securities (bid and offer prices)
- generate income by the profits they make from the difference (or 'spread') between the bid and offer prices.

Countries may have more than one stock market. In the UK, there are three:

- (1) London Stock Exchange – main market
- (2) Alternative Investment Market (AIM) – for smaller companies
- (3) Ofex (off exchange) – trade via specialist brokers.



Stock markets

A country's stock market is the institution that embodies many of the processes of the capital market. Essentially, it is the market for the issued securities of public companies, government bonds, loans issued by local authority and other publicly-owned institutions, and some overseas stocks. Without the ability to sell long-term securities easily, few people would be prepared to risk making their money available to business or public authorities.

A stock market assists the allocation of capital to industry; if the market thinks highly of a company, that company's shares will rise in value and it will be able to raise fresh capital through the new issue market at relatively low cost. On the other hand, less popular companies will have difficulty in raising new capital. Thus, successful firms are helped to grow and the unsuccessful will contract.

The role of speculation

Any consideration of a stock market has to face up to the problem of speculation, i.e. gambling. It is suggested that speculation can perform the following functions:

It smoothes price fluctuations. Speculators, to be successful, have to be a little ahead of the rest of the market.

Bulls believe that prices will rise and so will buy today in the expectation of selling later at a higher price.

Bears believe that prices will fall and so will sell today in the expectation of buying back later at a lower price.

The skilled speculator will be buying when others are still selling and selling when others are still buying. The speculator, therefore, removes the peaks and troughs of inevitable price fluctuations and so makes price changes less violent.

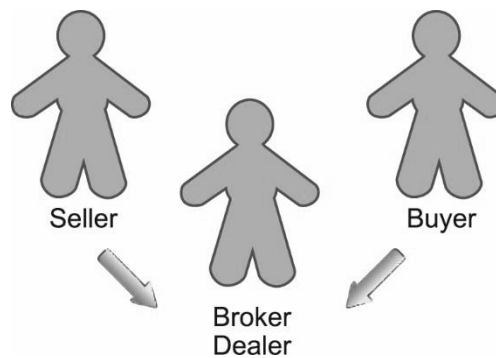
Speculation ensures that shares are readily marketable. Almost all stock can be quickly bought and sold, at a price. Without the chance of profit, there would be no professional operator willing to hold stock or agree to sell stock that is not immediately available. The fact that there are always buyers and sellers is of considerable importance to the ordinary individual investor, who may have to sell unexpectedly at any time with little warning.

Stock markets help in the determination of a fair price for assets, and ensure that assets are readily marketable.

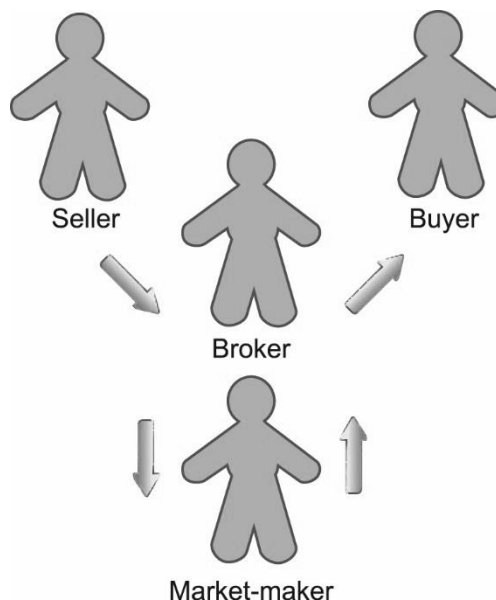
Buying and selling shares or loan notes

An investor will contact a broker in order to buy and sell securities. The broker may act as agent for the investor by contacting a market-maker (see below) or may act as principal if he or she makes a market in them (i.e. buys and sells on their own behalf). In the latter case, they are acting as a broker-dealer.

Trading via a broker-dealer



Trading via a broker and a market-maker



Market-makers maintain stocks of securities in a number of quoted companies, appropriate to the level of trading in that security, and their income is generated by the profits they make by dealing in securities. A market-maker undertakes to maintain an active market in shares that it trades, by continually quoting prices for buying and prices for selling (bid and offer prices).

If security prices didn't move, their profits would come from the difference (or 'spread') between the bid and offer prices. This profit is approximately represented by the difference between the 'bid' and 'offered' price for a given security – the price at which a market-maker is prepared to buy the stock and the price at which he or she would be prepared to sell it.

For example, assume a quotation in respect of the shares of a fictitious company, Clynch Co.

The dealer might quote 145, 150.

This means they will buy the shares at \$1.45 each and sell at \$1.50 each.

Quotations are based on expectations of the general marketability of the shares and, as such, will probably vary constantly. For example, if an investor wants to buy 10,000 shares, the dealers might decide to raise their prices to encourage people to sell and thereby ensure that they will have sufficient shares to meet the order they have just received. Thus, their next quote might be 150, 155.

Conversely, if the investor wants to sell 10,000 shares, the dealers will be left with that number of additional shares, and may wish to reduce their quotation to encourage people to buy; thus they may quote 140, 145.

If sufficient numbers of people wish to buy and sell the shares of Clynch Co, eventually a price will be found at which only marginal transactions are taking place.

If the general economic climate is reflected by each company's shares, it follows that there will be times when in general people wish to sell shares and hence prices drop, and other times when in general people are buying shares and prices rise.

Share prices are dictated by the laws of supply and demand. If the future return/risk profile of the share is anticipated to improve, the demand for that share will be greater and the price higher.

Broker-dealers in the UK act as both brokers for clients and trade on their own account as dealers. In the UK, their activities in share dealing are restricted largely to listed companies whose shares are traded on SETS, the electronic 'trading book' of the London Stock Exchange.

Types of stock market

A country may have more than one securities market in operation. For illustration, the UK has the following:

The London Stock Exchange – the main UK market for securities, on which the shares of large public companies are quoted and dealt. Costs of meeting entry requirements and reporting regulations are high.

The AIM – a separate market for the securities of smaller companies. Entry and reporting requirements are significantly less than those for the main market of the London Stock Exchange.

An 'Ofex' (off exchange) market in which shares in some public companies are traded, but through a specialist firm of brokers and not through a stock exchange.

3 The role of financial institutions

Faced with a desire to lend or borrow, there are three choices open to the end-users of the financial system:

- (1) Lenders and borrowers contact each other directly.

This is rare due to the high costs involved, the risks of default and the inherent inefficiencies of this approach.

- (2) Lenders and borrowers use an organised financial market.

For example, an individual may purchase corporate bonds from a recognised bond market. If this is a new issue of bonds by a company looking to raise funds, then the individual has effectively lent money to the company.

If the individual wishes to recover their funds before the redemption date on the bond, then they can sell the bond to another investor.

- (3) Lenders and borrowers use financial institutions as intermediaries.

In this case the lender obtains an asset which cannot usually be traded but only returned to the intermediary. Such assets could include a bank deposit account, pension fund rights, etc.

The borrower will typically have a loan provided by an intermediary.



Intermediation refers to the process whereby potential borrowers are brought together with potential lenders by a third party, the intermediary.

Financial intermediaries have a number of important roles.

- Risk reduction
- Aggregation
- Maturity transformation
- Liquidity
- Hedging
- Financial intermediation



Intermediaries – What they do

Risk reduction

By lending to a wide variety of individuals and businesses financial intermediaries reduce the risk of a single default resulting in total loss of assets.

Aggregation

By pooling many small deposits, financial intermediaries are able to make much larger advances than would be possible for most individuals.

Maturity transformation

Most borrowers wish to borrow in the long term whilst most savers are unwilling to lock up their money for the long term. Financial intermediaries, by developing a floating pool of deposits, are able to satisfy both the needs of lenders and borrowers.

Liquidity

Intermediaries provide liquidity through the processes of aggregation and maturity transformation, meaning that cash may be sourced / invested quickly.

Hedging

Organisations wishing to reduce their exposure to risk may turn to intermediaries to provide hedging products, such as forwards, futures, options, etc.

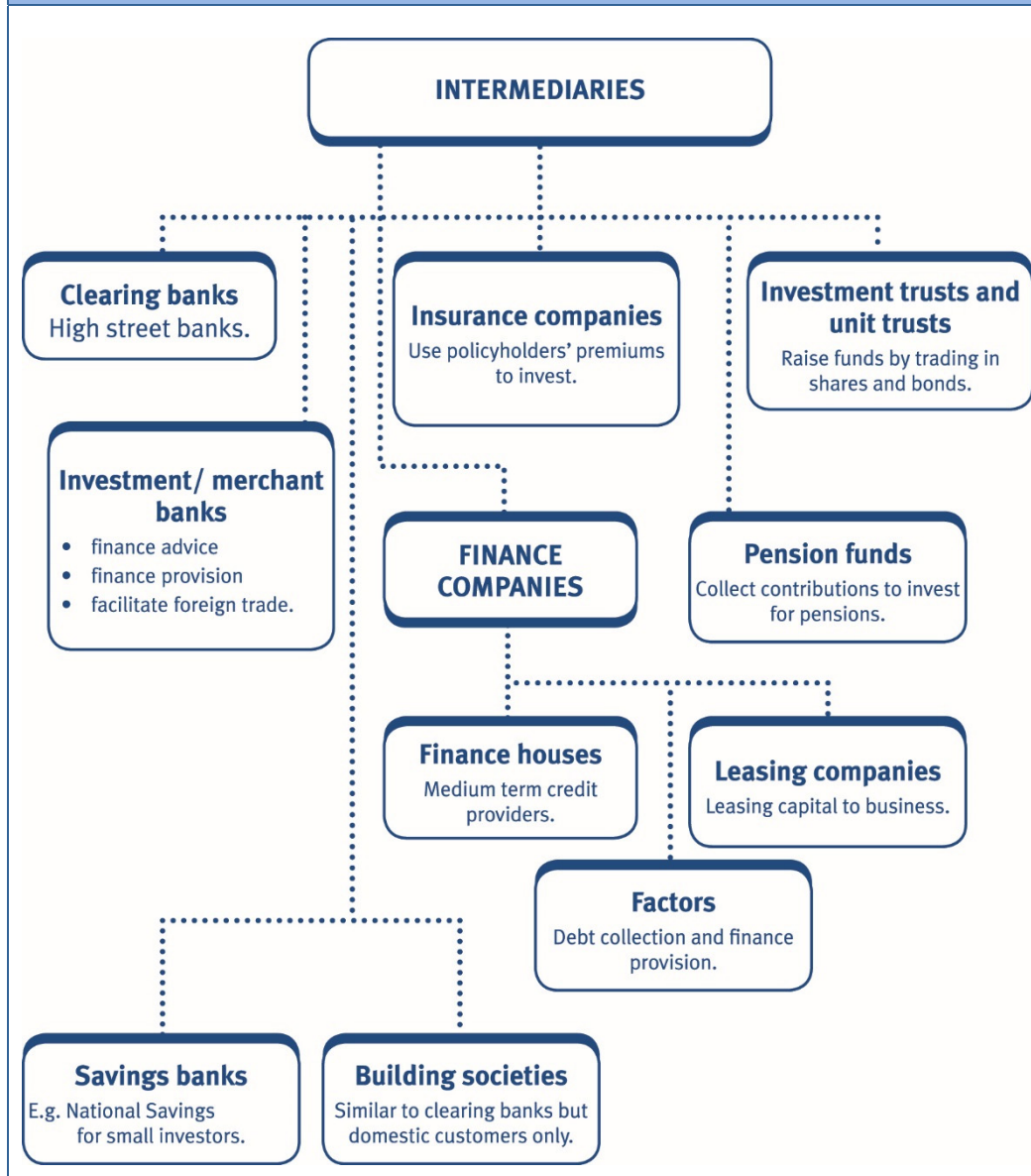
Financial intermediation

Financial intermediaries bring together lenders and borrowers through a process known as financial intermediation.

There are many types of institutions and other organisations that act as intermediaries in matching firms and individuals who need finance with those who wish to invest.



Intermediaries



Clearing banks

The familiar high street banks provide a payment and cheque clearing mechanism. They offer various accounts to investors and provide large amounts of short- to medium-term loans to the business sector and the personal sector. They also offer a wide range of financial services to their customers.

Investment banks

Investment banks, sometimes called merchant banks, concentrate on the following.

i) Financial advice to business firms

Few manufacturing or commercial companies of any size can now afford to be without the advice of a merchant bank. Such advice is necessary in order to obtain investment capital, to invest surplus funds, to guard against takeover, or to take over others.

Increasingly, the merchant banks have themselves become actively involved in the financial management of their business clients and have had an influence over the direction these affairs have taken.

ii) Providing finance to business

Merchant banks also compete in the services of leasing, factoring, hire-purchase (HP) and general lending. They are also the gateway to the capital market for long-term funds because they are likely to have specialised departments handling capital issues as 'issuing houses'.

iii) Foreign trade

A number of merchant banks are active in the promotion of foreign trade by providing marine insurance, credits, and assistance in appointing foreign agents and arranging foreign payments.

Not all merchant banks are large and not all offer a wide range of services: the term is now rather misused. However, it is expected that a merchant bank will operate without the large branch network necessary for a clearing bank. It will work closely with its business clients, and will be more ready to take business risks and promote business enterprise than a clearing bank. It is probably fair to say that a merchant bank is essentially in the general business of creating wealth and of helping those who show that they are capable of successful business enterprise.

Savings banks

Public sector savings banks (e.g. the National Savings Bank in the UK) are used to collect funds from the small personal saver, which are then mainly invested in government securities.